Lecture Notes in Mathematics

E. E. Odell Jr.
H. P. Rosenthal (Eds.)

Functional Analysis

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Lecture Notes aim to report new developments - quickly, informally and at a high level. The following describes criteria and procedures for multi-author volumes. For convenience we refer throughout to "proceedings" irrespective of whether the papers were presented at a meeting.

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Further remarks and relevant addresses at the back of this book.

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E. Odell H. Rosenthal (Eds.)

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Editors

Edward E. Odwell, jr.
Haskell P. Rosenthal
Department of Mathematics
The University of Texas at Austin
Austin, TX 78712, USA

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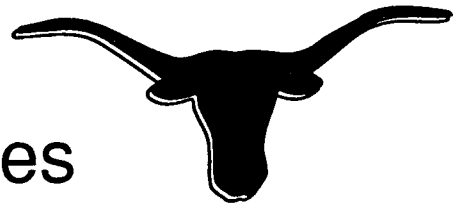
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Longhorn Notes



Preface

This is the sixth annual proceedings of our Functional Analysis Seminar at The University of Texas, and the second one to be published in the Springer-Verlag Lecture Notes. All the articles that appear are based on talks given in the seminar during the years 1987-89. We thank the participants for their effort in communicating mathematical ideas in both spoken and written form.

We again wish to express our deep appreciation to Margaret Combs for her expert craftsmanship and considerable patience in typesetting this issue. Thanks are also due to The University of Texas for supporting the publication of the Longhorn Notes.

Ted Odell
Haskell Rosenthal
August 1990

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On certain classes of Baire-1 functions with applications to Banach space theory

R. HAYDON, E. ODELL and H. ROSENTHAL

R. Haydon
Brasenose College
Oxford OX1 4AJ
England

E. Odell, H. Rosenthal
The University of Texas at Austin
Austin, Texas 78712
U.S.A.

0. Introduction.

Let X be a separable infinite dimensional Banach space and let K denote its dual ball, $Ba(X^*)$, with the weak* topology. K is compact metric and X may be naturally identified with a closed subspace of $C(K)$. X^{**} may also be identified with a closed subspace of $A_\infty(K)$, the Banach space of bounded affine functions on K in the sup norm. Our general objective is to deduce information about the isomorphic structure of X or its subspaces from the topological nature of the functions $F \in X^{**} \subseteq A_\infty(K)$. A classical example of this type of result is: X is reflexive if and only if $X^{**} \subset C(K)$.

A second example is the following theorem. ($B_1(K)$ is the class of bounded Baire-1 functions on K and $DBSC(K)$ is the subclass of differences of bounded semicontinuous functions on K. The precise definitions appear below in §1.) We write $Y \hookrightarrow X$ if Y is isomorphic to a subspace of X.

Theorem A. *Let X be a separable Banach space and let $K = Ba(X^*)$ with the weak* topology.*

a) [**35**] $\ell_1 \hookrightarrow X$ iff $X^{**} \setminus B_1(K) \neq \emptyset$.

b) [**7**] $c_0 \hookrightarrow X$ iff $[X^{**} \cap DBSC(K)] \setminus C(K) \neq \emptyset$.

Theorem A provides the motivation for this paper: What can be said about X if $X^{**} \cap [B_1(K) \setminus DBSC(K)] \neq \emptyset$? To study this problem we consider various subclasses of $B_1(K)$ for an arbitrary compact metric space K. J. Bourgain has also used this approach and some of our results and techniques overlap with those of [**8,9,10**]. In a different direction, generalizations of $B_1(K)$ to spaces where K is not compact metric with ensuing applications to Banach space theory have been developed in [**22**].

In §1 we consider two subclasses of $B_1(K)$ denoted $B_{1/4}(K)$ and $B_{1/2}(K)$ satisfying

$$C(K) \subseteq DBSC(K) \subseteq B_{1/4}(K) \subseteq B_{1/2}(K) \subseteq B_1(K)\ . \tag{0.1}$$

Research partially supported by NSF Grant DMS-8601752.

Our interest in these classes stems from Theorem B (which we prove in §3).

Theorem B. *Let K be a compact metric space and let (f_n) be a uniformly bounded sequence in $C(K)$ which converges pointwise to $F \in B_1(K)$.*

a) If $F \notin B_{1/2}(K)$, then (f_n) has a subsequence whose spreading model is equivalent to the unit vector basis of ℓ_1.

b) If $F \in B_{1/4}(K) \setminus C(K)$, there exists (g_n), a convex block subsequence of (f_n), whose spreading model is equivalent to the summing basis for c_0.

Theorem B may be regarded as a local version of Theorem A (see Corollary 3.10). In fact the proof is really a localization of the proof of Theorem A. In Theorem 3.7 we show that the converse to a) holds and thus we obtain a characterization of $B_1(K) \setminus B_{1/2}(K)$ in terms of ℓ_1 spreading models. We do not know if the condition in b) characterizes $B_{1/4}(K)$ (see Problem 8.1).

Given that our main objective is to deduce information about the subspaces of X from the nature of $F \in X^{**} \cap B_1(K)$, it is useful to introduce the following definition.

Let $\mathcal{C}$ be a class of separable infinite-dimensional Banach spaces and let $F \in B_1(K)$. F is said to *govern* $\mathcal{C}$ if whenever $(f_n) \subseteq C(K)$ is a uniformly bounded sequence converging pointwise to F, then there exists a $Y \in \mathcal{C}$ which embeds into $[(f_n)]$, the closed linear span of (f_n). We also say that F *strictly governs* $\mathcal{C}$ if whenever $(f_n) \subseteq C(K)$ is a uniformly bounded sequence converging pointwise to F, there exists a convex block subsequence (g_n) of (f_n) and a $Y \in \mathcal{C}$ with $[(g_n)]$ isomorphic to Y.

Theorem A (b) can be more precisely formulated as: if $F \in DBSC(K) \setminus C(K)$, then F governs $\{c_0\}$. (In fact Corollary 3.5 below yields that $F \in B_1(K) \setminus C(K)$ strictly governs $\{c_0\}$ if and only if $F \in DBSC(K)$.) In §4 we prove that the same result holds if $F \in DSC(K) \setminus C(K)$. (A more general result, with a different proof, has been obtained by Elton [**13**].) We also note in §4 that there are functions that govern $\{c_0\}$ but are not in $DSC(K)$.

In §6 we give a characterization of $B_{1/4}(K)$ (Theorem 6.1) and use it to give an example of an $F \in B_{1/4}(K) \setminus C(K)$ which does not govern $\{c_0\}$. Thus Theorem B (b) is best possible.

In §7 we note that there exists a K and an $F \in B_{1/2}(K)$ which governs $\{\ell_1\}$. We also give an example of an $F \in B_{1/2}(K)$ which governs $\mathcal{C} = \{X : X \text{ is separable and } X^* \text{ is nonseparable}\}$ but does not govern $\{\ell_1\}$.

§1 contains the definitions of the classes $DBSC(K)$, $DSC(K)$, $B_{1/2}(K)$ and $B_{1/4}(K)$. At the end of §1 we briefly recall the notion of spreading model. In §2 we recall some ordinal indices which are used to study $B_1(K)$. A detailed study of such indices can be found in [**25**]. Our use of these indices and many of the results of this paper have been motivated by [**8,9,10**].

Proposition 2.3 precisely characterizes $B_{1/2}(K)$ in terms of our index.

In §5 we show that the inclusions in (0.1) are, in general, proper. We first deduce this from a Banach space perspective. Subsequently, we consider the case where K is countable. Proposition 5.3 specifies precisely how large K must be in order for each separate inclusion in (0.1) to be proper.

In §8 we summarize some problems raised throughout this paper and raise some new questions regarding $B_{1/4}(K)$.

We are hopeful that our approach will shed some light on the central problem: if X is infinite dimensional, does X contain an infinite dimensional reflexive subspace or an isomorph of c_0 or ℓ_1? A different attack has been mounted on this problem in the last few years by Ghoussoub and Maurey. The interested reader should also consult their papers (*e.g.*, [**18,19,20,21**]). Another fruitful approach has been via the theory of types ([**26**], [**24**], [**38**]). We wish to thank S. Dilworth and R. Neidinger for useful suggestions.

1. Definitions.

In this section we give the basic definitions of the Baire-1 subclasses in which we are interested. Let K be a compact metric space. $B_1(K)$ shall denote the class of bounded Baire-1 functions on K, *i.e.*, the pointwise limits of (uniformly bounded) pointwise converging sequences $(f_n) \subseteq C(K)$. $DBSC(K) = \{F : K \to \mathbb{R} \mid$ there exists $(f_n)_{n=0}^{\infty} \subseteq C(K)$ and $C < \infty$ such that $f_0 \equiv 0$, (f_n) converges pointwise to F and

$$\sum_{n=0}^{\infty} |f_{n+1}(k) - f_n(k)| \leq C \quad \text{for all} \quad k \in K\Big\} \,. \tag{1.1}$$

If $F \in DBSC(K)$ we set $|F|_D = \inf\{C \mid$ there exists $(f_n)_{n=0}^{\infty} \subseteq C(K)$ converging pointwise to F satisfying (1.1) with $f_0 \equiv 0\}$. $DBSC(K)$ is thus precisely those F's which are the "difference of bounded semicontinuous functions on K." Indeed if (f_n) satisfies (1.1), then $F = F_1 - F_2$ where $F_1(k) = \sum_{n=0}^{\infty}(f_{n+1} - f_n)^+(k)$ and $F_2(k) = \sum_{n=0}^{\infty}(f_{n+1} - f_n)^-(k)$ are both (lower) semicontinuous. The converse is equally trivial.

It is easy to prove that $(DBSC(K), |\cdot|_D)$ is a Banach space by using the series criterion for completeness. The fact that $\|F\|_\infty \leq |F|_D$ but the two norms are in general not equivalent on $DBSC(K)$, leads naturally to the following two definitions.

$$\begin{aligned} B_{1/2}(K) = \big\{F \in B_1(K) \mid & \text{ there exists a sequence} \\ & (F_n) \subseteq DBSC(K) \text{ converging uniformly to } F\big\} \text{ and} \\ B_{1/4}(K) = \big\{F \in B_1(K) \mid & \text{ there exists } (F_n) \\ & \text{converging uniformly to } F \text{ with } \sup_n |F_n|_D < \infty\big\} \,. \end{aligned}$$

It can be shown that $DBSC(K)$ is a Banach algebra under pointwise multiplication, and hence $B_{1/2}(K)$ can be identified with $C(\Omega)$, where Ω is the "structure space" or "maximal ideal space" of Ω. Thus $B_{1/4}(K)$ also has a natural interpretation in the general context of commutative Banach algebras.

There is a natural norm on $B_{1/4}(K)$ given by

$$|F|_{1/4} = \inf\{C : \text{ there exists } (F_n) \text{ converging uniformly with } \sup_n |F_n|_D \leq C\}$$

Furthermore $(B_{1/4}(K), |\cdot|_{1/4})$ is a Banach space. One way to see this is to use the following elementary

Lemma 1.1. *Let (M, d_1) be a complete metric space and let d_2 be a metric on M with $d_1(x,y) \leq d_2(x,y)$ for all $x, y \in M$. If all d_2-closed balls in M are also d_1-closed, then (M, d_2) is complete.*

The hypotheses of the lemma apply to $M = \{F : |F|_{1/4} \leq 1\}$ and d_1, d_2 given, respectively, by $\|\cdot\|_\infty$ and $|\cdot|_{1/4}$.

Remark 1.2. While we shall confine our attention to $B_{1/2}$ and $B_{1/4}$, one could of course continue the game, defining

$$\begin{aligned} B_{1/8}(K) = \{F \in B_1(K) \mid & \text{ there exists } (F_n) \subseteq DBSC(K) \\ & \text{with } |F_n - F|_{1/4} \to 0\} \text{ and} \\ B_{1/16}(K) = \{F \in B_1(K) \mid & \text{ there exists } F_n \\ & \text{with } \sup_n |F_n|_D < \infty \text{ and } |F_n - F|_{1/4} \to 0\} . \end{aligned}$$

This could be continued obtaining

$$DBSC(K) \subseteq \cdots \subseteq B_{1/2^{2n}}(K) \subseteq B_{1/2^{2n-1}}(K) \subseteq \cdots \subseteq B_{1/2}(K)$$

with $B_{1/2^{2n}}(K)$ having a norm $|\cdot|_{1/2^{2n}}$ which, using Lemma 1.1, is easily seen to be complete.

There is another class of Baire-1 functions that shall interest us, the differences of (not necessarily bounded) semi-continuous functions on K.

$$\begin{aligned} DSC(K) = \{F : K \to \mathbb{R} \mid & \text{ there exists a uniformly bounded sequence} \\ & (f_n)_{n=0}^\infty \subseteq C(K) \text{ converging pointwise to } F \text{ with} \\ & \sum_{n=0}^{\infty} |f_{n+1}(k) - f(k)| < \infty \text{ for } k \in K\} . \end{aligned}$$

An interesting subclass of $DSC(K)$, is $PS(K)$, the pointwise limits of *pointwise stabilizing* (pointwise ultimately constant) sequences.

$$PS(K) = \{F \in B_1(K) \mid \text{ there exists a uniformly bounded sequence } (f_n) \subseteq C(K) \text{ with the property that for all } k \in K \text{ there exists } m \in \mathbb{N} \text{ such that } f_n(k) = F(k) \text{ for } n \geq m\} .$$

Remark 1.3. We discuss $PS(K)$ in Proposition 4.9. Both of these classes were considered in [**10**], and as noted there, if an indicator function $\mathbf{1}_A \in B_1(K)$, then $\mathbf{1}_A \in PS(K)$. Indeed A must be both F_σ and G_δ (cf. Proposition 2.1 below) and so we can write $A = \bigcup_n F_n = \bigcap_n G_n$ where $F_1 \subseteq F_2 \subseteq \cdots$ are closed sets and $G_1 \supseteq G_2 \supseteq \cdots$ are open sets. Then by the Tietze extension theorem, for each n choose $f_n \in Ba(C(K))$ with f_n identically 1 on F_n and identically 0 on $K \setminus G_n$. Thus for all $k \in K$, $(f_n(k))_n$ is ultimately $\mathbf{1}_A(k)$.

The *summing basis* (s_n) for (an isomorph of) c_0 is characterized by

$$\|\sum a_n s_n\| = \sup_k |\sum_{i=1}^{k} a_i| .$$

Let (x_n) be a seminormalized basic sequence. A basic sequence (e_n) is said to be a *spreading model* of (x_n) if for all $k \in \mathbb{N}$ and all $\varepsilon > 0$ there exist N so that if $N < n_1 < n_2 < \cdots < n_k$ and $(a_i)_1^k \subseteq \mathbb{R}$ with $\sup_i |a_i| \leq 1$, then

$$\left| \|\sum_{i=1}^{k} a_i x_{n_i}\| - \|\sum_{i=1}^{k} a_i e_i\| \right| < \varepsilon .$$

For further information on spreading models see [**4**].

We recall that if $(f_n) \subseteq Ba(C(K))$ converges pointwise to $F \in B_1(K) \setminus C(K)$ then there exists a $C = C(F)$ such that (f_n) has a basic subsequence (f_n') with basis constant C which C-dominates (s_n). Thus $C\|\sum a_n f_n'\| \geq \|\sum a_n s_n\|$, for all $(a_n) \subseteq \mathbb{R}$ (see *e.g.*, [**31**]). Furthermore (f_n') can be taken to have a spreading model [**4**]. The constant C depends only on $\sup\{\mathrm{osc}(F,k) \mid k \in K\}$ (see §2 for the definition of $\mathrm{osc}(F,k)$).

Finally we recall that a sequence (g_n) in a Banach space is a *convex block subsequence* of (f_n) if $g_n = \sum_{i=p_n+1}^{p_{n+1}} a_i f_i$ where (p_n) is an increasing sequence of integers, $(a_i) \subseteq \mathbb{R}^+$ and for each n, $\sum_{i=p_n+1}^{p_{n+1}} a_i = 1$.

2. Ordinal Indices for $B_1(K)$.

Let (K,d) be a compact metric space and let $F : K \to \mathbb{R}$ be a bounded function. The Baire characterization theorem [**3**] states that $F \in B_1(K)$ iff for all closed nonempty $L \subseteq K$, $F|_L$ has

a point of continuity (relative to the compact space (L,d)). This leads naturally to an ordinal index for Baire-1 functions which we now describe.

For a closed set $L \subseteq K$ and $\ell \in L$ let the *oscillation of* $F|_L$ at ℓ be given by $\operatorname{osc}_L(F,\ell) = \lim_{\varepsilon \downarrow 0} \sup\{f(\ell_1) - f(\ell_2) \mid \ell_i \in L \text{ and } d(\ell_i,\ell) < \varepsilon \text{ for } i = 1,2\}$. We define the *oscillation of* F *over* L by $\operatorname{osc}_L F = \sup\{F(\ell_1) - F(\ell_2) \mid \ell_1, \ell_2 \in L\}$.

For $\delta > 0$, let $K_0(F,\delta) = K$ and if $\alpha < \omega_1$ let

$$K_{\alpha+1}(F,\delta) = \left\{k \in K_\alpha(F,\delta) \mid \operatorname{osc}_{K_\alpha(F,\delta)}(F,k) \geq \delta\right\} .$$

For limit ordinals α, set

$$K_\alpha(F,\delta) = \bigcap_{\beta<\alpha} K_\beta(F,\delta) .$$

Note that $K_\alpha(F,\delta)$ is always closed and $K_\alpha(F,\delta) \supseteq K_\beta(F,\delta)$ if $\alpha < \beta$. The index $\beta(F,\delta)$ is given by

$$\beta(F,\delta) = \inf\left\{\alpha < \omega_1 \mid K_\alpha(F,\delta) = \emptyset\right\}$$

provided $K_\alpha(F,\delta) = \emptyset$ for some $\alpha < \omega_1$ and $\beta(F,\delta) = \omega_1$ otherwise. Since K is separable, the transfinite sequence $(K_\alpha(F,\delta))_{\alpha<\omega_1}$ must stabilize: there exists $\beta < \omega_1$ so that $K_\alpha(F,\delta) = K_\beta(F,\delta)$ for $\beta \geq \alpha$.

The Baire characterization theorem yields that $\beta(F,\delta) < \omega_1$ for all $\delta > 0$ iff $F \in B_1(K)$. In fact we have the following proposition. In its statement $\mathcal{A}$ denotes the algebra of *ambiguous* subsets of K. Thus $A \in \mathcal{A}$ iff A is both F_σ and G_δ. Also we write $[F \leq a]$ for the set $\{k \in K \mid F(k) \leq a\}$.

Proposition 2.1. *Let $F : K \to \mathbb{R}$ be a bounded function on the compact metric space K. The following are equivalent.*

1) *$F \in B_1(K)$.*
2) *$\beta(F,\delta) < \omega_1$ for all $\delta > 0$.*
3) *For a and b real, $[F \leq a]$ and $[F \geq b]$ are both G_δ subsets of K.*
4) *For U an open subset of $\mathbb{R}$, $F^{-1}(U)$ is an F_σ subset of K.*
5) *For $a < b$, $[F \leq a]$ and $[F \geq b]$ may be separated by disjoint sets in $\mathcal{A}$. Equivalently, there exists $A \in \mathcal{A}$ with $[F \leq a] \subseteq A$ and $A \cap [F \geq b] = \emptyset$.*
6) *F is the uniform limit of a sequence of $\mathcal{A}$-simple functions ($\mathcal{A}$-measurable functions with finite range).*
7) *F is the uniform limit of a sequence $(g_n) \subseteq DSC(K)$.*
8) *F is the uniform limit of a sequence $(g_n) \subseteq PS(K)$.*

The proof is standard and can be compiled from [23]. We are more interested in an analogous characterization of $B_{1/2}(K)$. Before stating that proposition we need a few more definitions.

$\mathcal{D}$ shall denote the algebra of all finite unions of differences of closed subsets of K. $\mathcal{D}$ is easily seen to be a subalgebra of $\mathcal{A}$.

One of the statements in our next proposition involves another ordinal index for Baire-1 functions, $\alpha(F;a,b)$, which as we shall see is closely related to our index. For $a < b$, let $K_0(F;a,b) = K$ and for any ordinal α, let

$$\begin{aligned} K_{\alpha+1}(F;a,b) = \{k \in K_\alpha(F;a,b) \mid & \text{ for all } \varepsilon > 0 \text{ and } i = 1,2, \\ & \text{there exist } k_i \in K_\alpha(F;a,b) \text{ with } d(k_i,k) \le \varepsilon \ , \\ & F(k_1) \ge b \text{ and } F(k_2) \le a\} \ . \end{aligned}$$

Equivalently, $K_{\alpha+1} = \overline{K_\alpha \cap [F \le a]} \cap \overline{K_\alpha \cap [F \ge b]}$. At limit ordinals α we set

$$K_\alpha(F;a,b) = \bigcap_{\beta<\alpha} K_\beta(F;a,b) \ .$$

As before these sets are closed and decreasing. We let $\alpha(F;a,b) = \inf\{\gamma < \omega_1 \mid K_\gamma(F;a,b) = \emptyset\}$ if $K_\gamma(F;a,b) = \emptyset$ for some $\gamma < \omega_1$ and let $\alpha(F;a,b) = \omega_1$ otherwise.

Remark 2.2. The index $\alpha(F;a,b)$ is only very slightly different from the index $L(F,a,b)$ considered by Bourgain [8]. $L(F;a,b) = \inf\{\eta < \omega_1 \mid$ there exists a transfinite increasing sequence of open sets $(G_\alpha)_{\alpha\le\eta}$ with $G_0 = \emptyset$, $G_\eta = K$, $G_{\alpha+1} \setminus G_\alpha$ is disjoint from either $[F \le a]$ or $[F \ge b]$ for all $\alpha < \eta$ and $G_\gamma = \bigcup_{\alpha<\gamma} G_\alpha$ if $\gamma \le \eta$ is a limit ordinal$\}$. In fact one can show that if $\alpha(F;a,b) = \eta + n$ where η is a limit ordinal and $n \in \mathbb{N}$, then $L(F,a,b) \in \{\eta+2n, \eta+2n-1\}$. In Proposition 2.3 we shall show that $\alpha(F;a,b) < \omega$ for all $a < b$ iff $\beta(F,\delta) < \omega$ for all $\delta > 0$. We note that a more general result has subsequently been obtained in [25]. Indeed if we define $\beta(F) = \sup\{\beta(F;\delta) \mid \delta > 0\}$ and $\alpha(F) = \sup\{\alpha(F;a,b) \mid a < b \text{ rational}\}$ then Kechris and Louveau have shown that $\beta(F) \le \omega^\xi$ iff $\alpha(F) \le \omega^\xi$.

Also we note that the following result follows from [8]. Let X be a separable Banach space not containing ℓ_1. Let $K = Ba(X^*)$ in its weak* topology. Then

$$\sup\{\beta(x^{**}|_K) : x^{**} \in X^{**}\} < \omega_1 \ .$$

Proposition 2.3. *Let $F : K \to \mathbb{R}$ be a bounded function on the compact metric space K. The following are equivalent*

1) $F \in B_{1/2}(K)$.

2) F is the uniform limit of $\mathcal{D}$-simple functions on K.

3) For $a < b$, $[F \le a]$ and $[F \ge b]$ may be separated by disjoint sets in $\mathcal{D}$.

4) $\beta(F) \le \omega$.

5) $\alpha(F; a, b) < \omega$ for all $a < b$.

Proof.

4) $\Rightarrow$ 5). This follows from the elementary observation that for all ordinals α and reals $a < b$, $K_\alpha(F; a, b) \subseteq K_\alpha(F, b - a)$, and the fact that 4) holds if and only if $\beta(F, \delta) < \omega$ for all $\delta > 0$.

5) $\Rightarrow$ 3). Let $K_i = K_i(F; a, b)$. Thus $K = K_0 \supseteq K_1 \supseteq \cdots \supseteq K_n = \emptyset$ where $n = \alpha(F; a, b)$. Let

$$D = \bigcup_{i=1}^{n} \overline{(F \le a \cap K_{i-1})} \setminus \overline{([F \ge b] \cap K_{i-1})} \in \mathcal{D} .$$

Since $K_i = \overline{([F \le a] \cap K_{i-1})} \cap \overline{([F \ge b] \cap K_{i-1})}$,

$$\begin{aligned} D &= \bigcup_{i=1}^{n} \big(\overline{[F \le a] \cap K_{i-1}} \setminus K_i \big) \\ &\supseteq \bigcup_{i=1}^{n} \Big[\big([F \le a] \cap K_{i-1} \big) \setminus K_i \Big] \\ &= \bigcup_{i=1}^{n} \big([F \le a] \cap (K_{i-1} \setminus K_i) \big) = [F \le a] . \end{aligned}$$

Since K_{i-1} is closed,

$$\begin{aligned} D &\subseteq \bigcup_{i=1}^{n} \big(K_{i-1} \setminus \overline{[F \ge b] \cap K_{i-1}} \big) \\ &\subseteq \bigcup_{i=1}^{n} \Big[K_{i-1} \setminus \big([F \ge b] \cap K_{i-1} \big) \Big] \\ &= \bigcup_{i=1}^{n} \big(K_{i-1} \setminus [F \ge b] \big) = K \setminus [F \ge b] . \end{aligned}$$

3) $\Rightarrow$ 2). This is a standard exercise in real analysis.

2) $\Rightarrow$ 1). Since every $\mathcal{D}$-simple function can be expressed in the form $\sum_{i=1}^{k} a_i \mathbf{1}_{L_i}$ where the L_i's are closed sets and $DBSC(K)$ is a linear space it suffices to recall that $\mathbf{1}_L \in DBSC(K)$ whenever L is closed. In fact $\mathbf{1}_L$ is upper semicontinuous.

1) $\Rightarrow$ 4). Let F be the uniform limit of $(F_n) \subseteq DBSC(K)$. For $\delta > 0$ and n sufficiently large, $\beta(F, 2\delta) \le \beta(F_n, \delta)$ and thus is suffices to prove that for $G \in DBSC(K)$, $\beta(G, \delta) < \omega$ for $\delta > 0$. This is immediate from the following

Lemma 2.4. *If $m \in \mathbb{N}$, $\delta > 0$ and $G : K \to \mathbb{R}$ is such that $K_m(G, \delta) \neq \emptyset$, then $|G|_D \geq m\delta/4$.*

Proof. Let $(g_n) \subseteq C(K)$ converge pointwise to G. It suffices to show that there exist integers $n_1 < n_2 < \cdots < n_{m+1}$ and $k \in K$ such that $|g_{n_{i+1}}(k) - g_{n_i}(k)| > \delta/4$ for $1 \leq i \leq m$.

Let $n_1 = 1$, $k_0 \in K_m(G, \delta)$ and let U_0 be a neighborhood of k_0 for which $\operatorname{osc}_{U_0} g_{n_1} < \delta/8$. Choose k_0^1 and k_0^2 in $U_0 \cap K_{m-1}(G, \delta)$ with $G(k_0^1) - G(k_0^2) > 3\delta/4$. Then choose $n_2 > n_1$ such that $g_{n_2}(k_0^1) - g_{n_2}(k_0^2) > 3\delta/4$. Thus there is a nonempty neighborhood $U_1 \subset U_0$ of either k_0^1 or k_0^2 such that for $k \in U_1$, $|g_{n_2}(k) - g_{n_1}(k)| > \delta/4$.

Similarly we can find a neighborhood $U_2 \subseteq U_1$ of a point in $K_{m-1}(G, \delta)$ and $n_3 > n_2$ so that for $k \in U_2$, $|g_{n_3}(k) - g_{n_2}(k)| > \delta/4$, etc. ∎

Remarks 2.5. 1. Of course by using a bit more care one can show that $|G|_D \geq m\delta/2$ whenever $K_m(G, \delta) \neq \emptyset$.

2. Following [**25**] we say that for $F \in B_1(K)$, $F \in B_1^\xi(K)$ iff $\beta(F) \leq \omega^\xi$. Thus $B_{1/2}(K) \equiv B_1^1(K)$ by Proposition 2.3, a result also observed in [**25**].

3. We do not yet have an index characterization of $B_{1/4}(K)$, however we have a necessary condition (which may be sufficient). To describe this we first must generalize our index above. Let $F : K \to \mathbb{R}$ and let $(\delta_i)_{i=1}^\infty$ be positive numbers. Set $K_0(F, (\delta_i)) = K$ and for $0 \leq i$ set

$$K_{i+1}\big(F, (\delta_j)\big) = \left\{k \in K_i\big(F, (\delta_j)\big) \mid \operatorname{osc}_{K_i(F,(\delta_j))}(F, k) \geq \delta_{i+1}\right\}.$$

Proposition 2.6. *Let $F \in B_{1/4}(K)$. Then there exists an $M < \infty$ so that if $K_n(F, (\delta_i)) \neq \emptyset$, then $\sum_{i=1}^n \delta_i \leq M$.*

Proof. Let F be the uniform limit of (G_n) with $|G_n|_D \leq C < \infty$ for all n. Suppose that $K_n(F, (\delta_i)) \neq \emptyset$ for some sequence $(\delta_i)_{i=1}^\infty \subseteq \mathbb{R}^+$. Since $K_n(F, (\delta_i)) \subseteq K_n(G_m, (\delta_i/2))$ for large m, the latter set is non-empty as well. The proof of Lemma 2.4 yields

$$\text{(2.1)} \qquad \begin{cases} \text{If } G : K \to \mathbb{R} \text{ and } (\delta_i)_{i=1}^\infty \subseteq \mathbb{R}^+ \text{ is such that } K_n(G, (\delta_i)) \neq \emptyset, \\ \text{then } |G|_D \geq 4^{-1} \sum_{i=1}^n \delta_i. \end{cases}$$

Thus by (2.1) we have, for large m, $C \geq |G_m|_D \geq 4^{-1} \sum_{i=1}^n \delta_i$ and so $\sum_{i=1}^n \delta_i \leq 4C$. ∎

We shall explore in greater detail in §3 and §8 some questions related to the problem of an index characterization of Baire-1/4. The following proposition gives a sufficient index criterion for a function to be Baire-1/4. It also shows (via Proposition 2.3) that if $F \in B_{1/2}(K) \backslash B_{1/4}(K)$, then $\beta(F) = \omega$.

Proposition 2.7. *Let $F \in B_1(K)$. If $\beta(F) < \omega$, then $F \in B_{1/4}(K)$.*

Proof. Without loss of generality let $F : K \to [0,1]$ with $\beta(F) \leq n$. Thus $\alpha(F; a, b) \leq n$ for all $a < b$. It follows from the proof of 5) $\Rightarrow$ 3) in Proposition 2.3 that for all $0 < a < b < 1$ there exists a $D \in \mathcal{D}$ with $|\mathbf{1}_D|_D \leq 2n$, $[F \leq a] \subseteq D$ and $[F \geq b] \cap D = \emptyset$. Thus for all $m < \infty$ there exist sets $D_1 \supseteq D_2 \supseteq \cdots \supseteq D_m$ in $\mathcal{D}$ with $[F \geq i/m] \subseteq D_i$, $[F \leq (i-1)/m] \cap D_i = \emptyset$ and $|\mathbf{1}_{D_i}|_D \leq 2n$ for $i \leq m$. In particular if $G = \sum_{i=1}^m m^{-1}\mathbf{1}_{D_i}$, then $\|F - G\|_\infty \leq m^{-1}$ and $|G|_D \leq 2n$. ∎

The following proposition is related to work of A. Sersouri [**39**]. It is of interest to us because it shows that a separable Banach space X can have functions of large index in X^{**} and yet be quite nice. In fact it shows there are Baire-1 functions of arbitrarily large index which strictly govern the class of quasireflexive (order 1) Banach spaces. Our proof was motivated by discussions with A. Pełczyński.

Proposition 2.8. *For all $\gamma < \omega_1$ there exists a quasireflexive (of order 1) Banach space Q_γ such that $Q_\gamma^{**} = Q_\gamma \oplus \langle F_\gamma \rangle$ where $\beta(F_\gamma) > \gamma$.*

(The index $\beta(F_\gamma)$ is computed with respect to $Ba(Q_\gamma^*)$.)

Remark 2.9. In §6 we shall show the existence of a quasireflexive space whose new functional (in the second dual) is Baire-1/4.

Proof of Proposition 2.8. We use interpolation, namely the method of [**12**]. (This has also been used in [**19**] in a slightly different manner to produce a quasireflexive space from a weak* convergent sequence.)

To begin let $\gamma < \omega_1$ be any ordinal and choose a compact metric space K containing an ambiguous set A_γ with $\alpha(\mathbf{1}_{A_\gamma}; \frac{1}{4}, \frac{3}{4}) > \gamma$. (For example $\mathbf{1}_{A_\alpha}$ could be taken to be one of the functions F_δ described in §5 with $\delta > \omega^\gamma +$.) Choose a sequence $(f_n) \subseteq Ba(C(K))$ converging pointwise to $\mathbf{1}_{A_\gamma}$ such that $(\mathbf{1}_{A_\gamma}, f_1, f_2, \ldots)$ is basic in $C(K)^{**}$. Let W be the closed convex hull of $\{\pm f_n\}_{n=1}^\infty$ in $C(K)$. Let Q_γ be the Banach space obtained from $W \subseteq Ba(C(K))$ by [DFJP]-interpolation. Thus for all $n \in \mathbb{N}$, $\|\cdot\|_n$ is the gauge of $U_n = 2^n W + 2^{-n} Ba(C(K))$, and $Q_\gamma = \{x \in C(K) : |\!|\!| x |\!|\!| \equiv (\sum_n \|x\|_n^2)^{1/2} < \infty\}$. Following the notation of [**12**], we let $C = Ba(Q_\gamma) = \{x \in C(K) : |\!|\!| x |\!|\!| \leq 1\}$ and let $j : Q_\gamma \to C(K)$ be the natural semiembedding.

We first observe that Q_γ is quasireflexive of order 1. Indeed it is easy to check that $\widetilde{W}$, the weak* closure of W in $C(K)^{**}$ is just

$$\widetilde{W} = \left\{ \sum_{i=1}^\infty a_i f_i + a_\infty \mathbf{1}_{A_\gamma} : |a_\infty| + \sum_{i=1}^\infty |a_i| \leq 1 \right\}.$$

Furthermore $\widetilde{C} \subseteq [\widetilde{W}]$ ([12], Lemma 1(v)) which has the basis $(\mathbf{1}_{A_\gamma}, f_1, f_2, \ldots)$. Now $j^{**} : Q_\gamma^{**} \to C(K)^{**}$ is one-to-one and $(j^{**})^{-1}(C(K)) = Q_\gamma$ (Lemma 1(iii)). Thus if $F_\gamma \in Q_\gamma^{**}$ satisfies $j^{**}F_\gamma = \mathbf{1}_{A_\gamma}$, then $Q_\gamma^{**} = Q_\gamma \oplus (F_\gamma)$. Of course F_γ must be the weak* limit of $(j^{-1}(f_n))_n$ in Q_γ^{**}.

It remains to show that $\beta(F_\gamma) \geq \gamma$. We shall prove

$$\overline{\alpha}(F_\gamma; \tfrac{1}{4}, \tfrac{3}{4}) \geq \alpha(\mathbf{1}_{A_\gamma}; \tfrac{1}{4}, \tfrac{3}{4}) \tag{2.2}$$

where $\overline{\beta}$ is the index computed with respect to $F_\gamma \in B_1(3Ba(Q_\gamma^*))$. Since $\beta(F_\gamma) \geq \alpha(F_\gamma; \frac{1}{12}, \frac{1}{4}) \geq \overline{\alpha}(F_\gamma; \frac{1}{4}, \frac{3}{4})$, the result follows.

Since $\|j\| \leq 3$, if $K_0 = 3Ba(Q_\gamma^*)$ and $H_0 = Ba(C(K)^*)$, then $j^*H_0 \subseteq K_0$. More generally if $K_{\beta+1} = \{y^* \in K_\beta \mid$ for all non-empty relative weak* neighborhoods U of y^* in K_β there exists $y_1^*, y_2^* \in U$ with $F_\gamma(y_1^*) \geq \frac{3}{4}$ and $F_\gamma(y_2^*) \leq \frac{1}{4}\}$ and $H_{\beta+1}$ is defined similarly in terms of $\mathbf{1}_{A_\gamma}$, then $j^*H_{\beta+1} \subseteq K_{\beta+1}$ for all β, since j^* is ω^*-continuous and $F_\gamma(j^*x^*) = (j^{**}F_\gamma)x^* = \mathbf{1}_{A_\gamma}(x^*)$. This proves (2.2). ∎

3. Theorem B.

For the proof of Theorem (B) (a) we need a lemma. Recall that a collection of pairs of subsets of K, $(A_i, B_i)_{i=1}^n$, is said to be *(Boolean) independent* if for all $I \subseteq \{1, \ldots, n\}$, $\bigcap_{i \in I} A_i \cap \bigcap_{i \notin I} B_i \neq \emptyset$.

Lemma 3.1. *Let $F : K \to \mathbb{R}$ be the pointwise limit of $(f_n) \subseteq C(K)$. If $K_m(F; a, b) \neq \emptyset$ for some $m \in \mathbb{N}$ and $a < b$, then for $a < a' < b' < b$ there exists a subsequence (f_n') of (f_n) so that if $n_1 < \cdots < n_m$, then $(A'_{n_i}, B'_{n_i})_{i=1}^m$ are independent where $A'_{n_i} = [f'_{n_i} \leq a']$ and $B'_{n_i} = [f'_{n_i} \geq b']$.*

Proof. The proof is similar to that of Lemma 2.4 and is actually a local version of the proof of the main result of [35] (see [8] for a more general discussion of the consequences of $K_\beta(F; a, b) \neq \emptyset$).

We first show how to choose a finite subsequence $(f_{n_i})_{n=1}^m$ of (f_n) so that $(A_{n_i}, B_{n_i})_{i=1}^m$ is independent, where $A_{n_i} = [f_{n_i} \leq a']$ and $B_{n_i} = [f_{n_i} \geq b']$. Let $k_\phi \in K_m(F; a, b)$. Thus there exist k_0 and k_1 in $K_{m-1}(F; a, b)$ with $F(k_0) \leq a$ and $F(k_1) \geq b$. Choose n_1 and neighborhoods U_0 and U_1 of k_0 and k_1, respectively, so that $f_{n_1} < a'$ on U_0 and $f_{n_1} > b'$ on U_1. Let $k_{\varepsilon_1, \varepsilon_2} \in U_{\varepsilon_1} \cap K_{m-2}(F; a, b)$ for $\varepsilon_1, \varepsilon_2 \in \{0, 1\}$ with $F(k_{\varepsilon_1, 0}) \leq a$ and $F(k_{\varepsilon_1, 1}) \geq b$ for $\varepsilon_1 \in \{0, 1\}$. Choose $n_2 > n_1$ and neighborhoods $U_{\varepsilon_1, \varepsilon_2} \subseteq U_{\varepsilon_1}$ of $k_{\varepsilon_1, \varepsilon_2}$ so that $f_{n_2} < a'$ on $U_{\varepsilon_1, 0}$ and $f_{n_2} > b'$ on $U_{\varepsilon_1, 1}$ (for $\varepsilon_1, \varepsilon_2 \in \{0, 1\}$). Continue up to f_{n_m}. The sets $(A_{n_i}, B_{n_i})_1^m$ are then independent since for $I \subseteq \{1, \ldots, m\}$, $\bigcap_{i \in I} A_{n_i} \cap \bigcap_{i \notin I} B_{n_i} \supseteq U_{\varepsilon_1, \ldots, \varepsilon_m} \neq \emptyset$ where $\varepsilon_i = 0$ if $i \in I$ and $\varepsilon_i = 1$ if $i \notin I$.

Now the existence of an infinite subsequence (f'_n) satisfying the conclusion of 3.1 follows immediately from Ramsey's theorem. Indeed, by the latter, there exists (f'_n) a subsequence of (f_n) so that (f'_n) satisfies the conclusion, or such that for *all* $n_1 < \cdots < n_m$, $(A'_{n_i}, B'_{n_i})_{i=1}^m$ *is not* independent. But we have proved that the second alternative is impossible.

Proof of Theorem B(a). (f_n) is a bounded sequence in $C(K)$ converging pointwise to $F \notin B_{1/2}(K)$. By Proposition 2.3 there exists $a < b$ so that $K_m(F; a, b) \neq \emptyset$ for all $m \in \mathbb{N}$. By passing to a subsequence we may assume (f_n) has a spreading model. Furthermore by Lemma 3.1, passing to subsequences and diagonalization we may assume that for some $a < a' < b' < b$, $(A_{n_i}, B_{n_i})_{i=1}^m$ is independent whenever $m \leq n_1 < n_2 < \cdots < n_m$ and $A_{n_i} = [f_{n_i} \leq a']$, $B_{n_i} = [f_{n_i} \geq b']$. By Proposition 4 of [**36**] it follows that there exists $C < \infty$ so that $(f_{n_i})_{i=1}^m$ is C-equivalent to the unit vector basis of ℓ_1^m whenever $m \leq n_1 < \cdots < n_m$. ∎

The proof of Theorem B(b) will require a more precise version of Theorem A(b) and the following elementary lemma (which follows easily from the Hahn-Banach theorem). If C is a subset of a Banach space X, $\widetilde{C}$ denotes the w^*-closure of C in X^{**}.

Lemma 3.3. *Let C and D be convex subsets of X. Then $md(C, D) = md(\widetilde{C}, \widetilde{D})$. By $md(C, D)$ we mean the minimum distance,*

$$\inf\{\|c - d\| \mid c \in C \ , \ d \in D\} \ .$$

The variant of Theorem A(b) which we need is

Lemma 3.4. *Let $F : K \to \mathbb{R}$ be bounded and let $(f_n) \subseteq C(K)$ converge pointwise to F with $\sum_{n=0}^{\infty} |f_{n+1}(k) - f_n(k)| \leq M$ for all $k \in K$ ($f_0 \equiv 0$). Suppose $\operatorname{osc}(F, k) > \delta$ for some $\delta > 0$. Then there exists a subsequence (f'_n) of (f_n) which is $C = C(M, \delta)$ equivalent to the summing basis.*

Let $F \in B_1(K) \setminus C(K)$. It is evident that if F strictly governs $\{c_0\}$, then $F \in DBSC(K)$. The next result shows that the converse is true.

Corollary 3.5. *Let $F \in DBSC(K)$ and let (f_n), M and δ be as in the hypothesis of Lemma 3.4. Let $(g_n) \subseteq C(K)$ converge pointwise to F with $\sup_n \|g_n\|_\infty < \infty$. Then there exists (h_n), a convex block subsequence of (g_n), which is $C(M, \delta)$-equivalent to the summing basis.*

The proof is straightforward from Lemmas 3.3 and 3.4.

Proof of Theorem B(b). Let $F \in B_{1/4}(K) \setminus C(K)$ and let $(f_n) \subseteq C(K)$ be a bounded sequence converging pointwise to F. Choose $(F_n) \subseteq DBSC(K)$ which converges uniformly to F so that

$\sup_n |F_n|_D < M < \infty$. For each $n \in \mathbb{N}$, choose $(f_i^n)_{i=0}^\infty \subseteq C(K)$, $f_0^n \equiv 0$, which converges pointwise to F_n and satisfies $\sum_{i=0}^\infty |f_{i+1}^n(k) - f_i^n(k)| \leq M$ for $k \in K$.

Since $F \notin C(K)$ we may assume there exists $\delta > 0$ so that for all n, $\operatorname{osc}_K(F_n, k) > \delta > 0$ for some $k \in K$. Thus, by Lemma 3.4, we may suppose for all n, $(f_i^n)_{i=1}^\infty$ is $C = C(M, \delta)$-equivalent to the summing basis. We may also assume $\|F_n - F_{n+1}\|_\infty < \varepsilon_n$ where $\varepsilon_n \downarrow 0$ and for all $n \in \mathbb{N}$, $\sum_{i=n+1}^\infty \varepsilon_i < \varepsilon_n$.

By induction and Lemma 3.3 we may replace each sequence $(f_i^n)_{i=1}^\infty$ by a convex block subsequence $(g_i^n)_{i=1}^\infty$ such that for $n > 1$,

$$(*) \qquad \begin{cases} \text{there exists a convex block subsequence } (h_i^n)_{i=1}^\infty \text{ of } (g_i^{n-1})_{i=1}^\infty \\ \text{with } \|g_i^n - h_i^n\|_\infty < \varepsilon_{n-1} \text{ for } i \in \mathbb{N}. \end{cases}$$

Let $(g_n^n)_{n=1}^\infty$ be the diagonal sequence. Clearly (g_n^n) converges pointwise to F. Also by $(*)$ for $n > k$, $md(g_n^n, co(g_j^k)_{j=1}^\infty) < \sum_{j=k}^n \varepsilon_j < \varepsilon_{k-1}$. In fact for k fixed, there exists a convex block subsequence $(d_n^k)_{n>k}$ of $(g_j^k)_{j=1}^\infty$ with $\|g_n^n - d_n^k\|_\infty < \varepsilon_{k-1}$ for $n > k$. Thus for any k, $(g_n^n)_{n>k}$ is an ε_{k-1}-perturbation of a sequence $(d_n^k)_{n>k}$ which is C'-equivalent to the summing basis where C' depends solely on C.

By Lemma 3.3 applied to $co(f_n)$ and $co(g_n^n)$, there are convex block subsequences (g_n) of (f_n) and $(\bar{g}_n)$ of (g_n^n) with $\|g_n - \bar{g}_n\|_\infty \to 0$. Since $(\bar{g}_n)_{n>i}$ is an ε_{i-1}-perturbation of a sequence which is C'-equivalent to the summing basis, $(\bar{g}_n)$ and hence (g_n) has a subsequence which has spreading model equivalent to the summing basis. ∎

Remark 3.6. The constant of equivalence of the spreading model of (g_n) with the summing basis depends solely upon $\sup_{k \in K} \operatorname{osc}_K(F, k)$ and $|F|_{1/4}$.

Our next theorem is a converse to Theorem B(a).

Theorem 3.7. *Let $F \in B_1(K)$. Assume that whenever $(f_n) \subseteq C(K)$ is a uniformly bounded sequence converging pointwise to F, then any spreading model of (f_n) is equivalent to the unit vector basis of ℓ_1. Then $F \notin B_{1/2}(K)$.*

Lemma 3.8. *Let $F \in B_{1/2}(K) \setminus C(K)$, $\|F\|_\infty \leq 1$. Then there exists $(f_n) \subseteq C(K)$ converging pointwise to F with spreading model (e_n) and a function $M : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying*

$$(3.1) \qquad \|\sum a_n e_n\| \leq M(\varepsilon) \|\sum a_n s_n\| + \varepsilon \sum |a_n|$$

for all $(a_n) \subseteq \mathbb{R}$ and $\varepsilon > 0$.

Proof. Let $(g_n) \subseteq Ba(C(K))$ converge pointwise to F and let $\varepsilon_n \downarrow 0$. By the proof of Theorem B(b) we can choose (f_n), a convex block subsequence of (g_n) such that for all m,

$(f_n)_{n=m}^{\infty}$ is an ε_m-perturbation of a sequence which is $M(\varepsilon_m, F)$-equivalent to the summing basis. ∎

Proof of Theorem 3.7. This is immediate from Lemma 3.8, since if (e_n) satisfies (3.1), then

$$\lim_{n\to\infty} \frac{1}{n}\Big\|\sum_{i=1}^{n}(-1)^i e_i\Big\| = 0 .$$

In particular (e_i) is not equivalent to the unit vector basis of ℓ_1. ∎

The proof of Theorem 3.7 combined with Theorem B(a) yields the following result. *Let $F \in B_1(K)$. Then $F \notin B_{1/2}(K)$ if and only if there exists $(f_n) \subseteq C(K)$, a uniformly bounded sequence converging pointwise to F, so that if (g_n) is a convex block subsequence of (f_n), then some subsequence of (g_n) has the unit vector basis of ℓ^1 as a spreading model.*

We do not know if the converse to Theorem B(b) is valid.

Problem 3.9. Let $F \in B_1(K)$ and $C < \infty$ be such that whenever (f_n) is a uniformly bounded sequence in $C(K)$ converging pointwise to F, then there exists (g_n), a convex block subsequence of (f_n) with spreading model C-equivalent to the summing basis. Is $F \in B_{1/4}(K)$?

We now turn to the Banach space implications of Theorem B. Let K be compact metric and let X be a closed subspace of $C(K)$. For example, K could be $Ba(X^*)$ but we do not require this. X^{**} is naturally isometric to $X^{\perp\perp} \subseteq C(K)^{**}$. In this setting it can be shown (see [**35**]) that if $B_1(X) = \{x^{**} \in X^{**} :$ there exists $(x_n) \subseteq X$ with (x_n) converging weak* in X^{**} to $x^{**}\}$, then $B_1(X) \subseteq B_1(C(K))$ and $B_1(C(K))$ is naturally identified with $B_1(K)$.

Corollary 3.10. *Let K be compact metric and let X be a closed subspace of $C(K)$.*

a) *If $X^{**} \cap [B_1(K) \setminus B_{1/2}(K)] \neq \emptyset$, then X contains a basic sequence with spreading model equivalent to the unit vector basis of ℓ_1.*

b) *If $[X^{**} \cap B_{1/4}(K)] \setminus X \neq \emptyset$ then X contains a basic sequence with spreading model equivalent to the summing basis.*

Remark 3.11. This corollary has immediate purely local consequences. Thus if X and K are as above and X does not contain ℓ_n^∞'s uniformly, then $X^{**} \cap B_{1/4}(K) \subset X$. Moreover if X is B-convex, *i.e.*, does not contain ℓ_n^1's uniformly, then $X^{**} \setminus X \subset B_{1/2}(K) \setminus B_{1/4}(K)$.

4. $DSC(K)$.

Theorem 4.1. *Let K be compact metric and let $F \in DSC(K) \setminus C(K)$. Then F governs $\{c_0\}$.*

Remark 4.2. If X is a separable Banach space, $K = Ba(X^*)$ in its weak* topology and $F \in X^{**}$, then if $F \in DSC(K)$, $F \in DBSC(K)$ (and hence for such functions Theorem 4.1

follows from Theorem A). To see this assertion, first choose (f_n) uniformly bounded in $C(K)$ so that $f_n \to F$ pointwise and $\sum_{n=1}^{\infty} |f_{n+1}(k) - f_n(k)| < \infty$ for all $k \in K$. Now since $F \in B_1(X)$, we may choose (g_j) a convex block subsequence of (f_j) and (x_j) a sequence in X with $\|g_j - x_j\| < 2^{-j}$ for all j. But then it follows that $x_j \to F$ pointwise and moreover $\sum_{j=1}^{\infty} |x_{j+1}(k) - x_j(k)| < \infty$ for all $k \in K$. Thus by the uniform boundedness principle,

$$\sup_{k \in K} \sum_{j=1}^{\infty} |x_{j+1}(k) - x_j(k)| < \infty ,$$

so $F \in DBSC(K)$.

Theorem 4.1 follows from the stronger result of Elton [13] which was motivated by work of Fonf [16].

Theorem. [13]. Let X be a Banach space and let $\mathcal{E}$ be the set of extreme points of $Ba(X^*)$. Let (x_i) be a normalized basic sequence in X such that $\sum_{i=1}^{\infty} |x^*(x_i)| < \infty$ for all $x^* \in \mathcal{E}$. Then $c_0 \hookrightarrow [(x_i)]$.

Theorem 4.1. can be phrased in this way provided $\mathcal{E}$ is replaced by $\overline{\mathcal{E}}$. However we wish to present a separate proof of our weaker result which seems to be of interest in its own right. The main step is given by the following lemma. If Y is a subspace of $C(K)$, $U \subseteq K$ and $r > 0$, we say U *r-norms* Y if $\|y|_U\|_\infty \geq r\|y\|$ for all $y \in Y$.

Lemma 4.3. *Let L be a compact metric space and let (f_i) be a normalized basic sequence in $C(L)$. If $c_0 \not\hookrightarrow [(f_i)]$, then there exists a nonempty compact set $K \subseteq L$ and a normalized block basis (g_i) of (f_i) so that*

$$(4.1) \qquad \begin{cases} \text{for any nonempty relatively open subset } U \text{ of } K \text{ there are an} \\ r > 0 \text{ and an } n_0 \in \mathbb{N} \text{ such that } U \text{ } r\text{-norms } [(g_n)_{n=n_0}^{\infty}]. \end{cases}$$

Remark 4.4. It can be deduced from [36] that $[(x_n)]$ contains an isomorph of ℓ_1 iff there exists a compact set $K \subseteq L$ such that (4.1) holds for some fixed $r > 0$ independent of U.

Proof of Lemma 4.3. Let $(U_m)_{m=1}^{\infty}$ be a base of open sets for L. We inductively construct for each m a normalized block basis $(f_i^m)_{i=1}^{\infty}$ of (f_i) and a certain subsequence M of $\mathbb{N}$.

Let $(f_i^0) = (f_i)$ and suppose $(f_i^m)_{i=1}^{\infty}$ has been chosen. There are two possibilities.

(i) There is a normalized block basis (g_i) of $(f_i^m)_{i=1}^{\infty}$ with $\|g_i|_{U_m}\|_\infty \to 0$ as $i \to \infty$.

(ii) There exists no such sequence.

If (i) holds, choose $(f_i^{m+1})_{i=1}^{\infty}$ to be a normalized block basis of $(f_i^m)_{i=1}^{\infty}$ with

$$(4.2) \qquad \|f|_{U_m}\|_\infty < 2^{-k}\|f\|_\infty \text{ for all } f \in [(f_i^{m+1})_{i=k}^{\infty}]$$

and put m in M. If (ii) holds let $(f_i^{m+1})_{i=1}^\infty = (f_i^m)_{i=1}^\infty$ and put m in $\mathbb{N} \setminus M$. Let $K = L \setminus \bigcup_{m\in M} U_m$ and for all $n \in M$ let $g_n = f_{n+1}^{n+1}$. We may assume M is infinite or else the conclusion of the lemma is satisfied with $K = L$ and $g_i = f_i^m$ ($m = \max M$ or 0 if $M = \emptyset$).

First we check that $K \neq \emptyset$. If $K = \emptyset$, then $L \subseteq \bigcup_{n\in M} U_n$. By compactness there exists $n_1 \in M$ so that $L \subseteq \bigcup_{n\in M,\ n\leq n_1} U_n$. But then since $\|g_{n_1}|_{U_n}\|_\infty < 2^{-(n_1+1)}$ for $n \in M$ with $n \leq n_1$, we have $\|g_{n_1}\|_\infty < 1$, a contradiction.

We claim that K and (g_n) satisfy (4.1). If not there exist (h_n), a normalized block basis of (g_n) and a U_m such that $K \cap U_m \neq \emptyset$ and so $m \notin M$ yet $\|h_i|_{K\cap \overline{U}_m}\| < 2^{-i}$ for all i. Indeed there must exist $m' \in M$ with $K \cap U_{m'} \neq \emptyset$ and (h_i), a normalized block basis of (g_n), with $\|h_i|_{K\cap U_{m'}}\| < 2^{-i}$. Then choose $m \in \mathbb{N}$ so that $\overline{U}_m \subseteq U_{m'}$ and $K \cap U_m \neq \emptyset$. Let $j_0 = m$ and if j_i is defined choose $j_{i+1} > j_i$ so that

$$\overline{U}_m \cap [h_{j_i} \geq 2^{-i}] \subseteq \bigcup_{\substack{n\in M\\ n\leq j_{i+1}}} U_n \,.$$

This can be done since $\overline{U}_m \cap [h_{j_i} \geq 2^{-i}] \subseteq \overline{U}_m \cap [h_{j_i} \geq 2^{-j_i}] \subseteq L \setminus K = \bigcup_{n\in M} U_n$. This completes the definition of $j_1, j_2, \ldots$. Now for $t \in U_m$, $|h_{j_i}(t)| \geq 2^{-i}$ for at most one i. Indeed let i_0 be the first integer such that $|h_{j_{i_0}}(t)| \geq 2^{-i_0}$ (if such an i_0 exists). Then $t \in \bigcup_{n\in M,\ n\leq j_{i_0+1}} U_n$ and for $i > i_0$, h_{j_i} is a normalized element in $[(g_j)_{j\geq j_i, j\in M}] = [(f_{j+1}^{j+1})_{j\geq j_i, j\in M}] \subseteq [(f_p^{j_i+1})_{p\geq j_i+1}]$. Thus if $t \in U_n$ with $n \leq j_{i_0+1}, n \in M$, then $h_{j_i} \in [(f_p^{n+1})_{p\geq j_i+1}]$ and so by (4.2), $|h_{j_i}(t)| \leq \|h_{j_i}|_{U_n}\| < 2^{-j_i} \leq 2^{-i}$.

Thus $\sum_{i=1}^\infty |h_{j_i}(t)| \leq 2$ for all $t \in \overline{U}_m$. Since $\overline{U}_m$ norms $[h_{j_i}]$, it follows from [7] that $c_0 \hookrightarrow [h_{j_i}]$, a contradiction. ∎

Proof of Theorem 4.1. Let (f_n) be a bounded sequence in $C(K)$ converging pointwise to F. By Lemma 3.3 and passing to a convex block subsequence of (f_n), if necessary, we may suppose that $\sum_{n=1}^\infty |f_{n+1}(k) - f_n(k)| < \infty$ for all $k \in K$. Also since $F \notin C(K)$, by passing to a subsequence $(f_n') \subseteq (f_n)$ we may assume that $(h_n) \equiv (f_{2n}' - f_{2n+1}')$ is a seminormalized basic sequence satisfying $\sum_{n=1}^\infty |h_n(k)| < \infty$ for all $k \in K$. If $c_0 \not\hookrightarrow [(h_n)]$, then by Lemma 4.3 there exist (g_n), a normalized block basis of (h_n), and a closed nonempty set $K_0 \subseteq K$ satisfying (4.1) (with K replaced by K_0).

For $m \in \mathbb{N}$ set $K_m = \{k \in K_0 : \sum_{n=1}^\infty |g_n(k)| \leq m\}$. Since (g_n) is a normalized block basis of (h_n), $\sum_{n=1}^\infty |g_n(k)| < \infty$ for all $k \in K$ and thus $\bigcup_{m=1}^\infty K_m = K_0$. By the Baire category theorem there exists m_0 so that K_{m_0} has nonempty interior U (relative to K_0). Choose n_0 and $r > 0$ so that U r-norms $[(g_n)_{n\geq n_0}]$. Since $\sum |g_n| \leq m_0$ on U, (g_n) is equivalent to the unit vector basis of c_0 [7], a contradiction. ∎

A natural problem is to classify those functions $F \in B_1(K)$ which govern $\{c_0\}$. We do not know how to do this, but it is easy to see that this class is strictly larger than $DSC(K)$.

Example 4.5. Let L be a countable compact metric space, large enough so that there exists an $F \in B_1(L) \setminus DBSC(L)$ (see Proposition 5.3). Choose a bounded sequence $(f_n) \subseteq C(L)$ which converges pointwise to F and let $X = [(f_n)]$. $C(L)$ is c_0-saturated (every infinite dimensional subspace of $C(L)$ contains c_0 isomorphically) and thus X is c_0-saturated. Thus F governs $\{c_0\}$ by Lemma 3.3. Let $K = Ba(X^*)$. $F \notin DSC(K)$ or otherwise (Remark 4.2) $F \in DBSC(K)$ and hence $F \in DBSC(L)$. Using this example, it can be shown that if K is any uncountable compact metric space, there exists an $F \in B_1(K) \setminus DSC(K)$ which governs $\{c_0\}$.

Question 4.6. Let $F \in B_1(K)$. If F governs $\{c_0\}$ does there exist a bounded sequence $(f_n) \subseteq C(K)$ converging pointwise to F and a w^*-closed set $L \subseteq Ba[(f_n)]^*$ such that L norms $[(f_n)]$ and $F|_L \in DSC(L)$? (Could L be taken to be countable?)

Question 4.7. Let $F \in B_1(K)$. Suppose there exists $(f_n) \subseteq C(K)$, a bounded sequence converging pointwise to F and satisfying $\sum_{n=1}^{\infty} |f_{n+1}(k) - f_n(k)| < \infty$ for all k in some residual set (complement of a first category set). Does F govern $\{c_0\}$?

We should also mention the following result of Bourgain which gives some global information about the class $DSC(K)$.

Proposition 4.8. [10] Let $F \in DSC(K) \setminus C(K)$ and let (f_n) be a bounded sequence in $C(K)$ converging pointwise to F with $\sum |f_{n+1}(k) - f(k)| < \infty$ for all $k \in K$. Then there exists a subsequence (f'_n) of (f_n) with $[(f'_n)]^*$ separable.

It follows that if $F \in DSC(K) \setminus C(K)$, then F strictly governs the class $\mathcal{C}$ of infinite dimensional Banach spaces with separable duals. However we don't know that if F governs $\{c_0\}$, then F strictly governs $\mathcal{C}$. (A negative answer, of course, would give a negative answer to 4.6.)

We give a somewhat different proof than that of [10].

Proof. We may assume that $\|f_n\| = 1$ for all n. As mentioned in the introduction there exists a subsequence (f'_n) of (f_n) which is basic and C_1-dominates the summing basis for some $C_1 < \infty$. It follows that $(h_n)_1^\infty$ is seminormalized basic where $h_1 = f'_1$ and $h_n = f'_n - f'_{n-1}$ for $n > 1$. [Indeed let $(a_i)_1^m$ be given and let $1 \leq n < m$ with $\|\sum_1^n a_i h_i\| = 1$. $\sum_{i=1}^n a_i h_i = (a_1 - a_2)f'_1 + \cdots + (a_{n-1} - a_n)f'_{n-1} + a_n f'_n \equiv f + a_n f'_n$. If $\|f\| \geq \frac{1}{2}$, then $\|\sum_1^m a_i h_i\| \geq C_2^{-1}\|f\| \geq 2^{-1}C_2^{-1}$ where C_2 is the basis constant of (f'_n). Otherwise $|a_n| \geq \frac{1}{2}$

and so

$$\Big\|\sum_{i=1}^{m} a_i h_i\Big\| = \Big\|\sum_{i=1}^{m-1}(a_i - a_{i+1})f_i' + a_m f_m'\Big\|$$

$$\geq (C_2+1)^{-1}\Big\|\sum_{i=n}^{m-1}(a_i - a_{i+1})f_i' + a_m f_m'\Big\|$$

$$\geq (C_2+1)^{-1}C_1^{-1}\Big\|\sum_{i=n}^{m-1}(a_i - a_{i+1})s_i + a_m s_m\Big\|$$

$$\geq (C_2+1)^{-1}C_1^{-1}\Big|\sum_{i=n}^{m-1}(a_i - a_{i+1}) + a_m\Big|$$

$$= (C_2+1)^{-1}C_1^{-1}|a_n| \geq 2^{-1}(C_2+1)^{-1}C_1^{-1}\ .]$$

Also for $k \in K$, $\sum_{n=1}^{\infty}|h_n(k)| < \infty$. Thus (h_n) is shrinking. Indeed if (h_n) has basis constant C and $g_n = \sum_{i=p_n+1}^{p_{n+1}} a_i h_i$ is a normalized block basis, then for $k \in K$

$$|g_n(k)| \leq \Big(\max_{p_n+1\leq i\leq p_{n+1}}|a_i|\Big)\sum_{i=p_n+1}^{p_{n+1}}|h_i(k)|$$

$$\leq (C+1)\min_i\|h_i\|^{-1}\sum_{i=p_n+1}^{p_{n+1}}|h_i(k)|$$

which goes to 0 as $n \to \infty$. ∎

The following proposition characterizes the subclass $PS(K)$ of $DSC(K)$ which was defined in §1.

Proposition 4.9. *Let $F \in B_1(K)$. The following are equivalent.*

a) *$F \in PS(K)$.*

b) *For all closed $L \subseteq K$, $F|_L$ is continuous on a relatively open dense subset of L.*

c) *There exists $\eta < w_1$ and a family $(K_\alpha)_{\alpha\leq\eta}$ of closed subsets of K with $K_0 = K$, $K_\eta = \emptyset$, $K_\gamma = \bigcap_{\alpha<\gamma} K_\alpha$ if γ is a limit ordinal and $K_\alpha \supseteq K_\beta$ if $\alpha < \beta$, such that $F|_{K_\alpha\setminus K_{\alpha+1}}$ is continuous for all α.*

d) *There exists a sequence (K_n) of closed subsets of K with $K_n \subseteq K_{n+1}$ for all n such that $K = \bigcup_n K_n$ and $F|_{K_n}$ is continuous for all n.*

Remark 4.10. Property (c) suggests the following index for $PS(K)$:

$$I(F) = \inf\{\eta < w_1 : \exists\ (K_\alpha)_{\alpha\leq\eta} \text{ satisfying (c)}\}\ .$$

Proof of 4.9. d) ⇒ a): Let (K_n) be as in d) and for $n \in \mathbb{N}$ let $f_n \in C(K_n)$ be given by $f_n = F|_{K_n}$. By the Tietze extension theorem there exists an extension of f_n, $\widetilde{f}_n \in C(K)$, with $\|\widetilde{f}_n\|_\infty \leq \|F\|_\infty$. Clearly $(\widetilde{f}_n)$ is pointwise stabilizing and has limit F.

a) $\Rightarrow$ b): For $n \in \mathbb{N}$ set

$$L_n = \{k \in L : f_m(k) = F(k) \text{ for } m \geq n\}$$

where $(f_n) \subseteq C(K)$, $\|f_n\| \leq \|F\|$ and (f_n) is pointwise stabilizing with limit F. Let $G = \bigcup_n \operatorname{int}(L_n)$. Thus G is open in L. Also by the Baire Category theorem, G is dense in L.

b) $\Rightarrow$ c): Let $K_0 = K$ and let $K_1 = \sim G_0$ where G_0 is a dense open subset of K and F is continuous on G_0. Now if K_α is defined choose G_α, a dense open subset of K_α, so that $F|_{K_\alpha}$ is continuous on G_α and set $K_{\alpha+1} = K_\alpha \setminus G_\alpha$. At limit ordinals γ, set $K_\gamma = \bigcap_{\alpha<\gamma} K_\alpha$. Since K is a separable metric space, $K_\eta = \emptyset$ for some $\eta < w_1$.

c) $\Rightarrow$ d): Let $(K_\alpha)_{\alpha \leq \eta}$ be as in c). Let $\mathcal{E}_n \downarrow 0$ and for each n set $K_{\alpha,n} = \{k \in K_\alpha : d(k, K_{\alpha+1}) \geq \mathcal{E}_n\}$ where d is the metric on K. Let $K_n = \bigcup_{\alpha<\eta} K_{\alpha,n}$. We note that K_n is closed. Indeed let $(k_i) \subseteq K_n$ converge to k. Then there exists $\alpha < \eta$ so that $k \in K_\alpha$ but $k \notin K_{\alpha+1}$. We claim that $k_i \in K_{\alpha,n}$ for sufficiently large i and thus $k \in K_{\alpha,n}$ since $K_{\alpha,n}$ is closed. To see this note first that if $k_i \notin K_\alpha$, then $d(k_i, k) \geq \mathcal{E}_n$. Thus for large i, $k_i \in K_\alpha$ and (since $k \notin K_{\alpha+1}$) $k_i \notin K_{\alpha+1}$. Hence $k_i \in K_{\alpha,n}$ for large i (since the $K_{\alpha,n}$'s are disjoint in n).

Finally $F|_{K_n}$ is continuous, for if $(k_i) \subseteq K_n$ and (k_i) converges to $k \in K_{\alpha,n}$, then by the above argument $k_i \in K_{\alpha,n}$ for large i and $F|_{K_{\alpha,n}}$ is continuous. ∎

We end this section with an improvement of Proposition 4.8 in a special case.

Proposition 4.11. *Let K be a compact metric space and let F be a simple Baire-1 function on K. Then there exists $(f_n) \subseteq C(K)$ converging pointwise to F such that $[(f_n)]$ embeds into $C(L)$ for some countable compact space L.*

Proof. First we consider the case where K is totally disconnected. Choose $\mathcal{E}_0 > 0$ so that if $F(k_1) \neq F(k_2)$, then $|F(k_1) - F(k_2)| > \mathcal{E}_0$. Let $K_\alpha = K_\alpha(F, \mathcal{E}_0)$ for $\alpha \leq \eta$ with $K_\eta = \emptyset$. By our choice of $\mathcal{E}_0$, $F|_{K_\alpha \setminus K_{\alpha+1}}$ is continuous (with respect to K_α) for all $\alpha < \eta$.

Choose a countable partition (D_j) of K into closed sets with the following properties.

a) $\operatorname{diam} D_j \to 0$

b) for each j, D_j is a relatively clopen subset of $K_\alpha \setminus K_{\alpha+1}$ for some $\alpha < \eta$ such that $F|_{D_j}$ is constant.

This can be done as follows. For each α choose a finite partition of relatively clopen subsets of $K_\alpha \setminus K_{\alpha+1}$ such that F is constant on each set of the partition. Each such set is relatively open in K_α and thus may be in turn partitioned into a countable number of relatively clopen subsets of K_α. List all the sets thus obtained for all $\alpha < \eta$ as $(C_i)_{i=1}^\infty$. Each C_i is closed in K and thus may in turn be partitioned into a finite number of closed subsets of diameter not exceeding $1/i$. We list all these sets as $(D_j)_{j=1}^\infty$.

Let $L = K/\{D_j\}$ be the quotient space of K. Since each D_j is closed and diam $D_j \to 0$, L is compact metric. For $n \in \mathbb{N}$ choose $\hat{f}_n \in C(L)$ with $\|\hat{f}_n\|_\infty \leq \|F\|_\infty$ and $\hat{f}_n(D_j)$ equal to the constant value of $F|_{D_j}$ for $j \leq n$. Let $\phi : K \to L$ denote the quotient map and let $f_n = \hat{f}_n \circ \phi$. Clearly $f_n \in C(K)$, $\|f_n\| \leq \|F\|$ and (f_n) converges pointwise to F. Also $[(f_n)]$ is isometric to $[(\hat{f}_n)] \subseteq C(L)$.

For the general case let $\phi : \Delta \to K$ be a continuous surjection and let F be a simple Baire-1 function on K. By the first part of the proof there exist $(f_n) \subseteq C(\Delta)$ converging pointwise to $F \circ \phi$ and a countable compact metric space L such that $[(f_n)] \hookrightarrow C(L)$. Let (g_n) be a bounded sequence in $C(K)$ converging pointwise to F. By Lemma 3.3 there exist convex block subsequences (h_n) and (d_n) of (g_n) and (f_n), respectively, such that $\sum \|g_n \circ \phi - d_n\| < \infty$. Thus $[(g_n)] \cong [(g_n \circ \phi)] \hookrightarrow C(L)$. ∎

Question 4.12. Does Proposition 4.11 remain true if we only assume $F \in PS(K)$ or even $F \in DSC(K)$? Note that if F satisfies the conclusion of 4.11, F strictly governs the class of c_0-saturated spaces, while it is not clear that DSC functions have this property.

5. The Baire-1 Solar System.

In this section we shall examine the relationships between the various classes of Baire-1 functions which we have defined. We begin with a result which follows easily from the Banach space theory — that developed above and some examples presented in later sections.

Proposition 5.1. *Let K be an uncountable compact metric space. Then*

$$C(K) \subsetneqq DBSC(K) \subsetneqq B_{1/4}(K) \subsetneqq B_{1/2}(K) \subsetneqq B_1(K) \,. \tag{5.1}$$

Proof. Since $C(K)$ and $C(K')$ are isomorphic whenever K and K' are both uncountable compact metric spaces [29], it suffices to separately consider each of the inclusions in (5.1). Thus if we show $C(K') \neq DBSC(K')$ for some uncountable compact metric space K', then $C(K) \neq DBSC(K)$ as well. Indeed if $j : C(K) \to C(K')$ is an onto isomorphism, then $\tilde{j} = j^{**}|_{B_1(K)} : B_1(K) \to B_1(K')$. is an onto isomorphism satisfying $\tilde{j}(DBSC(K)) = DBSC(K')$, $\tilde{j}(B_{1/4}(K)) = B_{1/4}(K')$ and $\tilde{j}(B_{1/2}(K)) = B_{1/2}(K')$.

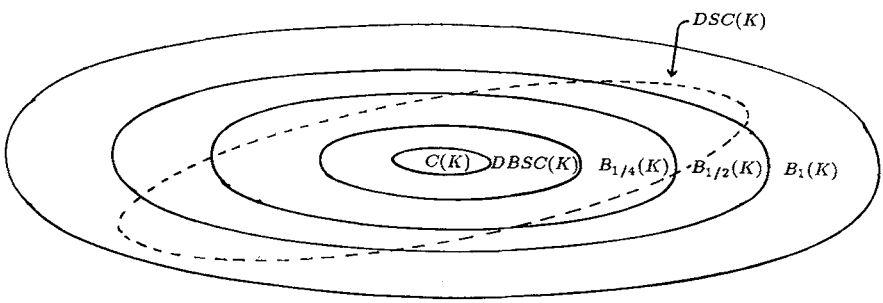
For the first inclusion, $C(K) \subsetneqq DBSC(K)$, let $X = c_0$. Then $K = (Ba(X^*), w^*)$ is uncountable compact metric and, as is well known, $X^{**} \subseteq DBSC(K)$. In particular if $F \in X^{**} \setminus X$, then $F \in DBSC(K) \setminus C(K)$.

The fact that $B_{1/4}(K) \supsetneqq DBSC(K)$ follows from Theorem A(b) and our example in §6 where we produce a nonreflexive separable Banach space X not containing c_0 such that $X^{**} \subseteq B_{1/4}(K)$, where $K = Ba(X^*)$.

For the next inclusion let $X = J$, the James space. J is not reflexive and has no spreading model isomorphic to c_0 or ℓ_1 [1]. Thus if $K = (Ba(J^*), w^*)$, then $X^{**} \setminus X \subseteq B_{1/2}(K) \setminus B_{1/4}(K)$ by virtue of Theorem B.

For the last inclusion let Y be the quasi-reflexive space of order 1 (see the proof of Proposition 6.3) whose dual is $J(e_i)$, where (e_i) is the unit vector basis of Tsirelson's space. It is proved in [**32**] that the only spreading models of Y are isomorphic to ℓ_1. Thus by Theorem 3.7, if $Y^{**} = Y \oplus \langle F \rangle$ and $K = Ba(Y^*)$, then $F \notin B_{1/2}(K)$. An alternative method would be to consider the quasi-reflexive spaces Q_γ constructed in Proposition 2.8. ∎

Remark 5.2. How does the class $DSC(K)$ relate to the classes in (5.1)? Of course we always have $DBSC(K) \subseteq DSC(K)$ and in fact for K an uncountable compact metric space we have the following diagram.



Thus $DSC(K)$ is an asteroid in the Baire-1 solar system. Indeed our proof of Proposition 5.1 along with Theorem 4.1 yields that $B_{1/4}(K) \setminus DSC(K) \neq \emptyset$, $B_{1/2}(K) \setminus [DSC(K) \cup B_{1/4}(K)] \neq \emptyset$ and $B_1(K) \setminus [DSC(K) \cup B_{1/2}(K)] \neq \emptyset$. The fact that $DSC(K) \cap B_1(K) \setminus B_{1/2}(K)$, $DSC(K) \cap B_{1/2}(K) \setminus B_{1/4}(K)$ and $DSC(K) \cap B_{1/4}(K) \setminus DBSC(K)$ are all nonempty follows from Proposition 5.3 below.

We now turn to the case where K is a *countable* compact metric space. In this setting we have, of course, $DSC(K) = B_1(K)$. However if K is large enough, the classes in (5.1) are still distinct. Since every countable compact metric space is homeomorphic to some countable ordinal, given the order topology [**30**], we confine ourselves to this setting.

Proposition 5.3.

a) If $K = \omega^{\omega^2}+$, *then* $B_{1/4}(K) \setminus DBSC(K) \neq \emptyset$.

b) If $K = \omega^{\omega}+$, *then* $B_{1/2}(K) \setminus B_{1/4}(K) \neq \emptyset$.

c) If $K = \omega^{\omega}+$, *then* $B_1(K) \setminus B_{1/2}(K) \neq \emptyset$.

d) If $K = \omega^{+}$, *then* $DBSC(K) \setminus C(K) \neq \emptyset$.

Before proving this proposition we need some terminology. Recall that an indicator function $\mathbf{1}_A$ is Baire-1 iff A is ambiguous (simultaneously F_σ and G_δ). Thus if $A \subseteq K$ where K is countable compact metric, then $\mathbf{1}_A \in B_1(K)$. We begin with a discussion of such functions.

Let δ be a countable compact ordinal space (in its order topology). Recursively we define $I_0 = \emptyset$, $I_1 = \{x \in \delta : x \text{ is an isolated point of } \delta\}$, and for $\alpha > 1$, $I_\alpha = \{x \in \delta \setminus \bigcup_{\beta<\alpha} I_\beta : x$ is an isolated point of $\delta \setminus \bigcup_{\beta<\alpha} I_\beta\}$. The I_α's are just the relative complements of the usual derived sets.

Let us say an ordinal is *even* if it is of the form $\gamma + 2n$ for some $n \in \mathbb{N}$ where $\gamma = 0$ or γ is a limit ordinal. Let $F_\delta = \mathbf{1}_{A_\delta}$ where $A_\delta = \bigcup_{\alpha \text{ even}} I_\alpha$. We have

°1) $\|F_{\omega^n+}\|_\infty = 1$ and $|F_{\omega^n+}|_D = n$.

°2) $|F_\delta|_D = \infty$ if $\delta \geq \omega^\omega+$.

°1) implies °2) trivially. To see °1), one first notes that $K_n(F_{\omega^n+}, 1) \neq \emptyset$. Indeed, $K_\alpha(F_\delta, 1)$ is just the αth derived set of δ. Hence $|F_{\omega^n+}|_D \geq n$ by the proof of Lemma 2.4. We leave the reverse inequality to the reader.

Definition. We say that a function $F : \omega^n+ \to \mathbb{R}$ is of *type* 0 if $F = n^{-1}F_{\omega^n+}$. The domain of F, ω^n+, is called a *space of type* 0.

Thus if F is a function of type 0 with domain ω^n+, $|F|_D = 1$ and $\|F\|_\infty = n^{-1}$.

More generally for $n \in \mathbb{N}$ we have the

Definition. A class of real valued functions $\mathcal{F}_n$ defined on countable compact metric spaces is said to be of *type* n if

a) For $F \in \mathcal{F}_n$, $|F|_D \geq n$.

b) For $F \in \mathcal{F}_n$, F is the uniform limit of (F_m) with $\sup_m |F_m|_D \leq 1$.

c) For each $\varepsilon > 0$, there is an $F \in \mathcal{F}_n$ with $\|F\|_\infty < \varepsilon$.

The domain of $F \in \mathcal{F}_n$ is called a *space of type* n.

Lemma 5.4. *For $n \in \mathbb{N} \cup \{0\}$ there exists a class $\mathcal{F}_n$ of functions of type n.*

Proof. We have seen that $\mathcal{F}_0$ exists. Suppose $\mathcal{F}_n$ exists. To obtain functions $F \in \mathcal{F}_{n+1}$ we begin with a function $G \in \mathcal{F}_0$ defined on a set K. Let $(t_i)_{i=1}^\infty$ be a list of the isolated points of K. We enlarge K as follows. To each t_i we adjoin a sequence of disjoint clopen sets $K_1^i, K_2^i, \ldots$ clustering only at t_i. Each of the K_j^i's is a space of type n supporting a function F_j^i of type n with $\|F_j^i\|_\infty \leq (i+j+m)^{-1}$. Here $m \in \mathbb{N}$ is arbitrary but fixed. K_{n+1}, the new space of type $n+1$, is this enlarged space. Set

$$F(t) = \begin{cases} G(t)\,, & t \in K \\ F_j^i(t)\,, & t \in K_{ij}\,. \end{cases}$$

Let $\mathcal{F}_{n+1}$ be the set of all such F's thusly obtained. We must check that $\mathcal{F}_{n+1}$ satisfies a) and b) with n replaced by $n+1$ (c) is immediate). b) holds since F is the uniform limit of (F_k) where

$$F_k(t) = \begin{cases} G(t)\,, & t \in K \\ F_j^i(t)\,, & t \in K_j^i \text{ with } i+j \leq k \\ 0\,, & \text{otherwise} \end{cases}$$

and each F_k is the uniform limit of $(F_{k,n})_{n=1}^{\infty}$ where $|F_{k,n}|_D \leq 1$ for all n.

Finally we check a). Let $(f_m)_0^{\infty} \subseteq C(K_{n+1})$, $f_0 \equiv 0$, converge pointwise to F. Since $F\big|_K = G$ and $|G|_D = 1$, for $\varepsilon > 0$ there exist $t_{i_0} \in K$ and $k \in \mathbb{N}$ with $\sum_{i=0}^{k-1} |f_{i+1}(t_{i_0}) - f_i(t_{i_0})| > 1 - \varepsilon$. Moreover by the nature of G we may assume $|G(t_{i_0})| < \varepsilon$. Since the $K_j^{i_0}$'s cluster at t_{i_0} and each f_i is continuous there exists $j_0 \in \mathbb{N}$ so that for $t \in K_{j_0}^{i_0}$, $\sum_{i=0}^{k-1} |f_{i+1}(t) - f_i(t)| > 1 - \varepsilon$ and $|f_k(t)| < \varepsilon$. But on $K_{j_0}^{i_0}$, (f_m) converges pointwise to $F_{j_0}^{i_0}$ and $|F_{j_0}^{i_0}|_D \geq n$. Thus there exists $t \in K_{j_0}^{i_0}$ with

$$|f_{k+1}(t)| + \sum_{i>k} |f_{i+1}(t) - f_i(t)| > n - \varepsilon\,.$$

It follows that

$$\sum_{i=0}^{\infty} |f_{i+1}(t) - f_i(t)| > n + 1 - 3\varepsilon$$

which proves a). ∎

Remark 5.4. Our proof yields that the spaces of type-n can be constructed within $\omega^{\omega\cdot(n+1)}+$.

Proof of Proposition 5.3. a) Let $K = \omega^{\omega^2}+$ and choose (by Remark 5.4) a sequence $(K_n)_{n=0}^{\infty}$ of disjoint clopen subspaces of K with K_n of type-n. Let F_n be a function of type-n supported on K_n with $\|F_n\|_{\infty} \to 0$ and let F be the sum of the F_n's. Clearly $|F|_D = \infty$ since $|F_n|_D \geq n$. Yet F is the uniform limit of a sequence of functions with $|\cdot|_D$ not exceeding 1.

b) Let $(K_n)_{n\in\mathbb{N}}$ be a sequence of disjoint clopen subspaces of type-0 of $\omega^{\omega}+ = K$ such that K_n supports a function F_n, which is a multiple of a function of type-0, with $\|F_n\|_{\infty} \leq n^{-1}$ and $|F_n|_D \geq n$. Define

$$F(t) = \begin{cases} F_n(t) & \text{if } t \in K_n \\ 0 & \text{otherwise.} \end{cases}$$

Clearly $F \in B_{1/2}(K) \setminus B_{1/4}(K)$.

c) The type-0 function $F_{\omega^{\omega}+}$ is not Baire-1/2.

d) $F_{\omega+}$ is $DBSC$. ∎

It is easy to check that the results of Proposition 5.3 are best possible.

6. A Characterization of $B_{1/4}(K)$ and an Example.

In this section we give an example which shows that functions of class Baire-1/4 need not govern

$\{c_0\}$. Thus Theorem B(b) is best possible. Before giving the example we give a sufficient (and necessary) criterion for a function to be Baire-1/4.

Theorem 6.1. *Let K be a compact metric space and let $F \in B_1(K)$. Then $F \in B_{1/4}(K)$ iff there exists a $C < \infty$ such that for all $\varepsilon > 0$ there exists a sequence $(S_n)_{n=0}^{\infty} \subseteq C(K)$, $S_0 \equiv 0$, with $S_n(k) \to F(k)$ for all $k \in K$ and such that for all subsequences (n_i) of $\{0\} \cup \mathbb{N}$ and $k \in K$,*

$$\sum_{j \in B((n_i),k)} |S_{n_{j+1}}(k) - S_{n_j}(k)| \leq C \ . \tag{6.1}$$

Here $B((n_i),k) = \{j : |S_{n_{j+1}}(k) - S_{n_j}(k)| \geq \varepsilon\}$.

Proof. First assume $F \in B_{1/4}(K)$, let $\varepsilon > 0$ and let $\varepsilon_n \downarrow 0$. By the proof of Theorem B(b) there exists $(f_n)_{n=0}^{\infty} \subseteq C(K)$, $f_0 \equiv 0$, converging pointwise to F with the following property. For each $m \in \mathbb{N}$, there exists $(h_j^m)_{j=0}^{\infty} \subseteq C(K)$ with $h_0^m \equiv 0$ and

$$\sum_{j=0}^{\infty} |h_{j+1}^m(k) - h_j^m(k)| \leq M \equiv 2|F|_{1/4} \ , \quad \text{for} \quad k \in K \ . \tag{6.2}$$

Furthermore $\|h_j^m - f_j\|_\infty \leq \varepsilon_m$ for $j \geq m$.

Let $\varepsilon > 0$ and fix m with $4\varepsilon_m < \varepsilon$. Let $(S_n)_{n=0}^{\infty} = (0, f_m, f_{m+1}, \ldots)$, and let (n_i) be a subsequence of $\{0\} \cup \mathbb{N}$ and let $k \in K$ be fixed. Then

$$\sum_{j \in B((n_i),k)} |S_{n_{j+1}}(k) - S_{n_j}(k)| \leq \sum_{j=0}^{\infty} |h_{j+1}^m(k) - h_j^m(k)| + 2\varepsilon_m \#B((n_i),k) \ . \tag{6.3}$$

Since $|f_p(k) - f_q(k)| \geq \varepsilon$ implies for $p > q \geq m$ or $q = 0$ that $|h_p^m(k) - h_q^m(k)| \geq \varepsilon - 2\varepsilon_m > \varepsilon/2$, (6.2) yields that $\#B((n_i),k) \leq 2M/\varepsilon$. Thus (6.3) yields (6.1) with $C = 2M = 4|F|_{1/4}$.

For the converse, let $C > \varepsilon > 0$ and let $(S_n)_0^{\infty} \subseteq C(K)$, $S_0 \equiv 0$, converge pointwise to F and satisfy (6.1) for any subsequence (n_i) of $\{0, 1, 2, \ldots\}$ and any $k \in K$. For $k \in K$ we linearly extend the sequence $(S_n(k))_{n=0}^{\infty}$ to $(S_r(k))_{r \geq 0}$. Precisely, if $r = \lambda n + (1-\lambda)(n+1)$ we set $S_r(k) = \lambda S_n(k) + (1-\lambda)S_{n+1}(k)$. Since the S_n's are continuous, $S_r \in C(K)$ as well. Furthermore, if $0 \leq r_1 < r_2 < r_3 < \cdots$, $k \in K$ and $B = B((r_i),k) = \{j : |S_{r_{j+1}}(k) - S_{r_j}(k)| \geq \varepsilon\}$, then

$$\sum_{j \in B} |S_{r_{j+1}}(k) - S_{r_j}(k)| \leq 3C \ . \tag{6.4}$$

Indeed if $J_n = \{j \in B : n \leq r_j < r_{j+1} \leq n+1\} \neq \emptyset$, then $\varepsilon \leq \sum_{j \in J_n} |S_{r_{j+1}}(k) - S_{r_j}(k)| \leq |S_{n+1}(k) - S_n(k)|$. If $j \in B \setminus \bigcup_n J_n$, there exists integers ℓ_j and m_j with $\ell_j - 1 \leq r_j < \ell_j \leq$

$m_j < r_{j+1} \le m_{j+1}$. Thus by linearity for some choice of $p_j \in \{\ell_j - 1, \ell_j\}$ and $q_j \in \{m_j, m_j+1\}$ we have $\varepsilon \le |S_{r_{j+1}}(k) - S_{r_j}(k)| \le |S_{q_j}(k) - S_{p_j}(k)|$. Thus

$$\sum_{j\in B} |S_{r_{j+1}}(k) - S_{r_j}(k)| \le \sum_{\{n: J_n \neq \emptyset\}} |S_{n+1}(k) - S_n(k)| + \sum_{2j\in B\setminus \bigcup_n J_n} |S_{q_{2j}}(k) - S_{p_{2j}}(k)| + \sum_{2j+1\in B\setminus \bigcup_n J_n} |S_{q_{2j+1}}(k) - S_{p_{2j+1}}(k)| \le 3C .$$

We shall construct a sequence $(f_n)_{n=0}^{\infty} \subseteq C(K)$, $f_0 \equiv 0$, such that for $k \in K$,

$$\sum_{n=0}^{\infty} |f_{n+1}(k) - f_n(k)| \le 4C \quad \text{and} \tag{6.5}$$

$$\text{if } H \text{ is the pointwise limit of } (f_n) \text{ then } \|H - F\|_\infty \le 5\varepsilon . \tag{6.6}$$

This will complete the proof.

Each f_n shall be an average of functions S_t where $t : K \to [0,\infty)$ is continuous and $S_t(k) \equiv S_{t(k)}(k)$ for $k \in K$. Let $f_0 = S_0 \equiv 0$. Let $\alpha_1^1 : [0,\infty) \to [0,1]$ be identically 0 on $[0,\varepsilon]$, identically 1 on $[3\varepsilon/2, \infty)$ and linear on $[\varepsilon, 3\varepsilon/2]$. Let $\alpha_2^1 : [0,\infty) \to [0,1]$ be identically 0 on $[0, 3\varepsilon/2]$, identically 1 on $[2\varepsilon, \infty)$ and linear on $[3\varepsilon/2, \varepsilon]$. For $i = 1,2$ let $t_i(k) = \alpha_i^1(|S_1(k)|)$. Let $f_1 = 2^{-1}(S_{t_1} + S_{t_2})$. We next define continuous functions $t_{i,j}$ for $i = 1,2$ and $j = 1,2,3,4$ by $t_{i,j}(k) = t_i(k) + \alpha_j^2(|S_2(k) - S_{t_i}(k)|)(2 - t_i(k))$. Here $\alpha_j^2 : [0,\infty) \to [0,1]$ is identically 0 on $[0,(4+j-1)\varepsilon/4]$, identically 1 on $[(4+j)\varepsilon/4, \infty)$ and linear on $[(4+j-1)\varepsilon/4, (4+j)\varepsilon/4]$. Set $f_2 = 8^{-1}\sum_{i=1}^2 \sum_{j=1}^4 S_{t_{i,j}}$.

In general if $f_n = 2^{-1}2^{-2}\cdots 2^{-n}\sum S_{t_{i_1,\ldots,i_n}}$, where the indices of summation range over $\{(i_1,\ldots,i_n) : 1 \le i_j \le 2^j\}$. We define $t_{i_1,\ldots,i_n}$ for $1 \le i_{n+1} \le 2^{n+1}$ by

$$t_{i_1,\ldots,i_{n+1}}(k) = t_{i_1,\ldots,i_n}(k) + \alpha_{i_{n+1}}^{n+1}\big(|S_{n+1}(k) - S_{t_{i_1,\ldots,i_n}}(k)|\big)\big(n+1-t_{i_1,\ldots,i_n}(k)\big) .$$

The functions α_j^{n+1} for $1 \le j \le 2^{n+1}$ are defined as before to be identically 0 on $[0, \varepsilon + (j-1)\varepsilon 2^{-n-1}]$, identically 1 on $[\varepsilon + j\varepsilon 2^{-n-1}]$ and linear on $[\varepsilon + (j-1)\varepsilon 2^{-n-1}, \varepsilon + j\varepsilon 2^{-n-1}]$.

The point of the construction is this. For $k \in K$ and $(i_1,\ldots,i_n)$ fixed, $|S_{t_{i_1,\ldots,i_{n+1}}}(k) - S_{t_{i_1,\ldots,i_n}}(k)|$ is either 0 or a number exceeding ε for all but perhaps one choice of i_{n+1}. [This is because the nonconstant parts of the α_j^{n+1}'s are disjointly supported.] Also except for at most one value of i_{n+1}, if $|S_{t_{i_1,\ldots,i_{n+1}}}(k) - S_{t_{i_1,\ldots,i_n}}(k)| \ge \varepsilon$ then $S_{t_{i_1,\ldots,i_{n+1}}}(k) = S_{n+1}(k)$.

We next check (6.5). Fix $k \in K$ and $m \in \mathbb{N}$. A simple calculation using the triangle inequality shows that

$$\sum_{n=0}^{m} |f_{n+1}(k) - f_n(k)| \le \text{AVE}\sum_{n=0}^{m} |S_{t_{i_1,\ldots,i_{n+1}}}(k) - S_{t_{i_1,\ldots,i_n}}(k)| \tag{6.7}$$

where the average is taken over $\{(i_1,\dots,i_m) : 1 \le i_j \le 2^j \text{ for all } j\}$. If we fix $(i_1,\dots,i_m)$ and let

$$B = \{n \le m \mid |S_{t_{i_1,\dots,i_{n+1}}}(k) - S_{t_{i_1,\dots,i_n}}(k)| \ge \varepsilon\}$$

then

$$\sum_{n\in B} |S_{t_{i_1,\dots,i_{n+1}}}(k) - S_{t_{i_1,\dots,i_n}}(k)| \le 3C$$

by (6.4).

Now for $1 \le n \le m$ fixed, the percentage of terms in the "AVE" of (6.7) for which $0 < |S_{t_{i_1,\dots,i_{n+1}}}(k) - S_{t_{i_1,\dots,i_n}}(k)| < \varepsilon$ is at most 2^{-n-1}. It follows that

$$\text{AVE} \sum_{n=0}^{m} |S_{t_{i_1,\dots,i_{n+1}}}(k) - S_{t_{i_1,\dots,i_n}}(k)| \le 3C + 2^{-1}\varepsilon + \cdots + 2^{-m-1}\varepsilon$$

and (6.5) follows from this since $\varepsilon < C$.

(6.5) implies (f_n) is pointwise convergent to some function H. For fixed $k \in K$ choose $m \in \mathbb{N}$ so that $2^{-m}C < \varepsilon$, $|S_m(k) - F(k)| < \varepsilon$ and $|f_m(k) - H(k)| < \varepsilon$. We claim that $|f_m(k) - S_m(k)| < 3\varepsilon$, which proves (6.6). Indeed

$$f_m(k) = \text{AVE } S_{t_{i_1,\dots,i_m}}(k) \quad \text{and} \quad C \ge |S_{t_{i_1,\dots,i_m}}(k) - S_m(k)| \ge 2\varepsilon$$

for at most $2^{-m}\#\{(i_1,\dots,i_m) : i_j \le 2^j\}$ choices of $(i_1,\dots,i_m)$. Thus $|f_m(k) - S_m(k)| \le 2\varepsilon + 2^{-m}C < 3\varepsilon$. ∎

Remark 6.2. Let $F \in B_1(K)$. Our proof shows that $F \in B_{1/4}(K)$ iff there exists $C < \infty$ and $(S_n)_{n=0}^{\infty} \subseteq C(K)$, $S_0 \equiv 0$, converging pointwise to F such that for all $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that if (n_i) is any subsequence of $\{0, m, m+1, \dots\}$ then (6.1) holds.

Proposition 6.3. *There exists a compact metric space K and $F \in B_{1/4}(K)$ which does not govern $\{c_0\}$.*

Proof. Let (e_i) be the unit vector basis of the Tsirelson space T constructed in [17] (see also [11]) and let $X = J(e_i)$ be its "Jamesification" as described in [6]. For completeness we recall the definition of X. Let c_{oo} be the linear space of all finitely supported functions $x : \mathbb{N} \to \mathbb{R}$ and for $n \in \mathbb{N}$ define $S_n : c_{oo} \to \mathbb{R}$ by $S_n(x) = \sum_{i=1}^{n} x(i)$. Let $S_0 \equiv 0$. For $x \in c_{oo}$ let

$$\|x\| = \sup\left\{ \Big\| \sum_{i=1}^{m} (S_{n_i} - S_{p_i-1})(x) e_{p_i} \Big\|_T \;\Big|\; 1 \le p_1 \le n_1 < p_2 \le n_2 < \cdots < p_m \le n_m \right\}.$$

Let X be the completion of $(c_{oo}, \|\cdot\|)$.

As shown in [6], the unit vectors (u_i) form a boundedly complete normalized basis for X. Thus $X = Y^*$ (where $Y = [(u_i^*)] \subseteq X^*$). Furthermore it was shown that Y is quasi-reflexive and Y^{**} has a basis given by $\{S, u_1^*, u_2^*, \ldots\}$, where

$$S\Big(\sum a_i u_i\Big) = \sum_1^\infty a_i \ .$$

Of course (u_i^*) are the biorthogonal functionals to (u_i) and S is the weak* limit in Y^{**} of (S_n).

Let $K = Ba(X) = Ba(Y^*)$ in the weak* topology (of Y^*). Since Y does not contain c_0, our example will be complete if we can prove that $S \in B_{1/4}(K)$. By Theorem 6.1 it suffices to prove that if $\varepsilon > 0$ then for $m \in \mathbb{N}$ with $m > 2/\varepsilon$, if $x \in Ba(X)$ and (n_i) is a subsequence of $\{m, m+1, m+2, \ldots\}$, then

$$\sum_{j \in B} |S_{n_{j+1}}(x) - S_{n_j}(x)| \leq 2$$

where

$$B = \{j : |S_{n_{j+1}}(x) - S_{n_j}(x)| \geq \varepsilon\} \ .$$

We first note that $\#B < m$. Indeed if $\#B \geq m$, then by the properties of T,

$$\begin{aligned} 1 \geq \|x\| &\geq \Big\| \sum_{j \in B} (S_{n_{j+1}}(x) - S_{n_j}(x)) e_{n_j} \Big\|_T \\ &\geq 2^{-1} m\varepsilon \ , \end{aligned}$$

a contradiction. The last inequality is due to the fact that $\| \sum_A a_i e_i \|_T \geq 2^{-1} \sum_A |a_i|$ provided $\min A \leq \#A$.

Thus $m \leq \min B \leq \#B$ and so

$$\begin{aligned} \sum_{j \in B} |S_{n_{j+1}}(x) - S_{n_j}(x)| &\leq 2 \Big\| \sum_{j \in B} (S_{n_{j+1}}(x) - S_{n_j}(x)) e_{n_j} \Big\| \\ &\leq 2\|x\| \leq 2 \ . \end{aligned}$$

∎

Remark 6.4. Our proof of Proposition 6.3 shows that there exists a quasi-reflexive (of order one) Banach space Y such that if $K = Ba(Y^*)$ then $Y^{**} \setminus Y \subseteq B_{1/4}(K)$. In particular, it follows that there exists an $F \in B_{1/4}(K) \setminus C(K)$ which strictly governs the class of quasi-reflexive Banach spaces.

7. Some Bad Baire-1/2 Functions.

In this section we show that functions of class Baire-1/2 need not be that nice.

Proposition 7.1. *There exists a compact metric space K and $F \in B_{1/2}(K)$ which governs $\{\ell_1\}$.*

Remark 7.2. The first example of an $F \in B_1(K)$ which governs $\{\ell_1\}$ was due to Bourgain [**9,10**]. His ingenious construction forms the motivation behind our next example (Proposition 7.3). Another example of such an F appears in [**2**]. While the example of [**2**] can be shown to be Baire-1/2, we prefer to present a very slight modification.

Proof. Let (e_n) be the unit vector basis of a Lorentz sequence space $d_{w,1}$ (see *e.g.*, [**27**]). Let $J(e_i)$ be the Jamesification of (e_n) (see [**6**]) and let (u_i) be the unit vector basis of $J(e_i)$. Thus

$$\Big\|\sum_{i=1}^{k} a_i u_i\Big\| = \sup\left\{\Big\|\sum_{i=1}^{p}\Big(\sum_{j=n_i}^{m_i} a_j\Big)e_i\Big\|_{d_{w,1}} \;\Big|\; 1 \le n_1 \le m_1 < n_2 \le m_2 < \cdots < n_p \le m_p\right\}.$$

(u_i) is a normalized spreading basis for $J(e_i)$ which is not equivalent to the unit vector basis of ℓ_1 and thus by [**36**], (u_i) is weak Cauchy. Furthermore by standard block basis arguments one can show that $J(e_i)$ is hereditarily ℓ_1. Also if F is defined by $u_i \to F$ weak* then $F \in B_{1/2}(K)$ where $K = Ba(J(e_i)^*)$. But this is immediate by Theorem B(a) since (u_i), being its own spreading model, does not have ℓ_1 as a spreading model. The fact that F governs ℓ_1 follows from Lemma 3.3. Indeed if (f_n) is a bounded sequence in $C(K)$ converging pointwise to F, then some convex block subsequence of (f_n) is a basic sequence equivalent to some convex block subsequence of (u_i). Since $[(u_i)]$ is hereditarily ℓ_1, $\ell_1 \hookrightarrow [(f_n)]$. ∎

Proposition 7.3. *There exists a compact metric space K and $F \in B_{1/2}(K)$ such that F does not govern $\{\ell_1\}$ yet F strictly governs $\{X : X$ is separable and X^* is not separable$\}$.*

Remark 7.4. In [**33**] a function $F \in B_1(K)\backslash B_{1/2}(K)$ was constructed satisfying the conclusion of Proposition 7.3. The construction we now present will be a modification of that example.

Proof of Proposition 7.3. We begin by defining a Banach space Y. (The space Y was first defined in [**34**]) Let $\mathcal{D} = \{\phi\} \cup \bigcup_n \{0,1\}^n$ be the dyadic tree with its natural order (see Remark 4.2) and let $(K_\alpha)_{\alpha\in\mathcal{D}}$ be the natural clopen base for the Cantor set Δ. For $f \in C(K_\alpha)$ we let $\widetilde{f} \in C(\Delta)$ be given by $\widetilde{f}(t) = f(t)$ for $t \in K_\alpha$ and $\widetilde{f}(t) = 0$ otherwise. Let

$$Y = \Big\{(f_\alpha)_{\alpha\in\mathcal{D}} \mid f_\alpha \in C(K_\alpha) \text{ for all } \alpha \in \mathcal{D} \text{ and}$$

$$\|(f_\alpha)\|_Y \equiv \sup\Big\{\Big(\sum_{k=1}^{\ell}\Big\|\sum_{\alpha\in S_k}\widetilde{f}_\alpha\Big\|_\infty^2\Big)^{1/2} : (S_k)_{k=1}^{\ell} \text{ are disjoint segments in } \mathcal{D}\Big\} < \infty\Big\}.$$

Y is a Banach space under the given norm.

We shall construct a weak Cauchy sequence $(g_n) \subseteq Y$ with weak* limit F such that

$$\ell_1 \not\hookrightarrow [(g_n)] \ , \tag{7.1}$$

$$\begin{cases} [(h_n)]^* \text{ is nonseparable for every convex block} \\ \quad \text{subsequence } (h_n) \text{ of } (g_n) \text{ and} \end{cases} \tag{7.2}$$

$$\begin{cases} \text{there exists a weak* closed set } K \subseteq Ba(Y^*) \text{ such that} \\ \quad K \text{ norms } [(g_n)] \text{ and } F\big|_K \in B_{1/2}(K) \ . \end{cases} \tag{7.3}$$

The proposition follows immediately from (7.1)–(7.3). Indeed to see that F governs $\{X : X$ is separable and X^* is nonseparable$\}$, let (f_n) be a bounded sequence in $C(K)$ converging pointwise to F. By Lemma 3.3 there exist convex block subsequences (d_n) and (h_n) of (f_n) and (g_n), respectively, such that $\|d_n - h_n\|_{C(K)} \to 0$. Since $[(h_n)]^*$ is nonseparable, so is $[(d_n)]^*$.

Our construction of (g_n) depends upon the following (which in turn follows from our discussion of functions of type-0 in §5): for $n \in \mathbb{N}$ there exists $F_n \in B_1(\Delta)$ such that

$$\|F_n\|_\infty = 1 \ \text{ and} \tag{7.4}$$

$$\begin{cases} |F_n|_D = n \ . \text{ Moreover if } (h_i) \subseteq C(\Delta) \text{ converges pointwise to } F_n \text{ then} \\ \text{there exists } k \in \Delta, \text{ integers } \ell_1 < \ell_2 < \cdots \ell_{n+1} \text{ and } \varepsilon_i = \pm 1 \ (1 \le i \le n) \\ \text{such that } \sum_{i=1}^{n} \varepsilon_i (h_{\ell_{i+1}} - h_{\ell_i})(k) > n - 1. \end{cases} \tag{7.5}$$

Actually our F_n's are indicator functions whose domains are countable compact metric spaces K. Of course one can embed K into Δ and the corresponding extended indicator functions have the desired properties (7.4) and (7.5).

We use "$<_L$" for the natural linear order on $\mathcal{D}$. Thus $\phi < 0 < 1 < 00 < 01 < 10 < 11 < 000 < \cdots$. For each $\alpha \in \mathcal{D}$ choose $n_\alpha \in \mathbb{N}$ and $c_\alpha \in \mathbb{R}^+$ satisfying the following seven properties:

i) $\sum_{\beta \in \mathcal{D}} c_\beta \le 1$.

ii) $c_\alpha^{-1} n_\alpha^{-1} \sum_{\beta <_L \alpha} n_\beta < 1/10$.

iii) $2c_\alpha^{-1} \sum_{\beta >_L \alpha} c_\beta < 1/10$.

iv) $1 - n_\alpha^{-1} > 9/10$.

v) $2c_{\alpha_0} c_\alpha^{-1} < 1/10$ if $\alpha <_L \alpha_0$.

vi) $c_{\alpha_0} c_\alpha^{-1} n_{\alpha_0} n_\alpha^{-1} < 1/10$ if $\alpha_0 <_L \alpha$.

vii) $\sum_{\beta \in \mathcal{D}} b_\beta^2 < \infty$ where $b_\beta = \sum_{\gamma \ge_L \beta} c_\gamma$.

Of course we could trim this list somewhat, but we prefer to list the properties in the form in which they are used. The c_α's and n_α's can be chosen as follows. Let $\{\alpha_1, \alpha_2, \alpha_3, \ldots\}$ be

a listing of $\mathcal{D}$ in the linear order. Let $c_{\alpha_j} = (22)^{-j}$. It is quickly checked that properties i), iii), v) and vii) hold. We then choose n_{α_j} inductively to be an increasing sequence of positive integers with $n_{\alpha_1} = 11$ (so that iv) holds). If n_{α_j} is picked, choose $n_{\alpha_{j+1}}$ to satisfy ii) and vi) for $\alpha = \alpha_{j+1}$. For each $\alpha \in \mathcal{D}$, let $F_{n_\alpha} \in B_1(K_\alpha)$ satisfy (7.4) and (7.5) (with Δ replaced by K_α and n replaced by n_α).

For each $\alpha \in \mathcal{D}$ choose $(f_\alpha^n)_{n=1}^\infty \subseteq C(K_\alpha)$, $f_\alpha^n \geq 0$ and $\|f_\alpha^n\| = 1$, so that $(f_\alpha^n)_{n=1}^\infty$ converges pointwise to F_{n_α} and is equivalent to (s_n) with

$$|f_\alpha^1(k)| + \sum_{n=1}^{\infty} |f_\alpha^{n+1}(k) - f_\alpha^n(k)| \leq n_\alpha \quad \text{for all} \quad k \in K_\alpha \,. \tag{7.6}$$

Let $g_n = (c_\alpha f_\alpha^n)_{\alpha \in \mathcal{D}}$. Clearly $g_n \in Y$ since $\|g_n\| \leq \sum_{\alpha \in \mathcal{D}} c_\alpha \leq 1$ by i). Furthermore $\ell_1 \not\hookrightarrow [(g_n)]$ by the following lemma and the fact that for all α, $\ell_1 \not\hookrightarrow [f_\alpha^n : n \in \mathbb{N}]$.

Lemma 7.5. *For all $\alpha \in \mathcal{D}$, let Y_α be a closed subspace of $C(K_\alpha)$ which does not contain ℓ_1. Let*

$$\widetilde{Y}_\alpha = \left\{(h_\beta)_{\beta \in \mathcal{D}} \in Y : h_\alpha \in Y_\alpha \ \text{ and } \ h_\beta \equiv 0 \ \text{ if } \ \alpha \neq \beta\right\} .$$

Let Z be the closed linear span of $\{\widetilde{Y}_\alpha : \alpha \in \mathcal{D}\}$. Then Z does not contain ℓ_1.

Proof. It is shown in [**34**] that Y does not contain a sequence $(h_n)_{n=1}^\infty = ((h_\alpha^n)_{\alpha \in \mathcal{D}})_{n=1}^\infty$ which is both equivalent to the unit vector basis of ℓ_1 and has the following property: for all $\alpha_0 \in \mathcal{D}$ there exists $m_0 \in \mathbb{N}$ so that for $m \geq m_0$ and $\alpha \leq_L \alpha_0$, $h_\alpha^m \equiv 0$.

But if Z contains ℓ_1, then Y must contain such a sequence (h_n). This follows easily from the fact that if $(f_n)_{n=1}^\infty$ is an ℓ_1-basis in Z, then for all $\varepsilon > 0$ and $\alpha_0 \in \mathcal{D}$, there exists a normalized block basis $(d_n)_{n=1}^\infty = ((d_\alpha^n)_{\alpha \in \mathcal{D}})_{n=1}^\infty$ of (f_n) with $\|d_{\alpha_0}^n\|_{C(K_{\alpha_0})} < \varepsilon$ for all n. ∎

Thus by [**36**] we may pass to a subsequence of (g_n) which is weak Cauchy. By relabeling we assume that (g_n) itself is weak Cauchy and converges weak* to $F \in Y^{**}$.

We next verify (7.2). Let (h_n) be a convex block subsequence of (g_n). For $k \in \Delta$ and $h = (h_\alpha)_{\alpha \in \mathcal{D}} \in Y$, define $\delta_k(h) = \sum_{\alpha \in \gamma_k} \widetilde{h}_\alpha(k)$ where $\gamma_k = \{\alpha \in \mathcal{D} : k \in K_\alpha\}$. Clearly δ_k is a normalized element of Y^*. We shall show that

$$\begin{cases} \text{for all } \alpha \in \mathcal{D} \text{ there exists } k_\alpha \in K_\alpha \text{ and } h = (h_\beta) \in Ba[(h_n)] \\ \text{such that } \delta_{k_\alpha}(h) > 7/10 \text{ and } \delta_k(h) < 3/10 \text{ if } k \in \Delta \setminus K_\alpha. \end{cases} \tag{7.7}$$

As in [**33**] this implies $[(h_n)]^*$ is nonseparable. Indeed by (7.7) we can choose $(h^\alpha)_{\alpha \in \mathcal{D}} \subseteq Ba[(h_n)]$ and a collection of basic clopen sets $(K'_\alpha)_{\alpha \in \mathcal{D}}$ in Δ such that for all $\alpha \in \mathcal{D}$,

a) $K'_{\alpha,0} \cap K'_{\alpha,1} = \emptyset$,

b) $K'_{\alpha,\varepsilon} \subseteq K'_\alpha$ for $\varepsilon = 0, 1$ and

c) $\delta_k(h_\alpha) > 7/10$ for $k \in K'_\alpha$ and
$\delta_k(h_\alpha) < 3/10$ for $k \notin K'_\alpha$.

For each branch (a maximal subset linearly ordered by $<$) γ in $\mathcal{D}$ choose $k_\gamma \in \bigcap_{\alpha\in\gamma} K'_\alpha$. By a) and b) k_γ is well defined and $k_\gamma \neq k_{\gamma'}$ if $\gamma \neq \gamma'$. By c), $\|(\delta_{k_\gamma} - \delta_{k_{\gamma'}})|_{[(h_n)]}\| > 2/5$ if $\gamma \neq \gamma'$.

We return to the proof of (7.7). Fix $\alpha \in \mathcal{D}$ and set $h_n = (h^n_\beta)_{\beta\in\mathcal{D}}$. Since $(h^n_\alpha)_{n=1}^\infty$ is a convex block subsequence of $(c_\alpha f^n_\alpha)_{n=1}^\infty$, $(h^n_\alpha)_{n=1}^\infty$ converges pointwise to $c_\alpha F_{n_\alpha}$. Thus by (7.5) and (7.6) we may assume (by passing to a subsequence and relabeling, if necessary) that there exist $\varepsilon_i = \pm 1$ $(1 \le i \le n_\alpha)$ and $k_\alpha \in K_\alpha$ such that

$$n_\alpha \ge \sum_{i=1}^{n_\alpha} c_\alpha^{-1}\varepsilon_i(h^{i+1}_\alpha - h^i_\alpha)(k_\alpha) > n_\alpha - 1 \ .$$

Let $h = n_\alpha^{-1}c_\alpha^{-1}\sum_{i=1}^{n_\alpha}\varepsilon_i(h_{i+1} - h_i) \equiv (h_\beta)_{\beta\in\mathcal{D}}$. Thus $1 \ge h_\alpha(k_\alpha) > 1 - n_\alpha^{-1} > 9/10$ by iv).

Furthermore by applying (7.6) to each $\beta < \alpha$ we have from ii)

$$\begin{aligned}\sum_{\beta<\alpha}\tilde{h}_\beta(k_\alpha) &\le \sum_{\beta<\alpha} n_\alpha^{-1}c_\alpha^{-1}c_\beta n_\beta \\ &\le c_\alpha^{-1}n_\alpha^{-1}\sum_{\beta<\alpha} n_\beta < 1/10 \ .\end{aligned}$$

By the triangle inequality and the definition of h,

$$\begin{aligned}\sum_{\beta>\alpha}\tilde{h}_\beta(k_\alpha) &\le c_\alpha^{-1}n_\alpha^{-1}\sum_{\beta>\alpha} 2c_\beta n_\alpha \\ &= 2c_\alpha^{-1}\sum_{\beta>\alpha} c_\beta < 1/10 \ \text{ (by iii)).}\end{aligned}$$

Thus $\delta_{k_\alpha}(h) > 9/10 - 2/10 = 7/10$ which proves the first part of (7.7).

Let $k \in \Delta \setminus K_\alpha$ be fixed. There exists a unique $\alpha_0 \in \mathcal{D}$ $(\alpha_0 \neq \alpha)$ with the same length as α_0, $|\alpha| = |\alpha_0|$, such that $k \in K_{\alpha_0}$. The calculations above yield $\sum_{\beta<\alpha_0}\tilde{h}_\beta(k) + \sum_{\beta>\alpha_0}\tilde{h}_\beta(k) < 2/10$. If $\alpha_0 <_L \alpha$ then by (7.6)

$$\begin{aligned}0 \le h_{\alpha_0}(k) &= n_\alpha^{-1}c_\alpha^{-1}\sum_{i=1}^{n_\alpha}\varepsilon_i(h^{i+1}_{\alpha_0} - h^i_{\alpha_0})(k) \\ &\le n_\alpha^{-1}c_\alpha^{-1}c_{\alpha_0}n_{\alpha_0} \le 1/10 \ \text{ (by vi)).}\end{aligned}$$

If $\alpha <_L \alpha_0$ then we have (from the equality above) that

$$0 \le h_{\alpha_0}(k) \le n_\alpha^{-1}c_\alpha^{-1}c_{\alpha_0}2n_\alpha = 2c_{\alpha_0}c_\alpha^{-1} < 1/10$$

by v)). It follows that $\delta_k(h) < 3/10$ which completes the proof of (7.7).

Finally, we verify (7.3). Let $S = [\alpha, \beta] \equiv \{\gamma \in \mathcal{D} \mid \alpha \leq \gamma \leq \beta\}$ be a *finite segment* in $\mathcal{D}$. For $k \in K_\beta$ and $f \in Y$ we set $\delta_{S,k}(f) = \sum_{\gamma \in S} \tilde{f}_\gamma(k)$. $\delta_{S,k}(f)$ is defined similarly if $S = [\alpha, \infty) \equiv \{\gamma \in \mathcal{D} : \alpha \leq \gamma\}$ is an infinite segment and $k \in \bigcap_{\beta \in S} K_\beta$. Define

$$K = \Big\{ \sum_{i=1}^{\infty} a_i \delta_{S_i,k_i} : (a_i)_1^\infty \in Ba(\ell_2) \ , \ (S_i)_1^\infty \text{ are disjoint segments and } k_i \in \bigcap_{\beta \in S_i} K_\beta \text{ for every } i \Big\} .$$

From the definition of the norm in Y it is clear that $K \subseteq Ba(Y^*)$. Furthermore it is easy to check that K is weak* closed and K 1-norms Y.

It remains to show that $F\big|_K \in B_{1/2}(K)$. For $m, n \in \mathbb{N}$ let $g(n,m) \in Y$ be given by $g(n,m) = (g(n,m)_\beta)_{\beta \in \mathcal{D}}$ where

$$g(n,m)_\beta = \begin{cases} g_\beta^n & \text{if } |\beta| \leq m \\ 0 & \text{otherwise.} \end{cases}$$

Let $y^* = \sum_{i=1}^\infty a_i \delta_{S_i,k_i} \in K$. Then for m fixed,

$$\begin{aligned} \sum_{n=1}^{\infty} \Big| \big[g(n+1,m) - g(n,m)\big](y^*) \Big| &= \sum_{n=1}^{\infty} \Big| \sum_{i=1}^{\infty} a_i \sum_{\substack{\gamma \in S_i \\ |\gamma| \leq m}} \big[\tilde{g}_\gamma^{n+1}(k_i) - \tilde{g}_\gamma^n(k_i)\big] \Big| \\ &\leq \sum_{i=1}^{\infty} |a_i| \sum_{\substack{\gamma \in S_i \\ |\gamma| \leq m}} \sum_{n=1}^{\infty} |\tilde{g}_\gamma^{n+1}(k_i) - \tilde{g}_\gamma^{n+1}(k_i)| \\ &\leq \sum_{i=1}^{\infty} |a_i| \sum_{\substack{\gamma \in S_i \\ |\gamma| \leq m}} c_\gamma n_\gamma \quad \text{(by (7.6))} \\ &\leq \sum_{|\gamma| \leq m} c_\gamma n_\gamma < \infty \ . \end{aligned}$$

In particular $(g(n,m))_{n=1}^\infty$ converges pointwise on K to a function $G_m \in DBSC(K)$.

All that remains is to show that $\|G_m - F\big|_K\|_{C(K)} \to 0$ as $m \to \infty$. Let $m \in \mathbb{N}$ be fixed and let $y^* = \sum_{i=1}^\infty a_i \delta_{S_i,k_i} \in K$. Then

$$\begin{aligned} |G_m(y^*) - F(y^*)| &= \Big| \sum_{i=1}^{\infty} a_i \sum_{\substack{\gamma \in S_i \\ |\gamma| > m}} c_\gamma \tilde{F}_{n_\gamma}(k_i) \Big| \\ &\leq \sum_{i=1}^{\infty} |a_i| \Big(\sum_{\substack{\gamma \in S_i \\ |\gamma| > m}} c_\gamma \Big) . \end{aligned}$$

For each i set

$$b_i^m = \sum_{\substack{\gamma \in S_i \\ |\gamma| > m}} c_\gamma \ .$$

Thus

$$|G_m(y^*) - F(y^*)| \leq \left(\sum_{i=1}^{\infty}(b_i^m)^2\right)^{1/2}$$

by Hölder's inequality. The latter goes to 0 as $m \to \infty$ by vii). ∎

8. Problems.

We have previously raised two problems concerning $B_{1/4}(K)$.

Problem 8.1. Let $F \in B_1(K)$ and $C < \infty$ be such that if $(f_n) \subseteq C(K)$ is a bounded sequence converging pointwise to F, then there exists (g_n), a convex block subsequence of (f_n), with spreading model C-equivalent to the summing basis. Is $F \in B_{1/4}(K)$?

Problem 8.2. Let $F \in B_1(K)$ and assume there exists a $C < \infty$ such that if $(\varepsilon_i) \subseteq \mathbb{R}^+$ and $K_n(F,(\varepsilon_i)) \neq \emptyset$, then $\sum_1^n \varepsilon_i \leq C$. Is $F \in B_{1/4}(K)$?

These problems lead naturally to the following definitions. Let $F \in B_1(K)$.

$$|F|_I = \max\left\{\sup\left\{\sum_{i=1}^{m}\delta_i : K_m(F,(\delta_i)) \neq \emptyset\right\},\ \|F\|_\infty\right\}.$$

$$|F|_{I'} = \max\left\{\sup\{m\delta : K_m(F,\delta) \neq \emptyset\},\ \|F\|_\infty\right\}.$$

$$|F|_S = \inf\Bigg\{C : \text{ there exist } (f_n) \subseteq C(K) \text{ converging pointwise to } F$$
$$\text{with for all } (a_i)_1^k \subseteq \mathbb{R},\ \lim_{\substack{n_1\to\infty \\ n_1<\cdots<n_k}} \left\|\sum_{i=1}^{k} a_i f_{n_i}\right\| \leq C\left\|\sum_1^k a_i s_i\right\|\Bigg\}.$$

Remark 8.3. We do not know if $|F|_I$ or $|F|_{I'}$ are norms. It is clear that $|F|_S$ is a norm and also that

$$\|F\|_\infty \leq |F|_{I'} \leq |F|_I \leq |F|_S \leq |F|_{1/4} \leq |F|_D$$

($|F|_S \leq |F|_{1/4}$ follows from the proof of Theorem B.) Furthermore, using the series criterion for completeness, it is easy to show that $(\{F \in B_1(K) : |F|_S < \infty\}, |\cdot|_S)$ is a Banach space.

Problem 8.4. Are $|\cdot|_I$ and $|\cdot|_S$ equivalent? Are $|\cdot|_S$ and $|\cdot|_{1/4}$ equivalent?

The solution of Problem 8.4 would of course solve Problems 8.1 and 8.2. Furthermore an affirmative answer to Problem 8.2 would yield an affirmative answer to Problems 8.1 and 8.4.

Proposition 8.5. *$|\cdot|_I$ and $|\cdot|_{I'}$ are not (in general) equivalent.*

Proof. Define $F : [0,1]^\omega \to \mathbb{R}$ as follows:

If $t_0 \neq 0$ let

$$F(t_0, t_1, \ldots) = \sin t_0^{-1}.$$

If $t_0 = t_1 = \cdots = t_r = 0 \neq t_{r+1}$, set

$$F(t_0, t_1, \ldots) = \frac{1}{r+2} \sin t_r^{-1} .$$

It's easy to see that $\operatorname{osc}(F; (0, t_1, t_2, \ldots)) = 2$ for all $t_1, t_2, \ldots \in [0,1]$ and so

$$K_1(F, \varepsilon) = \{0\} \times [0,1]^{\omega \setminus \{0\}}$$

whenever $0 < \delta < 2$. Similar calculations show that if $r = [\![\frac{2}{\varepsilon}]\!]$ then

$$K_r(F, \varepsilon) = \{0\}^r \times [0,1]^{\omega \setminus r}$$

and $K_{r+1}(F, \varepsilon) = \emptyset$. Thus $K_m(F, \varepsilon) \neq \emptyset$ implies $m\varepsilon \leq 2$. On the other hand, for $m \geq 1$,

$$K_m\left(F, \left(2, 1, \frac{2}{3}, \cdots, \frac{2}{m}\right)\right) = \{0\}^m \times [0,1]^{\omega \setminus m} . \qquad \blacksquare$$

We conclude by mentioning some further problems for study, some of which have been raised above.

Problem 8.6. Classify (or give useful sufficient conditions) for a function $F \in B_1(K)$ to govern $\{X : X^*$ is separable and $\dim X = \infty\}$. In particular is $F \in B_{1/4}(K) \setminus C(K)$ a sufficient condition?

Problem 8.7. Classify those $F \in B_1(K)$ which govern $\{\ell_1\}$, which govern $\{c_0\}$, which govern $\{X : X$ is reflexive$\}$ or which govern $\{X : X$ is quasi-reflexive$\}$.

We note that if X is a Polish Banach space (i.e., $Ba(X)$ is Polish in the weak topology) then Edgar and Wheeler [14] have shown that X is hereditarily reflexive (see also [37] and [18]). Bellenot [5] and Finet [15] have independently extended this result by showing that whenever X is Polish, if $x^{**} \in X^{**} \setminus X$ then $x^{**}|_{Ba(X^*)}$ strictly governs the class of quasi-reflexive spaces of order 1.

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Normed spaces with a weak-Gordon-Lewis property

KEITH BALL

Trinity College
Cambridge
and
Texas A&M University
College Station, TX

Introduction.

Much interest has centered recently on so-called "weak" properties of Banach spaces, initiated by the paper [**10**] and depending strongly on the results from [4] that are quoted below. Weak properties provide a language in which to couch elegant theorems relating the subspace structure of Banach spaces to the integral geometry of their unit balls. The purpose of this paper is the discussion of some geometric properties of spaces with a weak version of the Gordon-Lewis property [**6**] and the clarification of present knowledge concerning the hyperplane conjecture for finite-dimensional convex bodies (stated in Section 2).

Several well-known definitions and theorems will be needed.

Definition. If X is an n-dimensional normed space represented on $\mathbf{R}^n$ with unit ball B_X then the volume ratio, $vr(X)$ of X is

$$vr(X) = \left(\frac{|B_X|}{|\mathcal{E}|}\right)^{1/n}$$

where $\mathcal{E}$ is the ellipsoid of maximal volume included in B_X (and $|\cdot|$ denotes volume).

Theorem. (Bourgain-Milman, [4]) *There is an absolute constant M such that for every finite-dimensional subspace X of L_1, $vr(X) \leq M$.*

(Bourgain and Milman actually proved that the volume ratio of a finite-dimensional space can be bounded solely in terms of its cotype-2 constant: the special case stated, is all that is needed here. *Remark*: this author recently showed that in the case stated above, M can be taken to be $\sqrt{2e/\pi}$, which is best possible.)

Theorem. (Pietsch, [12]) *Let X and Y be Banach spaces, $T : X \to Y$ linear and $1 \le p < \infty$. Then the p-absolutely summing norm, $\pi_p(T)$ of T is the least K (if such exists) for which there is a probability P on B_{X^*}, the unit ball of the dual of X, so that for every $x \in X$,*

$$\|Tx\| \le \left(\int_{B_X^*} |\phi(x)|^p \, dP(\phi) \right)^{1/p} . \tag{1}$$

For the purpose of this paper, inequality (1) can be regarded as the definition of $\pi_p(T)$.

It is a consequence of Lewis's duality theory for operator norms, [9] that, up to some absolute constant, the volume ratio $vr(X)$ of an n-dimensional space X is the least K so that for every operator $T : X \to \ell_2^n$,

$$|T(B_X)|^{1/n} \le \frac{K}{n} \pi_2(T) . \tag{2}$$

Thus, the preceding theorem of Bourgain and Milman states that there is an absolute constant M so that for an n-dimensional subspace X of L_1 and $T : X \to \ell_2^n$

$$|T(B_X)|^{1/n} \le \frac{M}{n} \pi_2(T) . \tag{3}$$

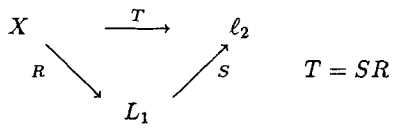
Although it is not strictly needed, the reverse Santalo inequality of Bourgain and Milman will also be used.

Theorem. ([4]) *There is an absolute constant $\delta > 0$ so that if C is a symmetric convex body in $\mathbf{R}^n$ and $C^* = \{y \in \mathbf{R}^n : |\langle x, y \rangle| \le 1 \text{ for } x \in C\}$ is its polar, then*

$$|C|^{1/n} |C^*|^{1/n} \ge \frac{\delta}{n} . \tag{4}$$

1. A weakening of the Gordon-Lewis property.

The Gordon-Lewis constant $gl_2(X)$ of a Banach space X (for which the constant is defined) is the least K with the property that for any $T : X \to \ell_2$ with $\pi_1(T)$ finite, there is a factorisation of T through an L_1-space



with $\|S\| \cdot \|R\| \le K\pi_1(T)$. The simplest examples of spaces with the Gordon-Lewis property (finite gl_2 constant) are those with unconditional bases and those that are subspaces of L_1 or (their duals) quotients of L_∞.

The following proposition demonstates a weak property possessed by spaces with small gl_2 constant.

Proposition 1. *There is an absolute constant M' so that if X is an n-dimensional normed space and $T : X \to \ell_1^n$ is linear then*

$$|T(B_X)|^{1/n} \leq \frac{M'}{n} gl_2(X)\pi_1(T) .$$

Proof. Choose a factorisation of T as

$$\begin{array}{ccc} X & \xrightarrow{T} & \ell_2^n \\ {\scriptstyle R}\downarrow & & \uparrow{\scriptstyle S} \\ R(X) & \xrightarrow[j]{} & L_1 \end{array}$$

with j the identical inclusion of the subspace $R(X)$ into L_1, $\|R\| = 1$ and $\|S\| \leq gl_2(X)\pi_1(T)$.

Since $\|R\| = 1$, $R(B_X) \subset B_{R(X)}$, the unit ball of $R(X)$, and hence

$$|T(B_X)| \leq |Sj(B_{R(X)})| .$$

Since $R(X)$ is a subspace of L_1,

$$|Sj(B_{R(X)})|^{1/n} \leq \frac{M}{n} \pi_2(Sj) \leq \frac{M}{n} \pi_2(S)$$

where M is the constant appearing in (3) and the second inequality is a consequence of the ideal property of π_2 and the fact that $\|j\| = 1$. By the "little" Grothendieck theorem (see [**13**]),

$$\pi_2(S) \leq \sqrt{\frac{\pi}{2}} \|S\|$$

since S acts from an L_1-space into a Hilbert space. So the proposition is established with

$$M' = \sqrt{\frac{\pi}{2}} M . \qquad \blacksquare$$

The following definition is motivated by Proposition 1 and the form of inequality (2).

Definition. Let X be an n-dimensional normed space. The weak-right-hand-Gordon-Lewis constant $wrgl_2(X)$ of X is the least K so that for any linear map $T : X \to \ell_2^n$,

$$|T(B_X)|^{1/n} \leq \frac{2K}{n} \pi_1(T) .$$

Remarks.

a) The ungainly terminology is used because a weak-gl_2 constant differing from that above was defined in Pisier's paper [**14**]. The distinction is explained in more detail in the appendix.

b) The factor 2 in the definition is included only for convenience. With the above definition $wrgl_2(\ell_\infty^n) = 1$ for each n; and ℓ_∞^n can be regarded as the canonical example of a space with small $wrgl_2$ constant.

c) Proposition 1 and inequality (2) show that there is a constant M' so that for any (finite-dimensional) X,

$$wrgl_2(X) \leq M' gl_2(X)$$

and

$$wrgl_2(X) \leq M' vr(X) \ . \tag{5}$$

d) Inequality (5) can be strengthened in the following way:

$$wrgl_2(X) \leq \left(\frac{|B_X|}{|Q|}\right)^{1/n} \tag{6}$$

where Q is an affine cube (parallelipiped) of maximal volume contained in B_X. For, if $T : X \to \ell_2^n$ is linear and $S : \ell_\infty^n \to X$ is a linear map taking the unit ball of ℓ_∞^n onto Q,

$$\begin{aligned} |T(B_X)|^{1/n} &= \left(\frac{|B_X|}{|Q|}\right)^{1/n} |T(Q)|^{1/n} \\ &= \left(\frac{|B_X|}{|Q|}\right)^{1/n} |TS(B_\infty^n)|^{1/n} \\ &\leq \left(\frac{|B_X|}{|Q|}\right)^{1/n} \frac{2wrgl_2(\ell_\infty^n)}{n} \pi_1(TS) \\ &\leq \frac{2}{n} \left(\frac{|B_X|}{|Q|}\right)^{1/n} \pi_1(T) \ . \end{aligned}$$

The $wrgl_2$ constant is an isomorphic invariant of finite-dimensional normed spaces and so if $\dim X = n$, $wrgl_2(X)$ is at most a fixed multiple of $\sqrt{n}$ (in fact $wrgl_2(X) \leq \sqrt{en}$). It is worth noting that, apart from the constant factor, this is all that can be said in general.

Example 2. *There is an absolute constant $\delta > 0$ and, for each n, an n-dimensional space X with $wrgl_2(X) \geq \delta\sqrt{n}$.*

Proof. A lemma of Figiel and Johnson, [5] states that there are constants α and $\beta > 0$ so that if Y is m-dimensional and $T : Y \to \ell_2^m$ is linear then there is a subspace X of Y with $\dim X \geq \alpha m$ and $\pi_1(T|_X) \leq \beta^{-1}\pi_2(T)$.

Consider the formal identity $I : \ell_\infty^m \to \ell_2^m$. $\pi_2(I) = \sqrt{m}$. Let X be a subspace of ℓ_∞^m of dimension $n \geq \alpha m$ on which

$$\pi_1(I|_X) \leq \beta^{-1}\sqrt{m} \leq \beta^{-1}\alpha^{-1/2}\sqrt{n} \ .$$

$I|_X$ is a linear map from X into an isometric copy of ℓ_2^n and

$$|I|_X(B_X)|^{1/n} = |X \cap B_\infty^m|^{1/n}$$

where B_∞^m is the cube $[-1,1]^m \subset \mathbf{R}^m$. A result of Vaaler, [17] states that $|X \cap B_\infty^m|^{1/n} \geq 2$. Hence

$$wrgl_2(X) \geq n(\beta^{-1}\alpha^{-1/2}\sqrt{n})^{-1}$$
$$= \sqrt{\alpha}\,\beta\sqrt{n}\ .$$

∎

2. Geometric properties of weak-Gordon-Lewis spaces.

The hyperplane conjecture for finite-dimensional convex bodies states that there is a $\delta > 0$ so that if C is a symmetric convex body in $\mathbf{R}^n$, $n \geq 1$, then there is a 1-codimensional subspace H of $\mathbf{R}^n$ for which

$$|H \cap C| \geq \delta |C|^{(n-1)/n}\ . \tag{7}$$

A result of Hensley, [8] shows that if this conjecture is true, any normed space X of dimension $n \geq 1$ may be represented on $\mathbf{R}^n$, with unit ball B (say) so that all 1-dimensional sections of B satisfy (7) (with a slightly worse constant). Thus, it is natural to regard the conjecture as a statement about normed spaces, *i.e.*, classes of affinely equivalent convex bodies, rather than individual bodies.

A collection of finite-dimensional normed spaces will be said to satisfy the hyperplane conjecture uniformly if there is a $\delta > 0$ so that for every space X in the collection and every representation of X on $\mathbf{R}^n$, $n = \dim X$, there is some hyperplane H for which

$$|H \cap B_X|^{1/n} \geq \delta |B_X|^{(n-1)/n}\ .$$

Uniform bounds are known for certain collections of normed spaces. Milman observed such an estimate for the spaces with 1-unconditional basis and Pajor for classes of spaces whose duals have uniformly bounded volume ratio. These two cases are included in the more general case of spaces whose dual balls contain large (in the volume ratio sense) affine cubes: estimates for the volumes of sections of balls of these spaces were obtained by Bourgain. In view of inequality (6) above, Bourgain's result is a consequence of the following theorem.

Theorem 3. *There is an absolute constant $\delta > 0$ such that if X is an n-dimensional normed space represented on $\mathbf{R}^n$ with unit ball B_X, then there is a 1-codimensional subspace H of $\mathbf{R}^n$ for which*

$$|H \cap B_X| \geq \delta \big(wrgl_2(X^*)\big)^{-1} |B_X|^{(n-1)/n}\ .$$

Proof. For a symmetric convex body C and unit vector θ in $\mathbf{R}^n$, consider the function $f : \mathbf{R} \to [0,\infty)$ obtained by scanning C with translates of $\langle\theta\rangle^\perp$ — the 1-codimensional space orthogonal to θ — *i.e.*, $f(x) = |\big(\langle\theta\rangle^\perp + x\big) \cap C|$. It follows from the Brunn-Minkowski inequality

that f attains its maximum at 0: the central section of a symmetric convex body is largest among all parallel sections. It is simple to check that under this condition

$$f(0)\int_{\mathbf{R}} f(x)|x|\,dx \geq \frac{1}{4}\left(\int_{\mathbf{R}} f(x)\,dx\right)^2$$

with equality if f is a multiple of the characteristic function of a symmetric interval $[-t,t]$. (This fact is essentially due to Hensley.) In terms of C and θ this says

$$|\langle\theta\rangle^{\perp} \cap C| \int_C |\langle x,\theta\rangle|\,d\lambda(x) \geq \frac{1}{4}|C|^2$$

where λ is Lebesgue measure on $\mathbf{R}^n$. So it suffices, for the theorem, to find a unit vector θ with

$$\frac{1}{|B_X|}\int_{B_X} |\langle x,\theta\rangle|\,d\lambda(x) \leq K wrgl_2(X^*)|B_X|^{1/n}$$

for some absolute constant K.

Let $T : X^* \to \ell_2^n$ be the formal identity map and P the probability on B_X which is normalised Lebesgue measure. Then (by the Pietsch Theorem, (1)) there is a vector θ of Euclidean norm 1 with

$$\begin{aligned} 1 = |\theta| &\geq \pi_1(T)\int_{B_X} |\langle x,\theta\rangle|\,dP(x) \\ &= \pi_1(T)\frac{1}{|B_X|}\int_{B_X} |\langle x,\theta\rangle|\,d\lambda(x)\ . \end{aligned}$$

But $|B_{X^*}|^{1/n} \leq (2wrgl_2(X^*)/n)\,\pi_1(T)$ so that

$$\begin{aligned} \frac{1}{|B_X|}\int_{B_X} |\langle x,\theta\rangle|\,d\lambda(x) &\leq \frac{2wrgl_2(X^*)}{n}\,|B_{X^*}|^{-1/n} \\ &\leq K wrgl_2(X^*)|B_X|^{1/n} \end{aligned}$$

where $2/K$ is the constant appearing in the reverse Santalo inequality (4). ∎

Remark. Probably the most interesting consequence of Theorem 3 is that the hyperplane conjecture holds uniformly for spaces with uniformly bounded Gordon-Lewis constants. Although the argument demonstrating this, made use of the reverse Santalo inequality of Bourgain and Milman, an inspection shows that it would have been enough to use such an inequality for the unit balls of subspaces of L_1 (and apply it during the proof of Proposition 1 instead of Theorem 3). For subspaces of L_1 the best possible constant in (4) is known (Reisner [**15**]) and the inequality now has a very short proof, [**7**].

The next result shows that spaces with small $wrgl_2$ constants have a geometric property that is intuitively stronger than that implied by Theorem 3. The isoperimetric inequality in

$\mathbf{R}^n$ states that the surface area of a convex body of given volume is mimimal if that body is an Euclidean ball. So if C is a (symmetric) convex body in $\mathbf{R}^n$

$$\frac{|\partial C|}{|C|^{(n-1)/n}} \geq n v_n^{1/n} \sim \sqrt{2\pi e n}$$

where v_n is the volume of the Euclidean unit ball in $\mathbf{R}^n$. While it is impossible to obtain an upper estimate for $|\partial C|/|C|^{(n-1)/n}$ for symmetric convex bodies, the affine invariant formulation of this problem can be solved. It is not hard to see that every n-dimensional normed space can be represented on $\mathbf{R}^n$ with a unit ball C satisfying

$$\frac{|\partial C|}{|C|^{(n-1)/n}} \leq \sqrt{2\pi e}\, n \,. \tag{8}$$

This estimate cannot be essentially improved: if C is the cube in $\mathbf{R}^n$ (the unit ball of the "best" representation of ℓ_∞^n) then

$$\frac{|\partial C|}{|C|^{(n-1)/n}} = 2n \,.$$

(It follows from Theorem 3 of [1] that in fact ℓ_∞^n is exactly the worst case: inequality (8) can be improved to $\left(|\partial C|/|C|^{(n-1)/n}\right) \leq 2n$.)

Now for any convex body C,

$$|\partial C| = \frac{n v_n}{v_{n-1}} \int_{S^{n-1}} |P_\theta C|\, d\sigma(\theta)$$

where σ is normalised rotation invariant measure on the sphere and $P_\theta C$ is the orthogonal projection of C onto the 1-codimensional space $\langle\theta\rangle^\perp$. Since $(n v_n / v_{n-1}) \sim \sqrt{2\pi n}$ it is to be expected that the projections of the cube $Q_n = [-1,1]^n$ should have volume about $\sqrt{n}$ times the homogenised volume $|Q_n|^{(n-1)/n}$. Most do, but the projections onto the coodinate hyperplanes have volume equal to $|Q_n|^{(n-1)/n}$. This property of having at least a few small projections is shared by the unit balls of other spaces with small $wrgl_2$ constants.

Proposition 4. *Let X be an n-dimensional normed space represented on $\mathbf{R}^n$ with unit ball B_X. Then there is a unit vector θ in $\mathbf{R}^n$ for which*

$$|P_\theta B_X| \leq wrgl_2(X) |B_X|^{(n-1)/n} \,.$$

Proof. It is well-known (and easily checked, for example by considering polytopes) that there is a probability measure P on the polar body B_{X^*} of B_X so that for any unit vector $\theta \in \mathbf{R}^n$,

$$|P_\theta B_X| = \frac{n|C|}{2} \int_{B_{X^*}} |\langle x, \theta\rangle|\, dP(x) \,.$$

Applying the Pietsch theorem to the formal identity: $X \to \ell_2^n$ as in the preceding proof yields the desired assertion. ∎

Just as the constant in inequality (2) for a normed space X is essentially the volume ratio of X, so the $wrgl_2$ constant of a space may be interpreted in "volume ratio" terms. Recall that a zonoid in $\mathbf{R}^n$ is the unit ball of an n-dimensional quotient of ℓ_∞.

Theorem 5. *Let X be a n-dimensional space. Then*

$$wrgl_2(X) \approx \left(\frac{|B_X|}{|Z|}\right)^{1/n}$$

where Z is a zonoid of maximal volume included in B_X. (The $\approx$ sign indicates that the ratio of the two expressions is bounded between positive constants independently of the dimension and the particular space X.)

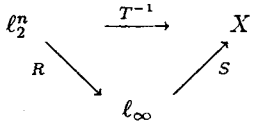
Proof. Since quotients of ℓ_∞ have uniformly bounded gl_2 constants, they have uniformly bounded $wrgl_2$ constants. The argument of inequality (6) now shows that $wrgl_2(X)$ is bounded above by a fixed multiple of

$$\left(\frac{|B_X|}{|Z|}\right)^{1/n} .$$

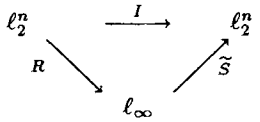
The inequality in the opposite direction uses the duality between the π_1 norm and the ℓ_∞ factorisation norm. It is possible to choose an Euclidean structure on X so that if T and T^{-1} are the formal identities

$$\ell_2^n \xrightarrow{T^{-1}} X \xrightarrow{T} \ell_2^n$$

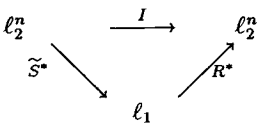
then $\pi_1(T) = n$ and T^{-1} factorises through ℓ_∞



with $\|S\| = \|R\| = 1$. On the one hand this gives that $|B_X|^{1/n} = |T(B_X)|^{1/n} \leq 2wrgl_2(X)$. On the other hand, the image $S(B_\infty)$ of the unit ball of ℓ_∞ is a zonoid included into B_X. So it remains to estimate the volume of $S(B_\infty)$ from below. Consider the factorisation of the identity $I = TT^{-1}$ through ℓ_∞



where $\widetilde{S} = TS$, and the dual factorisation



$|S(B_\infty)| = |\widetilde{S}(B_\infty)|$: so, by the reverse Santalo inequality for subspaces of L_1, it suffices to find a constant K (independent of n and X) so that the polar $\widetilde{S}(B_\infty)^*$ satisfies

$$|\widetilde{S}(B_\infty)^*|^{1/n} \leq \frac{K}{n} .$$

But $\widetilde{S}(B_\infty)^* = R^*(B_{\widetilde{S}^*\ell_2^n})$. Since $\pi_2(R^*) \leq \sqrt{\pi/2}\,\|R^*\| = \sqrt{\pi/2}$ by the "little" Grothendieck Theorem, and $\widetilde{S}^*\ell_2^n$ is a subspace of ℓ_1,

$$|R^*(B_{\widetilde{S}^*\ell_2^n})|^{1/n} \leq \frac{\sqrt{\pi/2}\,M}{n}$$

as in the proof of Proposition 1. ∎

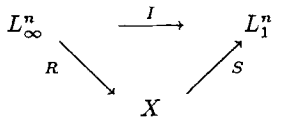
In view of inequality (6) it is natural to ask whether all spaces with small $wrgl_2$ constant have unit balls containing large affine cubes rather than, merely, large zonoids. By Theorem 5, the answer is yes if all zonoids contain large affine cubes.

Conjecture. *There is an absolute constant K so that if Z is an n-dimensional zonoid then*

$$\left(\frac{|Z|}{|Q|}\right)^{1/n} \leq K$$

where Q is an affine cube of maximal volume included in Z.

The results of [3] together with a simple combinatorial argument show that this conjecture is true apart from a power of $\log n$. The above conjecture weakens the conjecture implicit in [3] that there is an absolute constant γ so that any n-dimensional quotient of ℓ_∞ is γ-isomorphic to a quotient of $\ell_\infty^{\gamma n}$. It also weakens a conjecture of Pełczyński and Schütt, [11] stating that there is an absolute constant K so that if X is an n-dimensional normed space, there is a factorisation of the natural injection $I : L_\infty^n \to L_1^n$



with $\|S\| \cdot \|R\| \leq K gl_2(X)$.

3. Random unconditional systems.

The proposition below exhibits a link between a one-sided unconditionality property considered by Billard, Kwapien, Pełczyński and Samuel [2] and the conjecture discussed at the end of the last section.

Definition. Let X be a finite-dimensional space. The random unconditional constant $\mathrm{ruc}(X)$ of X is the least K for which X possesses a basis $x_1, \ldots, x_n$ satisfying

$$\left(\int \Big\| \sum_i \varepsilon_i a_i x_i \Big\|^2 d\varepsilon\right)^{1/2} \leq K \Big\| \sum_i a_i x_i \Big\|$$

for any sequence $(a_i)_1^n$ of scalars, the integral being the average over all choices of n signs $\varepsilon_1, \ldots, \varepsilon_n$.

Proposition 6. *Let X be an n-dimensional space. Then B_X contains an affine cube Q with*

$$\left(\frac{|B_X|}{|Q|}\right)^{1/n} \leq \frac{\pi e}{2} \operatorname{ruc}(X^*) .$$

Proof. Let $r = \operatorname{ruc}(X^*)$.

It is well-known that the unit ball B_Y of an n-dimensional space Y with 1-unconditional basis contains an affine cube Q with

$$\left(\frac{|B_Y|}{|Q|}\right)^{1/n} \leq \frac{n}{(n!)^{1/n}} ,$$

the extreme case being $Y = \ell_1^n$ (see for example [11]). By Santalo's inequality and the reverse Santalo inequality for spaces with 1-unconditional basis, due to Saint-Raymond [16],

$$|B_X|\,|B_{X^\bullet}| \leq v_n^2$$

and

$$|B_Y|\,|B_{Y^\bullet}| \geq \frac{4^n}{n!}$$

for such a space Y. Since $\left[(n/(n!)^{1/n})(v_n^{2/n}(n!)^{1/n}/4)\right] < (\pi e/2)$ it suffices to show that the unit ball $B_{X^\bullet}$ of X^* is included into the ball of a space Y^* with 1-unconditional basis with

$$\left(\frac{|B_{Y^\bullet}|}{|B_{X^\bullet}|}\right)^{1/n} \leq \operatorname{ruc}(X^*) = r .$$

Assume that X^* is represented on $\mathbf{R}^n$ so that the standard basis $e_1, \ldots, e_n$ constitutes a normalised sequence in X^* satisfying

$$\left(\int \left\| \sum \varepsilon_i a_i e_i \right\|_{X^\bullet}^2 d\varepsilon\right)^{1/2} \leq r \left\| \sum a_i e_i \right\|_{X^\bullet}$$

for every sequence of scalars (a_i). Let Y^* be $\mathbf{R}^n$ equipped with the norm

$$\left\| \sum a_i e_i \right\| = r^{-1} \left(\int \left\| \sum \varepsilon_i a_i e_i \right\|_{X^\bullet}^2 d\varepsilon\right)^{1/2} .$$

Then $B_{X^\bullet} \subset B_{Y^\bullet}$ and Y^* has a 1-unconditional basis. Moreover, for any sequence (a_i) of scalars

$$\begin{aligned} \left\| \sum a_i e_i \right\| &= r^{-1} \left(\int \left\| \sum \varepsilon_i a_i e_i \right\|^2 d\varepsilon\right)^{1/2} \\ &\geq r^{-1} \left(\int \left\| \sum \varepsilon_i a_i e_i \right\|^{-n} d\varepsilon\right)^{-1/n} \end{aligned}$$

by Hölder's inequality. Hence

$$\left\|\sum a_i e_i\right\|^{-n} \leq r^n \int \left\|\sum \varepsilon_i a_i e_i\right\|^{-n} d\varepsilon \ .$$

Integration over $a = (a_1, \ldots, a_n) \in S^{n-1}$ makes redundant the average over choice of signs, so

$$\int_{S^{n-1}} \left\|\sum a_i e_i\right\|^{-n} d\sigma(a) \leq r^n \int_{S^{n-1}} \left\|\sum a_i e_i\right\|^{-n} d\sigma(a)$$

and so $|B_{Y^*}|^{1/n} \leq r|B_{X^*}|^{1/n}$. ∎

Remark. Proposition 6, Example 2 and inequality (6) together answer the following question from [2] — can the ruc constant of an n-dimensional normed space be as large as (a fixed multiple of) $\sqrt{n}$? The above results show that the answer is yes for the duals of "random" subspaces of ℓ_∞^m, n a proportion of m.

Acknowledgement. I would like to thank Professor G. Pisier for his stimulating and helpful advice concerning the material presented in this paper.

Appendix.

In his paper [14], Pisier described a canonical method for weakening Banach space invariants and, in particular, defined the following weak-Gordon-Lewis property. A Banach space X has weak-gl_2 if there is a constant K so that for all n and any two operators

$$S : \ell_2^n \to X \quad , \quad T : X \to \ell_2^n$$

the composition TS satisfies

$$\|TS\|_{c_{1,\infty}} \leq K\pi_1(S^*)\pi_1(T) \ ,$$

i.e.,

$$\sup_k k\lambda_k(|TS|) \leq K\pi_1(S^*)\pi_1(T) \ , \tag{9}$$

where λ_k is the kth largest eigenvalue (according to multiplicity) of the operator $|TS|$. An argument similar to that of Proposition 1 shows that the $wrgl_2$ constant is dominated by (a fixed multiple of) the constant appearing in (9). Like the gl_2 property itself, Pisier's weak-gl_2 property is self-dual. The structural property measured by the $wrgl_2$ constant is far from self-dual: the duals of the spaces in Example 2 have $wrgl_2$ constants at most a power of $\log n$, n being the dimension. The $wrgl_2$ constant is effectively a one-sided weak-gl_2 constant. Working backwards it would be possible to define one-sided gl_2 properties using the ℓ and ℓ^* norms (see *e.g.*, [14] for definitions). Whether such properties would be useful is unclear.

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On the geometry of the Banach-Mazur compactum

S.J. SZAREK*

Case Western Reserve University
and
Institut des Hautes Etudes Scientifiques

The purpose of this note is to present some results concerning the geometry of the set of the n-dimensional compact convex symmetric bodies, *the Banach-Mazur compactum*, obtained recently by the author and his collaborators (J. Bourgain and M Talagrand; [3], [17], [19]).

Recall that if X, Y are normed spaces, we define *the Banach-Mazur "distance"* $d(X,Y)$ by

$$d(X,Y) \equiv \inf\{\|T\| \cdot \|T^{-1}\| \ : \ T : X \to Y \text{ an isomorphism}\} \ .$$

If we set $\mathcal{B}_n \equiv \{X : X\text{-normed space, } \dim X = n\}$, then $(\mathcal{B}_n, \log d)$ is a compact metric space, usually called *the Banach-Mazur compactum* (or *the Minkowski compactum*). The link to the set of the n-dimensional convex symmetric bodies is of course provided by associating with each X its unit ball B_X.

It is a well-known result due to F. John [9] that, for every compact symmetric convex body $B \subset \mathbb{R}^n$, there is an affine transformation T such that $D \subset T(B) \subset n^{1/2}D$, where D is the Euclidean unit ball. In the language of the local theory of Banach spaces as introduced above this means that, for *any* n-dimensional normed space X, $d_X \equiv d(X, \ell_2^n) \leq n^{1/2}$. The constant $n^{1/2}$ is clearly the best possible, as is seen by considering $X = \ell_\infty^n$ or ℓ_1^n. One may say that the "radius" of $\mathcal{B}_n$ relative to the center ℓ_2^n is $n^{1/2}$.

It follows right away from the John's result that the "*diameter*" of $\mathcal{B}_n$ is $\leq n$, while, on the other hand, by the remarkable result of Gluskin [6], that "diameter" is $\geq cn$, where $c > 0$ is a numerical constant.

The above facts lead to a number of natural questions. First, the problem of *uniqueness* of the center (in the asymptotic sense); *i.e.*, given $\lambda \geq 1$, does there exist a constant $C(\lambda)$ (independent of n) such that whenever, for some $X \in \mathcal{B}_n$, $\max\{d(X,Y) : Y \in \mathcal{B}_n\} \leq \lambda n^{1/2}$, then $d(X, \ell_2^n) \leq c(\lambda)$? In particular, could other "standard" spaces like *e.g.*, the unit balls of

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ℓ_∞^n or ℓ_1^n be "*asymptotic centers*"; *i.e.*, denoting

$$\mathcal{R}_\infty^n \equiv \max\{d(X, \ell_\infty^n) : X \in \mathcal{B}_n\} ,$$

do we have $\mathcal{R}_\infty^n \leq Cn^{1/2}$, where C is a universal constant? If not, what is the order of growth of $\mathcal{R}_\infty^n$ as $n \to \infty$? Some of these problems "were around" for quite a few years, all of them were stated by A. Pełczyński in his 1983 Congress talk [**14**]. The two theorems that follow answer the first two questions negatively and shed some light on the third one.

Theorem 1. *We have*

$$cn^{1/2} \log n \leq \mathcal{R}_\infty^n \leq Cn^{7/8}$$

where $C, c > 0$ are universal constants.

Theorem 2. *If X is a normed space with $\dim X = n$ and $s = [n/2]$, then the Banach-Mazur distance of $Y = \ell_2^s \oplus \ell_1^{n-s}$ (or $\ell_2^s \oplus \ell_\infty^{n-s}$) to X does not exceed $Cn^{1/2}$, C – a universal constant. Consequently, Y is an "asymptotic center" of the Banach-Mazur compactum $\mathcal{B}_n$ satisfying $d(Y, \ell_2^n) \geq (n/2)^{1/2}$.*

Some of the tools used to prove the Theorems are interesting in their own right. This applies in particular to Lemma 6, which is an "isomorphic" variant of *the Sauer-Shelah lemma* from combinatorics ([**15**], [**16**]), and to Lemma 8, which deals with the distribution of eigenvalues (and singular numbers) of random matrices and is motivated by *the Wigner's Semicircle Law* [**22**] (established in nuclear physics and widely used in statistics).

Remarks. 1) The lower estimate from Theorem 1 was shown in [**17**]. The upper estimate was stated in [**19**]; a somewhat weaker upper bound $Cn \exp(-c(\log n)^{1/2})$ was obtained earlier in the paper [**3**], which also contains Theorem 2. Besides the known value of $\mathcal{R}_\infty^2 = 3/2$ (attained when B_X is a regular hexagon) [**1**], only the trivial estimates $n^{1/2} \leq \mathcal{R}_\infty^n \leq n$ were apparently available before the work described here.

2) The proof of Theorem 1 can be easily modified to give a somewhat more precise estimate: if $d_X \equiv d(X, \ell_2^n)$ $(\leq n^{1/2})$, then

$$\max\{d(X, \ell_\infty^n), d(X, \ell_1^n)\} \leq Cn^{1/2} d_X^{3/4} .$$

One can also similarly obtain a nontrivial estimate for $d(X, \ell_p^n)$ (for any $p \in [1, \infty]$) of the form $Cn^\alpha d_X^{1-\alpha/2}$, where $\alpha = |1/p - 1/2|$.

3) The hypothesis of symmetry of the convex body in question cannot be omitted. Indeed, a simple geometric argument shows that if Σ is an n-dimensional simplex, $K \subset \mathbb{R}^n$ –

any symmetric convex body (in particular the Euclidean ball, cube or a ball in any ℓ_p^n) and $K \subset \Sigma \subset \lambda K$, then $\lambda \geq n$.

4) There is still a significant gap between the lower and upper estimates in Theorem 1. Most of the "evidence" suggests that the actual order of $\mathcal{R}_\infty^n$ is close to the lower one. However, even though it is quite clear that the upper one is not sharp, any progress going further than, say, reducing the exponent from 7/8 to 3/4 (which can possibly be done by organizing the present argument somewhat better) would require a new approach like *e.g.*, a properly set up iteration. On the other hand, any improvement in the lower estimate, if possible, would presumably involve modifications to the argument from [17] in the spirit of Gordon's approach to the Dvoretzky's theorem on almost spherical sections of convex bodies [7], [8]; see also a comment at the end of our discussion of the proof of the present result.

The next theorem, which is closely related to Theorem 2 and to the upper estimate in Theorem 1, improves the result of Dvoretzky-Rogers [5]. One may call it "*the proportional Dvoretzky-Rogers factorization.*"

Theorem 3. ([19], see also [3]) *Given $\delta \in (0,1)$ and a finite dimensional normed space X, there is an integer $m > (1-\delta)\dim X$ and vectors $x_1, \ldots, x_m \in X$ such that, for any scalars $t_1, \ldots, t_m$,*

$$\max_{j \leq m} |t_j| \leq \Big\| \sum_{j \leq m} t_j x_j \Big\| \leq C\delta^{-2} \Big(\sum_{j \leq m} |t_j|^2 \Big)^{1/2},$$

where C is a universal constant. In other words, the formal identity $i_{2,\infty} : \ell_2^m \to \ell_\infty^m$ can be written as $i_{2,\infty} = \alpha \circ \beta$, $\beta : \ell_2^m \to X$, $\alpha : X \to \ell_\infty^m$, with $\|\alpha\| \cdot \|\beta\| \leq C\delta^{-2}$. The same holds for the formal identity $i_{1,2} : \ell_1^m \to \ell_2^m$.

However, in general, the assertion of Theorem 3 doesn't hold when $m = \dim X$; we have

Theorem 4. ([17]) *Given $m \in \mathbb{N}$, there exists an m-dimensional normed space X such that, for any α, β as in Theorem 3, one has $\|\alpha\| \cdot \|\beta\| \geq c(m/\log m)^{1/10}$, where $c > 0$ is a universal constant.*

Remarks. 5) [5] contained a version of Theorem 3 with $m = [(\dim X)^{1/2}]$, used later in essentially all proofs of the Dvoretzky's theorem. We do not know what is the optimal dependence on ε in Theorem 3 (it follows from Theorem 4 that it must be of power type) and the optimal dependence on m in Theorem 4; it may conceivably be even $O(m^{1/2})$.

6) The question about the existence of a "good" Dvoretzky-Rogers factorization was also asked in another language. We say that a sequence of (say, norm one) vectors (x_j) in a

normed space is *Besselian* (resp. *Hilbertian*) with constant C if, for any sequence of scalars (t_j), $C\|\sum_j t_j x_j\| \geq (\sum_j |t_j|^2)^{1/2}$ (resp. $\leq$ etc.). One asks then

Does there exist $C < \infty$, such that every finite dimensional Banach space admits a normalized basis, which is Besselian with constant C?

The author heard this question from P. Enflo about ten years ago; of course Theorem 4 provides a negative answer to it, as well as to the appropriate dual problem about Hilbertian bases.

7) It follows immediately from Theorem 3 that, given $\delta \in (0,1)$, there is a subspace E of X with $\dim E = m \geq (1-\delta)\dim X$ and $d(E, \ell_\infty^m) \leq C\delta^{-2}m^{1/2}$ (resp. $d(E, \ell_1^m)$). In the case of ℓ_1^m, this was proved, up to a logarithmic factor, in [2]; a version with "there exists δ" was shown, independently of [3] and approximately at the same time, by K. Ball.

In what follows we present some of the ideas and sketch some of the arguments involved in the proofs of the Theorems. Concerning the "positive" results (Theorem 3, Theorem 2 and the upper estimate from Theorem 1), we derive them from a single auxiliary statement, Proposition 5, and sketch the proof of the latter; this is going to include the "isomorphic" version of the Sauer-Shelah lemma mentioned before. For the lower bounds on the quantities in question (Theorem 4 and the lower estimate from Theorem 1), we will restrict outselves to describing the spaces realizing them; to stating the version of the Wigner's Semicircle Law, on which the argument is based and, finally, to some heuristic remarks concering the actual proof.

We use the standard notation from the local theory of Banach spaces as can be found *e.g.*, in [13] or [20]. Let us only mention the following: by $|\cdot|_p$ we denote the norm on ℓ_p^n, by $|\cdot|$ – the norm on a generic Hilbert space (and also the cardinality of a set). (e_j) is the standard vector basis of $\mathbb{R}^n$. $[S]$ denotes the linear span of a set S. We consider only the case of real spaces; however, all the statements hold also in, and all the arguments carry over (with minor modifications) to the complex case. The letters C, c, c_0, c' etc. are reserved for universal numerical constants; identical symbols may represent different numbers in different places.

As was mentioned above, Theorem 3, Theorem 2 and the upper estimate from Theorem 1 are consequences of the following result.

Proposition 5. *Let $X = (\mathbb{R}^n, \|\cdot\|)$ be a normed space and let $\varepsilon \in (0,1)$. Then there exist a Euclidean norm $|\cdot|$ on X, an integer $k \geq (1-\varepsilon)n$ and $x_1, x_2, \ldots, x_k \in X$, $\|x_j\| \leq 1$ for all j, satisfying*

(1) $|\cdot| \leq \|\cdot\| \leq 2^{1/2} d_X |\cdot|$, where $d_X \equiv d(X, \ell_2^n)$

(2) for any scalars $t_1, \ldots, t_k$,

$$\Big|\sum_{j\leq k} t_j x_j\Big| \geq c_0 \varepsilon^{3/2} n^{-1/2} \sum_{j\leq k} |t_j|$$

where $c_0 > 0$ is a universal constant.

We now show how Proposition 5 is used to derive the Theorems.

Proposition 5 $\Rightarrow$ Theorem 1 (upper estimate). We prove the stronger version indicated in the remark 2) following the Theorem. By duality, it is enough to estimate $d(X, \ell_1^n)$. Apply Proposition 5 to X (with ε to be specified later) to obtain an Euclidean structure on X and $x_1, \ldots, x_k$. Let $y_1, \ldots, y_{n-k}$ (note that $n - k \leq \varepsilon n$) be an orthogonal basis of the orthogonal complement of $[x_1, \ldots, x_k]$ (the linear span of x_j's), normalized so that $|y_j| = (2^{1/2} d_X)^{-1}$ and so $\|y_j\| \leq 1$ for all j. Consequently, for any sequences of scalars $(t_j)_{j \leq k}, (s_j)_{i \leq n-k}$,

$$\begin{aligned}
\sum_{j \leq k} |t_j| + \sum_{i \leq n-k} |s_i| &\geq \Big\| \sum_j t_j x_j + \sum_i s_i y_i \Big\| \\
&\geq \Big| \sum_j t_j x_j + \sum_i s_i y_i \Big| = \left(\Big| \sum_j t_j x_j \Big|^2 + \Big| \sum_i s_i y_i \Big|^2 \right)^{1/2} \\
&\geq 2^{-1/2} \left(\Big| \sum_j t_j x_j \Big| + \Big| \sum_i s_i y_i \Big| \right) \\
&\geq 2^{-1/2} \left\{ c_0 \varepsilon^{3/2} n^{-1/2} \sum_j |t_j| + (2^{1/2} d_X)^{-1} \left(\sum_{i \leq n-k} |s_i|^2 \right)^{1/2} \right\} \\
&\geq 2^{-1/2} \left\{ c_0 \varepsilon^{3/2} n^{-1/2} \sum_j |t_j| + (2^{1/2} d_X)^{-1} (\varepsilon n)^{-1/2} \sum_i |s_i| \right\} .
\end{aligned}$$

Now choose $\varepsilon = (2^{1/2} c_0 d_X)^{-1/2}$ to get the assertion. The case of the general p (from remark 2)) requires only standard modifications and so we omit the proof.

Proposition 5 $\Rightarrow$ Theorem 3 and Theorem 4. First observe that of the three versions given in the statement of Theorem 3, the first two are equivalent by the extension property of the space ℓ_∞ and the last two by duality. We will prove the last statement (about the factorization of $i_{1,2}$ through X). First apply Proposition 5 with $\varepsilon = \delta/2$; this shows that the linear operator defined by $x_j \to e_j$, $1 \leq j \leq k$ ($k \geq (1 - \delta/2)n$), considered as acting from $[x_1, \ldots, x_k] \subset (\mathbb{R}^n, |\cdot|)$ into ℓ_1^k is of norm $\leq (c_0 \varepsilon^{3/2} n^{-1/2})^{-1}$ and consequently, by the "little" Grothendieck Theorem (applied in the same context in the proof of Theorem 1.2 [4] or in [10]), it "well factors" through a diagonal operator acting from ℓ_2^k into ℓ_1^k. More precisely, there exist $(\lambda_j)_{j \leq k}$, $\sum_{j \leq k} |\lambda_j|^2 \leq 1$, such that, for any scalars $t_1, \ldots, t_k$,

$$(\pi/2)^{1/2} (c_0 \varepsilon^{3/2} n^{-1/2})^{-1} \Big| \sum_{j \leq k} t_j z_j \Big| \geq \left(\sum_{j \leq k} |t_j / \lambda_j|^2 \right)^{1/2} .$$

Now let $\tau \subset \{1,2,\dots,k\}$ be such that $|\tau| \geq (1-\varepsilon/2)k$ (which is $> (1-\delta)n$) and $|\lambda_j| \leq (2/\varepsilon k)^{1/2}$ for $j \in \tau$. Then

$$\Big|\sum_{j\in\tau} t_j z_j\Big| \geq c_1\delta^2\Big(\sum_{j\in\tau}|t_j|^2\Big)^{1/2} .$$

Finally, identify τ with $\{1,2,\dots,m\}$, define $\beta : \ell_1^m \to X$ by $\beta(t_j) = \sum t_j x_j$, $\alpha_0 : [x_j] \to \ell_2^m$ by $\alpha_0(\sum t_j x_j) = (t_j)$ and $\alpha : X \to \ell_2^m$ by $\alpha = \alpha_0 P$, where P is the orthogonal projection of $(\mathbb{R}^n, |\cdot|)$ onto $[x_j]$, to get the required factorization $i_{1,2} = \alpha \circ \beta$ with $\|\beta\| \leq 1$ and $\|\alpha\| \leq (c_1\delta^2)^{-1}$.

Now, to get Theorem 2, set $\varepsilon = 1/2$, $s = n - m$ and let ρ be an isometry of ℓ_2^s onto $(\ker P, |\cdot|)$. Define $W : \ell_2^s \oplus \ell_1^m \to X$ by $W(\mathbf{u},\mathbf{v}) = n^{-1/2}\rho(\mathbf{u}) + \beta(\mathbf{v})$, then $\|W : \ell_2^m \oplus \ell_1^k \to X\| \leq C'$. On the other hand, since $W^{-1} = (n^{1/2}\rho^{-1}(I-P), \alpha)$,

$$\|W^{-1} : X \to \ell_2^m \oplus \ell_1^k\| \leq \|W^{-1} : (\mathbb{R}^n, |\cdot|) \to \ell_2^m \oplus \ell_1^k\| \leq C'' n^{1/2} ,$$

proving Theorem 2.

The proof of Proposition 5 is based on the following

Lemma 6. *Let $\mathcal{D} \subset \{-1,1\}^n \subset \mathbb{R}^n$ be such that $|\mathcal{D}| \geq 2^{n-1}$ and let $\varepsilon \in (0,1)$. Then there exists $\sigma \subset \{1,2,\dots,n\}$, $|\sigma| \geq (1-\varepsilon)n$, satisfying*

$$\operatorname{absconv}(R_\sigma \mathcal{D}) \supset c\varepsilon[-1,1]^\sigma ,$$

where R_σ denotes the restriction map $(a_j)_{1\leq j\leq n} \to (a_j)_{j\in\sigma} \in \mathbb{R}^\sigma$ and $c > 0$ is a numerical constant.

Lemma 6 (proved in [**19**] is, for our purposes, a "more efficient" substitute for the original Sauer-Shelah lemma ([**15**], [**16**]). Recall that the latter one asserts that, in the same notation,

> *If $0 \leq m < n$ and $\mathcal{D} \subset \{-1,1\}^n$ satisfies $|\mathcal{D}| > \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{m}$, then there exists $\sigma \subset \{1,2,\dots,n\}$, $|\sigma| > m$, such that $R_\sigma \mathcal{D} = \{-1,1\}^\sigma$.*

However, contrary to our Lemma 6, the above result is not easily applicable when m is close to n (because the binomial coefficients $\binom{n}{s}$ are highly concentrated around the value $s = n/2$). Since the proof of the Lemma is very simple, we sketch it here.

Proof of Lemma 6. The original Sauer-Shelah lemma implies that if $|\mathcal{D}| \geq 2^{n-1}$, then there exists $\sigma_1 \subset \{1,2,\dots,n\}$, $|\sigma_1| \equiv n_1 \geq n/2$, such that $R_{\sigma_1}\mathcal{D} = \{-1,1\}^{\sigma_1}$; this can be read as the assertion of Theorem 1 with $\varepsilon = 1/2$ and $c\varepsilon$ replaced by 1. The idea is to prove first the Theorem for $\varepsilon = 2^{-s}$, $s \in \mathbb{N}$, by induction with respect to s. More precisely, one shows that, for any $s \in \mathbb{N}$, there exists $\sigma_s \subset \{1,2,\dots,n\}$, $|\sigma_s| \equiv n_s \geq (1-2^{-s})n$, such that

$$(*) \qquad \operatorname{absconv}(R_{\sigma_s}\mathcal{D}) \supset (2^s-1)^{-1}[-1,1]^{\sigma_s} ;$$

from this the lemma easily follows (*e.g.*, with $c = 1/3$). To complete the induction assume that $(*)$ holds for some s. By the "pigeon hole principle," there exists $\mathbf{a} = (a_j) \in \{-1,1\}^{\sigma_s}$ such that if $\mathcal{D}_s \equiv R_{\sigma_s}^{-1}(\mathbf{a}) \cap \mathcal{D}$, then $|\mathcal{D}_s| \geq 2^{n-n_s-1}$. Consequently, we are in a position to apply the inductive step 1 (as explained above) with n replaced by $n - n_s$ to obtain $\tau \subset \{1,2,\ldots,n\} \setminus \sigma_s$, $|\tau| \geq (n-n_s)/2$, such that for any $\mathbf{b} = (b_j) \in \{-1,1\}^\tau$ there is $\mathbf{d}' = (d_j) \in \mathcal{D}$ with $R_{\sigma_s}(\mathbf{d}') = \mathbf{a}$ and $R_\tau(\mathbf{d}') = \mathbf{b}$ (resp. $\mathbf{d}'' \in \mathcal{D}$, $R_{\sigma_s}(\mathbf{d}'') = \mathbf{a}$, $R_\tau(\mathbf{d}'') = -\mathbf{b}$). Set $\sigma_{s+1} = \sigma_s \cup \tau$. Considering points of the type $(\mathbf{d}' - \mathbf{d}'')/2$, one shows that $\operatorname{absconv}(R_{\sigma_{s+1}}\mathcal{D}) \supset \mathbf{0} \times \{-1,1\}^\tau$, where $\mathbf{0} = (0,\ldots,0) \in \mathbb{R}^{n_s}$. From this and the inductive hypothesis $(*)$ the assertion for $s+1$ readily follows.

Lemma 6 $\Rightarrow$ Proposition 5. We use first the observation that when $K \subset \mathbb{R}^n$ (eventually K will equal B_X) is a compact convex symmetric body such that the Euclidean ball D is the ellipsoid of smallest volume containing K and $\delta \in (0,1)$, then there exist $x_1, x_2, \ldots, x_m \in K$, $m \geq (1-\delta)n$, satisfying

$$\operatorname{dist}\bigl(x_j, [x_k : i \neq j]\bigr) \geq \delta^{1/2}$$

for $j = 1,2,\ldots,m$ ("dist" means the Euclidean distance; take *e.g.*, $x_1, x_2, \ldots, x_m$ such that $\operatorname{conv}\{\pm x_i : 1 \leq i \leq m\}$ has the largest m-dimensional volume among all sets of that type contained in K). Apply this with $\delta = \varepsilon/2$ and let $u_1, u_2, \ldots, u_m \in \mathbb{R}^n$, $|u_i| \leq (\varepsilon/2)^{-1/2}$ be such that $\langle u_i, x_j \rangle = \delta_{ij}$ for $i,j \leq m$. Then clearly

$$\Bigl(\operatorname{Ave} \Bigl|\sum_{j\leq m} \pm u_j\Bigr|^2\Bigr)^{1/2} = \Bigl(\sum_{j\leq m} |u_j|^2\Bigr)^{1/2} \leq (2m/\varepsilon)^{1/2}$$

and so there exists $\mathcal{D} \subset \{-1,1\}^s \subset \mathbb{R}^s$, $|\mathcal{D}| \geq 2^{s-1}$, such that if $(\delta_i) \in \mathcal{D}$, then

$$\Bigl|\sum_{j\leq s} \delta_j u_j\Bigr| \leq (4n/\varepsilon)^{1/2} .$$

Apply now Lemma 6 to obtain $\sigma \subset \{1,2,\ldots,n\}$, $|\sigma| \geq (1-\varepsilon)n$, such that each $(\delta_j)_{j\in\sigma} \in \{-1,1\}^\sigma$ belongs to $(c\varepsilon/2)^{-1} \operatorname{conv}(R_\sigma \mathcal{D})$. For a sequence of scalars $(t_j)_{j\in\sigma}$, set $\delta_j = \operatorname{sgn} t_j$; then

$$\begin{aligned}
\sum_{j\in\sigma} |t_i| &= \Bigl\langle \sum_{j\in\sigma} t_j x_j \,,\, \sum_{j\in\sigma} \delta_j u_j \Bigr\rangle \\
&\leq \max\Bigl\{ \Bigl\langle \sum_{j\in\sigma} t_j x_j, u \Bigr\rangle : u \in (c\varepsilon/2)^{-1}\mathcal{D} \Bigr\} \\
&\leq \Bigl|\sum_{j\in\sigma} t_j x_j\Bigr| \cdot (c\varepsilon/2)^{-1} \cdot (4n/\varepsilon)^{1/2} \\
&= \Bigl|\sum_{j\in\sigma} t_j x_j\Bigr| \cdot (4c^{-1})\varepsilon^{-3/2} n^{1/2} .
\end{aligned}$$

This is exactly the assertion (ii) from the Proposition (with $c_0 = c/4$). Of course we could have assumed to begin with that $K = B_X$ and that the Euclidean ball D is the ellipsoid of smallest volume containing B_X; this would yield (in general) $|\cdot| \leq \|\cdot\| \leq n^{1/2}|\cdot|$ in (i): enough to prove the Theorems, but not their improvements stated in the remarks. To get the precise estimate from (i) we need to "combine" the "smallest volume ellipsoid" with the "distance ellipsoid"; see [3] or [19] for details.

We now turn to the discussion of *the lower estimates*, *i.e.*, Theorem 4 and the lower bound in Theorem 1. The spaces realizing these bounds will be constructed by a random procedure (the idea of which comes from Gluskin [6]), which we now describe.

Let (γ_j) be a sequence of independent (scalar) Gaussian random variables (G.r.v.), defined on some probability space $(\Omega, \mathbb{P})$ and distributed according to the $N(0,1)$ law, and let

$$(+) \qquad g = n^{-1/2} \sum_{j \leq n} \gamma_j e_j$$

(an $\mathbb{R}^n$-valued G.r.v.). Let (g_j) be a sequence of independent copies of g, there is no harm in thinking of g_j's as "more or less" evenly distributed on the Euclidean sphere. For $m \geq n$, we define a *random body*

$$B = B(\omega) = \operatorname{absconv}(\{g_1, \ldots, g_m\}) \subset \mathbb{R}^n$$

and identify it with the *random normed space*, whose unit ball is B. B can be alternatively described as a quotient of ℓ_1^m via the random quotient map Γ, where Γ is an $n \times m$ Gaussian matrix with independent $N(0, 1/n)$ entries (and $g_j = \Gamma e_j$). We have the following

Claim 7. *There exists a universal constant $c > 0$ such that, in the above notation, we have, for every $n \in \mathbb{N}$,*

(1) if $m = 2n$, then

$$\mathbb{P}(\{B^* \text{ satisfies the assertion of Theorem 4}\}) > 1 - \exp(-cn)$$

(2) if $m = [n^{5/4}]$, then

$$\mathbb{P}(\{d(B^*, \ell_\infty^n) > cn^{1/2} \log n\}) > 1 - \exp(-cn)\,.$$

It would be of interest to find *explicit* spaces satisfying the assertion of Theorem 4 (or $d(\cdot, \ell_\infty^n) > cn^{1/2} \log n$, or the conclusion of [6] etc.); the argument we sketch shows just their existence. Let us remark here that recently several phenomena, previously realizable only by random methods (random spaces, random graphs and random operators respectively), were shown to be possible to exhibit in concrete situations; see [11], [12], [21] and their references.

We shall not present the complete proof of Claim 7 here. The general scheme (say, to prove the part (1)) involves, roughly speaking, two steps:

(i) "The concentration argument": *for any fixed factorization of* $i_{1,2}$ *through* $\mathbb{R}^n$, *say* $i_{1,2} = \alpha \circ \beta$, *the product* $\|\beta : \ell_1^n \to B\| \cdot \|\alpha : B \to \ell_2^n\|$ *is "large" for "nearly all" spaces* $B(\omega)$ *(say, with probability* $> 1 - \delta$ *)*

(ii) "The ε-net argument": *there exists a finite set of factorizations* $\{\alpha_v, \beta_v : i_{1,2} = \alpha_v \circ \beta_v\}_{v \in I}$ *(with* $|I|$ *not "too large") such that, for "most of" the spaces* $B(\omega)$, *the order of* $\inf\{\|\alpha\| \cdot \|\beta\|\}$ (inf *over all factorizations), may be found by looking just at* $\{\alpha_v, \beta_v\}$.

Once proper statements of the type (i) and (ii) are shown, it follows that, still for "most of" $B(\omega)$'s, $\|\alpha\| \cdot \|\beta\|$ is "large" for *all* factorizations. Note, however, that for the above scheme to work, the product $\delta|I|$ must be "small" (say, $< 1/2$), and so we need *very* sharp estimates on both factors.

In our case (and typically in these types of problems) the step (ii) is relatively routine, once the right framework is figured out. For producing "ε-nets", one uses the Sudakov's inequality or its relatives, or even much simpler "volumic" arguments (see [**17**] for details). On the other hand, the step (i) is quite complicated here; as an illustration, we present calculations relevant to factorizations having some special form. For this, we need to describe the framework of our probabilistic argument somewhat more precisely.

Suppose that, for some $B = B(\omega)$, $\lambda \geq 1$ and some factorization $i_{1,2} = \alpha \circ \beta$, one has $\|\beta : \ell_1^n \to B\| \cdot \|\alpha : B \to \ell_2^n\| \leq \lambda$. Without loss of generality, one can assume that $\|\alpha\| \leq \lambda$, $\|\beta\| \leq 1$ and that $\beta = \Gamma \circ \overline{\beta}$, $\|\overline{\beta} : \ell_1^n \to \ell_1^m\| = \|\beta\| \leq 1$ (*i.e.*, $\overline{\beta}$ is a norm preserving lifting of β). In fact, rather than working directly with the factorizations, we shall investigate the existence (for a given Γ, hence B) of $\overline{\beta}$ as above giving rise to α satisfying $\|\alpha\| \leq \lambda$; note that, for a fixed Γ, α and β are uniquely determined. The condition $\|\alpha\| \leq \lambda$ means then that

$$(**) \qquad g_j = \Gamma e_j \in \lambda \beta(D) = \lambda \Gamma \overline{\beta}(D) \quad \text{for} \quad j = 1, 2, \ldots, m \ ,$$

where D is the Euclidean ball. Let us analyse that condition more closely for a particular $\overline{\beta}$ defined by $\overline{\beta} e_i = e_i$ for $i = 1, 2, \ldots, n$ (the general case is much more complicated, but carries similar features). As $\Gamma\overline{\beta}(D) = \{\sum_{j \leq n} t_j g_j : \sum_{j \leq n} |t_j|^2 \leq 1\}$ then, $(**)$ is automatically satisfied for $j \leq n$. On the other hand, for $j > n$, g_j's and $\Gamma\overline{\beta}(D)$ are independent; furthermore, $G \equiv \Gamma\overline{\beta}$ is an $n \times n$ Gaussian matrix with independent $N(0, 1/n)$ entries. We thus have, for

our particular $\overline{\beta}$,

$$\mathbb{P}\big(\{\|\alpha\| \le \lambda\}\big) = \mathbb{P}\Big(\big\{g_j \in \lambda G(D) \ \text{ for } \ n < j \le m\big\}\Big)$$
$$= \mathbb{P}\Big(\big\{g \in \lambda G(D)\big\}\Big)^{m-n} = \mathbb{P}\big(\{|G^{-1}g|_2 \le \lambda\}\big)^{m-n} ,$$

where g has the same meaning as in $(+)$ (and is independent from G). Accordingly, the problem reduces to analysis of the distribution of $|G^{-1}g|_2$. Observe that, for a fixed G,

$$(\#) \qquad (\mathbb{E}|G^{-1}g|_2^2)^{1/2} = n^{-1/2}\Big(\sum_{j\le n} s_j(G^{-1})^2\Big)^{1/2} = n^{-1/2} hs(G^{-1}) ,$$

where, for an $n\times n$ matrix A, $(s_j(A))_{1\le j\le n}$ are its *s-numbers* (*i.e.*, the eigenvalues of $|A|$), which we choose to arrange in the *nondecreasing* order, and $hs(A) \equiv \mathrm{tr}(|A|^2)^{1/2} = (\sum_{j\le n} s_j(A)^2)^{1/2}$ is the Hilbert-Schmidt norm. This is where we are going to need the information about the distribution of eigenvalues of random matrices (our version of *the Wigner's Semicircle Law*). We use the following

Lemma 8. *There exist universal constants $C, c, c_0 > 0$ such that,*

$$(\$) \qquad \mathbb{P}\big(\{cj/n \le s_j(G) \le Cj/n\}\big) > 1 - \exp(-c_0 j^2)$$

for all $j \le n/2$.

Remark. 8) In fact, the following more precise statement from [18] (used there to solve a problem from numerical analysis) holds.

There exist universal constants $C_1, c_1, C_2, c_2, C_0 > 0$ such that,

(a) $$\exp(-C_1\beta^2 j^2) \le \mathbb{P}\big(s_j(G) > \beta j/n\big) \le \exp(-c_1\beta^2 j^2)$$

for $\beta \ge C_0$ and $j \le n/2$

(b) $$(c_2\alpha)^{j^2} \le \mathbb{P}\big(s_j(G) < \alpha j/n\big) \le (C_2\alpha)^{j^2}$$

for $\alpha \ge 0$ and $j \le n$. In the complex case we need to replace j^2 by $2j^2$.

Lemma 8 says, roughly, that "by far the most likely" value of $s_j(G)$ is of order j/n. Taking into account the fact that, disregarding the order, the s-numbers of A^{-1} are just reciprocals of the s-numbers of A and substituting the "most likely" values of $s_j(G^{-1})$ into $(\#)$ we obtain that, "on the average," $|G^{-1}g|_2$ is of order

$$n^{-1/2}\Big(\sum_{j\le n}(n/j)^2\Big)^{1/2} \cong n^{1/2} .$$

The above means that, again on the average, the norm of the factorization arising out of $\overline{\beta}$ is of order $n^{1/2}$. This does not, however, allow us to conclude that, for a "typical" B, the norm of *any* factorization is of that order; the "concentration" is too weak (*i.e.*, one does not get (i) with δ small enough). It turns out that to get the right (at least for the "ε-net argument" we employ) estimates on $\mathbb{P}(\{\|\alpha\| \leq \lambda\})$, one need to take $\lambda = c(n/\log n)^{1/10}$. This is where we use (\$), as well as various isoperimetric inequalities. Roughly, this is how Theorem 4 is proved.

For the lower estimate in Theorem 1 (part (2) of Claim 7), the argument is somewhat more delicate, but its structure remains the same. Instead of $i_{1,2}$, we factor the identity on ℓ_1^n and, consequently, the role of the ℓ_2-norm on the range space of α is played now by the ℓ_1-norm. One gets the order $n^{1/2}\log n$ because, when $\Delta = (d_{ij})$ is an $n \times n$ diagonal matrix with $d_{jj} = n/j$, then

$$\mathbb{E}|\Delta g|_1 = n^{-1/2}\mathbb{E}\left(\sum_{j\leq n}(n/j)|\gamma_j|\right) \cong n^{1/2}\log n\ .$$

Filling the gaps in the argument sketched above is just technical, even if not necessarily easy. We note, however, that, as is easily computable (cf. [18]), the "most likely" value of $|G^{-1}g|_1$ is of order n and so it is imaginable that, differentiating carefully between the "worst case scenario" (the diagonal matrices) and the "typical case," and constructing appropriately the "ε-nets" in the set of $\overline{\beta}$'s, one can improve the lower estimate to n^θ for some $\theta \in (1/2, 7/8]$. However, we do not know, as of now, how (and whether) this can be done.

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Some remarks about the space of measures with uniformly bounded partial sums and Banach-Mazur distances between some spaces of polynomials

P. WOJTASZCZYK

Department of Mathematics
Texas A&M University
College Station, Texas 77843

A lot of work have been done on the space of uniformly convergent Fourier series. In this note we attempt to present what is known about the analogous space of measures with uniformly bounded partial sums. Unfortunately very little is known, and many natural problems remain open. Let us start with the definition of the norm. For a formal Fourier series $\sum_{-\infty}^{\infty} a_n e^{in\theta}$ we put

$$\Big\| \sum_{-\infty}^{\infty} a_n e^{in\theta} \Big\| = \sup_{N\in\mathbf{N}} \frac{1}{2\pi} \int_0^{2\pi} \Big| \sum_{-N}^{N} a_n e^{in\theta} \Big| \, d\theta \ .$$

We will be interested in various spaces equipped with this norm. Anytime we use the notation $\|.\|$ in this note we mean the above norm. The spaces we will be dealing with are

$$M^b = \Big\{ \mu \in M(\mathbf{T}) \ : \ \Big\| \sum_{-\infty}^{\infty} \hat{\mu}(n) e^{in\theta} \Big\| < \infty \Big\} \ ,$$

$$L_1^b = \Big\{ f \in L_1(\mathbf{T}) \ : \ \Big\| \sum_{-\infty}^{\infty} \hat{f}(n) e^{in\theta} \Big\| < \infty \Big\}$$

$$H_1^b = \Big\{ f \in H_1(\mathbf{T}) \ : \ \Big\| \sum_{n=0}^{\infty} \hat{f}(n) e^{in\theta} \Big\| < \infty \Big\} \ .$$

Clearly $H_1^b \subset L_1^b \subset M^b$ where the inclusions are isometric. S_N will always denote the symmetric partial sum projection *i.e.*, $S_N(\sum_{-\infty}^{\infty} a_j e^{ij\theta}) = \sum_{-N}^{N} a_j e^{ij\theta}$. This projection has norm one in each of the above spaces.

A few words about exposition. I will be rather informal (as is proper, I hope, in the seminar notes). In particular this will mean that I will not pay much attention to the constants and will assume that all numbers (like $\log N$ or $\sqrt{N}$) are integers, if needed.

We assume here the basic knowledge of harmonic analysis on the circle $\mathbf{T}$, in particular the F.M. Riesz Theorem and basic properties of the Dirichlet and Fejer kernels. Everything can

be found in Zygmund or in any other book about trigonometric series or harmonic analysis. We also assume the standard knowledge of Banach space theory.

Theorem 1. (Helson) *If $\mu \in M_1^b$, then $\hat{\mu}(n) \to 0$ as $|n| \to \infty$.*

Proof. Suppose that for some $\mu \in M_1^b$ and for some sequence of integers (n_j) tending to ∞ with j we have $|\hat{\mu}(n_j)| > \varepsilon > 0$. We can assume that μ is singular with respect to the Lebesque measure (because if $\mu = \mu_s + f dm$ then clearly $\hat{f}(n) \to 0$, so we can replace μ by μ_s). Take ε_j, $j = 1, 2, \ldots$ such that $|\varepsilon_j| = 1$ and $\varepsilon_j \hat{\mu}(n_j) > 0$. Let us call $\mu_j = \varepsilon_j e^{-in_j\theta}\mu$. Clearly $\hat{\mu}_j(0) > \varepsilon$. Let us consider also $\varphi_j = \varepsilon_j e^{-in_j\theta}\, S_{n_j}\mu$. We have $\|\varphi_j\|_1 \leq \|\mu\|$ and $\|\mu_j\|_1 \leq \|\mu\|$ for all j, so we can choose the subsequence such that (we will not change the numbering) φ_j converges weak* to ν and μ_j converges weak* to ξ. Note that $\hat{\xi}(0) > \varepsilon$ so ξ is not zero and also ξ is singular with respect to the Lebesque measure, since μ was. But looking at Fourier coefficients we realise that $\hat{\xi}(s) = \hat{\nu}(s)$ for $s < 0$ and $\hat{\nu}(s) = 0$ for $s > 0$. Now the F.M. Riesz Theorem yields that $\nu \ll m$ and also $\xi - \nu \ll m$ so $\xi \ll m$. This contradiction finishes the proof. ∎

Remark 1. More quantitative estimates have been obtained by Pigno-Smith and J.J.F. Fournier. They showed that there exists a function $N(\varepsilon)$ such that for $\mu \in M_1^b$ with $\|\mu\| \leq 1$ we have

$$\operatorname{card}\{n \in \mathbf{N} \,:\, |\hat{\mu}(n)| > \varepsilon\} \leq N(\varepsilon)\,.$$

The exact asymptotics of $N(\varepsilon)$ as $\varepsilon \to 0$ is unknown.

Remark 2. It was shown by Mary Weiss and Y. Katznelson that there exists a singular measure μ on $\mathbf{T}$ with $\|\mu\| < \infty$.

Remark 3. The Helson theorem is equivalent to saying that the map $T : L_1^b \to c_0$ given as $T(f) = (\hat{f}(n))_{n=-\infty}^{\infty}$ is weakly compact.

To see it note that if T is a weakly compact operator then $T^{**}((L_1^b)^{**}) \subset c_0$ and clearly $M^b \hookrightarrow (L_1^b)^{**}$. If T is not weakly compact, then there exists a sequence $(f_n)_{n=1}^{\infty} \subset L_1^b$ with $\|f_n\| \leq 1$ such that $T(f)$ converges pointwise to some $\eta \in \ell_\infty \setminus c_0$. But clearly any $\sigma(M(\mathbf{T}), C(\mathbf{T}))$ cluster point μ of $(f_n)_{n=1}^{\infty}$ is in M^b and $(\hat{\mu}(n)) = \eta$, which contradicts the Helson theorem.

Remark 4. The space M^b does not have the Dunford-Pettis property, because the sequence $(e^{in\theta}dm)_{n=1}^{\infty} \subset M^b$ is weakly null but $(T(e^{in\theta}))_{n=1}^{\infty}$ is not norm convergent in c_0.

Theorem 2. (Salem-Zygmund) *If $\mu \in M^b$, then*

$$\sup_n \frac{\log n}{n} \sum_{k=1}^{n} |\hat{\mu}(k)| < \infty\,.$$

Proof. Applying the Hardy's inequality we get

$$\frac{\hat{\mu}(n)}{1} + \frac{\hat{\mu}(n-1)}{2} + \cdots + \frac{\hat{\mu}(1)}{n} \leq C\|\mu\|$$
$$\frac{\hat{\mu}(n-1)}{1} + \cdots \frac{\hat{\mu}(1)}{n-1} \leq C\|\mu\|$$
$$\cdots$$

Adding those together we get

$$\sum_{k=1}^{n} \hat{\mu}(n-k+1)\Big(\sum_{j=1}^{k} \frac{1}{j}\Big) \leq Cn\|\mu\|$$

so

$$\sum_{k=1}^{n/2} \hat{\mu}(k) \log n \leq Cn\|\mu\| \,.$$

Since n was arbitrary this gives the claim. ∎

We can use the above Theorem to exhibit certain coefficient multipliers from M^b into ℓ_1.

Proposition 3. *If $0 < \alpha \leq 1$, then the sequence $\left((n+1)^{-1}(\log(n+2))^{-\alpha}\right)_{n=0}^{\infty}$ is a coefficient multiplier from M^b into ℓ_1 i.e., for every $\mu \in M^b$ we have*

$$\sum_{n=0}^{\infty} |\hat{\mu}(n)|\,(n+1)^{-1}\,(\log(n+2))^{-\alpha} \leq C\|\mu\| \,.$$

Proof. From the Salem-Zygmund Theorem we infer that for each N we have

$$\sum_{2^N}^{2^{N+1}} |\hat{\mu}(n)| \leq CN^{-1}2^N\|\mu\| \,.$$

So

$$\sum_{-\infty}^{\infty} |\hat{\mu}(n)|(n+1)^{-1}\,(\log(n+2))^{-\alpha} \leq \sum_{N} \sum_{2^N}^{2^{N+1}} |\hat{\mu}(n)|(n+1)^{-1}\,(\log(n+2))^{-\alpha}$$
$$\leq \sum_{N} \frac{1}{2^{N+1}N^{\alpha}} \sum_{2^N}^{2^{N+1}} |\hat{\mu}(n)| \leq \sum_{N} N^{-(1+\alpha)}\|\mu\| \leq C\|\mu\| \,.$$ ∎

Remark 5. It is known (cf. Zygmund, vol.I, pp.184-5) that $\sum_{n=2}^{\infty} \frac{\cos nx}{\log n}$ represents a function in L_1^b, so $\frac{1}{n}$ is not a multiplier from M^b into ℓ_1.

All spaces M^b, L_1^b and H_1^b are convolution ideals in the sense that if $\mu \in M(\mathbf{T})$ and $\nu \in M^b$ (or $\nu \in L_1^b$ or $\nu \in H_1^b$), then $\mu * \nu \in M^b$ (or L_1^b or H_1^b). Since the series mentioned in Remark 1 is in L_1^b we infer that for every $\mu \in M(\mathbf{T})$ the series

$$\sum_{n=-\infty,\, n\neq 0,\pm 1}^{\infty} \frac{\hat{\mu}(n)}{\log|n|} e^{in\theta}$$

is a Fourier series of a function in L_1^b.

Let us consider the space LB of sequences $(\alpha_n)_{n=1}^{\infty}$ such that there exists a measure $\mu \in M^b$ with $\hat{\mu}(2^n) = \alpha_n$. We put $\|(\alpha_n)\| = \inf \|\mu\|$. It seems to be an interesting problem to describe this space. The following observations can be made :

1) The sequence space ℓ_2 embeds continuously into LB.

2) The unit vectors are unconditional in LB. This follows from the well known fact that for every bounded sequence $(\xi_n)_{n=1}^{\infty}$ there exists a measure $\mu \in M(\mathbf{T})$ such that $\hat{\mu}(2^n) = \xi_n$ and $\|\mu\|_1 \leq C\|\xi\|_\infty$ and from the observation that M^b is a convolution ideal. It is clear that LB is a dual space. I do not know if it is reflexive or if the sequence of unit vectors is a basis in LB.

Since the norm $\|.\|$ is built from the L_1 norm and the L_∞ norm the following fact is not surprising.

Proposition 4. *The space H_1^b (so also M^b and L_1^b) contains the subspace isomorphic to ℓ_∞.*

Proof. Let us fix the sequence of polynomials $p_j = \sum_{n_j+1}^{n_j+1} a_k e^{ik\theta}$ for $j = 1, 2, \ldots$ such that $\|p_j\|_1 = 1$ but $\|p_j\| = \alpha_j$ with $\sum_{j=1}^{\infty} \frac{1}{\alpha_j} < \frac{1}{2}$. Then for any sequence of scalars $(\beta_j)_{j=1}^{\infty}$ with $|\beta_j| \leq 1$ and every integer N we have

$$\left\|S_N\left(\sum_j \frac{\beta_j}{\alpha_j} p_j\right)\right\|_1 \leq \sum_{j=1}^{s} \frac{|\beta_j|}{\alpha_j}\|p_j\|_1 + \left\|\sum_{n_s+1}^{N} \alpha_k e^{ik\theta}\right\|_1 \frac{|\beta_j|}{\alpha_j}$$

$$\leq \|\beta\|_\infty \sum_{j=1}^{s} \frac{1}{\alpha_j} + |\beta_j| \leq 2\|\beta\|_\infty$$

where s was chosen in such a way that $n_s \leq N < n_{s+1}$. For arbitrary s we fix l in such a way that $n_s \leq l < n_{s+1}$ and $\|\sum_{n_s+1}^{l} a_k e^{ik\theta}\|_1 = \alpha_s$. We have

$$\left\|\sum \frac{\beta_j}{\alpha_j} p_j\right\| \geq \left\|\sum_{j=1}^{s-1} \frac{\beta_j}{\alpha_j} p_j + \sum_{n_s+1}^{l} a_k \frac{\beta_s}{\alpha_s} e^{ik\theta}\right\|_1$$

$$\geq \frac{|\beta_s|}{\alpha_s}\left\|\sum_{n_s+1}^{l} a_k e^{ik\theta}\right\|_1 - \sum_{j=1}^{s-1} \frac{|\beta_j|}{\alpha_j}$$

$$\geq |\beta_s| - \tfrac{1}{2}\|\beta\|_\infty .$$

Since s was arbitrary we get $\|\sum_j \beta_j \frac{1}{\alpha_j} p_j\| \geq \frac{1}{2}\|\beta\|_\infty$. ∎

Now we will turn to the local problems. We consider the space $(W_N, \|.\|)$ consisting of all trigonometric polynomials $\sum_{-N}^{N} a_n e^{in\theta}$ equipped with the norm $\|.\|$. We are interested in distances from $(W_N, \|.\|)$ to ℓ_p^{2N+1} for various values of p. We will also investigate how big ℓ_1 and ℓ_∞ subspaces W_N contains.

Lemma 5. $d(W_N, \ell_2^{2N+1}) \leq C\sqrt{\frac{N}{\log N}}$.

Proof. The operator $I : \ell_2^{2N+1} \longrightarrow W_N$ defined as $I((\xi_j)_{j=-N}^N) = \sum_{j=-N}^N \xi_j e^{ij\theta}$ has clearly norm 1. On the other hand for $p = \sum_{j=-N}^N \xi_j e^{ij\theta} \in W_N$ with $\|p\| \leq 1$ we have

$$\left(\sum_{j=-N}^{N} |\xi_j|^2\right)^{1/2} \leq \left(\sum_{j=-n}^{n} |\xi_j|\right)^{1/2} (\sup |\xi_j|)^{1/2} .$$

Since $|\xi_j| \leq \|p\|$, the Salem-Zygmund theorem gives

$$\left(\sum_{j=-N}^{N} |\xi_j|^2\right)^{1/2} \leq C\left(\frac{N}{\log N}\right)^{1/2} \|p\| .$$

so $\|I^{-1}\| \leq C\sqrt{\frac{N}{\log N}}$. ∎

Since $d(\ell_1^N, \ell_2^N) = \sqrt{N}$ we get from this Lemma that if $X \subset W_N$ and $d(X, \ell_1^k) \leq C$, then $k \leq C\frac{N}{\log N}$. I do not know if W_N actually contains such big ℓ_1 subspaces. We have however

Proposition 6. *For every $\alpha < 1$ there exist constants C_1 and C_2 such that for every N the space W_N has a subspace X_N such that*

$$\dim X_N \geq C_1(2N+1)^\alpha$$

and

$$d(X_N, \ell_1^{\dim X_N}) \leq C_2 .$$

Proof. Let us take an integer K such that $(2N+1)^\alpha < \frac{2N+1}{K} \leq (2N+1)^\alpha + 1$ and let $D_N(t) = \sum_{j=-N}^N e^{ijt}$ be the usual Dirichlet kernel. Let us put $f_j = \frac{1}{\log N} \cdot D_N\left(t - \frac{2\pi Kj}{2N+1}\right)$ for $j = 0, 1, \ldots, \frac{2N+1}{K} = L$. Clearly $\| \sum_{j=0}^L \alpha_j f_j \| \leq C \sum |\alpha_j|$. On the other hand

$$A = \int_0^1 \left\| \sum_{j=1}^{L} r_j(t) f_j \right\| dt \geq \int_0^1 \left\| \sum_{j=0}^{L} r_j(t) f_j \right\|_1 dt \geq C \int_0^1 \left(\sum_{j=0}^{L} |f_j|^2 \right)^{1/2}$$

where r_j's are Rademacher functions.

If we consider disjoint intervals $I_j = (\frac{2\pi}{L}j - \frac{\pi}{L}, \frac{2\pi}{L}j + \frac{\pi}{L})$ and replace each f_j by $f_j \cdot \chi_{I_j}$ we get

$$A \geq \sum_{j=0}^{L} \int_{I_j} |f_j| = L \cdot \int_{I_o} |f_o| .$$

Elementary and classical estimates of the Dirichlet kernel give

$$\int_{I_o} |f_o| \geq \frac{C}{\log N} \int_{-\pi/L}^{\pi/L} |D_N| \geq \frac{C}{\log N} \log\left(\frac{N}{L}\right) \geq C .$$

This shows that $A \geq L \cdot C$ so by the theorem of Elton we can choose a subset of $(f_j)_{j=0}^L$ of cardinality proportional to L which spans ℓ_1 of the right dimension. This is our space X_N. ∎

Let us now discuss subspaces of W_N which are isomorphic to ℓ_∞^n. We have

Proposition 7. *Each space W_N contains a subspace X such that $\dim X = (\log N)^2$ and $d(X, \ell_\infty^{(\log N)^2}) < C$ where C is an absolute constant.*

Proof. Let F be the Fejer kernel of order $\sqrt{N}$ and let $f_j = \frac{1}{\log N} e^{ijkt} F$ for $j = 1, 2, \ldots, (\log N)^2$, where k is such that intervals $\operatorname{supp} \hat{f}_j$ are disjoint subsets of the set $\{0, 1, 2, \ldots, N\}$. Clearly $(f_j)_{j=1}^{(\log N)^2}$ is a basic sequence in W_N and $0 < c \leq \|f_j\| \leq C$ for some absolute constants c and C, so we have

$$\Big\| \sum_{j=1}^{(\log N)^2} a_j f_j \Big\| \geq C \sup |a_j| \ .$$

We want to show the converse inequality. We have

$$\Big\| \sum_{j=1}^{(\log N)^2} a_j f_j \Big\| \leq \sup_r \int \Big| \sum_{j=1}^{r-1} a_j f_j + a_r \frac{1}{\log N} e^{irkt} PF \Big| \leq \sup_r \int \Big| \sum_{j=1}^{r-1} a_j f_j \Big| + C|a_r|$$

where PF is some partial sum of the Fejer kernel F.

In order to estimate this integral let us observe that it follows from the classical estimates of the Fejer kernel that if I_l's are disjoint intervals covering the circle, each of the length $\frac{1}{\sqrt{N}}$ then $\sum_l \frac{1}{\sqrt{N}} \sup_{I_l} |F(t)| \leq C$. Thus we have

$$\begin{aligned} \int \Big| \sum_{j=1}^{r} a_j f_j \Big| &= \int \Big| \sum_{j=1}^{r} a_j \frac{1}{\log N} e^{ijkt} \Big| F(t)\, dt \\ &\leq \frac{1}{\log N} \sum_l \int_{I_l} \Big| \sum_{j=1}^{r} a_j e^{ijkt} \Big| dt \cdot \sup_{I_l} F(t) \ . \end{aligned}$$

Observe that the function $\sum_{j=1}^{r} a_j e^{ijkt}$ is periodic with the period $\frac{1}{k}$. Since $\frac{1}{k} < \frac{1}{\sqrt{N}}$ we get thet for each l

$$\int_{I_l} \Big| \sum_{j=1}^{r} a_j e^{ijkt} \Big| dt \leq |I_l| \int \Big| \sum_{j=1}^{r} a_j e^{ijkt} \Big| dt$$

so

$$\int \Big| \sum_{j=1}^{r} a_j f_j \Big| \leq \frac{C}{\log N} \Big\| \sum_{j=1}^{r} a_j e^{ijkt} \Big\|_1 \ .$$

Since $r \leq (\log N)^2$ and we have

$$\Big\| \sum_{j=1}^{r} a_j e^{ijkt} \Big\|_1 \leq \Big\| \sum_{j=1}^{r} a_j e^{ijkt} \Big\|_2 \leq \sqrt{r} \max_{j=1,2,\ldots,r} |a_j|$$

we get the claim. ∎

There is one rather curious and unexpected consequence of the last argument. If we consider $\operatorname{span}(f_j)_{j=1}^{(\log N)^2}$ in $(W_N, \|.\|)$ it is isomorphic to $\ell_\infty^{(\log N)^2}$ but if we consider the same span in the space $(W_N, \|.\|_1)$ then we have

$$\Big\| \sum_{j=1}^{(\log N)^2} a_j f_j \Big\|_1 \sim \Big\| \sum_{j=1}^{(\log N)^2} a_j e^{ijt} \Big\|_1 \cdot \frac{1}{\log N} \ .$$

This can be seen exactly like before using the fact that $\sum_{j=1}^{(\log N)^2} a_j e^{ijkt}$ oscillates fast compared to F. But this means that $\mathrm{span}(f_j)_{j=1}^{(\log N)^2}$ in $(W_N, \|.\|_1)$ is isomorphic (with constants independent of N) to $(W_{(\log N)^2}, \|.\|_1)$. But for $f \in W_N$ we have $\|f\|_1 \leq \|f\| \leq \log N \cdot \|f\|_1$, so $d\big((W_{(\log N)^2}, \|.\|_1), \ell_\infty^{(\log N)^2}\big) \leq \log N$. If we forget about logarithms, we get the upper estimate in the following

Corollary 8. $d\big((W_N, \|.\|_1), \ell_\infty^{2N+1}\big) \sim \sqrt{N}$.

The lower estimate is clear since $(W_N, \|.\|_1)$ is a subspace of L_1, so it has cotype 2.

Open problems. It follows from Proposition 7 that $d(W_N, X) \geq \log N$ for X any subspace of L_1, so in particular we have $d\big((W_N, \|.\|), (W_N, \|.\|_1)\big) \sim \log N$. It is not clear what is the correct order of the quantity $d\big((W_N, \|.\|), \ell_1^{2N+1}\big)$. From what we have said follows that $\log N \leq d\big((W_N, \|.\|), \ell_1^{2N+1}\big) \leq (\log N)^2$ because by the classical result of Marcinkiewicz (cf. Zygmund vol.II, p.30) we have $d\big((W_N, \|.\|_1), \ell_1^{2N+1}\big) \leq \log N$. Also the distance $d\big((W_N, \|.\|), \ell_2^{2N+1}\big)$ is unclear. One can easily check that it is greater than $\frac{\sqrt{N}}{\log N}$ but this and Lemma 5 leave an obvious gap.

Also I do not know if $(W_N, \|.\|)$ has local unconditional stucture. This is related to the question if M^b admits the lattice order. However if it has some, then it is not the natural one. Suppose that M^b is a Banach lattice with the natural order of measures. This would imply that multiplication by e^{int} would have uniformly bounded norms, so the maps $\sum_{-\infty}^{\infty} \hat{\mu}(n) e^{int} \mapsto \sum_{n=0}^{N} \hat{\mu}(n) e^{int}$ would be uniformly bounded. But this would imply that H_1^b is complemented in M^b, so by the F.M. Riesz Theorem we would get that $M^b \subset L_1^b$ as a set. This contradicts the existence of a singular measure in M^b.

I do not know if the following is true or false : if $\mu \in M^b$ and $\mu = \mu_s + f dm$ where μ_s is singular with respect to the Lebesque measure m, does $f \in L_1^b$ and $\mu_s \in M^b$?

Also the invariant projections in M^b and in H_1^b are unknown.

I do not know if ℓ_2 is isomorphic to a complemented subspace of M^b or L_1^b. This is not a local problem like in L_1 because W_N contains $\ell_2^{\log N}$ uniformly complemented. One has to take $\mathrm{span}(z^{N-2^k})_{k=1}^{\log N} \subset W_N$ and use the Paley's inequality to check that it is complemented. On the other hand the standard infinite dimensional argument also fail because M^b does not have the Dunford-Pettis property (cf. Remark 4).

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Operators which factor through Banach lattices not containing c_0

N. GHOUSSOUB and W.B. JOHNSON*

Department of Mathematics
University of British
Columbia, Vancouver, B.C., Canada

Department of Mathematics
Texas A&M University
College Station, Texas 77843, USA

In this supplement to [5], [7], we give an intrinsic characterization of (bounded, linear) operators on Banach lattices which factor through Banach lattices not containing a copy of c_0 which complements the characterization of [5], [7] that an operator admits such a factorization if and only if it can be written as the product of two operators neither of which preserves a copy of c_0. The intrinsic characterization is that the restriction of the second adjoint of the operator to the ideal generated by the lattice in its bidual does not preserve a copy of c_0. This property of an operator was introduced by C. Niculescu [10] under the name "strong type B." That this condition is stronger than the property that the operator itself does not preserve a copy of c_0 is exhibited by the example constructed in [4] and studied further in [6].

When we wrote [5], [7] we did not know that Niculescu [9], [10], [11] was independently considering some of the same problems; we now want to acknowledge his priority for discovering some results that overlap with ours. Also, in remark 1 below, we indicate how our results, when translated into Niculescu's language, answer some questions he raised.

Before stating the theorem of this note, we introduce some notation: Given a Banach lattice E, $I(E)$ (respectively, $Bd(E)$) denotes the closed order ideal (respectively, band) in E^{**} generated by E. The symbols E, F, G will be reserved for Banach lattices while X, Y, Z will be used for general Banach spaces. Terminology is as in [7], and the reader will probably need a copy of [7] on hand to follow the proof of the theorem.

Theorem. *Let E be a Banach lattice and let T be an operator from E into a Banach space X. The following are equivalent:*

(1) $T^{**}|_{I(E)}$ does not preserve a copy of c_0.

(2) $T^{**}|_{I(E)}$ does not preserve a positive disjoint copy of c_0.

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(3) $T^{**}[Bd(E)] \subset X$.

(4) *There exists a Banach space* Y *and bounded linear operators* $u: E \to Y$ *and* $v: Y \to X$ *such that neither* u *nor* v *preserves a copy of* c_0 *and* $T = vu$.

(5) *There exists a weakly sequentially complete Banach lattice* F, *an interval preserving lattice homomorphism* $R: I(E) \to F$, *and an operator* $S: F \to X$ *so that* $T^{**}{}_{|I(E)} = SR$.

Proof. That (1) and (2) are equivalent follows from Theorem I.3 of [7] (that the other conditions in Theorem I.3 are equivalent to (2) was proved earlier by Niculescu — see Theorem 3.4 in [10]). Niculescu [10, Proposition 4.2] stated the equivalence of (2) and (3); we follow his lead in leaving the proof as an exercise for the reader, but note that (2) is the statement we tie to the other conditions. The equivalence of (4) with (5) was proved in Corollary I.8 in [7] and (5) $\Rightarrow$ (2) is immediate, so we only need to check that (2) implies (5). In fact, the proof of this implication is essentially contained in the proof of Theorem I.7 of [7], but since only weaker implications are explicitly stated there we present here a detailed proof.

Let W be the absolute convex solid hull of $T^*Ball(X^*)$ in E^* and let G be the interpolation space (cf. [7]) corresponding to W in E^*. Let $J: G \to E^*$ be the canonical injection and $I: X^* \to G$ such that $T^* = JI$. By Lemma I.5 of [7] we have for each μ in E^{**} and $n = 1, 2, \ldots,$

$$(\dagger) \qquad \|J^*\mu\| \leq 2^n \sup\{\, \|T^{**}\nu\| \colon \nu \in E^{**}, |\nu| \leq |\mu| \,\} + 2^{-n}\|\mu\| \,.$$

Letting F be the norm closure of $J^*[I(E)]$ in G^* and setting $R = J^*{}_{|I(E)}$, $S = I^*{}_{|F}$, we have that $T^{**}{}_{|I(E)}$ factors through R and hence in order to complete the proof we only need to check that F does not contain a copy of c_0.

Since $I(E)$ is an ideal in E^{**} we have from $(\dagger)$ that for each $f \in I(E)$ and $n = 1, 2, \ldots,$

$$(\ddagger) \qquad \|Rf\| \leq 2^n \sup\{\, \|T^{**}g\| \colon g \in I(E), |g| \leq |f| \,\} + 2^{-n}\|f\| \,.$$

Now J^* is a lattice homomorphism since J is interval preserving ([1, p.90]). Since $T^{**}{}_{|I(E)}$ does not preserve a copy of c_0, neither does R by Lemma I.1.a of [7], hence $(\ddagger)$ implies that F has order continuous norm by Lemma I.1.b of [7].

We first prove that if $(e_m)_{m=1}^{\infty}$ is an increasing sequence in $I(E)_+$ such that $(J^*e_m) = (Re_m)$ is norm bounded in F then it is norm convergent in F. Indeed, let μ be the w^*-limit of $(J^*e_m)_{m=1}^{\infty}$ in G^*. Since $J^*[E^{**}]$ is norm dense in G^*, there exists for each $\epsilon > 0$ an element $\nu_\epsilon \in E^{**}$ such that $\|J^*\nu_\epsilon - \mu\| \leq \epsilon$.

Since $I(E)$ is an ideal in E^{**}, the sequence $(\nu_\epsilon \wedge e_m)_{m=1}^{\infty}$ is in $I(E)$; moreover, it is increasing and norm bounded by $\|\nu_\epsilon\|$, hence $J^*(\nu_\epsilon \wedge e_m)$ norm converges to some z_ϵ in F by

Theorem I.3.d in [7]. Using again the fact that J^* is a lattice homomorphism, we have that $J^*(\nu_\epsilon \wedge e_m) = J^*(\nu_\epsilon) \wedge J^*(e_m)$ hence $J^*(\nu_\epsilon \wedge e_m)$ w^*–converges in G^* to $J^*(\nu_\epsilon) \wedge \mu$. Therefore $J^*(\nu_\epsilon) \wedge \mu = z_\epsilon$ is in F. Thus we get

$$\lim_{\epsilon \to 0} \|z_\epsilon - \mu\| = \lim_{\epsilon \to 0} \|(J^*\nu_\epsilon) \wedge \mu - \mu\| \leq \lim_{\epsilon \to 0} \|J^*\nu_\epsilon - \mu\| = 0 \ .$$

It follows that μ is in F. Since the sequence $(J^*e_m)_{m=1}^\infty$ is in the order interval $[0, \mu]$ and F is order continuous, it follows that $(J^*e_m)_{m=1}^\infty$ norm converges to μ.

Suppose now that $(f_m)_{m=1}^\infty$ is any increasing sequence in F. To show that it is norm convergent, it is enough to show that for any $\epsilon > 0$ there exists $(e_m)_{m=1}^\infty$ increasing in $I(E)_+$ such that $\|f_m - J^*e_m\| \leq \epsilon$. For that, let $z_m \in I(E)_+$ be such that $\|J^*z_m - f_m\| \leq \epsilon 2^{-m}$. Set $e_m = \vee_{i=1}^m z_i$. Then $(e_m)_{m=1}^\infty$ is increasing and

$$|J^*e_m - f_m| = |J^*(\vee_{i=1}^m z_i) - f_m| = |\vee_{i=1}^m J^*z_i - \vee_{i=1}^m f_i| \leq \sum_{i=1}^m |J^*z_i - f_i| \ .$$

It follows that

$$\|J^*e_m - f_m\| \leq \sum_{i=1}^m \|f_i - J^*z_i\| \leq \epsilon \sum_{i=1}^m 2^{-i} \leq \epsilon \ .$$

By the classical version of Theorem I.3 in [FG3], F does not contain a copy of c_0 and hence F is a weakly sequentially complete Banach lattice ([8, p.34]). ∎

In the above proof of (5) $\Rightarrow$ (2), if X is Banach lattice and T is a positive operator, then every positive operator dominated by T also factors through R. Thus we get:

Corollary. *Let E be a Banach lattice, let T be a positive operator from E into a Banach lattice X, and assume that $T^{**}{}_{|I(E)}$ does not preserve a positive disjoint copy of c_0. Then there exists a weakly sequentially complete Banach lattice F and an interval preserving lattice homomorphism $R: I(E) \to F$ such that for any operator $0 \leq T_1 \leq T$ there is a positive operator $S_1: F \to X$ so that $T_1^{**}{}_{|I(E)} = S_1R$.*

Remark 1. The examples in [4] of positive quotient mappings onto c_0 which do not preserve a copy of c_0 were shown in [6] to fail property (4) of the theorem, hence are, in Niculescu's terminology [10], operators of type B which are not of strong type B. This answers problem 4.1 in [10]. (However, if E is order continuous then $I(E) = E$ and so operators from E of type B are of strong type B.) These operators also give a solution to problem 4.4 in [10]; namely, there exists a lattice homomorphism T on some Banach lattice which does not preserve a copy

of c_0 or of l_1, yet T^2 is not weakly compact, while T^3 must be weakly compact: see remark 2 in [7, p.164].

Remark 2. One of the operators mentioned in remark 1 ("T_1" in the terminology of [6]) maps weakly Cauchy sequences into weakly convergent sequences; this gives a counterexample to one part of Proposition 4.2 in [10].

Remark 3. Since a weakly sequentially complete Banach lattice has the so-called *analytic Radon-Nikodým-ARN-property* (cf. [2]), an operator which satisfies condition (3) of the theorem is an ARN-operator. This implication was recently proved in a direct way by Bukhvalov [3]. Although aware of [7], Bukhvalov mistakenly thought that his result could not follow from the factorizations in [7] (he applied the error from [10] mentioned in remark 2 to conclude that the operators mentioned in remark 1 satisfy condition (3) of the theorem).

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Remarks on Talagrand's deviation inequality for Rademacher functions

WILLIAM B. JOHNSON* AND GIDEON SCHECHTMAN

Department of Mathematics
Texas A&M University
College Station, Texas 77843

Department of Mathematics
Texas A&M University
College Station, Texas 77843
and
Department of Theoretical Mathematics
The Weizmann Institute of Science
Rehovot, Israel

Introduction.

Recently Talagrand [4] estimated the deviation of a function on $\{0,1\}^n$ from its median in terms of the Lipschitz constant of a convex extension of f to ℓ_2^n; namely, he proved that

$$P(|f - M_f| > c) \leq 4e^{-t^2/4\sigma^2}$$

where σ is the Lipschitz constant of the extension of f and P is the natural probability on $\{0,1\}^n$.

Here we extend this inequality to more general product probability spaces; in particular, we prove the same inequality for $\{0,1\}^n$ with the product measure $((1-\eta)\delta_0 + \eta\delta_1)^n$. We believe this should be useful in proofs involving random selections. As an illustration of possible applications we give a simple proof (though not with the right dependence on ε) of the Bourgain, Lindenstrauss, Milman result [1] that for $1 \leq r < s \leq 2$ and $\varepsilon > 0$, every n-dimensional subspace of L_s $(1+\varepsilon)$-embeds into ℓ_r^N with $N = c(r,s,\varepsilon)n$.

The main results.

For $i = 1, \ldots, n$ let $(X_i, \|\cdot\|_i)$, be normed spaces, let Ω_i be a finite subset of X_i with diameter at most one and let P_i be a probability measure on Ω_i. Define

$$X = \left(\sum_{i=1}^{n} \oplus X_i\right)_2$$

and

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$$\Omega = \Omega_1 \times \Omega_2 \times \cdots \times \Omega_n \subset X$$

and let

$$P = P^{(n)} = P_1 \times P_2 \times \cdots \times P_n$$

be the product probability measure on Ω. For a subset $A \subseteq \Omega$ and $t \in \Omega$ let

$$\phi_A(t) = d(t, \text{ conv } A)$$

be the distance in X from t to the convex hull of the set A.

Theorem 1. $\mathbf{E}e^{\frac{1}{4}\phi_A^2(t)} \leq \frac{1}{P(A)}$.

Remark 2. Talagrand's theorem is the special case of Theorem 1 when each Ω_i consists of two points and P_i gives weight $\frac{1}{2}$ to each of them. In the application below we use two point spaces for each Ω_i, but P_i does not assign the same mass to both points.

Proof. We repeat Talagrand's induction argument [4]; the difference is only on the calculus level. For $n = 1$

$$\mathbf{E}e^{\frac{1}{4}\phi_A^2(t)} \leq P(A) + (1 - P(A))e^{\frac{1}{4}} \leq \frac{1}{P(A)} ,$$

as the maximal value of $r(r + (1 - r)e^{\frac{1}{4}})$ for $0 \leq r \leq 1$ is 1. Assume the theorem holds for n and suppose that

$$A \subseteq \Omega_1 \times \cdots \times \Omega_n \times \Omega_{n+1} .$$

Set, for $w \in \Omega_{n+1}$,

$$A_w = \{t \in \Omega_1 \times \cdots \times \Omega_n : (t, w) \in A\} ,$$

where for $t = (t_1, \ldots, t_n) \in \Omega_1 \times \cdots \times \Omega_n$ and $w \in \Omega_{n+1}, (t, w)$ denotes $(t_1, \ldots, t_n, w)$. Let $v \in \Omega_{n+1}$ be such that

$$P^{(n)}(A_v) = \max_{w \in \Omega_{n+1}} P^{(n)}(A_w) .$$

We shall use the following two inequalities:

(1) $$\phi_A^2(t, v) \leq \phi_{A_v}^2(t) \quad \text{for all} \quad t \in \Omega_1 \times \cdots \times \Omega_n$$

(2) $$\phi_A^2(t, w) \leq \inf_{0 \leq \alpha \leq 1} \left[\alpha\phi_{A_w}^2(t) + (1 - \alpha)\phi_{A_v}^2(t) + (1 - \alpha)^2\right]$$
$$\text{for all} \quad t \in \Omega_1 \times \cdots \times \Omega_n \quad \text{and for} \quad w \neq v .$$

Using (1) and (2), Hölder's inequality and the induction hypothesis (in this order) we get

$$\mathbf{E}e^{\frac{1}{4}\phi_A^2(s)} = P_{n+1}\{v\}\mathbf{E}e^{\frac{1}{4}\phi_{A_v}^2(t)} + \sum_{w \neq v} P_{n+1}\{w\} \inf_{0 \leq \alpha \leq 1} \mathbf{E}\left(e^{\frac{1}{4}\phi_{A_w}^2(t)}\right)^{\alpha}\left(e^{\frac{1}{4}\phi_{A_v}^2(t)}\right)^{1-\alpha} e^{\frac{1}{4}(1-\alpha)^2}$$

$$(3) \qquad \leq P_{n+1}\{v\}\frac{1}{P^{(n)}(A_v)} + \sum_{w\neq v} P_{n+1}\{w\} \inf_{0\leq\alpha\leq 1}\left(\frac{1}{P^{(n)}(A_w)}\right)^{\alpha}\left(\frac{1}{P^{(n)}(A_v)}\right)^{1-\alpha} e^{\frac{1}{4}(1-\alpha)^2}$$

$$= \frac{1}{P^{(n)}(A_v)}\left[P_{n+1}(v) + \sum_{w\neq v} P_{n+1}(w) \inf_{0\leq\alpha\leq 1}\left(\frac{P^{(n)}(A_v)}{P^{(n)}(A_w)}\right)^{\alpha} e^{\frac{1}{4}(1-\alpha)^2}\right].$$

For $0 \leq \lambda \leq 1$, let $\alpha(\lambda)$ be the point where $\min_{0\leq\alpha\leq 1}\frac{1}{\lambda^{\alpha}}e^{\frac{1}{4}(1-\alpha)^2}$ in attained; *i.e.*,

$$\alpha(\lambda) = \begin{cases} 1 + 2\log\lambda & \text{if } 2\log\lambda > -1 \\ 0 & \text{otherwise} \end{cases}$$

and set

$$g(\lambda) = \frac{1}{\lambda^{\alpha(\lambda)}}e^{\frac{1}{4}(1-\alpha(\lambda))^2} = \begin{cases} e^{-\log\lambda-(\log\lambda)^2} & \text{if } 2\log\lambda > -1 \\ e^{\frac{1}{4}} & \text{otherwise.} \end{cases}$$

From (3) we get

$$(4) \qquad \mathbf{E}e^{\frac{1}{4}\phi_A^2(s)} \leq \frac{1}{P^{(n)}(A_v)}\left[P_{n+1}(v) + \sum_{w\neq v} P_{n+1}(w)g(\lambda_w)\right]$$

where

$$\lambda_w = \frac{P^{(n)}(A_w)}{P^{(n)}(A_v)}.$$

Claim. *$g(\lambda) \leq 2 - \lambda$ for $0 \leq \lambda \leq 1$.*

Using the claim we get from (4) that

$$(5) \qquad \mathbf{E}e^{\frac{1}{4}\phi_A^2(s)} \leq \frac{1}{P^{(n)}(A_v)}(q + (1-q)(2-t))$$

where $q = P_{n+1}(v)$ and $t = \frac{P(A)-P_{n+1}(v)P^{(n)}(A_v)}{(1-P_{n+1}(v))P^{(n)}(A_v)}$ (note that $0 \leq q, t \leq 1$). As

$$(6) \qquad \frac{1}{P(A)} = \frac{1}{P^{(n)}(A_v)}\,\frac{1}{q+(1-q)t},$$

it suffices to prove that

$$(7) \qquad q + (1-q)(2-t) \leq \frac{1}{q+(1-q)t}$$

for all $0 \leq q, t \leq 1$, which is easily checked.

The proof of the claim is elementary: Let

$$f(\lambda) = g(\lambda) + \lambda - 2\,.$$

Then $f(1) = f'(1) = 0$ and $f''(\lambda) \leq 0$, $0 \leq \lambda \leq 1$, which implies that $f(\lambda) \leq 0$ for $0 \leq \lambda \leq 1$.

Remark 3. Let $2 < p < \infty$ and consider Ω as a subset of $(\sum_{i=1}^n \oplus X_i)_p$. Set

$$\phi_{A,p}(t) = \inf\left\{\left(\sum_{i=1}^n \|t_i - s_i\|_i^p\right)^{1/p}\ ;\ s = (s_1,\ldots,s_n) \in \text{ conv } A\right\}.$$

Then, as pointed out by Talagrand, we also get

$$\mathbf{E}e^{\frac{1}{4}\phi_{A,p}^p(t)} \leq \frac{1}{P(A)}$$

because $\phi_{A,p}^p(t) \leq \phi_A^2(t)$.

Corollary 4. *Let $2 \leq p < \infty$ and let f be a real convex function on $(\sum_{i=1}^n \oplus X_i)_p$ (it suffices to assume that f is defined on* conv Ω*). Let σ_p be the Lipschitz constant of f. Then, for all $c > 0$,*

$$P(|f - M_f| > c) \leq 4e^{-c^p/4\sigma_p^p} \tag{8}$$

where M_f is the median of f. A similar inequality (with absolute constants replacing the two fours) holds with expectation replacing the median:

$$P(|f - \mathbf{E}f| > c) \leq Ke^{-\delta c^p/\sigma_p^p} . \tag{9}$$

(One can take $K = 8, \delta = \frac{1}{32}$.)

The proof of the first assertion is identical to the proof of Theorem 3 in Talagrand's paper [4]. The second assertion follows from the first; see [3], p. 142.

Remark 5. Inequality (9) easily extends to the more general setting where each P_i is a Radon probability on B_{X_i}.

Remark 6. In inequalities (8) and (9) the left hand side involves only the values of f on Ω while the right hand side involves, through σ_p, the values of f on conv Ω. Thus one can replace σ_p by the infinum of the Lipschitz constants of all convex extensions of $f|_\Omega$ to conv Ω. We do not know how to compute this infimum even in the original setup of Talagrand's theorem where each Ω_i is a two point set.

An application.

Lemma 7. *Let μ be a probability measure on $\{1, \ldots, N\}$. Let $0 < r < s \leq 2r$ and let X be an n-dimensional subspace of $L_r(\{1, \ldots, N\}, \mu)$ such that $\|x\|_s \leq K\|x\|_r$ for all $x \in X$ ($\|\cdot\|_s$ denotes the $L_s(\{1, \ldots, N\}, \mu)$ norm). Assume moreover that $\mu(i) \leq \frac{2}{N}$ for all $1 \leq i \leq N$. Then, for all $1 > \varepsilon > 0$ and all $k \geq c\varepsilon^{-r} r^{1/p} (\log \frac{2}{\varepsilon})^{1/p} K^r n^{1/p} N^{1/q}$, where $q = \frac{s}{r}$, $p = \frac{q}{q-1}$, there exists a subset $A \subseteq \{1, \ldots, N\}$ of cardinality k such that the restriction to A is a multiple of a $(1 + \varepsilon)$-isomorphism on X. In particular, X $(1 + \varepsilon)$-embeds into ℓ_r^k.*

Proof. Let $\delta_i, i = 1, \ldots, N$, be independent mean δ 0,1-valued random variables. Fix $x \in X, \|x\|_r = 1$ and define $f : \mathbf{R}^N \to \mathbf{R}$ by

$$f((a_i)_{i=1}^N) = \sum_{i=1}^N a_i \mu(i) |x(i)|^r .$$

Then

$$\begin{aligned}\sigma_p(f) &= \sup_{\sum_{i=1}^N |a_i|^p=1} \sum_{i=1}^N a_i\mu(i)|x(i)|^r \\ &= \left(\sum_{i=1}^N \mu(i)^q|x(i)|^s\right)^{1/q} \le \left(\frac{2}{N}\right)^{\frac{q-1}{q}} \left(\sum_{i=1}^N \mu(i)|x(i)|^s\right)^{\frac{r}{s}} \\ &\le \left(\frac{2}{N}\right)^{1/p} K^r \,.\end{aligned}$$

It follows from Corollary 4 that

$$P\left(\left|\sum_{i=1}^N \delta_i\mu(i)|x(i)|^r - \delta\right| > c\right) \le 4e^{-c^pN/8K^{rp}} \tag{10}$$

and, consequently, using the usual estimate on the size of an ε-net in ∂B_X; cf. [3] p. 7, that

$$P\left(\left|\sum_{i=1}^N \delta_i\mu(i)|x(i)|^r - \delta\right| \le \varepsilon^r\delta \text{ for all } x \in \partial B_X\right) \ge 1 - 4\exp\left(nr\log\frac{2}{\varepsilon} - \varepsilon^{rp}\delta^pN/8K^{rp}\right) .$$

Set $k = 2\delta N$ (= twice the average size of $\{i; \delta_i = 1\}$). Then, for $\eta = c\varepsilon^{rp}\left(r\log\frac{2}{\varepsilon}\right)^{-1}$ (c universal) and $n \le \eta\delta^pN/K^{rp}$, the probability above is larger then $\frac{1}{2}$, so we can find a set of cardinality k which satisfies the requirement. Eliminating δ from the two equations $k = 2\delta N$ and $n = \eta\delta^pN/K^{rp}$ we get

$$k \approx 2\eta^{-1/p}n^{1/p}N^{1/q}K^r \,.$$

Theorem 8. [1] Let $0 < r < t \le 2$ and let $T, \varepsilon > 0$. Then there exists a constant $C = C(\varepsilon, T, r, t)$ such that any n dimensional subspace X of L_r with type t constant K, $(1+\varepsilon)$-embeds into ℓ_r^N with $N \le C \cdot n$.

Proof. We may assume that $X \subseteq L_r^M$ for some finite M. By the Maurey-Nikishin-Rosenthal factorization theorem ([2] Theorem 8 and Proposition 44), there exists a probability measure μ on $\{1, \dots, M\}$ such that $\|x\|_s \le K\|x\|_r$ for all $x \in X$ where $s = \frac{r+t}{2}$ and K depends on r, t and T only. Splitting the large atoms of μ into ones with measure $\le \frac{1}{M}$ we get a new probability measure $\bar{\mu}$ on $\{1, \dots, N\}$ with $N \le 2M$ and $\bar{\mu}(i) \le \frac{1}{M} \le \frac{2}{N}$. $L_u(\mu)$ embeds isometrically (as a sublattice) into $L_u(\bar{\mu})$ for all u and the inequality $\|x\|_s \le K\|x\|_r$ for all $x \in X$ (in the new embedding) stays true. Applying Lemma 7 we get that X $(1+\delta)$-embeds into L_r^k where k is of order $n^{1/p}M^{1/q}$.

Using the Maurey-Nikishin-Rosenthal theorem again we may repeat the argument to get that X $(1+\delta)^2$-embeds into L_r^k for k of order $n^{\frac{1}{p}+\frac{1}{pq}}M^{\frac{1}{q^2}}$. Iterating one gets the result. This part is the same as in [1]. (One should be more careful than we have been above, taking the exact form of k into account, but it works.)

Remark 9. Both B. Maurey and M. Talagrand pointed out to us that versions of inequality (10) follow from known inequalities; in particular, (10) is an immediate consequence of the Azuma-Pisier inequality (see p. 45 in [3]) except that the exponent on the right side of (10) must be multiplied by a constant δ_p which tends to infinity with p. Since the degeneracy of this constant is unimportant for proving Theorem 8, we in fact do not have a good application of (our slight generalization of) Talagrand's isoperimetric inequality. On the other hand, it is possible that the approach outlined above can be used for general subspaces of L_r, in which case one expects to use a version of Lemma 7 with "s" close to "r", which forces "p" to infinity.

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A Global Approach to Certain Operator Extension Problems

M. ZIPPIN

The Hebrew University of Jerusalem
Jerusalem, Israel

Introduction

The purpose of this paper is to discuss a global approach to problems concerning the extension of bounded operators into $C(K)$ spaces. Our starting point is an old result of J. Lindenstrauss ([4] Theorem 6.1) which contains as a special case the following

Proposition 1. *Let X be a Banach space, let E be a subspace of X and let T be a compact operator from E into a $C(K)$ space Y. Then, for every $\epsilon > 0$ there is a compact extension $\tilde{T} : X \to Y$ of T with $\|\tilde{T}\| < (1+\epsilon)\|T\|$.*

The natural question arises under what conditions this result extends to bounded operators. Clearly, if Y itself is a subspace of X which is not complemented in X then the identity $I : Y \to Y$ does not extend to an operator from X into Y. On the other hand a few examples below will convince the reader that the phenomenon that every operator from E into a $C(K)$ space Y is extendable to an operator from X into Y is common enough to be worth a careful investigation.

As is well known (see [**1**, p.490]), if Z is a Banach space then every operator $T : Z \to C(K)$ determines a function $\varphi(T) : K \to \|T\| \cdot B(Z^*)$ defined by $(\varphi(T)(k))(z) = (Tz)(k)$ which is continuous in the ω^* topology of $B(Z^*)$ (as usual, $B(X)$ denotes the closed unit ball of the Banach space X). Conversely, every ω^* continuous function $\varphi : K \to \lambda \cdot B(Z^*)$ determines a linear operator $T(\varphi) : Z \to C(K)$ by $(T(\varphi)(z))(k) = \varphi(k)(z)$ and clearly, $\|T(\varphi)\| \leq \lambda$. As a consequence we get the following simple principle:

Proposition 2. *Let X be a Banach space, let E be a subspace of X and let Y be any $C(K)$ space. Then there exists a $\lambda \geq 1$ such that every operator $T : E \to Y$ has an extension $\tilde{T} : X \to Y$ with $\|\tilde{T}\| \leq \lambda\|T\|$ if and only if there is a ω^* continuous function $\psi : B(E^*) \to \lambda \cdot B(X^*)$ which extends functionals.*

This is quite obvious: if E and X enjoy the above extension property, let $J : E \to C(B(E^*))$ denote the natural embedding of E into $C(B(E^*))$ defined by $J(e)(e^*) = e^*(e)$, where $B(E^*)$ is equipped with the ω^* topology. Let $\tilde{J} : X \to C(B(E^*))$ be an extension of J with $\|\tilde{J}\| \leq \lambda$. Let $\psi = \tilde{J}^*|_{B(E^*)}$ then ψ is clearly a ω^* continuous function from $B(E^*)$ into $\lambda \cdot B(X^*)$ which extends functionals (indeed, for every $e^* \in B(E^*)$, $\psi(e^*)(e) = (\tilde{J}^* e^*)(e) = \tilde{J}(e)(e^*) = (Je)(e^*) = e^*(e)$). Conversely, if $\psi : B(E^*) \to \lambda B(X^*)$ is a ω^* continuous function which extends functionals then for every $T : E \to Y$ with $\|T\| = 1$, the function $\psi \circ \varphi(T) : K \to \lambda \cdot B(X^*)$ is ω^* continuous. It is easily checked that $\tilde{T} = T(\psi \circ \varphi(T))$ is the desired extension of T with $\|\tilde{T}\| \leq \lambda$. This proves Proposition 2.

This simple idea has been known for a long time; one version of it has been used in [4]. Note that, while the proof of [4] Theorem 6.1 uses local properties of $C(K)$, Proposition 2 provides us with a global tool. At first glance this tool seems to be too primitive to be of any significance, however, some progress made in [2] and [8] demonstrates that this global approach to extension problems may be fruitful.

Let us focus our attention on one specific type of extension problem.

Definition. *A Banach space X is said to have the "into $C(K)$ operator extension property" ($I.C(K)$ O.E.P. in short) if there is a constant $\lambda \geq 1$ such that for any subspace $E \subset X$ and any compact Hausdorff space K, every operator $T : E \to C(K)$ has an extension $\tilde{T} : X \to C(K)$ with $\|\tilde{T}\| \leq \lambda \|T\|$*

Example 1. The space ℓ_p, $1 < p < \infty$, has the $I.C(K)$ O.E.P. with $\lambda = 1$. Indeed, for every $E \subset \ell_p$, the unique Hahn Banach extension function $\psi : B(E^*) \to B(\ell_p^*)$ is easily checked to be ω^* continuous. Hence, by Proposition 2, ℓ_p has the desired property.

Example 2. Lindenstrauss and Pełczyński proved in [5] that for every $\epsilon > 0$ the sapce c_0 has the $I.C(K)$ O.E.P. with $\lambda = 1 + \epsilon$. The proof is of local nature: the extension of the operator is done step by step. In Section 3 we will give a "global" proof of a slightly weaker version of the Lindenstauss-Pełczyński result.

Example 3. (See [2] for details.) The global approach can be used to show that the space c_0 does not have the $I.C(K)$ O.E.P. with $\lambda = 1$.

Unlike the local approach, the global principle extends the Lindenstrauss-Pełczyński theorem mentioned in Example 2 to $c_0(\Gamma)$; namely, for every set Γ, the space $c_0(\Gamma)$ has the $I.C(K)$ O.E.P. for every $\lambda > 1$.

Example 4. The space ℓ_1 does not have the $I.C(K)$ O.E.P. with $\lambda < 2$.

Proof. Let $y^* = (1, 1, 1, \cdots) \in \ell_\infty$ and let E denote the annihilator of y^* in ℓ_1. Suppose

that ℓ_1 has the $I.C(K)$ O.E.P. with some $\lambda < 2$. Then, by Proposition 2, there is a ω^* continuous function $\psi : B(E^*) \to \lambda \cdot B(\ell_\infty)$ which extends functionals. Let $\psi(0) = \epsilon y^*$ for some $-\lambda \leq \epsilon \leq \lambda$ and let u_n^* denote the n-th unit vector of ℓ_∞. Put $f_n^* = (\sum_{i=1}^n u_i^* - u_{n+1}^*)|_E$ then $\omega^* \lim(\pm f_n^*) = 0$ and hence, the ω^* continuity implies that $\omega^* \lim \psi(\pm f_n^*) = \psi(0) = \epsilon y^*$. However, $\psi(f_n^*) = \sum_{i=1}^n u_i^* - u_{n+1}^* + \gamma_n y^*$ and $\psi(-f_n^*) = -\sum_{i=1}^n u_i^* + u_{n+1}^* + \delta_n y^*$ for some γ_n and δ_n. It follows that $\lim_n \gamma_n = \epsilon - 1$ and $\lim_n \delta_n = \epsilon + 1$. Since $\|\psi(x^*)\| \leq \lambda$ for all $x^* \in B(E^*)$, we get that for every n, $|1 \pm \gamma_n| \leq \lambda$ and $|-1 \pm \delta_n| \leq \lambda$. Hence $|1 \pm (\epsilon - 1)| \leq \lambda$ and $|-1 \pm (\epsilon + 1)| \leq \lambda$ which is absurd unless $\lambda \geq 2$.

Notations. For a Banach space X, $\mathring{B}(X)$ denotes the norm open unit ball. If $A \subset X$, $[A]$ is the norm closed linear span of A; $A^\perp$ is the annihilator of A in X^*.

2. The connection between the extension property and continuous selection theorems

Let X be a Banach space and let E be a subspace of E. Having established the usefulness of ω^* continuous functions $\psi : B(E^*) \to B(X^*)$ which extend functionals, the question arises how can we construct such functions? For every $e^* \in B(E^*)$ let $\epsilon(e^*) = \{x^* \in X^* : x^*(e) = e^*(e)$ for all $e \in E\}$. ϵ is a function from $B(E^*)$ into the set of ω^* closed convex subsets of X^*. Such a function is called by E. Michael a *Carrier* (see [6]). Every ω^* continuous selection ψ of ϵ (*i.e.*, a function $\psi : B(E^*) \to X^*$ with $\psi(e^*) \in \epsilon(e^*)$ for all $e^* \in B(E^*)$) is a ω^* continuous function from $B(E^*)$ to $\lambda \cdot B(X^*)$ (for some $\lambda \geq 1$) which extends functionals. One way of constructing the desired function ψ is thus to prove a proper continuous selection theorem. The well known Michael's selection theorem ([6] Theorem 3.2″) deals with lower semi continuous carriers ϵ (in Michael's terminology) from paracompact spaces into the closed convex subsets of a Banach space. Michael's proof works fine if "Banach space" is replaced by "Frechet space." However, completeness is crucial and, since we are interested in the ω^* topology, and dual Banach spaces under this topology are not complete, unfortunately Michael's selection theorem cannot be used automatically. It is shown in Section 5 of [8] that the carrier ϵ is ω^* lower semi continuous. For every $\lambda \geq 1$ the set $\lambda \cdot B(X^*)$ is ω^* compact, Michael's theorem can be used to construct a ω^* continuous selection of the carrier $\epsilon(e^*) \cap \lambda \cdot B(X^*)$ if one can show that $\epsilon(e^*) \cap \lambda B(X^*)$ is ω^* lower semi continuous. Unfortunately this is rarely the situation. But in certain cases something in this direction can be done, as is demonstrated in the next section.

3. The global approach to the case of c_0

In this section we will use the global approach to prove the following slightly weaker version of the Lindenstrauss-Pełczyński result.

Proposition 3. *Let $\lambda > 4$ and let E be a subspace of c_0. Then, for every compact Hausdorff space K, each operator $T : E \to C(K)$ has an extension $\tilde{T} : c_0 \to C(K)$ with $\|\tilde{T}\| \leq \lambda\|T\|$.*

Proof. Proposition 3 is a direct consequence of the following:

Theorem. *Let $\lambda > 4$ and let E be a subspace of c_0. Put $\theta(0) = \{0\}$ and for every $0 \neq e^* \in B(E^*)$ let $\theta(e^*) = \{x^* \in c_0^* : x$ is an extension of e^* and $\|x^*\| < \lambda\|e^*\|\}$. Then θ is a ω^* lower semi continuous carrier.*

Note that the carrier θ maps each $e^* \in B(E^*)$ onto a convex subset of the ω^* compact set $\lambda \cdot B(c_0^*)$. Assuming the Theorem, it follows from the proof of Michael's selection theorem that there is a ω^* continuous function $\psi : B(E^*) \to \lambda \cdot B(c_0^*)$ with $\psi(e^*)$ in the ω^* closure of $\theta(e^*)$ for all e^* in $B(E^*)$. This proves Proposition 3 in view of Proposition 2.

Proof of the Theorem. Let $\{u_n\}_{n=1}^{\infty}$ denote the usual unit vector basis of c_0 and let $\{u_n^*\}_{n=1}^{\infty}$ be the corresponding biorthogonal functionals in ℓ_1. Without loss of generality we may assume that the linear subspace $E_0 = E \cap$ (linear span $\{u_n\}_{n=1}^{\infty}$) is dense in E (if this is not the case a small perturbation argument is the remedy for the situation). Assuming $5 > \lambda > 4$ we let $\epsilon = \frac{1}{2}(\lambda - 4)$ and pick a decreasing sequence $\{\delta(n)\}_{n=1}^{\infty}$ of positive numbers with $\prod_{n=1}^{\infty}(1+\delta(n)) < 1+\epsilon$. Let $n(0) = 0$ and construct an increasing sequence $n(0) < n(1) < n(2) < \cdots$ of integers as follows: let $n(1)$ be so large that $E_1 = E_0 \cap [u_1, u_2, \cdots, u_{n(1)}] \neq \emptyset$. Let P_n denote the natural projection of c_0 onto $[u_1, u_2, \cdots, u_n]$. Pick a $\frac{1}{2}\delta(1)$-net $\{P_{n(1)}x_1^1, P_{n(1)}x_2^1, \cdots, P_{n(1)}x_{N(1)}^1\}$ in $B(P_{n(1)}(c_0))$. For each $1 \leq i \leq N(1)$ let $G_1(i) = \{y \in E_0 : \|P_{n(1)}y - P_{n(1)}x_i^1\| < \frac{1}{2}\delta(1)\}$ and pick $y_i^1 \in G_1(i)$ such that $\|y_i^1\| < \inf\{\|y\| : y \in G_1(i)\}(1 + \frac{1}{2}\delta(1))$. Choose $n(2) > n(1)$ so large that for each $1 \leq i \leq N(1), y_i^1 \in E_2 = E_0 \cap [u_1, u_2, \cdots, u_{n(2)}]$. Assume that $n(0) < n(1) < \cdots < n(k)$ have been determined. Pick a $\frac{1}{2}\delta(k)$-net $\{P_{n(k-1)}x_1^k, P_{n(k-1)}x_2^k, \cdots, P_{n(k-1)}x_{N(k)}^k\}$ in $B(P_{n(k)}(c_0))$ and, for each $1 \leq i \leq N(k)$ put $G_k(i) = \{y \in E_0 : \|P_{n(k)}y - P_{n(k)}x_i^k\| < \frac{1}{2}\delta(k)\}$. Pick $y_i^k \in G_k(i)$ with $\|y_i^k\| < \inf\{\|y\| : y \in G_k(i)\}(1 + \frac{1}{2}\delta(k))$.

Choose $n(k+1) > n(k)$ so that for each $1 \leq i \leq N(k)$, $y_i^k \in E_k = E_0 \cap [u_1, u_2, \cdots, u_{n(k+1)}]$. This concludes the inductive construction of $\{n(k)\}_{k=1}^{\infty}$.

As a consequence of our construction the following condition holds:

(1) for every $e \in E_0$ and $k \geq 2$ there is an $x \in E_k$ with $\|x\| \leq (1 + \delta(k))\|e\|$ such that $\|P_{n(k)}x - P_{n(k)}e\| \leq \delta(k)\|e\|$.

Indeed, assume that $\|e\| = 1$ and pick $1 \leq i \leq N(k)$ such that $\|P_{n(k)}x_i^k - P_{n(k)}e\| < \frac{1}{2}\delta(k)$. Our choice ensures that $\|y_i^k\| \leq \|e\|(1+\delta(k))$ and $\|P_{n(k)}y_i^k - P_{n(k)}x_i^k\| < \frac{1}{2}\delta(k)$. It follows that $\|P_{n(k)}y_i^k - P_{n(k)}e\| < \delta(k)\|e\| = \delta(k)$.

We now need the following

Lemma. *Let $\mu > 0$ and let $m \geq 1$ be an integer. Then there is an integer k satisfying the following condition*

(2) *whenever $e^* \in E^*$ and $e^*(e) = 0$ for all $e \in E_k$ there is an extension $x^* \in c_0^*$ of e^* such that $|x^*(u)| < \mu\|u\|$ for all $u \in [u_1, u_2, \cdots u_{n(m)}]$ and $\|x^*\| \leq 2\|e^*\|$.*

We will prove the Lemma at the end of this section and proceed now with the proof of the Theorem. In view of Proposition 2.1 of [6], it suffices to prove the following:

(3) *for every $e_0^* \in B(E^*), x_0^* \in \theta(e_0^*)$ and every w^* neighborhood V of 0 in $\lambda \cdot \mathring{B}(c_0^*)$ there is a ω^* neighborhood U of 0 in $B(E^*)$ such that for every $e \in U$ satisfying $\|e_0^* + e^*\| \leq 1$ we have that $\theta(e_0^* + e^*) \cap (x_0^* + V) \neq \emptyset$.*

For ω^* neighborhoods of 0 we will use sets of the form $V(\mu, k) = (\mu\mathring{B}(c_0^*) + [u_1, u_2, \cdots, u_{n(k)}]^\perp) \cap \lambda \cdot \mathring{B}(c_0^*)$ in $\lambda \cdot \mathring{B}(c_0^*)$ and $U(\delta, k) = (\delta\mathring{B}(E^*) + E_{n(k)}^\perp) \cap B(E^*)$ in $B(E^*)$ where $\mu, \delta > 0$ and $k \geq 1$. Clearly these collections form neighborhood bases at 0 in $\lambda\mathring{B}(c_0^*)$ and $B(E^*)$ respectively. To prove (3), let $0 \neq e_0^* \in B(E^*), x_0^* \in \theta(e_0^*)$ and let $V = V(\mu, m)$ where μ and m are determined as follows: let $x_0^* = \sum_{i=1}^\infty a_i u_i^*$ then $\|x_0^*\| = \sum_{i=1}^\infty |a_i| = \eta\|e_0^*\| < \lambda\|e_0^*\|$. Let $\gamma = max\{\frac{1}{2}(\eta + \lambda), \frac{1}{2}(4 + \lambda)\}$ and choose μ and m so that

$$\mu < 200^{-1}(\lambda - \gamma)\|e_0^*\| \quad \text{and} \quad \sum_{n(m)+1}^{\infty} |a_i| < \mu \,. \tag{4}$$

Use the Lemma to find an integer k satisfying (2). Now consider the ω^* neighborhood $U = U(\mu, k+1)$ of 0 in $B(E^*)$. Suppose that $e^* \in U$ and $\|e_0^* + e^*\| \leq 1$. We must show that $\theta(e_0^* + e^*) \cap (x_0^* + V) \neq \emptyset$. By the Lemma, there is an $x^* = \sum_{i=1}^\infty b_i u_i^*$ which is an extension of e^*, such that $\sum_{i=1}^{n(m)} |b_i| \leq \mu$ and $\|x^*\| \leq 2\|e^*\|$. Clearly,

$$x^* = \sum_{i=1}^{n(m)} b_i u_i^* + \sum_{i=n(m)+1}^{\infty} b_i u_i^* \in (\mu \cdot B(c_0^*) + [u_1, u_2, \cdots, u_{n(m)}]^\perp) \cap \lambda \cdot \mathring{B}(c_0^*) = V(\mu, m)$$

and $x_0^* + x^*$ is an extension of $e_0^* + e^*$. Hence, it remains to show that $\|x_0^* + x^*\| < \lambda\|e_0^* + e^*\|$. We may clearly assume that the integer k, chosen to satisfy (2), is so large that, in addition to (2) the following two conditions are satisfied

(5) $\delta(k) < \mu$ *and* $k - 1 > m$

(6) *There is $e_0 = \sum_{i=1}^\infty d_i u_i \in B(E_0)$ be such that $e_0^*(e_0) > (1 - \mu)\|e_0^*\|$ and $\|(I - P_{n(k)})(e_0)\| < \mu$.*

It follows from (1) that if $e \in B(E_0)$ is an element satisfying $e^*(e) \geq \|e^*\|(1-\mu)$ then there is an $x \in E_{k+1}$ such that

$$\|P_{n(k)}x - P_{n(k)}e\| \leq \mu\|e\| \leq 2\mu \quad \text{and} \quad \|x\| < (1+\mu)\|e\| \,.$$

Put $y = e - x = \sum_{i=1}^{\infty} c_i u_i$. Then $\|P_{n(k)}y\| \leq 2\mu$, $\|y\| \leq (2+\mu)\|e\|$ and $e^*(y) = e^*(e)$. Since e_0 and y are essentially disjointly supported, and $\|e_0^* + e^*\| \leq 1$, it follows from (5) and (6) that

$$\begin{aligned} &\|e_0^* + e^*\|(2+2\mu) \geq \|e_0^* + e^*\| \cdot \|2e_0 + y\| \geq \\ &(e_0^* + e^*)(2e_0 + y) \geq 2e_0^*(e_0) + e^*(y) - |e_0^*(y)| - 2|e^*(e_0)| \\ &\qquad \geq 2(1-\mu)\|e_0^*\| + (1-\mu)\|e^*\| - 4\mu \end{aligned} \tag{7}$$

and therefore the following inequality is satisfied

$$\|e^*\| \leq (1-\mu)^{-1}\left[(2+2\mu)\|e_0^* + e^*\| - 2(1-\mu)\|e_0^*\| + 4\mu\right] \tag{8}$$

Because $\|x^*\| \leq 2\|e^*\|$ we get from (8) and (4) the inequality

$$\begin{aligned} &\|x_0^* + x^*\| \leq \|x_0^*\| + \|x^*\| \leq \\ &\quad \eta\|e_0^*\| + 2(1-\mu)^{-1}\left[(2+2\mu)\|e_0^* + e^*\| - 2(1-\mu)\|e_0^*\| + 4\mu\right] \\ &\qquad \leq (\eta - 4)\|e_0^*\| + 2(1-\mu)^{-1}(2+2\mu)\|e_0^* + e^*\| + 8\mu \\ &\qquad \leq (\eta - 4)\|e_0^*\| + (4+4\mu)\|e_0^* + e^*\| + 20\mu \end{aligned} \tag{9}$$

Since $\|e_0^* + e^*\| \geq (e_0^* + e^*)(e_0) \geq (1-\mu)\|e_0^*\| - 2\mu$ we have that $\|e_0^*\| \leq (1+2\mu)\|e_0^* + e^*\| + 2\mu \leq (1+2\mu)\|e_0^* + e^*\| + 100^{-1}(\lambda - \gamma)\|e_0^*\|$ and so $\|e_0^*\| \leq (1 + 50^{-1}(\lambda - \gamma))(1+2\mu)\|e_0^* + e^*\|$. Since, in addition, $(\lambda - \gamma)\|e_0^*\| > 200\mu$ we obtain from (9) the inequality

$$\begin{aligned} \|x_0^* + x^*\| &\leq \Big[4 + 4\mu + (\eta - 4)\big(1 + 10^{-1}(\lambda - \gamma)\big) \\ &\qquad + 10^{-1}(\lambda - \gamma)\big(1 + 10^{-1}(\lambda - \gamma)\big)\Big]\|e_0^* + e^*\| \\ &< \lambda\|e_0^* + e^*\| \,. \end{aligned} \tag{10}$$

The argument for $e_0^* = 0$ is much simpler. This proves (3) and the proof of the Theorem is complete.

Proof of the Lemma. Suppose that the conclusion of the Lemma does not hold for any integer k then we can construct a sequence $\{e_k^*\}^{\infty} \subset E^*$ with $\|e_k^*\| = 1$ such that for each $k \geq 1$ $e_k^*(e) = 0$ for all $e \in E_k$ and every extension $x_k^* \in c_0^*$ of e_k^* with $\|x_k^*\| \leq 2$ will satisfy $\|P_{n(m)}^*(x_k^*)\| \geq \mu$. Clearly $\omega^* \lim e_k^* = 0$ hence by a well known argument of W.A. Veech [7] there exists a sequence $\{x_k^*\}_{m=1}^{\infty} \subset 2 \cdot B(c_0^*)$ such that x_k^* extends e_k^* and $\omega^* \lim x_k^* = 0$ — a contradiction. This proves the Lemma.

4. Remarks and open problems

Remark 1. It is easy to check that if λ is close to 1 then the carrier $\theta(e^*)$ is not ω^* lower semi continuous. Hence the global approach cannot yield the Lindenstrauss-Pełczyński result. However it provides an analogous isomorphic version and can be applied to particular subspaces E of certain spaces X where inequalities similar to (8), (9) and (10) can be proved. Note that the first part of the proof of the Theorem, including the Lemma, can be done in every space X with a monotone basis $\{u_n\}_{n=1}^{\infty}$.

We are interested in the following questions:

Problem 1. Does $L_p(1 < p < \infty, p \neq 2)$ have the $I.C(K)$ O.E.P. $\lambda > 1$? Recently W.B. Johnson showed that L_p does not have this property with $\lambda = 1$.

Problem 2. Does ℓ_1 have the $I.C(k)$ O.E.P. with some $\lambda \geq 2$?

Problem 3. Let $X = \ell_1$ and let $E \subset X$ be the subspace constructed in [3]. (E is a $\mathcal{L}_{1,2}$ space and it is "locally complemented" in ℓ_1 but not complemented). Can any operator $T : E \to C(K)$ be extended to an operator $\tilde{T} : \ell_1 \to C(K)$?

Remark 2. The existence problem of a ω^* continuous function $\psi : B(E^*) \to \lambda \cdot B(X^*)$ which extends functionals where $E \subset X$ seems to be very far from a "local" problem. In Problem 3 the subspace E behaves locally in a very pleasant way, however, so far it did not help us to construct the desired map ψ.

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Weakly null sequences with upper ℓ_p-estimates

HELMUT KNAUST* AND EDWARD ODELL

Department of Mathematics
The University of Texas at Austin
Austin, Texas 78712

1. Introduction

A Banach space X has property (S) if every weakly null sequence in $Ba\,X$, the unit ball of X, has a subsequence which is C-dominated by the unit vector basis of c_0 for some constant $C < \infty$. In [11] it was shown that if X has property (S), then the constant C can be chosen to be independent of the particular weakly null sequence in $Ba\,X$.

Here we generalize this result to the case of upper ℓ_p-estimates.

Definition 1.1. Let $1 < p < \infty$. A Banach space X has *property* (S_p) if every weakly null sequence (x_n) has a subsequence (y_n) such that for some constant $C < \infty$,

$$\left\|\sum_{n=1}^{\infty} \alpha_n y_n\right\| \leq C \text{ for all } (\alpha_n) \in \mathbb{R} \text{ with } \left(\sum_{n=1}^{\infty} |\alpha_n|^p\right)^{1/p} \leq 1\,. \tag{1}$$

X has *property* (US_p), if there is a constant C such that every normalized weakly null sequence in $Ba\,X$ admits a subsequence (y_n) so that (1) holds. We say that (y_n) has a *C-upper ℓ_p-estimate*, if (1) holds. ∎

Our main result is

Theorem 1. *A Banach space has property (S_p) if and only if it has property (US_p).*

Let us give some examples of Banach spaces which enjoy property (S_p): ℓ_p has property (S_p) $(1 < p < \infty)$. $L_p[0,1]$ has property (S_r), where $r = \min\{2, p\}$. (More generally, every Banach space which has type p and can be embedded into a Banach space with an unconditional basis has property (S_p).) The James space J and its tree version JT have property (S_2) [1]. It follows from the results of James [8] (see also [7]) that every superreflexive Banach space has property (S_p) for some $1 < p < \infty$. Let us note that every subspace of a Banach space X with

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property (S_p) has property (S_p). If X is in addition reflexive, quotients of X have property (S_p) as well.

A technique employed in the proof of Theorem 1 allows us to strengthen this last result in the following way:

Corollary 1. *Let X be a Banach space with property (S_p) and let Y be a subspace of X not containing ℓ_1. Then the quotient space X/Y has property (US_p).*

Our proof of Theorem 1 is strongly motivated by the arguments in [11]. In fact, the proof we present here is valid for the case of property (S) as well (with the usual changes of notation). The key proposition in our proof (Proposition 3.4) is an improvement to our construction in [11]. Furthermore we no longer have Johnson's lemma [11, Proposition 3.4] at our disposal.

The following remarks were made in [11] but seem worth recalling.

One might ask whether a result like Theorem 1 remains true, if one considers the property that every normalized weakly null sequence in X admits a subsequence which is equivalent to the unit vector basis of ℓ_p.

On the one hand Johnson and the second named author [9] have shown this to be false: for $1 < p < 2$ they construct a subspace of $L_p[0,1]$ where each normalized weakly null sequence has a subsequence equivalent to the unit vector basis of ℓ_p, but where the equivalence constant cannot be chosen uniformly for all sequences in question. Their construction can be carried out to produce counterexamples for $p \geq 2$ as well (not within $L_p[0,1]$).

We are indebted to H.P. Rosenthal for pointing out to us that under stronger conditions on the other hand, one obtains the following corollary of Theorem 1:

Corollary 2. *Let X be a Banach space such that X has property (S_p) and X^* has property (S_q) for some $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then the following properties hold for some constant $C < \infty$:*

(i) If ℓ_1 does not embed in X^*, then every normalized weakly null sequence in X contains a subsequence which is C-equivalent to the unit vector basis of ℓ_p and whose closed linear span is C-complemented in X.

(ii) If ℓ_1 does not embed in X, then every normalized weakly null sequence in X^* contains a subsequence which is C-equivalent to the unit vector basis of ℓ_q and whose closed linear span is C-complemented in X^*.

The proof of Theorem 1 will be presented in Section 3. In order to motivate the quite technical proof we present a version of Theorem 1 for spreading models in Section 2. We hope that the spreading-model version (which is quite easy and does not follow from Theorem 1) will

give the reader some insight into our approach to the proof of Theorem 1. Section 4 contains the proof of the corollaries; we also state a "weak Cauchy sequence" criterion for property (S_p), due to C. Schumacher [16].

Our notation is standard as can be found in [5] or [12]. If F is a subset in a Banach space X, then $[F]$ denotes the closed linear span of F in X. If L is an infinite subsequence of $\mathbb{N}$, we denote by $\mathcal{P}_\infty(L)$ the set of all infinite subsequences of L. We would like to thank Haskell Rosenthal for his useful suggestions.

2. The spreading model case

Let us recall that a semi-normalized basic sequence (x_n) in a Banach space X is said to have a *spreading model* (e_n), if (e_n) is basic in some Banach space such that for all $k \in \mathbb{N}$ and for all $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that for all $N < n_1 < n_2 < \ldots < n_k$ and for all scalars (a_i) with $\sup_i |a_i| \leq 1$

$$\left| \left\| \sum_{i=1}^{k} a_i e_i \right\| - \left\| \sum_{i=1}^{k} a_i x_{n_i} \right\| \right| < \varepsilon \ .$$

If (x_n) has a spreading model, the model is unique up to 1-equivalence. Every normalized weakly null sequence admits a subsequence with an unconditional spreading model [3], [4].

Definition 2.1. Let $1 < p < \infty$. A Banach space has *property* (M_p) if every normalized weakly null sequence in X admits a subsequence with a spreading model (e_n) which is C-dominated by the unit vector basis of ℓ_p, *i.e.*,

$$\left\| \sum_i a_i e_i \right\| \leq C \text{ for all } (a_i) \in Ba\,\ell_p \ .$$

We say X has *property* (UM_p), if the constant C can be chosen uniformly for all normalized weakly null sequences in X.

We will show the following analogue of Theorem 1:

Proposition 2.2. *A Banach space with property (M_p) has property (UM_p).*

For its proof we will need some technical definitions. Let $1 < p < \infty$ and let X be a Banach space. A normalized weakly null sequence in X is called *good* (respectively *C-good*), if it has a spreading model which satisfies an upper ℓ_p-estimate (respectively a C-upper ℓ_p-estimate). A good sequence in X is called *M-bad*, if its spreading model fails to have an M-upper ℓ_p-estimate. An array $(x_i^n)_{i,n=1}^\infty$ of elements in X is called a *bad array*, if each *column* $(x_i^n)_i$ is an M_n-bad sequence for all $n \in \mathbb{N}$ and $M_n \to \infty$ as $n \to \infty$. An array (y_i^n) is called a *subarray* of an array (x_i^n), if each column of (y_i^n) is a subsequence of $(x_i^{k_n})_i$ for some sequence

$k_1 < k_2 < \dots$. Finally let us say a bad array (x_i^n) satisfies the *ℓ_p-array procedure (for spreading models)* if it admits a subarray (y_i^n) and if there are positive scalars a_n with $\sum a_n \leq 1$ so that $y_i = \sum_{n=1}^{\infty} a_n y_i^n$ has no subsequence with a spreading model which has an upper ℓ_p-estimate. (Note that (y_i) is automatically weakly null.)

Proof of Proposition 2.2. Suppose a Banach space X has property (M_p) but fails property (UM_p). Then X contains a bad array. We will show that every bad array satisfies the ℓ_p-array procedure for spreading models, thus obtaining a contradiction. Indeed, a bad array (x_i^n) contains a (bad) subarray, which is basic in some order (see Lemma 3.7 below). The sequence (y_n) obtained from this subarray by the ℓ_p-array procedure is then semi-normalized and weakly null. The sequence $(y_n/\|y_n\|)$ then has no spreading model with an upper ℓ_p-estimate.

Let (x_i^n) be a bad array in X. We claim that we can find a subarray (y_i^n) of (x_i^n), positive scalars a_n with $\sum a_n \leq 1, k_N \in \mathbb{N}$ and constants C_N and M_N such that the following properties hold for all $N \in \mathbb{N}$:

(2) $\qquad (y_i^N)_i$ is a C_N-good sequence with its spreading model denoted by (e_i^N)

(3) $$\left\| \sum_{i=1}^{k_N} b_i^N e_i^N \right\| > 2M_N \text{ for some } (b_i^N) \in Ba\,\ell_p$$

(4) $$a_N M_N > N$$

(5) $$\sum_{n=1}^{N-1} a_n C_n < \tfrac{1}{8} a_N M_N$$

(6) $$k_N \sum_{n=N+1}^{\infty} a_n < \tfrac{1}{4} a_N M_N \ .$$

Once the claim has been established we proceed as follows. We set $y_i = \sum_{n=1}^{\infty} a_n y_i^n$ and let $(y_i)_{i \in L}, L \in \mathcal{P}_\infty(\mathbb{N})$, be a subsequence of (y_i) with spreading model (e_i). Let $C < \infty$ be given. We have to show that (e_i) does not have a C-upper ℓ_p-estimate. By (4) we can choose N so that $a_N M_N > 4C$. Next we choose $j_1 < j_2 < \dots < j_{k_N}$ in L with j_1 large enough, so that we obtain for all $n \leq N$ and all $(c_i) \subseteq \mathbb{R}$ with $|c_i| \leq 1$:

(7) $$2 \left\| \sum_{i=1}^{k_N} c_i e_i^n \right\| \geq \left\| \sum_{i=1}^{k_N} c_i y_{j_i}^n \right\| \geq \tfrac{1}{2} \left\| \sum_{i=1}^{k_N} c_i e_i^n \right\| \quad \text{and} \quad \left\| \sum_{i=1}^{k_N} c_i y_{j_i} \right\| \leq 2 \left\| \sum_{i=1}^{k_N} c_i e_i \right\| \ .$$

By (3) we can find $(b_i^N)_{i=1}^{k_N} \in Ba\,\ell_p$ with $\left\|\sum_{i=1}^{k_N} b_i^N e_i^N\right\| > 2M_N$. Thus

$$\begin{aligned}
\left\|\sum_{i=1}^{k_N} b_i^N y_{j_i}\right\| &= \left\|\sum_{i=1}^{k_N} b_i^N \left(\sum_{n=1}^{\infty} a_n y_{j_i}^n\right)\right\| \\
&\geq a_N \left\|\sum_{i=1}^{k_N} b_i^N y_{j_i}^N\right\| - \sum_{n=1}^{N-1} a_n \left\|\sum_{i=1}^{k_N} b_i^N y_{j_i}^n\right\| - \sum_{n=N+1}^{\infty} a_n \left\|\sum_{i=1}^{k_N} b_i^N y_{j_i}^n\right\| \\
&\geq \tfrac{1}{2} a_N \left\|\sum_{i=1}^{k_N} b_i^N e_i^N\right\| - \sum_{n=1}^{N-1} 2a_n \left\|\sum_{i=1}^{k_N} b_i^N e_i^n\right\| - \sum_{n=N+1}^{\infty} a_n \left\|\sum_{i=1}^{k_N} b_i^N y_{j_i}^n\right\| \quad \text{by (7)} \\
&\geq a_N M_N - 2\sum_{n=1}^{N-1} a_n C_N - k_N \sum_{n=N+1}^{\infty} a_n \quad \text{using (2)} \\
&\geq a_N M_N - \frac{a_N M_N}{4} - \frac{a_N M_N}{4} \geq \frac{a_N M_N}{2} \geq 2C \quad \text{by (5) and (6)} .
\end{aligned}$$

Consequently $\|\sum_{i=1}^{k_N} b_i^N e_i\| > C$ by (7).

It is left to present the construction of the subarray (y_i^n). Let $a_1 = 1/2$. Since (x_i^n) is a bad array we can find a column n such that the sequence $(x_i^n)_i$ has a subsequence (z_i) which has a spreading model (e_i^1) which fails to satisfy a $2M_1$-upper ℓ_p-estimate, where $M_1 > 2$, *i.e.*, $\|\sum_{i=1}^{k_1} b_i^1 e_i^1\| > 2M_1$ for some $k_1 \in \mathbb{N}$ and some $(b_i^1)_{i=1}^{k_1} \in Ba\,\ell_p$. We set $y_i^1 = z_i$. This defines the first column of the subarray (y_i^n) of (x_i^n). By choosing C_1 large enough we can assure that (y_i^1) is a C_1-good sequence.

We pick $a_2 < 1/4$ such that $a_2 < a_1 M_1/8k_1$, pick (y_i^2), a subsequence of some $(x_i^n)_i$, such that this column is $(2M_2)$-bad, where $a_1 C_1 < a_2 M_2/8$ and $a_2 M_2 > 2$. We choose C_2 so that (y_i^2) is a C_2-good sequence.

If the first $N-1$ columns of the subarray have been chosen in the way just described, we choose

$$0 < a_N < \min_{n=1,\ldots,N-1} \left\{2^{-N}, \frac{a_n M_n}{4 \cdot 2^{-N} k_n}\right\} ,$$

then we choose (y_i^N), again a subsequence of some column of (x_i^n), which is $2M_N$-bad where M_N satisfies (5). Finally k_N is chosen so that $\|\sum_{i=1}^{k_N} b_i^N e_i^N\| > 2M_N$ for some $(b_i^N) \in Ba\,\ell_p$. This completes the induction. It is straightforward to check that (2)–(6) are indeed satisfied. ∎

3. The proof of the main result

The proof for the spreading model version is quite easy, since we can control the length of the sum we consider in the Nth column. This enables us to gain control over the behavior of the columns following the Nth one. Similarly it is straightforward to prove Theorem 1, when we assume additionally that X is embedded into a space with an unconditional basis. The

projection onto the Nth column allows us then to preserve the badness of the Nth column without disturbances from the other columns. In the general case we do not know an easy way to similarly gain easy "access" to the Nth column. One step in the proof will be considerations very similar to the ones employed in the proof of the spreading model case (see Lemma 3.6).

We start with some technical definitions analogous to the ones employed in the proof in the last section.

Definitions 3.1. Let X be a Banach space.

(i) A sequence (x_n) in X is called a *$u\ell_p$-sequence*, if $\|x_n\| \leq 1$ for all $n \in \mathbb{N}$, (x_n) converges weakly to 0 and

$$\sup_{(\sum_{n=1}^{\infty}|\alpha_n|^p)^{1/p}\leq 1} \left\| \sum_{n=1}^{\infty} \alpha_n x_n \right\| < \infty \,. \tag{8}$$

(x_n) is called C-*$u\ell_p$-sequence*, if one can replace (8) by

$$\sup_{(\sum_{n=1}^{\infty}|\alpha_n|^p)^{1/p}\leq 1} \left\| \sum_{n=1}^{\infty} \alpha_n x_n \right\| \leq C \,. \tag{9}$$

(ii) A sequence (x_n) in X is called an *M-bad $u\ell_p$-sequence* for a constant $M < \infty$, if (x_n) is a $u\ell_p$-sequence, and no subsequence of (x_n) is an M-$u\ell_p$-sequence. Thus for all subsequences (y_n) there exist $k \in \mathbb{N}$ and $(\alpha_n)_{n=1}^k \in Ba\,\ell_p$ with

$$\left\| \sum_{n=1}^{k} \alpha_n y_n \right\| > M \,.$$

(iii) An array $(x_i^n)_{i,n=1}^{\infty}$ in X is called a *bad $u\ell_p$-array*, if each sequence $(x_i^n)_{i=1}^{\infty}$ is an M_n-bad $u\ell_p$-sequence for some constants M_n with $M_n \to \infty$ as $n \to \infty$.

(iv) (y_i^k) is called a *subarray* of (x_i^n), if there is a subsequence (n_k) of $\mathbb{N}$ such that every sequence $(y_i^k)_{i=1}^{\infty}$ is a subsequence of $(x_i^{n_k})_{i=1}^{\infty}$.

(v) A bad $u\ell_p$-array $(x_i^n)_{i,n=1}^{\infty}$ is said to satisfy the *ℓ_p-array procedure*, if there exists a subarray (y_i^n) of (x_i^n) and there exist $(a_n) \subseteq \mathbb{R}^+$ with $\sum_{n=1}^{\infty} a_n \leq 1$ such that the (weakly null) sequence (y_i) with $y_i := \sum_{n=1}^{\infty} a_n y_i^n$ has no $u\ell_p$-subsequence.

An immediate consequence of the definitions are the following observations: A subarray of a bad $u\ell_p$-array is a bad $u\ell_p$-array. A bad $u\ell_p$-array satisfies the ℓ_p-array procedure, if and only if it has a subarray satisfying the ℓ_p-array procedure. The "global" idea for the proof of Theorem 1 can be summarized as follows:

Lemma 3.2. *Let X have property (S_p). Then X has property (US_p) if and only if X satisfies the ℓ_p-array procedure.*

Proof. Clearly if X has property (US_p), X does not contain a bad $u\ell_p$-array and so the ℓ_p-array procedure is satisfied. Let us now assume that X has property (S_p) and satisfies the ℓ_p-array procedure, but fails property (US_p). Thus X contains a bad $u\ell_p$-array. Since X satisfies the ℓ_p-array procedure, we can find a sequence (y_i) in X (by the method described in the array procedure), which does not admit of a $u\ell_p$-subsequence. Observing that (y_i) converges weakly to 0 by the conditions imposed on the a_n's, we see that X fails property (S_p), a contradiction. ∎

The preceding lemma reduces the proof of Theorem 1 to the proof of

Theorem 2. *Every Banach space satisfies the ℓ_p-array procedure.*

The proof of Theorem 2 will be broken up into two major steps. First we show that Theorem 2 is valid for the special case of $C(K)$-spaces where K is a countable compact metric space:

Proposition 3.3. *Let K be a countable compact metric space. Then $C(K)$ satisfies the ℓ_p-array procedure.*

The second major step is to show that the case of a general Banach space can be reduced to the special case of Proposition 3.3:

Proposition 3.4. *Let $(x_i^n)_{i,n=1}^\infty$ be a bad $u\ell_p$-array in a Banach space X. Then there exist a subarray (y_i^n) of (x_i^n) and a countable ω^*-compact subset K of BaY^*, where we set $Y = [y_i^n]_{i,n=1}^\infty$, such that $(y_i^n \mid K)$ is a bad $u\ell_p$-array in $C(K)$.*

Let us observe that Theorem 2 is indeed an easy consequence of the Propositions 3.3 and 3.4. If (x_i^n) is a bad $u\ell_p$-array in X, Proposition 3.4 produces a subarray (y_i^n) and a countable ω^*-compact metric space K in BaY^* such that $(y_i^n \mid K)$ is a bad $u\ell_p$-array in $C(K)$. Thus there are $(a_n) \subseteq \mathbb{R}^+$ with $\sum_{n=1}^\infty a_n \leq 1$ so that the sequence (y_i') in $C(K)$ defined by $y_i' = \sum_{n=1}^\infty a_n y_{i\,|\,K}^n$ has no $u\ell_p$-subsequence. Since $K \subseteq BaY^*$ it follows that $y_i = \sum_{n=1}^\infty a_n y_i^n$ itself can have no $u\ell_p$-subsequence in X. Thus (x_i^n) satisfies the ℓ_p-array procedure.

We now present the proof of Proposition 3.3. This follows rather easily by induction from the following result.

Lemma 3.5. *Let (X_n) be a sequence of Banach spaces each satisfying the ℓ_p-array procedure. Then $(\sum_{n=1}^\infty X_n)_{c_0}$ satisfies the ℓ_p-array procedure.*

Before we present its proof we need another lemma:

Lemma 3.6. *Let (X_n) be a sequence of Banach spaces each satisfying the ℓ_p-array procedure and let (x_i^n) be a bad $u\ell_p$-array in some Banach space. Set $X = [x_i^n]_{i,n=1}^\infty$ and suppose that for all $m \in \mathbb{N}$ there is a bounded linear operator $T_m : X \to X_m$ with $\|T_m\| \leq 1$ such that $(T_m x_i^m)_{i=1}^\infty$ is an m-bad $u\ell_p$-sequence in X_m. Then (x_i^n) satisfies the ℓ_p-array procedure.*

Proof of Lemma 3.6. Let us first consider the (easy) case that there exist an $m \in \mathbb{N}$ and a subarray (y_i^n) of (x_i^n) such that $(T_m y_i^n)_{i,n}$ is a bad $u\ell_p$-array in X_m. Since X_m satisfies the ℓ_p-array procedure, so does $(T_m y_i^n)_{i,n}$. It follows that (y_i^n) satisfies the ℓ_p-array procedure. Thus (x_i^n) itself satisfies the ℓ_p-array procedure.

Now let us assume that the easy case does not apply. By passing to a subarray, if necessary, we can then assume that for all $m \in \mathbb{N}$ there is an integer M_m such that $(T_m x_i^n)_i$ is an M_m-$u\ell_p$-sequence for all $n \in \mathbb{N}$. Inductively we will choose $(m_n) \in \mathcal{P}_\infty(\mathbb{N})$, a subarray $(y_i^n) = (x_i^{m_n})$, $a_n > 0$ with $\sum_{n=1}^\infty a_n \leq 1$ and integers (N_n) such that the following properties hold for all $n \in \mathbb{N}$:

$$(T_{m_n}(y_i^n))_i \text{ is an } m_n\text{-bad } u\ell_p\text{-sequence in } X_{m_n} \tag{10}$$

$$(y_i^n)_i \text{ is an } N_n\text{-}u\ell_p\text{-sequence} \tag{11}$$

$$a_n m_n > n \tag{12}$$

$$\sum_{j=1}^{n-1} a_j N_j < a_n m_n/4 \tag{13}$$

$$\sum_{j=n+1}^{\infty} a_j M_{m_n} < a_n m_n/4 \tag{14}$$

$$\left(T_{m_n} y_i^\ell\right)_i \text{ is an } M_{m_n}\text{-}u\ell_p\text{-sequence for all } \ell\,. \tag{15}$$

Let us note that (10) is automatically fulfilled because of the hypotheses; (15) holds because of our assumption above.

We start the induction as follows. Let $a_1 = 1/2$; choose m_1 such that $a_1 m_1 > 1$ (thus (12) is satisfied for n=1). Since $(y_i^1)_i := (x_i^{m_1})_i$ is a $u\ell_p$-sequence, we can choose an N_1 so that (11) holds. If we require for future a_j's that

$$a_j M_{m_1} < 2^{-j} a_1 m_1/4 \text{ for all } j > 1\,,$$

condition (14) will hold for $n=1$. The condition (13) does not apply for $n = 1$.

Now let $n > 1$ and suppose that $(a_j)_{j=1}^{n-1}$, $(m_j)_{j=1}^{n-1}$ and $(N_j)_{j=1}^{n-1}$ have been chosen such that (11)–(13) hold for $n-1$ and additionally for all $2 \le j < n$

$$0 < a_j < \min\left\{2^{-j}, 2^{-j}\frac{a_k m_k}{4\, M_{m_k}} \;:\; 1 \le k < j\right\}$$

Choose a_n such that

$$0 < a_n < \min\left\{2^{-n}, 2^{-n}\frac{a_k m_k}{4\, M_{m_k}} \;:\; 1 \le k < n\right\}.$$

Then choose m_n large enough to satisfy (12) and (13). This defines $(y_i^n)_i = (x_i^{m_n})_i$. Next choose N_n so that (11) is fulfilled. The induction is complete. Because of the conditions imposed on the a_n's, (14) holds for all $n \in \mathbb{N}$ and $\sum a_n \le 1$.

We set $y_k = \sum_{j=1}^{\infty} a_j y_k^j$. Let (y_{k_i}) be a subsequence of (y_k). We have to show that (y_{k_i}) is not a $u\ell_p$-sequence, *i.e.*,

$$\sup_{\ell\in\mathbb{N}} \sup_{(\alpha_i)\in Ba\,\ell_p^{(\ell)}} \left\|\sum_{i=1}^{\ell} \alpha_i y_{k_i}\right\| = \infty\,.$$

To this end fix n and choose — using (10) — $k \in \mathbb{N}$ and $(\beta_i) \in Ba\,\ell_p$ such that

$$\left\|\sum_{i=1}^{k} T_{m_n}\left(\beta_i y_{k_i}^n\right)\right\| > m_n\,.$$

We obtain the following estimate:

$$\begin{aligned}
\left\|\sum_{i=1}^{k} \beta_i y_{k_i}\right\| &= \left\|\sum_{i=1}^{k}\sum_{j=1}^{\infty} \beta_i a_j y_{k_i}^j\right\| \\
&\ge \left\|\sum_{i=1}^{k}\sum_{j=n}^{\infty} T_{m_n}\left(\beta_i a_j y_{k_i}^j\right)\right\| - \left\|\sum_{i=1}^{k}\sum_{j=1}^{n-1} \beta_i a_j y_{k_i}^j\right\| \\
&\ge \left(a_n \left\|\sum_{i=1}^{k} T_{m_n}\left(\beta_i y_{k_i}^n\right)\right\| - \sum_{j=n+1}^{\infty} a_j \left\|\sum_{i=1}^{k} T_{m_n}\left(\beta_i y_{k_i}^j\right)\right\|\right) - \sum_{j=1}^{n-1} a_j \left\|\sum_{i=1}^{k} \beta_i y_{k_i}^j\right\| \\
&\ge a_n m_n - \sum_{j=n+1}^{\infty} a_j M_{m_n} - \sum_{j=1}^{n-1} a_j N_j \quad \text{by using (10), (15) and (11) resp.} \\
&\ge a_n m_n - a_n m_n/4 - a_n m_n/4 \quad \text{by (13) and (14)} \\
&= a_n m_n/2\,.
\end{aligned}$$

By (12) $a_n m_n \to \infty$ as $n \to \infty$ and the proof is complete. ∎

Proof of Lemma 3.5. Let (x_i^n) be a bad $u\ell_p$-array in $X = (\sum X_n)_{c_0}$. We denote by R_m the natural projections $R_m : X \to X_m$. Lemma 3.5 is an easy consequence of the following claim:

> For all $M < \infty$ there are $n, m \in \mathbb{N}$ and a subsequence (y_i) of (x_i^n) such that $(R_m y_i)_i$ is an M-bad $u\ell_p$-sequence.

Assume the claim is false. Thus we can find $M < \infty$ such that for all $m, n \in \mathbb{N}$ every subsequence of $(x_i^n)_{i=1}^\infty$ contains a further subsequence (y_i) such that $(R_m y_i)_i$ is an M-$u\ell_p$-sequence. Fix n such that (x_i^n) is an $(M+3)$-bad $u\ell_p$-sequence . We can find a subsequence (y_i) of (x_i^n) and $(m_i) \in \mathcal{P}_\infty(\mathbb{N})$ such that for all $i \in \mathbb{N}$

$$\sup_{m > m_i} \|R_m y_i\| \leq i^{-1} \tag{16}$$

$$(R_m y_j)_{j=i+1}^\infty \text{ is an } M\text{-}u\ell_p\text{-sequence for all } m \leq m_i \ . \tag{17}$$

Indeed, set $y_1 = x_1^n$ and choose $m_1 (= 1)$ such that $\|R_m y_1\| \leq 1$ for all $m > m_1$. Now pass to a subsequence $(y_{1;i})_{i=1}^\infty$ of $(x_i^n)_{i=2}^\infty$ such that $(R_m y_{1;j})_{j=1}^\infty$ is an M-$u\ell_p$-sequence for all $m \leq m_1$. Set $y_2 = y_{1;1}$. Choose m_2 such that $\|R_m y_2\| \leq 2^{-1}$ for $m > m_2$. Again by our assumption, we can find a subsequence $(y_{2;i})_{i=1}^\infty$ of $(y_{1;i})_{i=2}^\infty$ such that $(R_m y_{2;j})_{j=1}^\infty$ is an M-$u\ell_p$-sequence for all $m \leq m_2$. Set $y_3 = y_{2;1}$ and continue in the obvious fashion. The sequence (y_i) we have constructed clearly satisfies (16) and (17).

Since (x_i^n) is an $(M+3)$-bad $u\ell_p$-sequence , we can find $(\alpha_j) \in Ba\,\ell_p$ with

$$\left\| \sum_{j=1}^\infty \alpha_j y_j \right\| > M + 3 \ . \tag{18}$$

On the other hand, by (16) and (17) we obtain the following estimate for $i \in \mathbb{N}$ and $m \in (m_{i-1}, m_i]$ (where $m_0 := 0$) :

$$\left\| \sum_{j=1}^\infty R_m(\alpha_j y_j) \right\| \leq \left\| \sum_{j=1}^{i-1} R_m(\alpha_j y_j) \right\| + \|R_m(\alpha_i y_i)\| + \left\| \sum_{j=i+1}^\infty R_m(\alpha_j y_j) \right\|$$

$$\leq (i-1) \cdot (i-1)^{-1} + 1 + M = M + 2 \ .$$

Thus we obtain that

$$\left\| \sum_{j=1}^\infty \alpha_j y_j \right\| = \sup_m \left\| R_m \left(\sum_{j=1}^\infty \alpha_j y_j \right) \right\| \leq M + 2 \ ,$$

which contradicts (18).

The claim allows us to choose integers $N(1) < N(2) < \dots$ and $(M(n))_{n=1}^{\infty}$, and subsequences $(y_i^n)_{i=1}^{\infty}$ of $(x_i^{N(n)})_i$, such that $\left(R_{M(n)} y_i^n\right)_i$ is an n-bad $u\ell_p$-sequence for all $n \in \mathbb{N}$. We let

$$T_n = R_{M(n)} | [y_i^r]_{i,r=1}^{\infty}$$

and apply Lemma 3.6. This finishes the proof. ∎

Proof of Proposition 3.3. Recall that for every countable limit ordinal α we can find a sequence of ordinals $\beta_n < \alpha$, $\beta_n \nearrow \alpha$ such that $C(\alpha)$ is isomorphic to $(\sum C(\beta_n))_{c_0}$. Using induction and Lemma 3.5 we obtain that all $C(\alpha)$-spaces, where α is a countable limit ordinal, satisfy the ℓ_p-array procedure. Thus, in view of the isomorphic classification of $C(K)$-spaces for countable compact metric spaces K (see [2]), all $C(K)$-spaces for countable compact metric spaces K satisfy the ℓ_p-array procedure. ∎

This completes the first major step. In order to complete the proof of Theorem 2 (and thus the proof of Theorem 1) we have to prove Proposition 3.4.

What is the idea behind its proof? Let $(x_i^n)_{i,n=1}^{\infty}$ be the given bad $u\ell_p$-array. Let us focus on the nth column, $(x_i^n)_i$. Since $(x_i^n)_{i=1}^{\infty}$ is a M_n-bad $u\ell_p$-sequence for some constant M_n, we can find for each subsequence (y_i) of $(x_i^n)_{i=1}^{\infty}$ integers $k_1 < k_2 < \dots$ such that $\|\sum_{i=1}^{\infty} \alpha_i y_{k_i}\| > M_n$ for some choice of (α_i) with $(\sum_{i=1}^{\infty} |\alpha_i|^p)^{1/p} \leq 1$. This badness can be "witnessed" by a functional $f \in Ba\,X^*$: there is an $f \in Ba\,X^*$ such that $f(\sum_{i=1}^{\infty} \alpha_i y_{k_i}) > M_n$. The countable ω^*-compact metric space K will be formed from a subcollection of the functionals f, which witness the badness of a subsequence of the n-th column of a subarray (y_i^n) of (x_i^n) for some $n \in \mathbb{N}$. While it is not hard to make sure that only countably many functionals are required to sufficiently witness the badness of the array (x_i^n), major difficulties arise from the restriction that K has to be ω^*-compact.

The construction of the "right" functionals will be given in the proof of Proposition 3.8. Before we state this result and present the construction of the subarray (y_i^n) and K needed for Proposition 3.4 we will change the *shape* of the array (x_i^n) to facilitate the quite technical proofs which follow.

Instead of the *square* array $(x_i^n)_{i,n=1}^{\infty}$ we will use its *triangulated version* $(z_i^n)_{i,n=1}^{\infty}$, where $z_i^n = x_i^n$ if $n \leq i$ and $z_i^n = 0$ otherwise. It is easy to see that a square array satisfies the ℓ_p-array procedure if and only if its triangulated version does. From now on we will drop the zero-entries and just label (x_i^n) in a triangular fashion $(x_i^n)_{1 \leq n \leq i < \infty}$.

We use the triangular labeling to improve the structure of the given bad $u\ell_p$-array (x_i^n) :

Lemma 3.7. *A triangular bad $u\ell_p$-array $(x_i^n)_{n \leq i}$ admits a triangular subarray $(y_i^n)_{n \leq i}$, which*

is a basic sequence in its lexicographical order (where i is the first "letter" and n is the second "letter"): $y_1^1, y_2^1, y_2^2, y_3^1, y_3^2, y_3^3, y_4^1, \ldots$.

Note that we can assume that each column of (x_i^n) is semi-normalized. The proof is then an easy adaptation of the the proof that a normalized weakly null sequence has a basic subsequence, using each column of (x_i^n) is weakly null. We leave the details of the proof of this lemma to the reader.

In the sequel we will assume that the given bad $u\ell_p$-array (x_i^n) is labeled in a triangular fashion and that it is a *bimonotone* basic sequence in its lexicographical order. (Note that we can renorm the underlying Banach space X, since both properties "being a bad $u\ell_p$-array" and "satisfying the ℓ_p-array procedure" are invariant under isomorphisms.) We also assume that $(x_i^n)_i$ is a semi-normalized M_n-bad $u\ell_p$-sequence where $M_n \to \infty$. We will also have occasion to use arrays which are labeled in a trapezoidal manner. For this purpose we define the index set $T_{n_0} = \{(i,n) \mid n \leq i \text{ and } i \geq n_0\}$

Proposition 3.8. *Given $n_0 \in \mathbb{N}$ and a trapezoidal subarray of (x_i^n) indexed by T_{n_0} (and again denoted by (x_i^n)), there exist $(\ell_i)_{i=n_0}^{\infty} \in \mathcal{P}_\infty(\mathbb{N})$ and finite sets $F(i,n) \subseteq [-1,1]$ for $(i,n) \in T_{n_0}$ such that the following holds: If $y_i^n = x_{\ell_i}^n$ for $(i,n) \in T_{n_0}$ and if $n_0 \leq k_1 < k_2 < \ldots < k_q$ are given such that $\| \sum_{i=1}^q \beta_i y_{k_i}^{n_0} \| > M_{n_0}$ for some $(\beta_i) \in Ba\,\ell_p$, then there exists an $f \in 3\,Ba\,Y^*$ (where $Y = [y_i^n]_{(i,n)\in T_{n_0}}$) such that*

$$\sum_{i=1}^{q} f(\alpha_i y_{k_i}^{n_0}) > M_{n_0}/4 \quad \text{for some} \quad (\alpha_i) \in Ba\,\ell_p \tag{19}$$

$$f(y_i^n) \in F(i,n) \quad \text{for all} \quad (i,n) \in T_{n_0} \tag{20}$$

$$f(y_i^n) = 0 \quad \text{for all} \quad (i,n) \in T_{n_0} \quad \text{with} \quad i \notin \{k_1, k_2, \ldots, k_q\}\ . \tag{21}$$

We are now able to complete the proof of Theorem 1 "modulo" the proof of Proposition 3.8.

Proof of Proposition 3.4. We will define the subarray $(y_i^n)_{n\leq i}$ by a diagonal procedure using Proposition 3.8 repeatedly. First we apply Proposition 3.8 for $n_0 = 1$ and the array $(x_i^n)_{n\leq i}$ and obtain a certain subarray $(z_i^n)_{(i,n)\in T_1}$; we set $y_1^1 = z_1^1$. Next we apply the proposition to $(z_i^n)_{(i,n)\in T_2}$. In return we get a new subarray, which we denote by $(z_i^n)_{(i,n)\in T_2}$. The second row of (y_i^n) is defined by setting $y_2^1 = z_2^1$ and $y_2^2 = z_2^2$.

If the ℓth row of (y_i^n) has been defined (via the array $(z_i^n)_{(i,n)\in T_\ell}$), we apply Proposition 3.8 to the array $(z_i^n)_{(i,n)\in T_{\ell+1}}$. The first row of the array we obtain will be the $(\ell+1)$st row of

(y_i^n). This completes the induction. Clearly $(y_i^n)_{n\le i}$ is a subarray of $(x_i^n)_{n\le i}$. Moreover (y_i^n) in its lexicographical order is a subsequence of (x_i^n) in its lexicographical order and thus also bimonotone. Furthermore $(y_i^n)_i$ is a subsequence of $(x_i^n)_i$ for all $n \in \mathbb{N}$.

The set K is constructed next. Let $Y = [y_i^n]_{n\le i}$ and $m \in \mathbb{N}$, and define

$$\mathcal{K}_m = \Bigg\{ (k_1, k_2, \dots, k_q) \mid \quad m \le k_1 < k_2 < \dots < k_q,$$

$$\Big\| \sum_{i=1}^{r} \alpha_i y_{k_i}^m \Big\| \le M_m \text{ for all } (\alpha_i) \subseteq Ba\,\ell_p \text{ for all } r < q$$

$$\text{and } \Big\| \sum_{i=1}^{q} \alpha_i y_{k_i}^m \Big\| > M_m \text{ for some } (\alpha_i) \subseteq Ba\,\ell_p. \Bigg\}$$

Whenever $\vec{k} = (k_1, \dots, k_q) \in \mathcal{K}_m$, our application of Proposition 3.8 in the definition of the subarray (y_i^n) yields a functional f of norm not greater than 3, which is defined on a certain subspace of X, spanned by some elements of (x_i^n). By first restricting to $[(y_i^n)]_{(i,n)\in T_{n_0}}$ and then extending to Y in a trivial manner, we may assume that $f \in 3\ BaY^*$ with $\sum_{i=1}^{q} f(\alpha_i y_{k_i}^n) > M_n/4$ for some $(\alpha_i) \in Ba\,\ell_p$ (among its other properties (20) and (21)). It is here, where we use that (y_i^n) is a bimonotone basic sequence. We denote the functional $f/3$ by $f_{\vec{k}}$ and let

$$K_n = \{ Q_m^* f_{\vec{k}} \mid m \in \mathbb{N}, \vec{k} \in \mathcal{K}_n \}\ .$$

Here Q_m denotes the natural projection of norm 1 from Y onto $[y_i^n]_{1\le n\le i\le m}$. Finally we define

$$K = \bigcup_{n=1}^{\infty} K_n \cup \{0\}\ .$$

Let us first note that $(y_i^n{}_{\restriction K})_{n\le i}$ is a bad $u\ell_p$-array. Indeed, fix a column n_0. By our construction $(y_i^{n_0})_{i=1}^{\infty}$ is an M_{n_0}-bad $u\ell_p$-sequence . Consequently, given a subsequence $(y_{k_i}^{n_0})$ of $(y_i^{n_0})$, $\vec{k} := (k_1, \dots, k_q) \in \mathcal{K}_{n_0}$ for some $q \in \mathbb{N}$. By (21) $f_{\vec{k}} = Q_m^* f_{\vec{k}}$ for m large enough and thus $f_{\vec{k}} \in K_{n_0} \subseteq K$. By (19) we obtain that $(y_i^{n_0}{}_{\restriction K})_{i=1}^{\infty}$ is an $M_{n_0}/12$-bad $u\ell_p$-sequence in $C(K)$.

By the construction of K it is obvious that K is a countable subset of BaY^*. Since Y is separable, K is ω^*-metrizable. It remains to show that K is ω^*-closed.

Let $(g_j) \subseteq K$ and assume that (g_j) converges ω^* to some $g \in BaY^*$. We have to show that $g \in K$. Every g_j is of the form $Q_{m_j}^* f_{\vec{k}_j}$ for some $m_j \in \mathbb{N}$ and some $\vec{k}_j \in \mathcal{K}_{n_j}$ for some $n_j \in \mathbb{N}$. By passing to a subsequence of the sequence (g_j) we can assume that either $n_j \to \infty$ as $j \to \infty$, or there is an $n \in \mathbb{N}$ such that $n_j = n$ for all $j \in \mathbb{N}$.

Let us first deal with the first alternative. Let i_j be the first element of $\vec{k}_j$. Since $i_j \ge n_j$, $i_j \to \infty$ as $j \to \infty$. Moreover $f_{\vec{k}_j}(y_i^n) = 0$ for all $n \le i < i_j$ by (21). Thus we obtain that $f_{\vec{k}_j} \to 0$ in the ω^*-topology as $j \to \infty$, i. e. $g = 0 \in K$.

From now on we assume that there is an $n \in \mathbb{N}$ such that $\vec{k}_j \in \mathcal{K}_n$ for all $j \in \mathbb{N}$. $\mathcal{K}_n$ is relatively sequentially compact, if endowed with the relative product topology on $\{0,1\}^{\mathbb{N}}$; thus we can assume, by passing to a subsequence of (g_j) if necessary, that $\vec{k}_j \to \vec{k}$ for some $\vec{k} \in \overline{\mathcal{K}_n}$, the closure of $\mathcal{K}_n$.

We claim that $\vec{k}$ is finite. Suppose to the contrary that $\vec{k} = (k_i)_{i=1}^{\infty}$. Since $\vec{k} \in \overline{\mathcal{K}_n}$ we can find for every $q \in \mathbb{N}$ an element in $\mathcal{K}_n$ of the form $(k_1, \ldots, k_q, \ell_1, \ldots, \ell_r)$. By the definition of $\mathcal{K}_n$, $\left\| \sum_{i=1}^{q} \alpha_i y^n_{k_i} \right\| \leq M_n$ for all $(\alpha_i) \subseteq Ba\,\ell_p$. Thus $(y^n_{k_i})_{i=1}^{\infty}$ is an M_n-$u\ell_p$-sequence, a contradiction to the fact that $(y^n_i)_{i=1}^{\infty}$ is an M_n-bad $u\ell_p$-sequence .

Since $Ba\,Y^*$ is ω^*-sequentially compact, we may additionally assume that $f_{\vec{k}_j}$ converges in the ω^*-sense to some $f \in Ba\,Y^*$. We claim that $f \in K$.

First we observe that $Q^*_m f \in K$ for all $m \in \mathbb{N}$. Why? By (20) and (21) the set $\{Q^*_m f_{\vec{k}_j}(y^n_i) \mid j \in \mathbb{N}, 1 \leq n \leq i\}$ has only finitely many elements. Since $Q^*_m f_{\vec{k}_j} \overset{\omega^*}{\to} Q^*_m f$ as $j \to \infty$, we obtain that $Q^*_m f_{\vec{k}_j} = Q^*_m f$ for $j \in \mathbb{N}$ large enough; in particular $Q^*_m f \in K$. Next let $q = \max \vec{k}$. Since $\vec{k}_j \to \vec{k}$ and $\vec{k}$ is finite, $Q^*_q f = f$. Thus $f \in K$.

We want to show that $g \in K$. By passing to yet another subsequence of the (g_j) we can assure that either $m_j \to \infty$ as $j \to \infty$, or there is an $m \in \mathbb{N}$ such that $m_j = m$ for all $j \in \mathbb{N}$. If the first case occurs or if $m \geq q$ in the second case, $g_j = Q^*_{m_j} f_{\vec{k}_j}$ converges ω^* to f, hence $g = f \in K$. If on the other hand the second case applies and $m < q$, $g_j = Q^*_{m_j} f_{\vec{k}_j}$ converges ω^* to $Q^*_m f$. Since $f \in K$, $g = Q^*_m f \in K$. ∎

The major tool for the construction of the subarray (y^n_i) in Proposition 3.8 is the next lemma, which is a variation (and generalization) of a lemma employed by J. Elton in his thesis [6]. It allows us to "almost" achieve the condition in (21) for one row preceding the k_i's. Its versatility will allow repeated application to construct the array (y^n_i) row by row, carefully preserving properties obtained in previous steps.

Lemma 3.9. *Let $n_0 \in \mathbb{N}$ and let $(x^n_i)_{(i,n)\in T_{n_0}}$ be our bad $u\ell_p$-array indexed trapezoidally. Let $B \subseteq 2\,Ba\,X^*$, $C > 0$, $\varepsilon > 0$, $\delta > 0$, $N \in \mathcal{P}_\infty(\mathbb{N})$ and $n \geq n_0$ be given. Then there exists $L \in \mathcal{P}_\infty(N)$ such that if $(\ell_i)_{i=0}^{q} \subseteq L$ with $n \leq \ell_0 < \ell_1 < \ldots < \ell_q$ is given and if there exists an $f \in B$ with*

$$\sum_{j=1}^{q} f(\alpha_j x^{n_0}_{\ell_j}) > C \quad \text{for some } (\alpha_j) \text{ with } \left(\sum_{j=1}^{q} |\alpha_j|^p \right)^{1/p} \leq \delta \tag{22}$$

then there exists a $g \in B$ with

$$\begin{cases} \displaystyle\sum_{j=1}^{q} g(\beta_j x^{n_0}_{\ell_j}) > C \quad \text{for some } (\beta_j) \text{ with } \left(\sum_{j=1}^{q} |\beta_j|^p \right)^{1/p} \leq \delta \\ \text{and } \left| g(x^m_{\ell_0}) \right| < \varepsilon \text{ for } 1 \leq m \leq n\ . \end{cases} \tag{23}$$

Proof. We let

$$\mathcal{A}_q = \{I = (\ell_j)_{j=0}^{\infty} \in \mathcal{P}_\infty(N) \mid \ell_0 \geq n \text{ and if there is an } f \in B \text{ satisfying (22),}$$
$$\text{then there is a } g \in B \text{ satisfying (23)} \} .$$

Let $\mathcal{A} = \bigcap_{q=1}^{\infty} \mathcal{A}_q$. Every set $\mathcal{A}_q$ (and hence $\mathcal{A}$) is a Ramsey set, since the conditions imposed on I involve only the first $(q+1)$ elements in I. By Ramsey theory (see *e.g.*, [14] for a discussion of Ramsey theory) we can thus find an $L \in \mathcal{P}_\infty(N)$ such that either $\mathcal{P}_\infty(L) \subseteq \mathcal{A}$ or $\mathcal{P}_\infty(L) \subseteq \mathcal{P}_\infty(N) \setminus \mathcal{A}$. In the first case the proof is finished; we show that the second Ramsey alternative leads to a contradiction.

We write $L = \{\ell_0, \ell_1, \ldots\}$. For fixed $q \in \mathbb{N}$ and $1 \leq r \leq q$ we let $L_r = \{\ell_r, \ell_{q+1}, , \ell_{q+2} \ldots\}$. By our assumption $L_r \notin \mathcal{A}$, thus $L_r \notin \mathcal{A}_{s_r}$ for some $s_r \in \mathbb{N}$. Consequently there exist (α_j^r) with $(\sum_{j=1}^{s_r} |\alpha_j^r|^p)^{1/p} \leq \delta$ and a function $f_r \in B$ such that

$$f_r\left(\sum_{j=1}^{s_r} \alpha_j^r x_{\ell_{q+j}}^{n_0}\right) > C$$

and whenever $g(\sum_{j=1}^{s_r} \beta_j x_{\ell_{q+j}}^{n_0}) > C$ for some function $g \in B$ and some (β_j) with $(\sum_{j=1}^{s_r} |\beta_j|^p \leq \delta$, then $\left|g(x_{\ell_r}^m)\right| \geq \varepsilon$ for some $1 \leq m \leq n$.

Let r_0 be chosen such that $s_{r_0} = \min\{s_r \mid 1 \leq r \leq q\}$. We obtain for each $1 \leq r \leq q$

$$C < f_{r_0}\left(\sum_{j=1}^{s_{r_0}} \alpha_j^{r_0} x_{\ell_{q+j}}^{n_0}\right) = f_{r_0}\left(\sum_{j=1}^{s_r} \alpha_j^{r_0} x_{\ell_{q+j}}^{n_0}\right) ,$$

where we set $\alpha_j^{r_0} = 0$ for $s_{r_0} < j \leq s_r$. It follows that for each r we can find an $1 \leq m_r \leq n$ with $\left|f_{r_0}(x_{\ell_r}^{m_r})\right| \geq \varepsilon$.

Now we let q vary. Set $g_q = f_{r_0}$ and let g be a ω^*-clusterpoint of $\{g_q\}_{q \in \mathbb{N}}$. Due to our construction we can find for each $r \in \mathbb{N}$ a column $m_r', 1 \leq m_r' \leq n$, with $|g(x_{\ell_r}^{m_r'})| \geq \varepsilon$. Thus one of the sequences $(x_{\ell_i}^m)_{i=1}^{\infty}$, $1 \leq m \leq n$, is not weakly null, a contradiction. ∎

Proof of Proposition 3.8. In preparation for the diagonal procedure which follows we will introduce the following quantities and sets.

Let $\varepsilon = \min\{1, M_{n_0}/4\}$. Let (b_i^n) be the biorthogonal functionals associated with the bimonotone basic sequence $(x_i^n)_{(i,n) \in T_{n_0}}$ in its lexicographical order. For $(i,n) \in T_{n_0}$ we choose $(\varepsilon_i^n) > 0$ such that

$$\sum_{i=n_0}^{\infty} \sum_{n=1}^{i} \varepsilon_i^n \|b_i^n\| < \varepsilon . \tag{24}$$

We fix for each $(i,n) \in T_{n_0}$ a finite $\varepsilon_i^n/2$-net in $[-1,1]$ denoted by H_i^n. Let

$$B^1 := \{f \in 2\,Ba\,X^* \mid f(x_i^n) \in H_i^n \text{ for all } (i,n) \in T_{n_0}\}$$

Let us observe that by (24), whenever we can find a $g \in Ba\,X^*$, $(\alpha_i) \in Ba\,\ell_p$, and $(\ell_i)_{i=1}^{\infty}$, $\ell_1 > n_0$, with $g(\sum_{n=1}^{\infty} \alpha_i x_{\ell_i}^{n_0}) > M_{n_0}$, then there is an $f \in B^1$ with $f(\sum_{n=1}^{\infty} \alpha_i x_{\ell_i}^{n_0}) > \frac{3}{4} M_{n_0}$.

Next we choose $\varepsilon_m > 0$ for $m \geq n_0$ so that

$$\sum_{m=n_0}^{\infty} m\varepsilon_m \sup\{\|b_i^n\| \mid (i,n) \in T_{n_0}, n \leq m\} < \varepsilon \ . \tag{25}$$

The suprema above are finite, since each of the sequences $(x_i^n)_{i=1}^{\infty}$ was assumed to be semi-normalized.

For $m \geq n_0$ we let Γ_m be a finite ε_m-net in the interval $(0, M]$ and we let Δ_m be a set of positive reals such that the set $\{\delta^p : \delta \in \Delta_m\}$ is a (2^{-mp})-net for $[0,1]$, which contains 1. Furthermore we require that $\Delta_m \subseteq \Delta_{m+1}$ for all $m \geq n_0$.

We are ready to start the induction. Choose $C_1 \in \Gamma_{n_0}$ and $\delta_1 \in \Delta_{n_0}$ arbitrarily, apply Lemma 3.9 to $(B^1, \varepsilon_{n_0}, C_1, \delta_1, n_0, \mathbb{N})$ and obtain $L_1^1 \in \mathcal{P}_\infty(\mathbb{N})$. We pick $\delta_2 \in \Delta_{n_0}, \delta_1 \neq \delta_2$, and apply the lemma to $(B^1, \varepsilon_{n_0}, C_1, \delta_2, n_0, L_1^1)$, obtaining a new infinite subset L_2^1. We continue applying Lemma 3.9 successively until we have exhausted all combinations for which $(C, \delta) \in \Gamma_{n_0} \times \Delta_{n_0}$. If L_1 is the last infinite subset of $\mathbb{N}$ thus obtained, we let $\ell_1 = \min L_1$. This defines the first row of the (trapezoidal) subarray (y_i^n): $y_{n_0}^j = x_{\ell_1}^j$ for $1 \leq j \leq n_0$. We set $F(n_0, j) = H_{\ell_1}^j$ for $1 \leq j \leq n_0$.

For the second step of the induction we first partition the set B^1 into finitely many sets as follows: For $\vec{t} = (t_1, \ldots, t_{n_0}) \in \prod_{j=1}^{n_0} F(n_0, j)$ we let

$$B_{\vec{t}}^2 = \left\{ f \in B^1 \mid f(y_{n_0}^j) = t_j \ \text{ for all } \ 1 \leq j \leq n_0 \right\} \ .$$

Similarly to what we did in the first step we apply Lemma 3.9 successively to $(B_{\vec{t}}^2, \varepsilon_{n_0+1}, C, \delta, n_0 + 1, \cdot)$ beginning with the infinite set L_1, until we have exhausted all combinations $(\vec{t}, C, \delta) \in \prod_{j=1}^{n_0}(F(n_0, j) \times \Gamma_{n_0+1} \times \Delta_{n_0+1})$. Let L_2 denote the last sequence thus obtained. We choose as ℓ_2 an element in L_2 with $\ell_2 > \ell_1$. This defines the second row of the subarray: $y_{n_0+1}^j = x_{\ell_2}^j$ for $1 \leq j \leq n_0 + 1$. We set $F(n_0 + 1, j) = H_{\ell_2}^j$ for $1 \leq j \leq n_0 + 1$.

For the general induction step let us assume that $\ell_1 < \ell_2 < \ldots < \ell_m$ and L_m have been found in the way now described. This defines the first m rows of $(y_i^n)_{(i,n) \in T_{n_0}}$. We set $F(m', j) = H_{\ell_m}^j$ for $1 \leq j \leq m'$, where $m' = n_0 + m - 1$. We partition B^1 — as before — into finitely many sets: for $\vec{t} = (t_i^n) \in \prod \{F(i,n) \mid (i,n) \in T_{n_0} \text{ and } i \leq m'\}$ we let

$$B_{\vec{t}}^{m+1} = \left\{ f \in B^1 \mid f(y_i^n) = t_i^n \ \text{ for all } \ (i,n) \in T_{n_0} \ \text{ with } \ i \leq m' \right\} \ .$$

Now we apply Lemma 3.9, starting with the sequence L_m, successively to $(B_{\vec{t}}^{m+1}, \varepsilon_{m'+1}, C, \delta, m'+1, \cdot)$ where $(\vec{t}, C, \delta)$ ranges over all possible combinations in $\prod(F(i,n) \times \Gamma_{m'+1} \times \Delta_{m'+1})$. If

L_{m+1} is the last sequence we obtain, we choose $\ell_{m+1} \in L_{m+1}$ with $\ell_{m+1} > \ell_m$. The induction is complete.

From now on we will identify the functionals $f \in B^1$ with their restrictions to $Y = [y_i^n]_{(i,n)\in T_{n_0}}$.

The subarray $(y_i^n)_{(i,n)\in T_{n_0}}$ of $(x_i^n)_{(i,n)\in T_{n_0}}$ has been chosen such that the following holds:

Lemma 3.10. *Let $n_0 \le n < k_1 < k_2 < \ldots < k_r$ be given. If there exists an $f \in B^1$ such that $\sum_{i=1}^r f(\alpha_i y_{k_i}^{n_0}) > C$ for some (α_i) with $(\sum_{i=1}^r |\alpha_i|^p)^{1/p} \le \delta$, $\delta \in \Delta_n$, $C \in \Gamma_n$, then there exists $g \in B^1$ with*

$$\sum_{i=1}^r g(\beta_i y_{k_i}^{n_0}) > C \quad \text{for some } (\beta_i) \text{ with } \left(\sum_{i=1}^r |\beta_i|^p\right)^{1/p} \le \delta \tag{26}$$

$$g(y_i^m) = f(y_i^m) \in F(i,m) \quad \text{for all } (i,m) \in T_{n_0} \text{ with } i < n \tag{27}$$

$$|g(y_n^m)| < \varepsilon_n \quad \text{for all } 1 \le m \le n\ . \tag{28}$$

We claim that the proof of Proposition 3.8 is complete, once we prove

Lemma 3.11. *Let $n_0 < k_1 < \ldots < k_q$ be given. If $\|\sum_{i=1}^q \alpha_i y_{k_i}^{n_0}\| > M_{n_0}$ for some (α_i) with $(\sum_{i=1}^q |\alpha_i|^p)^{1/p} \le 1$, then there exists an $h \in B^1$ such that*

$$\sum_{i=1}^q h(\gamma_i y_{k_i}^{n_0}) > M_{n_0}/4 \quad \text{for some } (\gamma_i) \text{ with } \left(\sum_{i=1}^q |\gamma_i|^p\right)^{1/p} \le 1 \tag{29}$$

$$h(y_i^n) = 0 \quad \text{for all } n \ge n_0\ , \quad \text{if} \quad i > k_q \tag{30}$$

$$|h(y_i^n)| < \varepsilon_i \quad \text{for all } n \ge n_0\ , \quad \text{if} \quad i \notin \{k_1, k_2, \ldots, k_q\}\ . \tag{31}$$

Completion of the proof of Proposition 3.8. Indeed, by perturbing h, we can obtain an $f \in Y^*$ such that for $(i,n) \in T_{n_0}$, $f(y_i^n) = h(y_i^n)$ if $i \in \{k_1, k_2, \ldots, k_q\}$ and $f(y_i^n) = 0$ otherwise. Thus (21) is satisfied. Moreover it follows that $\sum_{i=1}^q f(\gamma_i y_{k_i}^{n_0}) = \sum_{i=1}^q h(\gamma_i y_{k_i}^{n_0}) > M_{n_0}/4$ and $f(y_i^n) \in F(i,n)$ for all $(i,n) \in T_{n_0}$. Thus (19) and (20) hold. By using (25) we can estimate $\|f\|$ as follows:

$$\begin{aligned}
\|f - h\| &\le \sum_{i=n_0}^{k_q} \varepsilon_i \sum_{n=1}^{i} \|b_{\ell_i}^n\| \\
&\le \sum_{m=n_0}^{\infty} m \cdot \varepsilon_m \cdot \sup\{\|b_i^n\| \mid (i,n) \in T_{n_0}, n \le m\} \\
&< \varepsilon \le 1\ .
\end{aligned}$$

Since $\|h\| \leq 2$, we have $\|f\| \leq 3$. This ends the proof of Proposition 3.8. ∎

Proof of Lemma 3.11. If $\|\sum_{i=1}^{q} \alpha_i y_{k_i}^{n_0}\| > M_{n_0}$ for some $q \in \mathbb{N}$, $n_0 \leq k_1 < \ldots < k_q$ and some (α_i) with $(\sum_{i=1}^{q} |\alpha_i|^p)^{1/p} \leq 1$, we can find by our earlier observation a function $g \in B^1$ with $g(\sum_{i=1}^{q} \alpha_i y_{k_i}^{n_0}) > \frac{3}{4} M_{n_0}$.

We will apply Lemma 3.10 (at most) $(k_q - n_0 + 1)$ times beginning with the function g and the row $n = n_0$. We choose $C_{n_0} \in \Gamma_{n_0}$ such that $0 < \frac{3}{4} M_{n_0} - C_{n_0} < \varepsilon_{n_0}$.

If $n_0 = k_1$, we set $h_{n_0} = g$ and $\alpha_{i,n_0} = \alpha_i$ for $1 \leq i \leq q$. We set $\gamma_1 = \alpha_{1,n_0}$, if $h_{n_0}(\alpha_{1,n_0} y_{n_0}^{n_0}) \geq 0$, and $\gamma_1 = -\alpha_{1,n_0}$ otherwise. We let $\beta_{n_0} = h_{n_0}(\gamma_1 y_{n_0}^{n_0})$ and choose a $\delta_1 \in \Delta_{n_0}$ such that $\delta_1^p - 2^{-p} \leq \sum_{j=2}^{q} |\alpha_{j,n_0}|^p \leq \delta_1^p$.

If on the other hand $n_0 < k_1$, we apply Lemma 3.10 to g, $n_0 = n < k_1 < \ldots < k_q$, $\delta = 1$ $(\in \Delta_{n_0})$ and $C = C_{n_0} \in \Gamma_{n_0}$. Note that $C < \frac{3}{4} M_{n_0}$ and that therefore the premise of the lemma is indeed fulfilled. The application yields a new functional $h_{n_0} \in B^1$ and a new $(\alpha_{i,n_0})_{i=1}^{q} \in Ba\ell_p^q$ with $\sum_{i=1}^{q} h_{n_0}(\alpha_{i,n_0} y_{k_i}^{n_0}) > C_{n_0}$ and $|h_{n_0}(y_{n_0}^m)| < \varepsilon_{n_0}$ for $1 \leq m \leq n_0$. We let $\beta_{n_0} = 0$ and $\delta_1 = 1$.

Let us now assume that $s \geq n_0$ and we have thus far constructed $h_s \in B^1$, $\alpha_{i,s}$ for $i = 1, \ldots, q$ and for $n_0 \leq r \leq s$, scalars $C_r \in \Gamma_r$, $\delta_r \in \Delta_r$ and $\beta_r \geq 0$, and furthermore for each $1 \leq k_i \leq s$ we have chosen $\gamma_i \in \mathbb{R}$ such that the following conditions hold:

$$0 < (C_{r-1} - \beta_{r-1}) - C_r < \varepsilon_r \quad \text{for all} \quad n_0 \leq r \leq s \tag{32}$$

$$\Big(\sum_{\{i | k_i > r\}} |\alpha_{i,r}|^p \Big)^{1/p} \leq \delta_j \quad \text{for all} \quad n_0 \leq r \leq s \quad \text{with} \quad k_j \leq r < k_{j+1} \tag{33}$$

$$\delta_j^p - 2^{-jp} \leq \sum_{i=j+1}^{q} |\alpha_{i,k_j}|^p \quad \text{for all} \quad n_0 \leq k_j \leq s \tag{34}$$

$$\sum_{\{i | k_i \geq s\}} h_s(\alpha_{i,s} y_{k_i}^{n_0}) > C_s \tag{35}$$

$$\beta_r = h_s(\gamma_i y_{k_i}^{n_0}) \quad \text{if} \quad r = k_i \quad \text{for some} \quad 1 \leq i \leq q\,, \quad \text{and} \quad \beta_r = 0 \quad \text{otherwise} \tag{36}$$

$$|h_s(y_r^m)| < \varepsilon_r \quad \text{for all} \quad 1 \leq m \leq r \leq s \quad \text{with} \quad r \notin \{k_1, k_2, \ldots, k_q\}\,. \tag{37}$$

Let us observe that all these conditions are satisfied for $s = n_0$, if we let $C_{n_0 - 1} = \frac{3}{4} M_{n_0}$, $\beta_{n_0 - 1} = 0$, $k_0 = n_0$ and $\delta_0 = 1$. (We also set $\alpha_{i,n_0 - 1} = \alpha_i$ for $1 \leq i \leq q$.)

Next we choose $C_{s+1} \in \Gamma_{s+1}$ with $0 < (C_s - \beta_s) - C_{s+1} < \varepsilon_{s+1}$ to satisfy (32) for $s+1$. Note that $C_{s+1} < C_s$. (If $C_s - \beta_s < \varepsilon_{s+1}$, we quit the procedure and set $h = Q_s^* h_s$. Estimates below will show that h satisfies the conclusion of Lemma 3.11. We set $\gamma_j = 0$ for $k_j > s$.)

If $s+1 = k_j$ for some $1 \le j \le q$ we set $h_{s+1} = h_s$ and $\alpha_{i,s+1} = \alpha_{i,s}$ for $1 \le i \le q$. We set $\gamma_j = \pm\alpha_{j,s+1}$ so that $\beta_{s+1} := h_{s+1}(\gamma_j y_{k_j}^{n_0}) \ge 0$ and choose $\delta_j \in \Delta_{s+1}$ so that (33) and (34) are satisfied for $s+1$. (36) is satisfied for $s+1$ by our construction; in (37) no new condition is imposed, so it remains to check (35). Indeed we have

$$\begin{aligned}\sum_{\{i|k_i \ge s+1\}} h_{s+1}(\alpha_{i,s+1} y_{k_i}^{n_0}) &= \sum_{\{i|k_i \ge s+1\}} h_s(\alpha_{i,s} y_{k_i}^{n_0}) \\ &\ge \sum_{\{i|k_i \ge s\}} h_s(\alpha_{i,s} y_{k_i}^{n_0}) - \beta_s \quad \text{by (36)} \\ &> C_s - \beta_s > C_{s+1} \quad \text{using (35) for } r = s \text{ and (32) for } r = s+1 \,.\end{aligned}$$

If $s+1 \notin \{k_1, k_2, \ldots, k_q\}$, say $k_{j-1} < s+1 < k_j$ for some j, we apply Lemma 3.10 to h_s, $n = s+1 < k_j < \ldots < k_q$, C_{s+1}, δ_{j-1} and $(\alpha_{i,s})$, $j \le i \le q$. Let us check that the lemma applies for these parameters: Clearly $C_{s+1} \in \Gamma_{s+1}$ and $\delta_{j-1} \in \Delta_s \subseteq \Delta_{s+1}$. Moreover

$$\left(\sum_{i=j}^{q} |\alpha_{i,s}|^p\right)^{1/p} = \left(\sum_{\{i|k_i > s\}} |\alpha_{i,s}|^p\right)^{1/p} \le \delta_{j-1} \quad \text{by (33)}.$$

Finally, as above, we obtain

$$\sum_{i=j}^{q} h_s(\alpha_{i,s} y_{k_i}^{n_0}) = \sum_{\{i|k_i \ge s+1\}} h_s(\alpha_{i,s} y_{k_i}^{n_0}) > C_{s+1} \,.$$

The application of Lemma 3.10 yields a new functional $h_{s+1} \in B^1$, and new $(\alpha_{i,s+1})$ for $j \le i \le q$ with $(\sum_{i=j}^{q} |\alpha_{i,s+1}|^p)^{1/p} \le \delta_{j-1}$, $\sum_{i=j}^{q} h_{s+1}(\alpha_{i,s+1} y_{k_i}^{n_0}) > C_{s+1}$ and $|h_{s+1}(y_{s+1}^m)| < \varepsilon_{s+1}$ for all $1 \le m \le s+1$; hence (37) is satisfied for $r = s+1$. Since h_{s+1} preserves the values of h_s on the rows prior to the $(s+1)$st row, h_{s+1} satisfies (37) also for $1 \le r \le s$. We set $\beta_{s+1} = 0$. By our construction (33), (35) and (36) are satisfied for $s+1$; (34) does not impose a new condition.

Unless we stopped the construction earlier, we quit after we have obtained h_{k_q} and let $h = Q_{k_q}^* h_{k_q}$. Observing that the conclusions (30) and (31) in Lemma 3.11 hold, it remains to

check (29). If h is defined to be $h = Q^*_{k_q} h_{k_q}$ we obtain

$$\sum_{i=1}^{q} h(\gamma_i y^{n_0}_{k_i}) = \sum_{i=1}^{q} \beta_{k_i} \qquad \text{by (36)}$$

$$= \beta_{k_q} + \sum_{r=n_0}^{k_q} \beta_{r-1}$$

$$\geq \beta_{k_q} + \sum_{r=n_0}^{k_q} (C_{r-1} - C_r - \varepsilon_r)$$

$$\geq (\beta_{k_q} - C_{k_q}) + \tfrac{3}{4} M_{n_0} - \sum_{r=n_0}^{k_q} \varepsilon_r$$

$$\geq \tfrac{3}{4} M_{n_0} - \varepsilon \;\geq\; \tfrac{1}{2} M_{n_0} \,.$$

(Note that $\beta_{k_q} - C_{k_q} \geq 0$ by (35) and (36).)

A similar estimate holds if $h = Q^*_s h_s$ for some $s < k_q$:

$$\sum_{i=1}^{q} h(\gamma_i y^{n_0}_{k_i}) \geq \beta_s + \sum_{r=n_0}^{s} (C_{r-1} - C_r - \varepsilon_r)$$

$$\geq (\beta_s - C_s) + \left(\tfrac{3}{4} M_{n_0} - \sum_{r=n_0}^{s} \varepsilon_r \right) \;\geq\; \tfrac{3}{4} M_{n_0} - \varepsilon \;\geq\; \tfrac{1}{2} M_{n_0} \,.$$

(This time $\beta_s - C_s \geq -\varepsilon_{s+1}$ by our stopping condition.)

The proof is complete, once we show that $(\sum_{j=1}^{q} |\gamma_j|^p)^{1/p} \leq 2$. Indeed, if $h = Q^*_{k_q} h_{k_q}$, we have for $1 \leq j < q$,

$$|\gamma_j|^p = |\alpha_{j,k_j - 1}|^p \leq \sum_{i=j}^{q} |\alpha_{i,k_j - 1}|^p - \sum_{i=j+1}^{q} |\alpha_{i,k_j}|^p$$

$$\leq \delta^p_{j-1} - (\delta^p_j - 2^{-jp}) \;\text{ by (33) and (34);}$$

$$|\gamma_q|^p = |\alpha_{q,k_q - 1}|^p \leq \delta^p_{q-1} \;\text{ by (33).}$$

Thus we obtain

$$\sum_{j=1}^{q} |\gamma_j|^p \leq \sum_{j=1}^{q-1} (\delta^p_{j-1} - \delta^p_j) + \delta^p_{q-1} + \sum_{j=1}^{q-1} 2^{-jp}$$

$$\leq 1 + \sum_{j=1}^{q-1} 2^{-jp} \leq 2 \,.$$

We obtain the same estimate, if we stopped the procedure before reaching the row k_q. ∎

4. Proofs of the corollaries

Proposition 3.8 can be phrased for a single weakly null sequence as follows:

Corollary 3. *Let (x_n) be a semi-normalized weakly null sequence in a Banach space X. Assume none of its subsequences has a C-upper ℓ_p-estimate. Then there exists a subsequence $(y_n) \subseteq (x_n)$, such that for all $(m_n) \in \mathcal{P}_\infty(\mathbb{N})$ there are a functional $f \in Ba([y_n]^*)$, an $\ell \in \mathbb{N}$ and $(\alpha_i)_{i=1}^{\ell} \in Ba\,\ell_p$ such that*

$$f\left(\sum_{i=1}^{\ell} \alpha_i y_{m_i}\right) > C/12$$

$$\text{and } f(y_i) = 0, \text{ if } i \notin \{m_1, \ldots, m_\ell\} .$$

C. Schumacher [**16**] uses this corollary to deduce the following weak-Cauchy criterion for property (S_p), which is a generalization of an analogous result in the c_0-case in [**11**]:

Proposition 4.1. *Let X be a Banach space. The following are equivalent:*

(i) X has property (S_p).

(ii) Every weak Cauchy-sequence in X has a subsequence (y_n) such that, for some constant $C < \infty$, all subsequences (y'_n) of (y_n) satisfy

$$\left\| \sum_{n=1}^{\infty} a_n (y'_n - y'_{n-1}) \right\| < C \quad \text{for all } (a_n) \in Ba\,\ell_p .$$

(Here $y_0 = 0$.)

Corollary 1 is an easy consequence of this proposition:

Proof of Corollary 1. Let (z_n) be a weak Cauchy-sequence in $Ba(X/Y)$ and let $q : X \to X/Y$ denote the quotient map. By a result of R.H. Lohman [**13**] we can find a weak Cauchy-sequence (x_n) in X such that its image under q is some subsequence of (z_n). Since X has property (S_p), (x_n) satisfies the conclusion of (ii) of Proposition 4.1; so does its image under q, which is still a subsequence of (z_n). ∎

Next we present the proof of the second corollary:

Proof of Corollary 2. We will only prove the second statement. The proof of the first statement is quite similar and will be left to the reader. By Theorem 1 we can find constants C_p and C^q such that every weakly null sequence in $Ba\,X$ (resp. in $Ba\,X^*$) admits a subsequence with a C_p-upper ℓ_p-estimate (resp. C^q-upper ℓ_q-estimate). Let (f_n) be a normalized weakly null sequence in X^*. By passing to a subsequence we may assume that (f_n) has a C^q-upper ℓ_q-estimate. We choose a separable subspace Y of X which isometrically norms all the f_n's and denote by g_n the restriction of f_n to Y. Using a result due to W.B. Johnson and H.P. Rosenthal [**10**], we can find a basic sequence $(x_n) \subseteq Y$ with $\|x_n\| \leq 3$ for all n, such that a subsequence of (g_n), which we still denote by (g_n), is biorthogonal to (x_n). Since ℓ_1 does not embed into Y, we can

assume that (x_n) is a weak Cauchy sequence. Furthermore, since Y has property (S_p) we may assume, by passing to a subsequence of (x_n), that $y_n := x_{2n} - x_{2n-1}$ has a $6C_p$-$u\ell_p$-estimate.

We claim that (f_{2n}) has a lower ℓ_q-estimate. Indeed, let (b_n) be given with $(\sum_{n=1}^{\infty} |b_n|^q)^{1/q} = 1$. Choose (a_n) with $(\sum_{n=1}^{\infty} |a_n|^p)^{1/p} = 1$ and $\sum_{n=1}^{\infty} a_n b_n = 1$. We obtain the following estimate:

$$\left\|\sum_{n=1}^{\infty} b_n f_{2n}\right\| \geq \left(\left\|\sum_{m=1}^{\infty} a_m y_m\right\|\right)^{-1} \cdot \left|\left(\sum_{n=1}^{\infty} b_n f_{2n}\right) \cdot \left(\sum_{m=1}^{\infty} a_m y_m\right)\right|$$

$$\geq (6C_p)^{-1} \cdot \left|\sum_{n=1}^{\infty} a_n b_n\right| \quad = \quad (6C_p)^{-1} \ .$$

The projection $Q : X^* \longrightarrow [f_{2n}]$ is defined by $Qf = \sum_{n=1}^{\infty} f(y_n) f_{2n}$. Note that f is only applied to the $u\ell_p$-sequence (y_n) in Y; thus Q is well defined. It is easy to check that Q is the identity map on $[f_{2n}]$ and that $\|Q\| \leq 6C_p C^q$. ∎

J. Elton obtained in [6] the following characterization: *Let (x_n) be a semi-normalized weakly null sequence in a Banach space without a subsequence equivalent to the unit vector basis of c_0. Then (x_n) has a subsequence (y_n) such that $\|\sum_{n=1}^{k} a_n y_n\| \to \infty$ as $k \to \infty$, whenever $(a_n) \notin c_0$.*

We conclude by showing that an analogous result fails in a strong way for the ℓ_p-case:

Proposition 4.2. *Let $1 < p < \infty$. There is a Banach space X with a 1-symmetric basis (e_n) such that the following properties hold:*

(i) (e_n) does not have an upper ℓ_p-estimate.

(ii) There is a sequence $(\alpha_n) \notin \ell_p$ such that $\sum_{n=1}^{\infty} a_n e_n$ converges.

Proof. Choose $1 < p_0 < p < p_1$. It is easy to construct a concave increasing function $\lambda : \mathbb{N} \to \mathbb{R}^+$ with $\lambda(1) = 1$, such that both $B := \{n : \lambda(n) \geq n^{1/p_0}\}$ and $L := \{n : \lambda_n \leq n^{1/p_1}\}$ are infinite subsets of $\mathbb{N}$. Let X be the Lorentz sequence space $d(w, 1)$, where $w_1 = \lambda(1)$ and $w_n = \lambda(n) - \lambda(n-1)$ for $n > 1$. X has a 1-symmetric basis (e_n) with the property $\lambda(n) = \|\sum_{i=1}^{n} e_i\|$ (see [12, I, p.120]). Since $\#B = \infty$, (e_n) does not have an upper ℓ_p-estimate. To see (ii) we proceed as follows: we choose an increasing sequence $(\ell_k) \subset L$ with

$$\ell_k^{1-(p/p_1)} > k^{2p} \ . \tag{37}$$

Set $m_0 = 0$, $m_j = \sum_{k=1}^{j} \ell_k$ and $F_j = (m_{j-1}, m_j]$ for $j \geq 1$. We consider $\sum_{n=1}^{\infty} a_n e_n$, where

$$a_n = j^{-2} \ell_j^{-1/p_1} \ , \quad \text{if} \quad n \in F_j \ .$$

Since $\ell_j \in L$, $\ell_j^{-1/p_1} \|\sum_{n \in F_j} e_n\| \leq 1$, and hence

$$\left\|\sum_{n=1}^{\infty} a_n e_n\right\| = \left\|\sum_{j=1}^{\infty} j^{-2} \ell_j^{-1/p_1} \left(\sum_{n \in F_j} e_n\right)\right\| < \sum_{j=1}^{\infty} j^{-2} < \infty \ .$$

On the other hand, for $N \in \mathbb{N}$,

$$\sum_{n=1}^{m_N} |a_n|^p = \sum_{j=1}^{N} \ell_j (j^{-2} \ell_j^{-1/p_1})^p = \sum_{j=1}^{N} \ell_j^{1-(p/p_1)} j^{-2p} \geq N \quad \text{by (37)} .$$

Thus $(a_n) \notin \ell_p$. ∎

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On tensor products of operators from L^p to L^q †

H.P. ROSENTHAL and S.J. SZAREK

Department of Mathematics
The University of Texas at Austin
Austin, Texas 78712

Department of Mathematics
Case Western Reserve University
Cleveland, Ohio 44106

Abstract. Let μ, μ', ν, ν' denote σ-finite measures on certain measurable spaces, let $1 \le q, p \le \infty$, and let $S : L^p(\mu) \to L^q(\mu')$, $T : L^p(\nu) \to L^q(\nu')$ be given bounded linear operators. It is proved that if $q < p$, then (in general) $S \otimes T$ extends to a bounded linear operator U from $L^p(\mu \times \nu)$ to $L^q(\mu' \times \nu')$ if and only if $1 \le q \le 2 \le p \le \infty$. In this case there are (best) constants $C_{p,q}$ depending only on p and q with $\|U\| \le C_{p,q}\|S\|\,\|T\|$. These constants are all larger than one. They are computed in certain cases. For example, $C_{2,1} = C_{\infty,2} = \sqrt{\pi/2}$. (This contrasts with a lemma due to W. Beckner: if $p \le q$, then $S \otimes T$ always extends to such a U with $\|U\| = \|S\|\,\|T\|$.) It is deduced as a corollary that if $\sum f_i$ and $\sum g_j$ are unconditionally converging series in $L^p(\mu)$ and $L^p(\nu)$, then $\sum_{i,j} f_i \otimes g_j$ unconditionally converges in $L^p(\mu \times \nu)$ provided $1 \le p \le 2$. However for every $2 < p < \infty$ there is an unconditionally converging series $\sum f_i$ in ℓ^p so that $\sum_{i,j} f_i \otimes f_j$ fails to converge unconditionally.

The results we give here are of local character. That is, it suffices to prove the statements quantitatively in the finite-dimensional case of the counting measures on finite sets. Nevertheless in the interest of possible applications, we wish first to formulate the results in their natural generality, which is somewhat greater than that given in the abstract. Suppose $(\Omega, \mathcal{S}, \mu)$ is a given measure space. (We shall ambiguously denote the measure space by the measure μ itself.) Say that μ is *nice* provided a subset A of Ω is measurable if and only if $A \cap S$ is measurable for all S in $\mathcal{S}$ with $\mu(S) < \infty$, and then $\mu(A) = \sup\{\mu(S) : S \text{ is measurable with } S \subset A \text{ and } \mu(S) < \infty\}$. Evidently σ-finite μ are nice. If μ and ν are nice, then the Fubini theorem holds for functions integrable with respect to the product measure $\mu \times \nu$; moreover one also has that $(L^1(\mu))^* = L^\infty(\mu)$. Given $1 \le p \le \infty$, we shall say that a Banach space X

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is an $L^p(\mu)$-space if there is a nice measure space μ with $X = L^p(\mu)$. Given μ, ν nice measure spaces, $1 \le p < \infty$, and closed linear subspaces X and Y of $L^p(\mu)$ and $L^p(\nu)$ respectively, we let $X \overset{p}{\otimes} Y$ denote the closed linear span in $L^p(\mu \times \nu)$ of all functions $f = g \otimes h$ with g in X and h in Y, where $(g \otimes h)(s,t) = g(s)h(t)$ $\mu \times \nu$-a.e. (Thus $L^p(\mu) \overset{p}{\otimes} L^p(\nu) = L^p(\mu \times \nu)$.) For arbitrary normed linear spaces X and Y, we let $X \overset{\infty}{\otimes} Y$ denote $X \overset{\vee}{\otimes} Y$, that is, the completion of $X \otimes Y$ in the least cross norm. (Thus if $X \subset L^\infty(\mu)$ and $Y \subset L^\infty(\nu)$, $X \overset{\infty}{\otimes} Y$ may be identified with the closure of $X \otimes Y$ in $L^\infty(\mu \times \nu)$.) For $1 \le p < \infty$ let $\mathcal{L}^p_{1+}$ denote the class of all Banach spaces isometric to $L^p(\mu)$ for some nice measure μ. Let $\mathcal{L}^\infty_{1+}$ denote the class of all Banach spaces *whose dual* is isometric to $L^1(\mu)$ for some nice measure μ.

Our main result may now be formulated as follows.

Theorem 1. *Let $1 \le q < p \le \infty$, X, Y, X', Y' be Banach spaces with $X, Y \in \mathcal{L}^p_{1+}$ and $X', Y' \in \mathcal{L}^q_{1+}$, and $S : X \to X'$, $T : Y \to Y'$ be given bounded linear operators. Then (in general) $S \otimes T$ extends to a bounded linear operator from $X \overset{p}{\otimes} Y$ to $X' \overset{q}{\otimes} Y'$ if and only if*

$$1 \le q \le 2 \le p \le \infty \,. \tag{1}$$

Of course this naturally divides into two parts.

Theorem 1.a. *Assume p and q satisfy (1) with $q < p$. There are constants $C_{p,q}$ so that for all $X, Y \in \mathcal{L}^p_{1+}$, $X', Y' \in \mathcal{L}^q_{1+}$, and $S : X \to X'$, $T : Y \to Y'$, $S \otimes T$ extends to a bounded linear operator from $X \overset{p}{\otimes} Y$ to $X' \overset{q}{\otimes} Y'$, also denoted $S \otimes T$, with*

$$\|S \otimes T\| \le C_{p,q} \|S\| \, \|T\| \,. \tag{2}$$

Henceforth we shall let $C_{p,q}$ denote the *best* constants satisfying the conclusion of Theorem 1.a.

For part b and the sequel, we let $L^p = L^p(m) = L^p([0,1], \mathcal{S}, m)$ with $\mathcal{S}$ the Lebesgue-measurable subsets of $[0,1]$ and m Lebesgue measure.

Theorem 1.b. *Let $1 \le q, p \le \infty$ fail (1) with $q < p$. There exists a bounded linear operator $T : L^p \to L^q$ so that $T \otimes T$ is unbounded.*

(Of course q, p satisfy the hypotheses of 1.b if and only if $1 \le q < p < 2$ or $2 < q < p \le \infty$.)

It is easily seen that the argument for Theorem 1.b yields the following more general result: *Given X, Y, X', Y' infinite-dimensional, satisfying the conditions of Theorem 1, then there are bounded linear operators $S : X \to X'$ and $T : Y \to Y'$ violating the conclusion of Theorem 1 provided $1 \le q < p \le \infty$ and q, p fail (1).*

We note that the terminology "$\mathcal{L}^p_{1+}$" is no accident. Indeed, an elegant known result [4] yields that if $1 \le p \le \infty$, then $X \in \mathcal{L}^p_{1+}$ if and only if X is an $\mathcal{L}^p_{1+\varepsilon}$ space for all $\varepsilon > 0$; that is,

for all finite-dimensional subspaces F of X and $\varepsilon > 0$, there exist an n and an n-dimensional subspace G of X with $F \subset G$ and $d(G, \ell_n^p) \leq 1 + \varepsilon$. (For Banach spaces X, Y, $d(X, Y)$ denotes the Banach-Mazur distance coefficient: $d(X, Y) = \inf\{\|T\| \, \|T^{-1}\| : T : X \to Y$ is an onto isomorphism$\}$, in case X and Y are isomorphic; otherwise $d(X, Y) = \infty$.) Throughout, we have attempted to make the exposition accessible to non-experts in Banach space theory. The reader is referred to [6] as a fundamental reference for this theory, and for certain concepts which we use and don't explicitly define here, such as that of a symmetric basis.

Before passing to the proofs, we first indicate the organization of the remainder of this article. The concept of the p-tensor $(X \overset{p}{\otimes} Y)$ clearly underlies much of our discussion. This concept was introduced by G. Schechtman in [8] and further elaborated by T. Figiel and W.B. Johnson in [2]. A lemma of W. Beckner (given shortly below as Lemma 2) shows the concept is isometrically and in fact isomorphically determined. (A quantitative formulation of the latter is given below in Corollary 3.) In Proposition 4, we give the known result that $\ell^2 \overset{p}{\otimes} \ell^2$ is isomorphic to ℓ^2 for all $p < \infty$. We also compute the exact value of $d(\ell^2 \overset{p}{\otimes} \ell^2, \ell^2)$ for all such p. This result seems to be new; its proof is somewhat delicate, and is also used in proving Theorem 1.a and in computing the exact values of certain of the constants $C_{p,q}$. By contrast, we note in Proposition 14 that $\ell^p \overset{q}{\otimes} \ell^p \sim \ell^p \overset{1}{\otimes} \ell^p \not\sim \ell^p$ for all $1 \leq q < p < 2$. ("$\sim$" means "is isomorphic to;" the above result (unpublished) was communicated to the first-named author some years ago by Professors Figiel and Johnson.)

In Proposition 9, we compute the exact value of $C_{2,p} = C_{p^*,2}$ (where $\frac{1}{p} + \frac{1}{p^*} = 1$) for $1 \leq p < 2$. We obtain that in fact $C_{2,p} = d(\ell^2 \overset{p}{\otimes} \ell^2, \ell^2)$.

The entire discussion up to this point, besides the basic concepts, involves only Beckner's Lemma and properties of Gaussian independent random variables.

The completion of the proof of Theorem 1.a is given after treating Proposition 9. This requires Grothendieck's inequality, or rather the factorization consequence: every operator from L^p to L^q factors through L^2, for $1 \leq q \leq 2 \leq p \leq \infty$. It would seem worthwhile to know (if possible) the exact constants arising in the quantitative version, especially for the case $p = \infty$, $q = 1$.

In Corollary 10 we show that if $1 \leq p \leq 2$ and $\sum f_i$, $\sum g_j$ are unconditionally converging series in L^p, then $\sum f_i \otimes g_j$ converges unconditionally in L^p. This sharply contrasts with the case $2 < p < \infty$; we exhibit in Proposition 13 an unconditionally converging series $\sum f_i$ in ℓ^p so that $\sum f_i \otimes f_j$ fails to converge unconditionally. (Also our Corollary 8 gives the result (cf. [7]) that if (f_i), (g_j) are unconditional basic sequences in L^p, then $(f_i \otimes f_j)$ is an unconditional basic sequence in L^p *for all* p *with* $1 \leq p < \infty$.) Finally, in Proposition 11, we show that

$C_{p,q} > 1$ for all $1 \le q \le 2 \le p \le \infty$ with $q < p$.

The remainder of the article is concerned with proving the "negative" assertions Theorem 1.b and Proposition 13. The main ingredient is given in Lemma 12, which really yields that if $1 \le q < p < 2$ and $T : \ell^p \to L^q$ is an (into) isomorphism, then $T \otimes T$ is unbounded (from $\ell^p \otimes \ell^p$ to $L^q \otimes L^q$). The quantitative local version of Theorem 1.b is formulated following this Lemma and the deduction of 1.b from it; we pose the problem of finding the correct order of magnitude of the (large) constants arising in this formulation.

We now return to the precise details and arguments beginning with the above-mentioned lemma of W. Beckner [1], which we formulate in our setting as follows:

Lemma 2. (W. Beckner) *Let $1 \le p \le q \le \infty$ and X, Y (resp. X', Y') be closed linear subspaces of $L^p(\mu)$-spaces (resp. $L^q(\mu)$-spaces). Then given bounded linear operators $S : X \to X'$ and $T : Y \to Y'$, $S \otimes T$ extends to a bounded linear operator from $X \overset{p}{\otimes} Y$ to $X' \overset{q}{\otimes} Y'$ (also denoted $S \otimes T$), with $\|S \otimes T\| = \|S\| \, \|T\|$.*

This lemma was formulated for X, Y $L^p(\mu)$-spaces themselves, in [1]. It was later observed by T. Figiel and W.B. Johnson [2], that the result holds in the above generality. An interesting generalization of Beckner's lemma has been given by T. Figiel, T. Iwaniec and A. Pełczyński ([3], Theorem 1.1). There are quite intriguing Banach-space structural consequences of the later formulation, which we will discuss shortly. For the sake of completeness, we first give the

Proof of Lemma 2. Let X, Y, X', Y', S, T be given as in the statement of 2. By just considering the definitions involved, one obtains that $\|S \otimes T\| \ge \|S\| \, \|T\|$. Suppose first that $q = \infty$. The conclusion of Lemma 2 then follows just because the L^p-norm on $X \otimes Y$ is a cross norm and the L^∞-norm on $X' \otimes Y'$ is the least cross norm. Thus, given $h = \sum x_i \otimes y_i$ in $X \otimes Y$, we may choose norm one elements $f \in (X')^*$ and $g \in (Y')^*$ with

$$\begin{aligned}
\|(S \otimes T)h\| &= \sum fS(x_i)gT(y_i) && \text{(by the definition of the least cross norm)} \\
&= \sum S^* f(x_i) T^* g(y_i) \\
&\le \|S^* f\| \, \|T^* g\| \left\| \sum x_i \otimes y_i \right\| && \text{(since the } L^p\text{-norm is a cross norm)} \\
&\le \|S\| \, \|T\| \left\| \sum x_i \otimes y_i \right\| .
\end{aligned}$$

Thus we may assume now that $1 \le p \le q < \infty$, and that nice measure spaces μ, μ', ν, ν' are chosen with $X \subset L^p(\mu)$, $Y \subset L^p(\nu)$, $X' = L^q(\mu')$, and $Y' = L^q(\nu')$. Rather than using general tensor product formulations, we now regard the elements of $X \otimes Y$ as linear combinations of the elements $f \otimes g$ in $L^p(\mu \times \nu)$, where $(f \otimes g)(s,t) = f(s)g(t)$ $\mu \times \nu$-a.e. Thus given

$h = \sum_{i=1}^n f_i \otimes g_i$ in $X \otimes Y$, we have $(S \otimes T)(h) = \sum_{i=1}^n Sf_i \otimes Tg_i$. To see this is indeed a well-defined linear map, it is enough to show that if $\sum f_i(s)g_i(t) = 0 \quad \mu \times \nu$-a.e., then $\sum Sf_i(s)Tg_i(t) = 0 \quad \mu' \times \nu'$-a.e. Thus assuming $\sum f_i \otimes g_i = 0 \quad \mu \times \nu$-a.e., then by Fubini's theorem, ν-a.e. t, $\sum f_i(s)g_i(t) = 0 \quad \mu$-a.e. s. For each such t, $\sum g_i(t)f_i = 0$ in $L^p(\mu)$. Hence $S(\sum g_i(t)f_i) = 0 = \sum g_i(t)Sf_i$ in $L^q(\mu')$. Thus μ'-a.e. s, $\sum (Sf_i)(s)g_i(t) = 0$. But again by Fubini, we can then say that μ' a.e. s, $\sum (Sf_i)(s)g_i(t) = 0 \quad \nu$-a.e. t. Thus fixing such an s, $\sum Sf_i(s)g_i = 0$ in $L^p(\nu)$, so $T(\sum Sf_i(s)g_i) = \sum Sf_i(s)Tg_i = 0$ in $L^q(\nu')$. At last, we have that μ'-a.e. s, $\sum Sf_i(s)Tg_i(t) = 0 \quad \nu'$-a.e. t, showing $\sum Sf_i \otimes Tg_i = 0$.

Now let h be as above. It suffices to show

$$I \overset{\text{df}}{=} \|S \otimes T(h)\|_q^q \le \|S\|^q \|T\|^q \|h\|_p^q \,. \tag{3}$$

Now a.e. t,

$$\begin{aligned} &\int \Big|\sum Sf_i(s)Tg_i(t)\Big|^q \, d\mu'(s) \\ &\quad = \Big\|S\Big(\sum Tg_i(t)f_i\Big)\Big\|_q^q \le \|S\|^q \Big\|\sum Tg_i(t)f_i\Big\|_p^q \\ &\quad = \|S\|^q \big(\psi(t)\big)^{q/p} \quad \text{where} \quad \psi(t) = \int \Big|\sum f_i(s)Tg_i(t)\Big|^p \, d\mu(s) \,. \end{aligned}$$

Thus we have

$$I \le \|S\|^q \int \big(\psi(t)\big)^{q/p} \, d\nu'(t) \,. \tag{4}$$

Since $\psi \in L^{q/p}(\nu')$, $\psi \ge 0$, and $(q/p) \ge 1$, we may choose $\varphi \ge 0$, $\varphi \in L^{(q/p)^*}(\nu')$ with $\|\varphi\|_{(q/p)^*} = 1$ and

$$\int \psi(t)\varphi(t) \, d\nu'(t) = \|\psi\|_{q/p} \tag{5}$$

where $(q/p)^*$ is the index conjugate to q/p. By (4) and (5), we obtain

$$\begin{aligned} I &\le \|S\|^q \Big(\int \psi(t)\varphi(t) \, d\nu'(t)\Big)^{q/p} \\ &= \|S\|^q \Big(\iint \Big|\sum f_i(s)Tg_i(t)\Big|^p \varphi(t) \, d\nu'(t) \, d\mu(s)\Big)^{q/p} \end{aligned} \tag{6}$$

where we have used Fubini's theorem in the last equality.

Now a.e. s,

$$\begin{aligned}
&\int \Big|\sum f_i(s)Tg_i(t)\Big|^p \varphi(t)\,d\nu'(t) \\
&\quad\le \Big(\int \Big|\sum f_i(s)Tg_i(t)\Big|^q d\nu'(t)\Big)^{p/q} \|\varphi\|_{(q/p)^*} \qquad \text{(by Hölder)} \\
&\quad= \Big\|T\Big(\sum f_i(s)g_i\Big)\Big\|_q^p \qquad\qquad \text{(since } \|\varphi\|_{(q/p)^*}=1\text{)} \\
&\quad\le \|T\|^p \Big\|\sum f_i(s)g_i\Big\|_p^p \\
&\quad= \|T\|^p \int \Big|\sum f_i(s)g_i(t)\Big|^p d\nu(t)\ .
\end{aligned}$$

Applying this in (6), we have

$$\begin{aligned}
I &\le \|S\|^q\|T\|^q \Big(\iint \Big|\sum f_i(s)g_i(t)\Big|^p d\nu(t)\,d\mu(s)\Big)^{q/p} \\
&= \|S\|^q\|T\|^q\|h\|_p^q\ , \quad \text{proving (3).}
\end{aligned}$$

∎

Remarks.

1. Beckner's lemma was preceded historically by a remarkable lemma of I. Segal [9]. A slight generalization of both results may be formulated as follows:

Theorem. *Let μ,μ',ν,ν' be nice measure spaces, $1\le q,p\le\infty$, and assume in case $p=q=\infty$ that $L^q(\mu'\times\nu')$ is a dual space. Let $S: L^p(\mu)\to L^q(\mu')$ and $T: L^p(\nu)\to L^q(\nu')$ be given bounded linear operators and suppose either that $p\le q$ or q,p are arbitrary but one of S or T is positive. Then $S\otimes T$ extends to a bounded linear operator U from $L^p(\mu\times\nu)$ to $L^q(\mu'\times\nu')$ with $\|U\|=\|S\|\,\|T\|$.*

If we don't assume that one of S or T is positive, then by our main result, the conclusion of this Theorem fails (in general) for all q,p with $1\le q<p\le\infty$. Of course if $(\Omega,\mathcal{S},\lambda)$ is a σ-finite measure space, $L^\infty(\lambda)=(L^1(\lambda))^*$ and hence $L^\infty(\lambda)$ is a dual space. More generally, this happens provided λ is a complete nice measure space and there exists a disjoint family $\mathcal{G}$ of measurable subsets of finite measure so that for all $S\in\mathcal{S}$, $\lambda(S)=0$ provided $\lambda(S\cap G)=0$ for all $G\in\mathcal{G}$. Thus if both μ' and ν' have this property, then we have that $L^\infty(\mu'\times\nu')=(L^1(\mu'\times\nu'))^*$. To see why the above Theorem is true, we have by Lemma 2 in case $p\le q$ or by Segal's lemma [9] in case S or T is positive, that $\|S\otimes T\|=\|S\|\,\|T\|$, where $L^p(\mu)\otimes L^p(\nu)$ is endowed with the $L^p(\mu\times\nu)$-norm. Evidently there is thus a unique linear extension $S\widetilde{\otimes}T$ of $S\otimes T$ to $L^p(\mu)\overset{p}{\otimes}L^p(\nu)$. In case $p<\infty$, we simply set $U=S\widetilde{\otimes}T$. However the case $p=\infty$ requires additional work, since then we essentially *never* have that $L^\infty(\mu)\overset{\infty}{\otimes}L^\infty(\nu)=L^\infty(\mu\times\nu)$. (I am indebted to T. Schlumprecht for this observation; as he shows in these Longhorn Notes,

in fact $L^\infty(\mu) \overset{\infty}{\otimes} L^\infty(\nu)$ is not usually even isomorphic to a dual Banach space.) Now letting $X = L^\infty(\mu) \overset{\infty}{\otimes} L^\infty(\nu)$ and $Y = L^q(\mu' \times \nu')$, we have that X belongs to $\mathcal{L}^\infty_{1+}$, which implies that X^{**} is a $\mathcal{P}_1$-space; that is, if B is a Banach space with $B \supset X^{**}$, there is a norm-one linear projection from B onto X^{**}. On the other hand, it follows from our assumptions that there is a norm-one projection π from Y^{**} onto Y. (We regard our Banach spaces as canonically embedded in their double duals.) Indeed, if $q = \infty$, this is so because we assume that $L^q(\mu' \times \nu')$ is a dual space; for $1 < q < \infty$ it's trivial because Y is reflexive, while for $q = 1$ this follows from a standard "band"-property of $L^1(\mu)$-spaces. But then we have that if Z is a Banach space with $Z \supset X$ and $V : X \to Y$ is a given bounded linear operator, there is a linear operator $U : Z \to Y$ extending V with $\|U\| = \|V\|$. Indeed just let $Q : Z^{**} \to X^{**}$ be a norm-one projection and set $U = \pi V^{**} Q | Z$; that is, U is obtained by composing the maps

$$Z \xrightarrow{\chi} Z^{**} \xrightarrow{Q} X^{**} \xrightarrow{V^{**}} Y^{**} \xrightarrow{\pi} Y .$$

Thus we set $Z = L^\infty(\mu \times \nu)$ and $V = S \widetilde{\otimes} T$ to obtain the existence of U when $p = \infty$.

2. Assuming μ, μ', ν, ν' are nice measure spaces, $2 \leq p < \infty$, and $S : L^\infty(\mu) \to L^p(\mu')$, $T : L^\infty(\nu) \to L^p(\nu')$ are given bounded linear operators, then by Theorem 1a, $S \otimes T$ uniquely extends to a bounded linear operator $S \widetilde{\otimes} T : L^\infty(\mu) \overset{\infty}{\otimes} L^\infty(\nu) \to L^p(\mu' \times \nu')$ with $\|S \widetilde{\otimes} T\| \leq C_{\infty,p} \|S\| \, \|T\|$. By the reasoning at the end of the above remark, we then obtain that there exists a linear operator $U : L^\infty(\mu \times \nu) \to L^p(\mu' \times \nu')$ extending $S \widetilde{\otimes} T$ with $\|U\| = \|S \widetilde{\otimes} T\|$. Thus we obtain the result formulated in the abstract in case $p = \infty$.

Now let $1 \leq p < \infty$, μ, ν nice, and $X \subset L^p(\mu)$, $Y \subset L^p(\nu)$ be given. As pointed out in [2], Lemma 2 yields that $X \overset{p}{\otimes} Y$ is isometrically determined, independently of *how* X and Y are embedded in $L^p(\mu)$ or $L^p(\nu)$, and even independently of the ambient measure spaces. Indeed, suppose we have also μ', ν' nice, $X' \subset L^p(\mu')$, $Y' \subset L^p(\nu')$, with X, X' and Y, Y' isometric. Let then $S : X \to X'$ and $T : Y \to Y'$ be surjective isometries. Then by Lemma 2, $\|S \otimes T\| = 1 = \|S^{-1} \otimes T^{-1}\|$ and clearly $S^{-1} \otimes T^{-1} = (S \otimes T)^{-1}$ (regarding $S \otimes T$ on $X \overset{p}{\otimes} Y$ and $S^{-1} \otimes T^{-1}$ on $X' \otimes Y'$). The most transparent instance of this phenomenon: let $S : \ell^2 \to L^2$ and $T : \ell^2 \to L^2$ be surjective isometries. Then $S \otimes T : \ell^2(\mathbb{N}^2) \to L^2(m^2)$ is a surjective isometry. (Of course $m^2 = m \times m$, the standard Lebesgue measure on the Lebesgue measurable subsets of $[0,1] \times [0,1)$; as usual $\ell^p = \ell^p(\mathbb{N})$.) This particular example can be reformulated quite simply; if (f_i) and (g_i) are orthonormal bases for L^2, then $(f_i \otimes g_i)_{i,j}$ is an orthonormal basis for $L^2(m^2)$.

As pointed out to the first-named author by Professors Figiel and Johnson several years ago, Lemma 2 (via the above reasoning) also yields that the "p-tensor" is *isomorphically*

determined. That is, if X, X', Y, Y' are closed linear subspaces of various $L^p(\mu)$'s with X isomorphic to X' and Y isomorphic to Y', then $X \overset{p}{\otimes} Y$ is isomorphic to $X' \overset{p}{\otimes} Y'$.

The proof of the following quantitative version of this result is left to the reader.

Corollary 3. *Let μ, ν, μ', ν' be nice measure spaces, $1 \le p \le \infty$, and X, X', Y, Y' closed linear subspaces of $L^p(\mu)$, $L^p(\nu)$, $L^p(\mu')$, and $L^p(\nu')$ respectively, with X isomorphic to X' and Y isomorphic to Y'. Then*

$$d(X \overset{p}{\otimes} Y \ , \ X' \overset{p}{\otimes} Y') \le d(X, X')d(Y, Y') \ .$$

Since ℓ^2 is isometric to a subspace of L^p, it makes sense to consider the Banach space $\ell^2 \overset{p}{\otimes} \ell^2$.

Proposition 4. *Let $1 \le p < \infty$. Then $\ell^2 \overset{p}{\otimes} \ell^2$ is isomorphic to ℓ^2. Moreover, defining C_p by*

$$C_p^{-p} = \pi^{-1/2} 2^{p/2} \Gamma\left(\frac{p+1}{2}\right) \ , \tag{7}$$

then $d(\ell^2 \overset{p}{\otimes} \ell^2, \ell^2) = C_p$ (resp. C_p^{-1}) if $1 \le p \le 2$ (resp. $2 \le p < \infty$).

Remarks.

1. Some other useful identities for C_p:

$$C_p^p = \frac{2^{p/2}\Gamma(p/2)}{2\Gamma(p)} \ ; \ C_p^{-p} = \sqrt{\frac{2}{\pi}} \int_0^\infty x^p e^{-x^2/2} \, dx \ .$$

In particular, $C_1 = \sqrt{\pi/2}$, $C_3^{-3} = 8\sqrt{2/\pi}$, $C_4^{-4} = 3$, $C_{2n}^{-2n} = (2n)!/2^n n!$. By Stirling's formula, $C_p^{-1} \sim e^{-1/2}\sqrt{p}$ as $p \to \infty$.

2. In fact for any infinite cardinal κ, $\ell^2(\kappa)$ is isometric to a subspace of $L^p([0,1]^\kappa)$, and we obtain that again $d(\ell^2(\kappa) \overset{p}{\otimes} \ell^2(\kappa), \ell^2(\kappa)) = C_p$ (resp. C_p^{-1}) if $1 \le p \le 2$ (resp. $2 \le p < \infty$).

For the proof, let (γ_i) be a sequence of independent standard Gaussian random variables (defined on $[0,1]$, say). Letting $\gamma = \gamma_1$ we thus have that

$$\frac{\|\gamma\|_p}{\|\gamma\|_2} = \|\gamma\|_p = C_p^{-1} \ \text{ for all } \ 1 \le p < \infty \ . \tag{8}$$

As is well-known, we have that $[\gamma_i]_p$ is isometric to ℓ^2 and again

$$\frac{\|x\|_p}{\|x\|_2} = C_p^{-1} \ \text{ for all non-zero } \ x \in [\gamma_i]_p \ . \tag{9}$$

(Given $x_1, x_2, \ldots$ in a Banach space B, $[x_i]$ denotes the closed linear span of the x_i's; $[\gamma_i]_p$ denotes the closed linear span of the γ_i's in L^p.)

The following lemma, besides being needed for Proposition 4, will be useful later on.

Lemma 5. *Let $1 \le p < \infty$, $X = [\gamma_i]_2$, $Y = Y_p = [\gamma_i]_p$, and $V_p : X \to Y$ the (formal) identity map. Then $V_p \otimes V_p : X \overset{2}{\otimes} X \to Y \overset{p}{\otimes} Y$ satisfies*

$$\|V_p \otimes V_p\| = C_p^{-1} \quad \text{and} \quad \|V_p^{-1} \otimes V_p^{-1}\| = C_p^2 = \|V_p^{-1}\|^2 \quad \text{if} \quad 1 \le p < 2 \tag{10}$$

while

$$\|V_p^{-1} \otimes V_p^{-1}\| = C_p^{-1} \quad \text{and} \quad \|V_p \otimes V_p\| = C_p^2 = \|V_p\|^2 \quad \text{if} \quad 2 \le p < \infty\ . \tag{11}$$

Proof. If $1 \le p < 2$, (9) yields that $\|V_p^{-1}\| = C_p$ and hence $\|V_p^{-1} \otimes V_p^{-1}\| = \|V_p^{-1}\|^2 = C_p^2$ by Lemma 2; by exactly the same reasoning, $\|V_p \otimes V_p\| = \|V_p\|^2 = C_p^2$ if $2 < p < \infty$.

Now suppose $1 \le p < 2$. We first show

$$\|V_p \otimes V_p\| \le C_p^{-1}\ . \tag{12}$$

Let scalars (c_{ij}) be given with only finitely many of them non-zero, and set $f = \sum_{i,j} c_{ij}\gamma_i \otimes \gamma_j$. It suffices to show

$$\|f\|_p \le C_p^{-1}\|f\|_2\ . \tag{13}$$

Now given any scalars (α_j) with only finitely many non-zero terms, (9) yields that

$$\int \Big|\sum \alpha_j\gamma_j(t)\Big|^p\, dt = C_p^{-p}\Big(\sum |\alpha_j|^2\Big)^{p/2} \tag{14}$$

since (γ_j) is an orthonormal sequence in L^2. Hence

$$\begin{aligned}
\|f\|_p^p &= \iint \Big|\sum_j \Big(\sum_i c_{ij}\gamma_i(s)\Big)\gamma_j(t)\Big|^p\, dt\, ds \\
&= C_p^{-p}\int \Big(\sum_j \Big|\sum_i c_{ij}\gamma_i(s)\Big|^2\Big)^{p/2} ds && \text{(by (14))} \\
&\le C_p^{-p}\Big(\int \sum_j \Big|\sum_i c_{ij}\gamma_i(s)\Big|^2 ds\Big)^{p/2} && \text{(by Hölder's inequality)} \qquad (15) \\
&= C_p^{-p}\Big(\sum_j\sum_i |c_{ij}|^2\Big)^{p/2} && \text{(by orthonormality of the } \gamma_i\text{'s)} \\
&= C_p^{-p}\|f\|_2^2 && \text{(by orthonormality of the } \gamma_i \otimes \gamma_j\text{'s.)}
\end{aligned}$$

Hence (13) and thus (12) are established. Next, we claim

$$\|V_p \otimes V_p\| \ge C_p^{-1}\ . \tag{16}$$

To see this, for each n, let $f_n = n^{-1/2}\sum_{i=1}^n \gamma_i \otimes \gamma_i$. Then for each n, $\|f_n\|_2 = 1$ and by the second equality of (15),

$$\|f_n\|_p^p = C_p^{-p}\int n^{-p/2}\left(\sum_{i=1}^n \gamma_i^2(s)\right)^{p/2} ds$$

(17)

$$= C_p^{-p}\int\left(\frac{1}{n}\sum_{i=1}^n \gamma_i^2(s)\right)^{p/2} ds\ .$$

But by the Kolmogoroff strong law of large numbers, $\frac{1}{n}\sum_{i=1}^n \gamma_i^2(s) \to 1$ a.e., since $\gamma_1^2, \gamma_2^2, \ldots$ are i.i.d. (independent identically distributed) with common mean equal to 1. Hence by Fatou's Lemma,

$$\|V_p \otimes V_p\| \geq \varliminf_{n\to\infty} \|f_n\|_p^p \geq C_p^{-p}\int \varliminf_{n\to\infty}\left(\frac{1}{n}\sum_{i=1}^n \gamma_i^2(s)\right)^{p/2} ds$$

$$= C_p^{-p}\ , \quad \text{proving (16).}$$

The entirely similar reasoning which establishes the remaining equality of (11) is left to the reader. ∎

We return to the proof of Proposition 4. Since $V_p^{-1} \otimes V_p^{-1} = (V_p \otimes V_p)^{-1}$, we evidently obtain immediately from Lemma 5 that

$$d(\ell^2 \otimes \ell^2, \ell^2) \leq C_p \ \ (\text{resp. } C_p^{-1}) \text{ if } \ 1 \leq p \leq 2 \ \ (\text{resp. } \ 2 \leq p < \infty)\ .$$

Now let $Z_p = [\gamma_i \otimes \gamma_i]_p$ and $T_p = V_p \otimes V_p | Z_2$. (Thus T_p^{-1} maps Z_p onto Z_2.) Evidently $(\gamma_i \otimes \gamma_i)$ is a symmetric basis for Z_p; in fact our proof of Lemma 5 yields that

$$\|T_p\|\ \|T_p^{-1}\| = C_p \ \text{ for all } \ 1 \leq p < \infty\ . \tag{18}$$

Indeed, if e.g. $1 \leq p < 2$, we have immediately that $\|T_p^{-1}\| \geq C_p^2$, since $1 = \|T_p^{-1}(\gamma_i \otimes \gamma_i)\|_2 = C_p^2\|\gamma_i \otimes \gamma_i\|_p$. Of course $\|T_p^{-1}\| \leq \|V_p^{-1} \otimes V_p^{-1}\| = C_p^2$, so $\|T_p^{-1}\| = C_p^2$. Again $\|T_p\| \leq \|V_p \otimes V_p\| \leq C_p^{-1}$. But our proof that $\|V_p \otimes V_p\| \geq C_p^{-1}$ really showed that $\|T_p\| \geq C_p^{-1}$. Analogous remarks apply to the case $p > 2$.

Now of course $T_p(\gamma_i \otimes \gamma_i)$ is an orthonormal sequence in L^2, and hence isometrically equivalent to the usual ℓ^2-basis. Thus (18) together with the following simple result completes the proof of Proposition 4.

Lemma 6. *Let (e_j) be the usual ℓ^2-basis, (b_j) an isometrically symmetric basis for a Banach space B with B isomorphic to ℓ^2, and $T : B \to \ell^2$ the operator with $Tb_j = e_j$ for all j. Then $d(B, \ell^2) = \|T\|\,\|T^{-1}\|$.*

Proof. Let $U : B \to \ell^2$ be an isomorphism. Let $\lambda = \|T\|\,\|T^{-1}\|$ and $\beta = \|U\|\,\|U^{-1}\|$. We shall show that $\beta \geq \lambda$. Let $\varepsilon > 0$ be given. Since $Ub_j \to 0$ weakly and U is an isomorphism,

we may choose a subsequence (b'_j) of (b_j) so that $\tau \overset{\text{df}}{=} \lim_j \|Ub'_j\|$ exists (and is positive) and

$$\frac{1}{1+\varepsilon}\Big(\sum |c_j|^2\Big)^{1/2} \le \Big\|\tau^{-1}\sum c_j U b'_j\Big\| \le (1+\varepsilon)\Big(\sum |c_j|^2\Big)^{1/2} \tag{19}$$

for all scalars (c_j) with $\sum |c_j|^2 < \infty$. By multiplying U by $1/\tau$ (which doesn't change β), we may thus assume that $\tau = 1$. Now define $V : [Ub'_j] \to \ell^2$ by $VUb'_j = e_j$ for all j. Since (b_j) is isometrically symmetric, (b'_j) is isometrically equivalent to (b_j) and hence

$$\|VU\|\,\|(VU)^{-1}\| = \lambda\,. \tag{20}$$

But

$$\begin{aligned}\|VU\|\,\|(VU)^{-1}\| &\le \|U\|\,\|U^{-1}\|\,\|V\|\,\|V^{-1}\| \\ &\le \|U\|\,\|U^{-1}\|(1+\varepsilon)^2 \qquad \text{by (19).}\end{aligned}$$

Hence by (20), $\beta \ge (1+\varepsilon)^{-2}\lambda$ for all $\varepsilon > 0$, so $\beta \ge \lambda$, completing the proof. ∎

Comment. Of course the argument (and hence the result) holds also for isometrically subsymmetric bases, but we have no need of this here.

We now pass to some quite simple but useful consequences of Lemma 2 and Proposition 4.

Corollary 7. *Let μ, ν be nice measure spaces, let $1 \le p < \infty$, and let (f_i) and (g_j) be sequences in $L^p(\mu)$ and $L^p(\nu)$ respectively, each equivalent to the usual ℓ^2-basis. Then in $L^p(\mu \times \nu)$, $(f_i \otimes g_j)_{i,j}$ is equivalent to the usual ℓ^2-basis.*

This result follows immediately from Lemma 2 and Proposition 4. We wish to give a quantitative reformulation. By definition a sequence (b_j) in a Banach space is *equivalent* to the usual ℓ^2-basis provided there are constants a and A so that

$$a\Big(\sum c_j^2\Big)^{1/2} \le \Big\|\sum c_j b_j\Big\| \le A\Big(\sum c_j^2\Big)^{1/2}$$

for all scalar-sequences (c_j) with $\sum c_j^2 < \infty$. Evidently if $T : \ell^2 \to B$ is the natural map with $Te_j = b_j$ for all j, then $A = \|T\|$, $a = \|T^{-1}\|^{-1}$ give the best constants a and A. For $\lambda \ge 1$, we say that (b_j) is λ-equivalent to the ℓ^2-basis if $\|T\|\,\|T^{-1}\| \le \lambda$, *i.e.*, if the constants a and A can be chosen with $\frac{A}{a} \le \lambda$. (Thus we allow for arbitrary "scaling" factors.) Corollary 7 now admits the following quantitative version. (*As above, (e_j) denotes the standard ℓ^2-basis.*)

Corollary 7′. *Let (f_i) and (g_j) be as in Corollary 7, $\lambda, \beta \ge 1$, and suppose (f_j) is λ-equivalent to (e_j) and (g_j) is β-equivalent to (e_j). Then $(f_i \otimes g_j)_{i,j}$ is $C_p\lambda\beta$-equivalent to $(e_i \otimes e_j)$.*

Proof. Let (γ_i) be independent standard Gaussian variables, and set $X = [\gamma_i]_p$. Let $S : [f_i] \to X$ and $T : [g_j] \to X$ be linear operators defined by $Sf_i = \gamma_i = Tg_i$ for all i. It follows since

(γ_i) is 1-equivalent to (e_i), that $\|S\|\,\|S^{-1}\| \le \lambda$ and $\|T\|\,\|T^{-1}\| \le \beta$. Lemma 2 now yields that $\|S \otimes T\|\,\|(S \otimes T)^{-1}\| \le \lambda\beta$, where $S \otimes T$ is regarded as an operator from $[f_i]_p \overset{p}{\otimes} [g_j]$ to $X \overset{p}{\otimes} X$. Thus $(f_i \otimes g_j)$ is $\lambda\beta$-equivalent to $(\gamma_i \otimes \gamma_j)$ (in $X \overset{p}{\otimes} X$). But Proposition 4 yields that $(\gamma_i \otimes \gamma_j)$ is C_p-equivalent to $(e_i \otimes e_j)$, *i.e.*, the usual ℓ^2-basis for $\ell^2(\mathbb{N} \times \mathbb{N})$. Hence Corollary 7′ follows. ∎

Of course the estimate in Corollary 7′ is best possible, in general. Nevertheless for certain sequences, other methods may give better estimates. For example, Corollary 7 (via the Khintchine inequalities) yields the old classical fact that if (r_i) is the sequence of mean-zero Rademacher functions, then $(r_i \otimes r_j)_{i,j}$, in $L^p(m^2)$, is equivalent to the usual ℓ^2-basis for all $1 \le p < \infty$. However suppose K_p denotes the "best" λ so that (r_i) in L^p is equivalent to (e_i) (*i.e.*, the best λ in the Khintchine inequalities). Lemma 2 yields directly that $(r_i \otimes r_j)_{i,j}$ is then K_p^2-equivalent to the usual ℓ^2-basis. Thus for example, it follows from [**10**] that $\sum(c_{ij}^2)^{1/2} \le 2\|\sum c_{ij} r_i \otimes r_j\|_1$ for all scalars c_{ij} and "2" is the best constant here. At any rate, these observations about the Rademacher functions have the following consequence, which also follows immediately from a result of C.A. McCarthy [**7**].

Corollary 8. *Let $1 \le p < \infty$, μ, ν be nice measure spaces, and (X_i), (Y_j) be unconditional decompositions in $L^p(\mu)$, $L^p(\nu)$ respectively. Then $(X_i \overset{p}{\otimes} Y_j)_{i,j}$ is an unconditional decomposition in $L^p(\mu \times \nu)$.*

Remark. A sequence (X_i) of non-zero closed linear subspaces of a Banach space B is called an *unconditional decomposition in B*, provided every element x in $[X_i]$, the closed linear span of the X_i's, admits a unique sequence (x_i) in $\prod X_i$ with $\sum x_i$ converging unconditionally to x. The special case of 1-dimensional X_i's deserves particular mention; if one has a sequence (b_i) so that $(X_i = \text{span } b_i)$ is an unconditional decomposition in B, one says (b_i) is an unconditional basic sequence. Thus Corollary 8 yields that *for all $1 \le p < \infty$, if (f_i) and (g_i) are unconditional basic sequences in L^p, $(f_i \otimes g_j)_{i,j}$ is an unconditional basic sequence in $L^p(m^2)$*. As noted above, the corresponding result for unconditionally converging series is valid (in general) only if $1 \le p \le 2$.

Proof of Corollary 8. We first note, by standard results, that (X_i) in B is an unconditional decomposition if and only if there is a constant λ so that

$$\Big\|\sum \varepsilon_i x_i\Big\| \le \lambda \Big\|\sum x_i\Big\| \quad \text{for all } (\varepsilon_i) \text{ with } \varepsilon_i = \pm 1$$

for all i and all finitely-non-null sequences (x_i) in $\prod X_i$. The *best* such λ is called the *unconditional constant* of (X_i).

Suppose then α and β are the unconditional constants of (X_i) and (Y_j) respectively.

We shall now prove that the unconditional constant γ of $(X_i \otimes Y_j)_{i,j}$ satisfies

$$\gamma \leq (K_p \alpha \beta)^2 \tag{21}$$

(where K_p is the constant in the Khintchine inequalities referred to above).

First fix (ε_i) and (ε'_j) sequences with $\varepsilon_i = \pm 1$, $\varepsilon'_j = \pm 1$ for all i and j. Also let $X = [X_i]$ and $Y = [Y_j]$. Define $S : X \to X$ and $T : Y \to Y$ by $S(x) = \sum \varepsilon_i x_i$ and $T(y) = \sum \varepsilon'_j y_j$ for $x \in X$ with $x = \sum x_i$, $((x_j) \in \prod X_i)$ and $y \in Y$ with $y = \sum y_j$, $(y_j) \in \prod Y_j$. We thus have that $\|S\| \leq \alpha$ and $\|T\| \leq \beta$. Hence $\|S \otimes T\| \leq \alpha\beta$ by Lemma 2. Of course on $X_i \overset{p}{\otimes} Y_j$, $S \otimes T = \varepsilon_i \varepsilon'_j I$ (I being the identity operator). Thus given $f_{ij} \in X_i \overset{p}{\otimes} Y_j$ with only finitely many f_{ij}'s non-zero, we have

$$\left\| (S \otimes T) \sum f_{ij} \right\| = \left\| \sum \varepsilon_i \varepsilon'_j f_{ij} \right\|_p \leq \alpha\beta \left\| \sum f_{ij} \right\|_p . \tag{22a}$$

Since $(S \otimes T)^2 = I$, of course we also have

$$\left\| \sum f_{ij} \right\|_p \leq \alpha\beta \left\| \sum \varepsilon_i \varepsilon'_j f_{ij} \right\|_p . \tag{22b}$$

Then if $p \geq 2$, (22a) yields that

$$\begin{aligned} \alpha\beta \left\| \sum f_{ij} \right\|_p &\geq \left(\iint \left| \sum r_i(w) r_j(w') f_{ij} \right|^p d(\mu \times \nu)\, d(w \times w') \right)^{1/p} \\ &= \left(\iint \left| \sum f_{ij}(s,t) r_i(w) r_j(w') \right|^p d(w \times w')\, d(\mu \times \nu) \right)^{1/p} \\ &\geq \left(\int \left(f_{ij}^2(s,t) \right)^{p/2} d(\mu \times \nu) \right)^{1/p} \\ &= \left\| \left(\sum f_{ij}^2 \right)^{1/2} \right\|_p . \end{aligned} \tag{23}$$

On the other hand, given ε_{ij} with $\varepsilon_{ij} = \pm 1$ for all j, (22b) and Lemma 2 (via the Khintchine inequalities) yield that

$$\begin{aligned} \left\| \sum \varepsilon_{ij} f_{ij} \right\|_p &\leq \alpha\beta \left(\iint \left| \sum r_i(w) r_j(w') \varepsilon_{ij} f_{ij} \right|^p d(\mu \times \nu)\, d(w \times w') \right)^{1/p} \\ &\leq \alpha\beta K_p^2 \left(\int \left(\sum (\varepsilon_{ij} f_{ij})^2 \right)^{p/2} d(\mu \times \nu) \right)^{1/p} \\ &= \alpha\beta K_p^2 \left\| \left(\sum f_{ij}^2 \right)^{1/2} \right\|_p . \end{aligned} \tag{24}$$

Evidently (23) and (24) give us that

$$\left\| \sum \varepsilon_{ij} f_{ij} \right\|_p \leq \alpha^2 \beta^2 K_p^2 \left\| \sum f_{ij} \right\|_p ,$$

proving (21) for $p \geq 2$. The proof for $1 \leq p < 2$ is similar and left to the reader. ∎

Remark. The result of C.A. McCarthy, mentioned above, has an operator-theoretic formulation. We wish to observe how his result yields our corollary, and also how it may be deduced in exactly the same way as our proof of Corollary 8. We may state his discovery as follows:

Theorem. (C.A. McCarthy [7]). *Let $1 \leq p < \infty$, μ a nice measure space, X a closed linear subspace of $L^p(\mu)$, and $\mathcal{A}$ and $\mathcal{B}$ uniformly bounded commuting Boolean algebras of projections on X be given. Then $\mathcal{G}$ is uniformly bounded, where $\mathcal{G}$ denotes the Boolean algebra of projections generated by the products of those in $\mathcal{A}$ and $\mathcal{B}$.*

(This was only claimed for $1 < p$ in [7], since the proof given there used duality and the case $p \geq 2$; however, the same reasoning (as we show below), works for all p.)

First let $\mu, \nu, (X_i)$ and (Y_i) be as in Corollary 8. Let $\lambda = \mu \times \nu$, $X = [X_i]$, $Y = [Y_j]$, and $B = X \overset{p}{\otimes} Y$. For A a subset of $\mathbb{N}$, let $P_A : X \to X$ be defined by: if $x \in X$ and $x = \sum_{i=1}^{\infty} x_i$ with $x_i \in X_i$ for all i, then $P_A x = \sum_{i \in A} x_i$. Similarly, define $Q_A : Y \to Y$ (where $P_A = Q_A = 0$ if $A = \emptyset$). Now let $\mathcal{A} = \{P_A \otimes I : A \subset \mathbb{N}\}$ and $\mathcal{B} = \{I \otimes Q_A : A \subset \mathbb{N}\}$. Then if a, b are the unconditional constants of (X_i) and (Y_j) respectively, we have that $\mathcal{A}$ and $\mathcal{B}$ are Boolean algebras of projections on B, bounded by a and b respectively. Of course $\mathcal{A}$ and $\mathcal{B}$ commute; the uniformly bounded algebra they generate contains all projections R_A where A is a finite subset of $\mathbb{N} \times \mathbb{N}$ and $R_A(\sum_{i,j} x_i \otimes y_j) = \sum_{(i,j) \in A} x_i \otimes y_j$. This yields Corollary 8. To go backwards, let $\mathcal{A}$ and $\mathcal{B}$ be as in McCarthy's Theorem, and let $a = \sup\{\|P\| : P \in \mathcal{A}\}$, $b = \sup\{\|Q\| : Q \in \mathcal{B}\}$. We shall show that then $\|g\| \leq (K_p ab)^2$ for all $g \in \mathcal{G}$. We may then fix m, n and disjoint projections $P_1, \ldots, P_m$ and $Q_1, \ldots, Q_n$ in $\mathcal{A}, \mathcal{B}$ respectively, with $\sum_{i=1}^m P_i = \sum_{j=1}^n Q_j = I_X$; and assume g is in the algebra generated by the P_i's and Q_j's. ("Disjoint" means $P_i P_{i'} = 0$ and $Q_\ell Q_{\ell'} = 0$ for all $i \neq i'$, $\ell \neq \ell'$.) Letting $X_i = P_i X$ and $Y_j = Q_j X$ for all i and j, we then have that $(X_i)_{i=1}^m$ and $(Y_j)_{j=1}^n$ are unconditional decompositions with constants at most a and b respectively. (For notational convenience we *drop in this Remark only*, our assumption that the "summands" of an unconditional decomposition are non-zero.) The assumption that the P_i's and Q_j's commute is then simply equivalent to the assertion that the spaces $\{(X_i \cap Y_j) : 1 \leq i \leq m, 1 \leq j \leq n\}$ form a *decomposition* of X. That is, for each x in X, there are *unique* x_{ij}'s with $x = \sum_{i=1}^m \sum_{j=1}^n x_{ij}$ and $x_{ij} \in X_i \cap Y_j$ for all i and j. To complete the proof, we thus need only show that $(X_i \cap Y_j)$ is an unconditional decomposition with unconditional constant at most $(K_p \alpha\beta)^2$. For then we get that $\|g\| \leq (K_p \alpha\beta)^2$. Now,

given sequences (ε_i) and (ε'_j) with $\varepsilon_i, \varepsilon'_j = \pm 1$, and elements $f_{ij} \in X_i \cap Y_j$ for all $1 \le i \le m$ and $1 \le j \le n$, we have for each i, $\sum_j \varepsilon'_j f_{ij}$ belongs to X_i and hence

$$\Big\| \sum_i \sum_j \varepsilon_i \varepsilon'_j f_{ij} \Big\|_p \le a \Big\| \sum_j \sum_j \varepsilon'_j f_{ij} \Big\|_p .$$

Similarly,

$$\Big\| \sum_j \sum_i \varepsilon'_j f_{ij} \Big\|_p \le b \Big\| \sum_i \sum_j f_{ij} \Big\|_p .$$

Thus letting $T(\sum f_{ij}) = \sum \varepsilon_i \varepsilon'_j f_{ij}$ for all such choices of f_{ij}, we have that $\|T\| \le ab$ and since $T^2 = I$, $\|T^{-1}\| \le ab$. (Thus the use of Lemma 1 in proving (22a,b) is really not necessary.) Now for variety, fix $1 \le p \le 2$, $f_{ij} \in X_{ij}$ and $\varepsilon_{ij} = \pm 1$ for all i and j. In the following, let all summations be taken over all i and j with $1 \le i \le m$ and $1 \le j \le n$. We then obtain by Fubini's theorem, the orthonormality of the $r_i \otimes r_j$'s, and the estimate on the norms of the operators T^{-1}, that

$$\begin{aligned} \Big\| \sum \varepsilon_{ij} f_{ij} \Big\|_p &\le \alpha\beta \Big(\iint \Big\| \sum r_i(w) r_j(w') \varepsilon_{ij} f_{ij} \Big\|_p^p \, dw dw' \Big)^{p'} \\ &\le \alpha\beta \Big\| \Big(\sum f_{ij}^2 \Big)^{1/2} \Big\|_p . \end{aligned}$$

On the other hand, we have by the estimate on the norms of the operators T and Lemma 2 that

$$\begin{aligned} \alpha\beta \Big\| \sum f_{ij} \Big\|_p &\ge \Big(\iint \Big\| \sum r_i(w) r_j(w') f_{ij} \Big\|_p^p \, dw dw' \Big)^{1/p} \\ &\ge K_p^{-2} \Big\| \Big(\sum f_{ij}^2 \Big)^{1/2} \Big\|_p . \end{aligned}$$

Evidently these two sets of inequalities give us again that $\| \sum \varepsilon_{ij} f_{ij} \|_p \le \alpha^2 \beta^2 K_p^2 \| \sum f_{ij} \|_p$, completing the proof. ∎

(Note that the argument is valid over the complex numbers as well; where here one lets the ε_{ij}'s vary over all complex numbers with $|\varepsilon_{ij}| = 1$; then one uses the complex Khintchine constant; alternatively (as is done in [7]), one can work with independent random variables $(\tilde{r}_i)$ uniformly distributed on the circle, rather than the Rademacher functions (r_i).)

We now proceed with the proof of Theorem 1.a. We first prove this for a special case, where we also obtain the best constants.

Proposition 9. *Let $1 \le p < 2$, p^* the conjugate to p. Then $C_{2,p} = C_{p^*,2} = C_p$ (where C_p is defined in Proposition 4.*

Proof. From what we have already obtained and duality, it suffices to prove that

$$C_{2,p} \le C_p . \tag{25}$$

Indeed, once this is shown, suppose $X, Y \in \mathcal{L}^{p^*}_{1+}$, X_1, $Y_1 \in \mathcal{L}^2_{1+}$, and $S : X \to X_1$, $T : Y \to Y_1$ are given bounded linear operators. But then $X^*, Y^* \in \mathcal{L}^p_{1+}$, $X_1^*, Y_1^* \in \mathcal{L}^2_{1+}$, and so $\|S^* \otimes T^*\| \leq C_p \|S^*\| \, \|T^*\|$. But then of course $S \otimes T$ is bounded, with $\|S \otimes T\| = \|(S \otimes T)^*\| = \|S^* \otimes T^*\| \leq C_p \|S^*\| \, \|T^*\| = C_p \|S\| \, \|T\|$. Thus $C_{p^*,2} \leq C_p$. But fixing p, letting (γ_i) be independent standard Gaussian random variables, and defining $V_p : \ell^2 \to L^p$ by $V_p(e_i) = \gamma_i$ for all i, we obtain by Lemma 5 and the remark preceding its proof that $\|V_p\| = C_p^{-1} = \|V_p \otimes V_p\|$. Hence since $C_p^{-1} = \|V_p \otimes V_p\| \leq C_{2,p} \|V_p\|^2 = C_{2,p} C_p^{-2}$, $C_{p,2} \geq C_p$. Of course then $V_p^* : L^{p^*} \to \ell^2$ satisfies $C_p^{-1} = \|V_p^* \otimes V_p^*\| \leq C_{p^*,2} \|V_p^*\|^2 = C_{p^*,2} C_p^{-1}$ so again $C_{p^*,2} \geq C_p$.

To prove (25), it is really enough to show this for *finite-dimensional* spaces and apply local approximations, or just do it for operators from ℓ^2 to L^p itself (in view of Lemma 2). However we will give the proof anyway in full generality, since only some notational changes are required. Thus it suffices to prove: *given κ a cardinal number, μ, ν nice measure spaces, and bounded linear operators $S : \ell^2(\kappa) \to L^p(\mu)$, $T : \ell^2(\kappa) \to L^p(\nu)$, then $S \otimes T$ extends to a bounded linear operator from $\ell^2(\kappa \times \kappa) \to L^p(\mu \times \nu)$ with $\|S \otimes T\| \leq C_p \|S\| \, \|T\|$.*

We let κ equal an initial ordinal, and in turn identify this with $\{\alpha : \alpha \text{ an ordinal}, \alpha < \kappa\}$. $\ell^2(\kappa)$ denotes the family of all functions $f : \kappa \to \mathbb{R}$ with $f(\alpha) = 0$ for all but countably many α and $(\sum_{\alpha<\kappa} |f(\alpha)^2|^{1/2} = \|f\| < \infty$. Let m_κ denote the product Lebesgue measure on $[0,1]^\kappa$ and for each $\alpha < \kappa$, let γ_α be a standard Gaussian variable defined on $[0,1]$ and "depending" only on α, *i.e.*, $\gamma_\alpha(w) = \gamma_\alpha(w')$ if $w_\alpha = w'_\alpha$ for all $w, w' \in [0,1]^\kappa$. Let also $e_\alpha(\beta) = \delta_{\alpha\beta}$ be the natural basis for $\ell^2(\kappa)$ and let $V : \ell^2(\kappa) \to L^p(m_\kappa)$ be the linear map with $Ve_\alpha = \gamma_\alpha$ for all $\alpha < \kappa$. Then of course V is a multiple of an isometry and Lemma 5 applies. That is, we obtain that setting $X = V\ell^2(\kappa)$ (and regarding $V : \ell^2(\kappa) \to X$), we have that $V \otimes V : \ell^2(\kappa \times \kappa) \to X \overset{p}{\otimes} X$ satisfies

$$\|V \otimes V\| = C_p^{-1} \quad \text{and} \quad \|V^{-1}\| = C_p \, . \tag{26}$$

Now define maps $U_1 : X \to L^p(\mu)$ and $U_2 : X \to L^p(\nu)$ by

$$U_1 = SV^{-1} \quad \text{and} \quad U_2 = TV^{-1} \, . \tag{27}$$

We then have the following diagram:

$$\begin{array}{ccccc} & \ell^2(\kappa) & \overset{2}{\otimes} & \ell^2(\kappa) & \\ & {\scriptstyle V}\downarrow & & \downarrow{\scriptstyle V} & \\ S & X & \overset{p}{\otimes} & X & T \\ & {\scriptstyle U_1}\downarrow & & \downarrow{\scriptstyle U_1} & \\ & L^p(\mu) & \overset{p}{\otimes} & L^p(\nu) & \end{array}$$

That is, $S \otimes T = (U_1 \otimes U_2) \circ (V \otimes V)$. By Lemma 2,

$$\begin{aligned} \|U_1 \otimes U_2\| &= \|U_1\| \, \|U_2\| \\ &= \|SV^{-1}\| \, \|TV^{-1}\| && \text{by (27)} \\ &\leq \|S\| \, \|T\| \, \|V^{-1}\|^2 \\ &= \|S\| \, \|T\| C_p^2 && \text{by (26).} \end{aligned}$$

Hence

$$\begin{aligned} \|S \otimes T\| &\leq \|U_1 \otimes U_2\| \, \|V \otimes V\| \\ &\leq \|S\| \, \|T\| C_p^2 C_p^{-1} = C_p \|S\| \, \|T\| \, , \end{aligned}$$

thus proving Proposition 9. ∎

The proof of Theorem 1.a may now be easily completed, via the following consequence of Grothendieck's inequality (cf. [5]): *Given $1 \leq q < 2 < p \leq \infty$, there exist constants $\gamma_{p,q}$ so that for all $X \in \mathcal{L}_{1+}^p$, $Y \in \mathcal{L}_{1+}^q$, and bounded linear operators $T : X \to Y$, there exist a $Z \in \mathcal{L}_{1+}^2$ and linear operators $U : X \to Z$ and $V : Z \to Y$ satisfying*

$$\|U\| \, \|V\| \leq \gamma_{p,q} \|T\| \quad \text{and} \quad T = VU \, . \tag{28}$$

(Moreover $\gamma_{p,q} \leq \gamma_{\infty,1} \leq K_G$ where K_G denotes Grothendieck's constant.)

Now Theorem 1.a has already been established except for the case $1 \leq q < 2 < p \leq \infty$. Thus for $j = 1, 2$, let $X_j \in \mathcal{L}_{1+}^p$, $Y_j \in \mathcal{L}_{1+}^q$, and $T_j : X_j \to Y_j$ be given bounded linear operators. Choose $Z_j \in \mathcal{L}_{1+}^2$ and operators U_j and V_j satisfying (28) for $U = U_j$, $V = V_j$, and $T = T_j$. We now have the following diagram:

$$\begin{array}{ccccc} & X_1 & \overset{p}{\otimes} & X_2 & \\ & {\scriptstyle U_1}\downarrow & & \downarrow{\scriptstyle V_1} & \\ T_1 & Z_1 & \overset{2}{\otimes} & Z_2 & T_2 \\ & {\scriptstyle U_2}\downarrow & & \downarrow{\scriptstyle V_2} & \\ & Y_1 & \overset{q}{\otimes} & Y_2 & \end{array}$$

That is,

$$T_1 \otimes T_2 = (U_2 \otimes V_2) \circ (U_1 \otimes V_1) \, . \tag{29}$$

Letting p^* be the conjugate index to p, we have by Proposition 9 that

$$\|U_1 \otimes V_1\| \leq C_{p,2} \|U_1\| \, \|V_1\| = C_{p^*} \|U_1\| \, \|V_1\| \tag{30a}$$

and

$$\|U_2 \otimes V_2\| \leq C_{2,q} \|U_2\| \, \|V_2\| = C_q \|U_2\| \, \|V_2\| \, . \tag{30b}$$

Hence by (28)–(30),

$$\begin{aligned}\|T_1 \otimes T_2\| &\le \|U_2 \otimes V_2\| \, \|U_1 \otimes V_1\| \\ &\le C_{p^*} C_q \|U_1\| \, \|U_2\| \, \|V_1\| \, \|V_2\| \\ &\le C_{p^*} C_q \gamma_{p,q} \|T_1\| \, \|T_2\| \ .\end{aligned}$$

That is, $C_{p,q} \le C_{p^*} C_q \gamma_{p,q}^2$. ∎

Remark. The proof of Theorem 1.a yields that for all $1 \le q < 2 < p \le \infty$, $C_{p,q} \le \frac{\pi}{2}\gamma_{\infty,1}^2 \le \frac{\pi}{2} K_G^2$ where K_G is Grothendieck's constant, and $\gamma_{\infty,1}$ is the "L^2-factorization constant" which follows from his inequality (see the statement containing (28) above). The proof of Proposition 11 below gives that $C_{p,q} \ge \max\{C_{p^*}, C_q\}$. It would be interesting to know the exact value of $C_{\infty,1}$ (and at a deeper level, of $\gamma_{\infty,1}$). By simply computing the norm of $T \otimes T$, for $T : \ell_2^\infty \to \ell_2^1$ the natural isometry, it follows that $C_{\infty,1} \ge 2$. Known estimates on K_G yield that in the real case, $C_{\infty,1} < 5$ while in the complex case, $C_{\infty,1} < 3.11$. (All of our results hold for real or complex Banach spaces. For interesting quite recent results concerning the value of complex K_G as well as a nice summary of known work on the value of real K_G, see [4].)

The next result is actually equivalent to the case in Theorem 1.a where the upper index is equal to infinity. (For another equivalence, see the second remark following the next proof.)

Corollary 10. *Let μ, ν be nice measure spaces, $1 \le p \le 2$, and $\sum f_i$, $\sum g_i$ be unconditionally converging series in $L^p(\mu)$ and $L^p(\nu)$ respectively. Then $\sum_{i,j} f_i \otimes g_j$ converges unconditionally in $L^p(\mu \times \nu)$.*

Proof. We first recall some standard results about unconditionally converging series in Banach spaces. Suppose then (x_j) is given in a Banach space B. Recall that $\sum x_j$ converges *unconditionally*, to an element x say, provided $\sum x_{\sigma(j)}$ converges to x for all permutations σ of the indices $1, 2, \ldots$. It turns out that $\sum x_j$ converges unconditionally if and only if for every $\varepsilon > 0$, there is an n so that $\sum_{j \ge n} |x^*(x_j)| < \varepsilon$ for all $x^* \in X^*$ with $\|x^*\| = 1$. If c_0 is not isomorphic to a subspace of X, this can be relaxed considerably, as discovered by Bessaga-Pełczyński: then $\sum x_j$ converges unconditionally if and only if there is a $u < \infty$ with

$$\sum_{j=1}^{\infty} |x^*(x_j)| \le u\|x^*\| \quad \text{for all} \quad x^* \in X^* \ . \tag{31}$$

The smallest u satisfying (31) is called the *unconditional constant* of the (formal) series $\sum x_j$. In fact, however, u is nothing but the *norm* of the natural linear operator $T : c_0 \to X$ with $Te_j = x_j$ for all j, where (e_j) is the standard basis for c_0.

Now it is also a standard fact that c_0 does not embed in $L^p(\mu)$ for any $1 \le p < \infty$. Thus Corollary 10 follows immediately from the "$p = \infty$" case of Theorem 1.a. Indeed, suppose

$\sum f_i$, $\sum g_j$ are as in the statement of Corollary 10, and let u, v be the unconditional constants of $\sum f_i$ and $\sum g_i$ respectively. Let then $S : c_0 \to L^p(\mu)$, $T : c_0 \to L^p(\nu)$ be the associated linear operators; $Se_i = f_i$, $Te_i = g_i$ for all i. Thus by Theorem 1.a, $\|S \otimes T\| \leq C_{\infty,p}\|S\|\,\|T\|$. That is, by the assertion containing (31), $\sum f_i \otimes g_j$ converges unconditionally, with constant $C_{\infty,p}uv$. ∎

Remarks.

1. In fact the corollary really yields the quantitative "$p = \infty$" case of Theorem 1.a; $C_{\infty,p}$ is the smallest constant C so that the unconditional constant of $\sum f_i \otimes g_j$ is at most Cuv for $\sum f_i$, $\sum g_j$, u, v as above.

2. The case of our main result for $p = \infty$ and $q = 1$ (*i.e.*, the existence of $C_{\infty,1}$), may be equivalently formulated as follows (cf. [6] for the relevant definitions). *Let X, Y, X', Y' be Banach spaces with X, Y $\mathcal{L}_\infty$-spaces and X', Y' $\mathcal{L}_1$-spaces. Then given $S : X \to X'$ and $T : Y \to Y'$ bounded linear operators, $S \otimes T$ extends to a bounded linear operator from $X \overset{\vee}{\otimes} Y$ to $X' \overset{\wedge}{\otimes} Y'$.* (As noted above, "$\overset{\vee}{\otimes}$" denotes the injective, or least cross norm; "$\overset{\wedge}{\otimes}$" denotes the projective, or greatest cross norm.) The validity of this reformulation rests on the fact that if X and Y are $\mathcal{L}_1$-spaces, each of which is isometric to a subspace of some $L^1(\mu)$-space, then $X \overset{\wedge}{\otimes} Y$ is (canonically) isomorphic to $X \overset{1}{\otimes} Y$. To see this, one uses the fact that X and Y are in turn isometric to "locally complemented" subspaces of $L^1(\mu)$-spaces; and then of course the standard fact that for nice measure spaces μ and ν, $L^1(\mu) \overset{\wedge}{\otimes} L^1(\nu) = L^1(\mu) \overset{1}{\otimes} L^1(\nu) = L^1(\mu \times \nu)$. It seems desireable to determine other pairs of classes of Banach spaces (besides $\mathcal{L}_\infty$ and $\mathcal{L}_1$-spaces), for which the conclusion of the above result holds.

We conclude our treatment of Theorem 1.a with a discussion of some lower bounds for the constants $C_{p,q}$.

Proposition 11. *Let $q < p$ with $1 \leq q \leq 2 \leq p \leq \infty$. Then $C_{p,q} > 1$.*

This is immediate from Proposition 9 for the case $q = 2$ or $p = 2$. Thus we may assume $1 \leq q < 2 < p \leq \infty$. In this case we claim

$$C_{p,q} \geq C_{p,2} = C_{p^*} \tag{32}$$

where p^* is the conjugate index to p.

Let (γ_i) be independent standard Gaussian variables, as before. Now let $V : \ell^2 \to L^q$ be the linear map defined by $Ve_i = C_q\gamma_i$ for all i. Thus V is an (into) isometric embedding and (10) of Lemma 5 yields that

$$\|V \otimes V(h)\| \geq \|h\| \quad \text{for all} \quad h \in \ell^2 \overset{2}{\otimes} \ell^2 \,. \tag{33}$$

(This is because essentially, $V = C_q V_q$ where V_q is the map given in Lemma 5, where q here equals "p" in Lemma 5.)

Again let $U : \ell^2 \to L^{p^*}$ be the linear map defined by $Ue_i = C_{p^*}\gamma_i$ for all i; thus U is an (into) isometry. Then by the more delicate part of (10), we have that

$$\|U \otimes U\| = C_{p^*} \,. \tag{34}$$

Now simply set $T = VU^*$. Then evidently $\|T\| = 1$ since U^* is an (isometric) quotient map, and $T : L^p \to L^q$. But for any $h \in L^p$,

$$\begin{aligned} \|T \otimes T(h)\| &= \|V \otimes V \circ U^* \otimes U^*(h)\| \\ &\geq \|U^* \otimes U^*(h)\| \quad \text{by (33).} \end{aligned}$$

Hence $\|T \otimes T\| \geq \|U^* \otimes U^*\| = \|(U^* \otimes U)^*\| = \|U \otimes U\| = C_{p^*}$ by (34). Thus (32) holds, finishing the proof of Proposition 11. (The following diagram summarizes this argument:

$$\begin{array}{ccccc} & & L^p & \overset{p}{\otimes} & L^p & & \\ & & {\scriptstyle U^*}\downarrow & & \downarrow{\scriptstyle U^*} & & \\ T & & \ell^2 & \overset{2}{\otimes} & \ell^2 & & T \\ & & {\scriptstyle V}\downarrow & & \downarrow{\scriptstyle V} & & \\ & & L^q & \overset{q}{\otimes} & L^q & & \end{array}$$

Since $V \otimes V$ is norm increasing, $T \otimes T$ has norm at least C_{p^*} by (34).) ∎

We now give the proof of Theorem 1.b and the "opposite" side of Corollary 10, mentioned in the Abstract. The essential ingredient of the proof is contained in the next result.

Lemma 12. *Let q, p be fixed with $1 \leq q < p < 2$. There exists an (into) isometry $T : \ell^p \to L^q$ with $T \otimes T$ unbounded, as an operator from $\ell^p \otimes \ell^p$ to $L^q \otimes L^q$.*

Remark. It follows from the invariants already introduced that if μ is a nice measure space and $T : \ell^p \to L^q(\mu)$ is an (into) isomorphism, then $T \otimes T$ is unbounded.

Proof of the Lemma. In a sense, the argument uses exactly the same principle as the proof of (14) in the argument for Lemma 5. Let γ be a p-stable random variable, normalized so that $\|\gamma\|_q = 1$. Let $\gamma_1, \gamma_2, \ldots$ be independent random variables on $[0,1]$, each with the same distribution as γ. We then have the standard facts:

$$(\gamma_i) \text{ is isometrically equivalent (in } L^q) \text{ to the usual } \ell^p\text{-basis} \tag{35}$$

and

$$\|\gamma\|_p = \infty \,. \tag{36}$$

Of course it follows from (36) and the independence of the γ_i's that $(|\gamma_i|^p)_{i=1}^\infty$ is an independent sequence of random variables with common distribution $|\gamma|^p$; thus by (36) and the Kolmogoroff strong law of large numbers, we have

$$\frac{1}{n}\sum_{i=1}^{n}|\gamma_i|^p \to \infty \quad \text{a.e.} . \tag{37}$$

Now let (e_i) be the usual basis for ℓ^p and let $T : \ell^p \to L^q$ be the linear map with $Te_i = \gamma_i$ for all i; (35) asserts that T is an into-isometry. To show that $T \otimes T$ is unbounded, it suffices to show that $T \otimes T \mid \mathrm{sp}\{e_i \otimes e_i : i = 1, 2, \ldots\}$ is unbounded. Given scalars $c_1, c_2, \ldots$, with only finitely many non-zero, we have that

$$\begin{aligned}\Big\|T \otimes T\Big(\sum c_i e_i \otimes e_i\Big)\Big\|^q &= \iint \Big|\sum c_i \gamma_i(t)\gamma_i(s)\Big|^q \, ds\, dt \\ &= \int \Big(\sum |c_i \gamma_i(t)|^p\Big)^{q/p} dt\end{aligned}$$

by (35) and Fubini's theorem. Thus fixing n and letting $c_i = 1/n^{1/p}$ for $1 \le i \le n$; $c_i = 0$ otherwise, we have that

$$\begin{cases} \Big\|T \otimes T\Big(n^{-1/p}\sum_{i=1}^{n} e_i \otimes e_i\Big)\Big\|^q \\ \qquad = \displaystyle\int \left(\frac{\sum_{i=1}^{n}|\gamma_i(t)|^p}{n}\right)^{q/p} dt \to \infty \\ \text{as } n \to \infty\ , \end{cases} \tag{38}$$

by Fatou's Lemma and (37). Since $\|n^{-1/p}\sum_{i=1}^n e_i \otimes e_i\| = 1$ for all n, (38) yields that $T \otimes T$ is unbounded. ∎

Theorem 1.b follows readily from Lemma 12. Indeed, first let q, p be as in Lemma 12, choose X a subspace of L^p with X isometric to ℓ^p, let $P : L^p \to X$ be a contractive linear projection onto X, and let $U : X \to \ell^p$ be a surjective isometry. It follows immediately that setting $S = TUP$, then $S : L^p \to L^q$ is a norm-one operator with $S \otimes S$ unbounded. Now setting $s = q^*$ and $r = p^*$ (with q^*, p^* the conjugate indices to q and p respectively), it follows by duality that $S^* : L^s \to L^r$ is a norm-one operator with $S^* \otimes S^* : L^s \otimes L^s \to L^r \otimes L^r$ unbounded. Since s, r are thus arbitrary numbers with $2 < r < s \le \infty$, the proof of Theorem 1.b is complete. ∎

In order to obtain the final assertion of the abstract, and to reveal the true "local" character of Theorem 1.b, we quantify the above discussion as follows:

Let $1 \le q < p \le \infty$ with q, p *not* satisfying (1), and let $n > 1$. We *define* $C_{p,q}^n$ to be the *smallest* constant λ so that $\|S \otimes T\| \le \lambda \|S\| \, \|T\|$ for all linear operators $S, T : \ell_n^p \to L^q$.

It is evident that $C_{p,q}^n = C_{q^*,p^*}^n$. Thus we may really assume $1 \le q < p < 2$ to compute $C_{p,q}^n$. (It is moreover evident that we may take the ranges of S and T to be ℓ_m^q for some m, rather than L_ℓ^q.) Now letting (γ_i) be p-stable i.i.d. random variables, *define* $\Gamma_{p,q}^n$ by

$$\Gamma_{p,q}^n = \frac{\left\| \left(\sum_{i=1}^n |\gamma_i|^p \right)^{1/p} \right\|_q}{n^{1/p} \|\gamma_1\|_q} \,. \tag{39}$$

Evidently $\Gamma_{p,q}^n \to \infty$ as $n \to \infty$, by (37) and Fatou's Lemma (as noted above). Of course our proof of Lemma 12 really amounts to the assertion that

$$C_{p,q}^n \ge \Gamma_{p,q}^n \quad \text{for all } n \text{ and } 1 \le q < p < 2. \tag{40}$$

Problem. Does $\Gamma_{p,q}^n$ give the correct order of magnitude for $C_{p,q}^n$ as n tends to infinity?

Those familiar with local theory considerations will no doubt see how the final assertion of the abstract follows directly from the fact that for $2 < r < \infty$, $C_{\infty,r}^n \to \infty$ as $r \to \infty$. In our proof, however, we wish to give a more or less explicit description of the desired series and also trace the argument through the natural invariants we have given so far.

Proposition 13. *Let $2 < r < \infty$. There exists an unconditionally convergent series $\sum f_i$ in ℓ^r so that $\sum_{i,j} f_i \otimes f_j$ fails to converge unconditionally in $\ell^r \overset{r}{\otimes} \ell^r$.*

Proof. Let $\frac{1}{p} + \frac{1}{r} = 1$. For simplicity in notation, let $\Gamma_p^n = \Gamma_{p,1}^n$. Now fix n. We may choose m $(= m(n))$ and vectors $b_1, \ldots, b_n$ in ℓ_m^1 so that

$$\left(\sum |c_i|^p \right)^{1/p} \le \left\| \sum c_i b_i \right\| \le 2 \left(\sum |c_i|^p \right)^{1/p} \tag{41}$$

for all scalars $c_1, \ldots, c_n$. Now let $\gamma_1, \ldots, \gamma_n$ be identically distributed independent p-stable variables with $\|\gamma_1\| = 1$. Thus $(\gamma_i)_{i=1}^n$ is isometrically equivalent in L^1, to the usual basis $(e_i)_{i=1}^n$ for ℓ_n^p. Define linear maps $T : \ell_n^p \to L^1$, $U : \ell_n^p \to \ell_m^1$, and $V : [b_i]_{i=1}^n \to L^1$ by $Te_i = \gamma_i$, $Ue_i = b_i$, and $Vb_i = \gamma_i$ for all i. Thus $T = VU$, T is an isometry and

$$\|U\| \le 2 \quad \text{and} \quad \|V\| \le 1 \quad \text{by (41).} \tag{42}$$

It follows of course that $T \otimes T = V \otimes V \circ U \otimes U$. But $\|V \otimes V\| \le 1$ by Lemma 2. Hence we deduce from this and the proof of Lemma 12 that

$$\|U \otimes U\| \ge \|T \otimes T\| \ge \Gamma_p^n \,. \tag{43}$$

Now regarding elements of ℓ^p and ℓ^r as functions on the positive integers, define elements $g_1^n, \ldots, g_m^n$ in ℓ_n^r by

$$g_i^n(j) = g_i(j) = b_j(i) \text{ for all } 1 \le i \le m \ , \ 1 \le j \le n \ .$$

We then have that

$$\text{(44a)} \qquad \sup \left\{ \left\| \sum_{i=1}^m \varepsilon_i g_i \right\|_r : \varepsilon_i = \pm 1 \text{ for all } i \right\} \le 2$$

and

$$\text{(44b)} \qquad \sup \left\{ \left\| \sum_{i,j=1}^m \varepsilon_{ij} g_i \otimes g_j \right\| : \varepsilon_{ij} = \pm 1 \text{ for all } i,j \right\} \ge \Gamma_p^n \ .$$

Indeed, let $u_1, \ldots, u_m$ be the usual basis for ℓ_m^∞. Now for $1 \le i \le m$ and $1 \le j \le n$,

$$g_i(j) = \langle U^* u_i, e_j \rangle = \langle u_i, U e_j \rangle = b_j(i) \ .$$

That is, $U^* u_i = g_i$. (44a) now follows immediately from (42) and the fact that $\|U^*\| = \|U\|$, while (44b) follows from (43) and the fact that $\|(U \otimes U)^*\| = \|U^* \otimes U^*\| = \|U \otimes U\|$ (and this *equals* the first side of the inequality (44b)).

Now let $n_0 = 0$ and choose positive integers $n_1 < n_2 < \cdots$ with $\Gamma_p^{n_k} \ge 4^k$ for all k. Set $\ell_k = n_0 + \cdots + n_k$ for $k = 0, 1, 2, \ldots$ and finally, for each fixed k, setting $s_k = m(n_k)$, let $f_1^k, \ldots, f_{s_k}^k$ "reproduce" $2^{-k} g_1^n, \ldots, 2^{-k} g_m^n$ on the interval of integers $(\ell_{k-1}, \ell_k]$ (where we let $n = n_k$ and $m = s_k$). That is, for each $1 \le i \le m$, $f_i^k = 0$ *off* $(\ell_{k-1}, \ell_k]$, and for $1 \le j \le n$, $f_i^k(\ell_{k-1} + j) = 2^{-k} g_i^n(j)$. Now we have that $\sum_{k=1}^\infty \sum_{i=1}^{s_k} f_i^k$ converges unconditionally. Indeed,

$$\sup \left\{ \left\| \sum_{k=N}^\infty \sum_{i=1}^{s_k} \varepsilon_i^k f_i^k \right\| : \varepsilon_i^k = \pm 1 \text{ for all } i \text{ and } k \right\}$$

$$\le \sum_{k=N}^\infty 2 \cdot 4^{-k} \to 0 \text{ as } N \to \infty \ ,$$

where the inequality follows from (44a). On the other hand, by (44b), for each k we may *choose* signs ε_{ij} so that

$$\left\| \sum_{i,j=1}^{s_k} \varepsilon_{ij} f_i^k \otimes f_j^k \right\| \ge 1 \ .$$

But then the series $\sum f_i^k \otimes f_{j'}^{k'}$ certainly fails to converge unconditionally since the series $\sum_{k=1}^\infty h_k$ fails to converge, where $h_k = \sum_{i,j=1}^{s_k} \varepsilon_{ij} f_i^k \otimes f_j^k$ for all k. ∎

We conclude with a discussion of the Banach spaces $\ell^p \overset{1}{\otimes} \ell^p$ (with $1 < p < 2$). As pointed out to the first-named author some years ago by T. Figiel and W.B. Johnson, these spaces are *not* isomorphic to ℓ^p. This fact can be used to obtain Theorem 1.b and Proposition 13. In fact our Lemma 12 essentially gives the proof. We now summarize this known result and give the details.

Proposition 14. *Let $1 < p < 2$. Then $\ell^p \overset{q}{\otimes} \ell^p$ is isomorphic to $\ell^p \overset{1}{\otimes} \ell^p$ for all $1 < q < p$, while $\ell^p \overset{1}{\otimes} \ell^p$ is not isomorphic to ℓ^p.*

Proof. Let (γ_i) be i.i.d. p-stable random variables. As we have noted above, letting X_q denote the closed linear span of the γ_i's in L^q, then X_q is isometric to ℓ^p for all $1 \leq q < p$. Moreover if we fix $1 < q < p$ and let $T : X_1 \to X_q$ be the "identity" injection, then $\|T\| \leq \|\gamma_1\|_q / \|\gamma_1\|_1$. But then by Lemma 2, we have that $T \otimes T : X_1 \overset{1}{\otimes} X_1 \to X_q \overset{q}{\otimes} X_q$ is an isomorphism with $\|T \otimes T\| = \|T\|^2$. It is just immediate that $(T \otimes T)^{-1}$ has norm at most equal to one; hence we obtain

$$d(\ell^p \overset{1}{\otimes} \ell^p \ , \ \ell^p \overset{q}{\otimes} \ell^q) \leq \frac{\|\gamma_1\|_q^2}{\|\gamma_1\|_1^2} \ .$$

To obtain that $\ell^p \overset{1}{\otimes} \ell^p$ is not isomorphic to ℓ^p, we prove the following

Claim: *$\ell^p \overset{1}{\otimes} \ell^p$ has an unconditional basis which dominates the usual ℓ^p-basis, but which has a subsequence which is symmetric and not equivalent to the usual ℓ^p-basis.*

It is a standard result that any symmetric basic sequence in ℓ^p is equivalent to the usual ℓ^p-basis; hence $\ell^p \overset{1}{\otimes} \ell^p$ is not isomorphic to ℓ^p.

To prove the claim, we take $(\gamma_i \otimes \gamma_j)_{i,j}$ as our basis (where for convenience, we assume $\|\gamma_1\|_1 = 1$). This is indeed an unconditional basis for $\ell^p \overset{1}{\otimes} \ell^p$ by Corollary 8. Letting (e_i) be the usual ℓ^p-basis and $T : X_1 \to \ell^p$ the linear isometry defined by $T\gamma_i = e_i$ for all i, we have that since T has norm one, $T \otimes T : X_1 \overset{1}{\otimes} X_1 \to \ell^p \overset{p}{\otimes} \ell^p$ also has norm one by Lemma 2. Thus $(\gamma_i \otimes \gamma_j)_{i,j}$ dominates the usual ℓ^p-basis with constant one. Finally, $(\gamma_i \otimes \gamma_i)_{i=1}^{\infty}$ is a 1-symmetric subsequence of our basis (since this is a sequence of independent identically distributed symmetric random variables). Of course our proof of Lemma 12 showed that $\|n^{-1/p} \sum_{i=1}^n \gamma_i \otimes \gamma_i\|_1 \to \infty$ as $n \to \infty$; hence $(\gamma_i \otimes \gamma_i)$ is not equivalent to the usual ℓ^p-basis. ∎

Remark. Let $1 \leq q < p < 2$. Then as is well-known, L^p is isometric to a subspace of L^q. By exactly the same argument as above, we obtain that $L^p \overset{q}{\otimes} L^p$ is isomorphic to $L^p \overset{1}{\otimes} L^p$. Again, $L^p \overset{1}{\otimes} L^p$ is not isomorphic to L^p, for in fact $\ell^p \overset{1}{\otimes} \ell^p$ is a complemented subspace of $L^p \overset{1}{\otimes} L^p$, but the above argument yields that $\ell^p \overset{1}{\otimes} \ell^p$ is not isomorphic to a complemented subspace of L^p. Indeed, were $\ell^p \overset{1}{\otimes} \ell^p$ isomorphic to a complemented subspace of L^p, $Y \overset{\text{df}}{=} [\gamma_i \otimes \gamma_i]$ would also be isomorphic to a complemented subspace of L^p. But this is false, for Y has a symmetric basis which is not equivalent to either the ℓ^2 or ℓ^p-basis, and hence Y is not isomorphic to either ℓ^2 or ℓ^p. However any complemented subspace of L^p with a symmetric basis must be isomorphic to either ℓ^2 or ℓ^p.

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Addendum. After this paper was written, we learned that our main result, Theorem 1, was already formulated and proved (for the case of operators from ℓ^p to ℓ^q) by G. Bennett (see Proposition 10.4 of his article *Schur multipliers*, Duke Mathematical Journal, **44** (1977), 603–639). As noted above, our Theorem 1, being of local character, may be deduced from his result, which yields the appropriate quantitative version for the case of operators from ℓ_n^p to ℓ_m^q. Moreover, Bennett obtains that for $1 \leq p < 2$, $C_p = C_{2,p}$ may be estimated by the value we give in (7), but does not show this is best possible.

Limited Sets in Injective Tensor Products

THOMAS SCHLUMPRECHT

Department of Mathematics
The University of Texas at Austin
Austin, Texas 78712

1. Introduction

Recall that a subset A of a Banach space X is called *limited in* X if each w*-zero sequence (x_n^*) in the dual X^* converges uniformly on A, *i.e.*, if $\lim_{n\to\infty} \sup_{x\in A} |\langle x, x_n^*\rangle| = 0$, or, equivalently, if for each sequence $(x_n) \subset A$ and each w*-zero sequence $(x_n^*) \subset X^*$ it follows that $\lim_{n\to\infty} |\langle x_n, x_n^*\rangle| = 0$.

It is easy to see that relatively (norm-) compact subset of X are limited and we say that X has the *Gelfand-Phillips property* (GP) if, conversely, each limited subset of X is relatively compact. For example, spaces having a w*-sequentially compact dual unit ball $B_1(X^*)$ or more generally, spaces admitting in $B_1(X^*)$ an $X-$norming and w*-sequentially precompact subset enjoy (GP) [**5,11**] (recall that $D \subset X^*$ is called w*-sequentially precompact if each sequence $(x_n^*) \subset D$ has a w*-Cauchy subsequence).

It is easy to observe that subsets, finite unions, finite sums, closed absolute convex hulls and images under bounded linear mappings of limited sets are limited. Limitedness of a set $A \subset X$ can also be described by sequences of pointwise convergent operators:

Proposition 1.1. *If $A \subset X$ is limited in the Banach space X, and (T_n) is a pointwise convergent sequence of bounded linear operators from X into a Banach space Y, then (T_n) converges uniformly on A.*

Proof. Note that, by the principle of uniform boundedness, the operator $T(x) := \lim_{n\to\infty} T_n(x)$, for $x \in X$, is linear and bounded and choose for each $n \in \mathbb{N}$ $x_n \in A$ and $y_n^* \in B_1(Y^*)$, such that

$$|\langle y_n^*, T(x_n) - T_n(x_n)\rangle| \geq \frac{1}{2} \sup_{x\in A} \|T(x) - T_n(x)\| \ .$$

Since T_n converges pointwise to T the sequence $(T^* - T_n^*)(y_n^*)$ is w*-zero convergent in X^* (where S^* denotes the adjoint of a bounded linear $S : X \to Y$), and thus by the assumption

that A is limited in X

$$\sup_{x\in A}\|T(x)-T_n(x)\|\leq 2|\langle y_n^*,T(x_n)-T_n(x_n)\rangle|\leq 2\sup_{x\in A}|\langle T'(y_n^*)-T_n'(y_n^*),x\rangle|\underset{n\to\infty}{\longrightarrow}0\,. \qquad \blacksquare$$

An other useful description of limitedness is the following analogue of Grothendieck's characterization of relatively weakly compact sets [**2**, Lemma 2, p.227].

Proposition 1.2. *Let $A\subset X$ and assume that for every $\varepsilon>0$ there exists an X-limited set A_ε with $A\subset A_\varepsilon+\varepsilon B_1(X)$.*

Then A is limited in X.

Proof. For an arbitrary w*-zero sequence $(x_n^*)\subset B_1(X^*)$ and an $\varepsilon>0$ we have

$$\limsup_{n\to\infty}\sup_{x\in A}|\langle x_n^*,x\rangle|\leq\limsup_{n\to\infty}\sup_{x\in A_\varepsilon+\varepsilon B_1(X)}|\langle x_n^*,x\rangle|\leq\limsup_{n\to\infty}\sup_{x\in A_\varepsilon}\varepsilon+|\langle x_n^*,x\rangle|=\varepsilon\,,$$

which implies the assertion. $\blacksquare$

J. Bourgain and J. Diestel [**1**] deduced from Rosenthal's ℓ_1-Theorem that limited sets are always weakly conditionally compact. If X has the Grothendieck property (G) as well as the Dunford-Pettis property (DP), where

(G) each w* convergent sequence in X converges weakly, and

(DP) if $(x_n)\subset X$ and $(x_n^*)\subset X$ converge both weakly to zero then $\lim_{n\to\infty}\langle x_n,x_n^*\rangle=0$,

then it is easily seen that each weakly conditionally compact subset of X is limited (note that (DP) implies that $\lim_{n\to\infty}\sup_{x\in A}|\langle x,x_n^*\rangle|=0$ whenever $A\subset X$ is weakly conditionally compact and $(x_n^*)\subset X^*$ converges weakly to zero). Since ℓ_∞ enjoys (G) and (DP) [**7**] each weakly conditionally compact subset of ℓ_∞ is limited, and in particular it follows that ℓ_∞ contains limited sets which are not relatively compact. For other basic properties of the notion of limited sets and the Gelfand-Phillips property we refer the reader to [**6,11**].

In this paper we are concerned with limitedness in injective tensor products. In particular we will deduce that the known characterizations of compact sets in injective tensor products (see Proposition (2.1)) cannot be transferred to limited sets.

We use the following notations:

For a set M, $\mathcal{P}(M)$ is the power set and $\mathcal{P}_\infty(M)$ is the set of all infinite subsets of M.

X and Y are Banach spaces over $\mathbb{R}$; $B_1(X)$ denotes the unit ball, X^* the dual of X and $\sigma(X,X^*)$ the weak topology on X. The w* topology on X^* is denoted by $\sigma(X^*,X)$. For a compact space K, $C(K,X)$ denotes the space of all continuous functions on K with values in X and $M(K,X)$ the space of all X-valued, regular Borel measures μ of bounded variation $|\mu|(\cdot)$ furnished with $\|\mu\|:=|\mu|(K)$. We recall that by Riesz' representation theorem,

$C(K,X)^* \equiv M(K,X^*)$ and write $C(K) = C(K,\mathbb{R})$ and $M(K) = M(K,\mathbb{R})$. For $\mu \in M(K,X)$, $\operatorname{supp}(\mu) = \cap\{A \mid A = \overline{A} \subset K, |\mu|(A) = |\mu|(K)\}$ is the support of μ and if $f \in C(K)$, $f.\mu$ is the element of $M(K,X)$ having the density f with respect to μ.

$X \otimes Y$ is the algebraic tensor product of X and Y while $X\check{\otimes}Y$ and $X\hat{\otimes}Y$ denote the injective and projective tensor product respectively [8]. The injective tensor norm is denoted by $\|\cdot\|_\vee$ and the projective tensor norm by $\|\cdot\|_\wedge$. For $x^* \in X^*$ and $y^* \in Y^*$ there are unique bounded linear operators $T_{x^*} : X\check{\otimes}Y \to Y$ and $T_{y^*} : X\check{\otimes}Y \to X$ with $T_{x^*}(x\otimes y) = \langle x^*, x\rangle y$ and $T_{y^*}(x\otimes y) = \langle y^*, y\rangle x$ for $x \in X$ and $y \in Y$. We put $z(x^*) := T_{x^*}(z)$ and $z^*(y^*) := T_{y^*}(z)$ and we observe that by this notation each $z \in X\check{\otimes}Y$ can be seen as a compact and $\sigma(X^*,X)-\sigma(Y,Y^*)$ continuous operator and that z^* is the adjoint of z.

In the sequel K_X and K_Y are compact sets for which X is isometrically embedded in $C(K_X)$ and Y in $C(K_Y)$ respectively (for example $K_X := B_1(X^*)$ furnished with $\sigma(X^*,X)$) and let Σ_X and Σ_Y denote the Borel sets of K_X and K_Y. We recall (*e.g.*, [13]) that $X\check{\otimes}Y$ is isometrically embeddable into $C(K_X,Y)$ $(\equiv C(K_X)\check{\otimes}Y)$ as well as into $C(K_Y,X)$ and that $C(K_X,Y)$ and $C(K_Y,X)$ are isometrically embeddable into $C(K_X\times K_Y)$ $(\equiv C(K_X)\check{\otimes}C(K_Y))$ which is a subspace of $L_\infty(\Sigma_X,\Sigma_Y)$, the space of all bounded functions on $K_X \times K_Y$ which are measurable corresponding to the product σ-algebra $\Sigma_X \times \Sigma_Y$.

2. Formulation of the main results

Let us first recall the following result stated in a more general setting in [12] which characterizes compactness in $X\check{\otimes}Y$ by compactness in X and Y respectively:

Proposition 2.1. *$A \subset X\check{\otimes}Y$ is relatively compact if and only if the sets $A(B_1(X^*)) := \{z(x^*) \mid x^* \in B_1(X^*), z \in A\}$ and $A^*(B_1(Y^*)) := \{z^*(y^*) \mid y^* \in B_1(Y^*), z \in A\}$ are relatively compact in Y and X respectively.*

For limitedness we get an analogous result only in special cases:

Proposition 2.2.

a) *If $A \subset X\check{\otimes}Y$ is limited, than $A(B_1(X^*))$ and $A^*(B_1(Y^*))$ are limited in Y and X respectively.*

b) *Assume that Y is (GP) and has the approximation property.*
Then, conversely, the limitedness of $A(B_1(X^*))$ and $A^*(B_1(Y^*))$ in Y and X respectively imply that A is limited in $X\check{\otimes}Y$.

Remark. We do not know if the approximation property is necessary for the second statement in (2.2). In §6 we will construct non limited subsets A of $\ell_\infty\check{\otimes}\ell_\infty$ having the property that

$A(\ell_\infty^*)$ and $A^*(\ell_\infty^*)$ are limited in ℓ_∞. This shows that in general the converse of the second statement of (2.2) is not true for arbitrary Banach spaces X and Y.

Proof of (2.2). Let $A \subset X \check{\otimes} Y$ be limited, $(z_n(x^*))$ an arbitrary sequence in $A(\mathrm{B}_1(X^*))$ (*i.e.*, $z_n \in A$ and $x_n^* \in \mathrm{B}_1(X^*)$ for $n \in \mathbb{N}$), and y_n^* a $\sigma(Y^*, Y)$-zero sequence. Since then $(x_n^* \otimes y_n^*)$ is $\sigma((X \check{\otimes} Y)^*, X \check{\otimes} Y)$-convergent to 0 it follows that

$$\limsup_{n\to\infty} |\langle y_n^*, z_n(x_n^*)\rangle| = \limsup_{n\to\infty} |\langle x_n^* \otimes y_n^*, z_n\rangle| = 0 \ .$$

Thus, $A(\mathrm{B}_1(X^*))$ is limited. The limitedness of $A(\mathrm{B}_1(Y^*))$ follows by symmetry.

Assume now that Y is a Gelfand-Phillips space with the approximation property and let $A \subset X \check{\otimes} Y$ have the property that $A(\mathrm{B}_1(X^*))$ is limited, thus relatively compact in Y and $A^*(\mathrm{B}_1(Y^*))$ is X-limited. From the assumption, that Y satisfies the approximation property we deduce that, for each $\varepsilon > 0$, there is a finite-dimensional projection $P^{(\varepsilon)} : Y \to Y$ $y \mapsto \sum_{i=1}^{n(\varepsilon)} y(\varepsilon, i)\langle y'(\varepsilon, i), y\rangle$, where $y(\varepsilon, 1), \ldots, y(\varepsilon, n(\varepsilon)) \in Y$, $y^*(\varepsilon, 1), \ldots, y^*(\varepsilon, n(\varepsilon)) \in Y^*$ and $n(\varepsilon) \in \mathbb{N}$ such that $\| P^{(\varepsilon)}(y) - y \| \leq \varepsilon$ for each $y \in A(\mathrm{B}_1(X^*))$. Thus, considering $z \in X \check{\otimes} Y$ as an operator $z : X^* \to Y$ we conclude $\| P^{(\varepsilon)} \circ z - z \| \leq \varepsilon$ whenever $z \in A$. By (1.2) it is enough to show that for an arbitrary $\varepsilon > 0$ the set $A(\varepsilon) := \{P^{(\varepsilon)} \circ z \mid z \in A\}$ is limited in $X \check{\otimes} Y$ (where $P^{(\varepsilon)} \circ z = (Id_X \otimes P^{(\varepsilon)})(z) \in X \check{\otimes} Y$ for z in $X \otimes Y$). To see this, we remark that, by assumption, the set $\tilde{A}(\varepsilon, i) := \{z^*(y'(\varepsilon, i)) \mid z \in A\}$ is limited in X for each $i \leq n(\varepsilon)$. Thus the set $A(\varepsilon, i) := \{z^*(y'(\varepsilon, i)) \otimes y(\varepsilon, i) \mid z \in A\}$ is limited in $X \check{\otimes} Y$ (it is the image of $\tilde{A}(\varepsilon, i)$ under the operator $X \ni x \mapsto x \otimes y(\varepsilon, i) \in X \check{\otimes} Y$). Since for each $x^* \in X^*$ and $z \in A$ we have

$$\begin{aligned} P^{(\varepsilon)} \circ z(x^*) &= \sum_{i=1}^{n(\varepsilon)} y(\varepsilon, i)\langle z(x^*), y'(\varepsilon, i)\rangle \\ &= \sum_{i=1}^{n(\varepsilon)} y(\varepsilon, i)\langle x^*, z^*(y'(\varepsilon, i))\rangle = (\sum_{i=1}^{n(\varepsilon)} z^*(y'(\varepsilon, i)) \otimes y(\varepsilon, i))(x^*) \ , \end{aligned}$$

we deduce that $A(\varepsilon) \subset \sum_{i=1}^{n(\varepsilon)} A(\varepsilon, i)$, and thus, that $A(\varepsilon)$ is limited in $X \check{\otimes} Y$. ∎

From (2.1) and the first assertion of (2.2) we deduce the following hereditary result for the Gelfand-Phillips property, proven directly in [6]:

Corollary. *$X \check{\otimes} Y$ is (GP) if and only if X and Y are (GP).*

Since the limitedness of $A(\mathrm{B}_1(X^*))$ and $A^*(\mathrm{B}_1(Y^*))$ is not enough to imply that $A \subset X \check{\otimes} Y$ is limited we are interested in additional properties of limited sets of $X \check{\otimes} Y$. For this we introduce the following two definitions.

Definition. A set $A \subset X\check{\otimes}Y$ is said to be *nearly bounded with respect to the projective tensor norm* if for each $\varepsilon > 0$ there is a $\|\cdot\|_\wedge$-bounded set $A_\varepsilon \subset X \otimes Y$ such that $A \subset A_\varepsilon + \varepsilon B_1(X\check{\otimes}Y)$.

Definition. A set $\mathcal{F} \subset C(K_X \times K_Y)$ is said to have the *countable partition property* if for each $\varepsilon > 0$ there are countable partitions $\Pi_1 \subset \Sigma_X$ and $\Pi_2 \subset \Sigma_Y$ of K_X and K_Y, respectively, such that for each $f \in \mathcal{F}$

$$\operatorname{osc}(f|_{A\times B}) = \sup_{\xi\in A,\zeta\in B} f(\xi,\zeta) - \inf_{\xi\in A,\zeta\in B} f(\xi,\zeta) < \varepsilon \ .$$

Remark. Since singletons $\{f\} \subset C(K_X \times K_Y)$ are nearly bounded in $C(K_X)\hat{\otimes}C(K_Y)$ and have the countable partition property, the same is true for relatively compact subsets of $C(K_X \times K_Y)$.

With these definitions we are in the position to state

Theorem 2.5. *Let $A \subset X\check{\otimes}Y$ be limited in $X\check{\otimes}Y$. Then*

(2.5.1) A is nearly bounded in $X\hat{\otimes}Y$.

(2.5.2) Each sequence $(z_n) \subset A$ contains a subsequence $(\tilde{z}_n)$ which has (as a subset of $C(K_X \times K_Y)$) the countable partition property.

If X and Y are $C(K)$-spaces having the Grothendieck property then also the converse is true:

Theorem 2.6. *Let K_1 and K_2 be compacts such that $C(K_1)$ and $C(K_2)$ have the Grothendieck property.*

If $A \subset C(K_1 \times K_2)$ $(= C(K_1)\check{\otimes}C(K_2))$

(2.6.1) is nearly bounded in $C(K_1)\hat{\otimes}C(K_2)$, and

(2.6.2) each subsequence of A contains a further subsequence with the countable partition property,

then A is limited in $C(K_1 \times K_2)$.

For general spaces X and Y we do not know any characterization of limited sets in $X\check{\otimes}Y$, but the following sufficient condition is easy to verify and the proof is left to the reader. In §6 it will be shown that this condition is not necessary.

Proposition 2.7. *If $A_1 \subset X$ and $A_2 \subset Y$ are limited, then their product $A_1 \otimes A_2 := \{x \otimes y \,|\, x \in A_1,\, y \in A_2\}$ is limited in $X\check{\otimes}Y$.*

In the next section we will state the main tool to prove the second part of Theorem (2.5).

3. A combinatorial result for vector-valued measurable functions

In this section (Ω, Σ, μ) is always a positive measure space and $L_\infty(\mu, X)$ denotes the space of all μ–measurable essentially bounded functions with values in X [**4**, p.161]. The subspace

of $L_\infty(\mu, X)$ consisting of all elements with essentially seperable range, *i.e.*, the closed linear span of

$$V := \left\{ \sum_{n\in\mathbb{N}} x_n \chi_{B_n} \;\middle|\; \begin{array}{c} (x_n) \subset X \text{ is bounded and} \\ (B_n) \subset \Sigma \text{ is pairwise disjoint} \end{array} \right\} ,$$

will be denoted by $L^c_\infty(\mu, X)$. We remark that $L_\infty(\mu, X) = L^c_\infty(\mu, X)$ whenever μ is σ-finite. If μ is the counting-measure on (Ω, Σ) ($\mu(A) = \infty$ for infinite A), we put $L^c_\infty(\Sigma, X) := L^c_\infty(\mu, X)$.

We denote by Π the set of all countable Σ-partitions of Ω, where a countable $\pi \subset \Sigma$ is called a *countable Σ-partition of Ω* if the elements of π are almost (always with respect to μ) pairwise disjoint and if their union is almost Ω. For $A \in \Sigma$ we set $\pi|_A := \{B \cap A \,|\, B \in \pi\}$ if $\pi \in \Pi$ and $\Pi(A) := \{\pi|_A \,|\, \pi \in \Pi\}$.

For two bounded subsets $A, B \subset X$, let $D(A)$ be the diameter of A, *i.e.*, $D(A) := \sup_{x,y\in A} \| x - y \|$, with $D(\emptyset) := 0$ and $d(A, B)$ the (minimal) distance between A and B, *i.e.*, $d(A, B) := \inf_{x\in A, y\in B} \| x - y \|$ with $d(\emptyset, \cdot) = d(\cdot, \emptyset) := \infty$.

If $f : \Omega \to X$ is bounded, measurable and has separable image and if $A, B \in \Sigma$, we define the *essential oscillation of f on A* by $\operatorname{ess\,osc}(f, A) := \inf_{\tilde{A}\equiv A} D(f(\tilde{A}))$ and the *essential distance between A and B under f* by: $\operatorname{ess\,dist}(f, A, B) := \sup_{\tilde{A}\equiv A, \tilde{B}\equiv B} d(f(\tilde{A}), f(\tilde{B}))$, where "$\inf_{\tilde{A}\equiv A}$" and "$\sup_{\tilde{A}\equiv A, \tilde{B}\equiv B}$" mean, that the infimum and the supremum are taken over all $\tilde{A}$ and $\tilde{B} \in \Sigma$ with $\mu((A \cup \tilde{A}) \setminus (A \cap \tilde{A})) = \mu((B \cup \tilde{B}) \setminus (B \cap \tilde{B})) = 0$. For two almost equal, measurable and bounded functions $f, \tilde{f} : \Omega \to X$ with separable image, there exists $\tilde{\Omega} \in \Sigma$, with $\tilde{\Omega} \equiv \Omega$ for which $f(\omega) = \tilde{f}(\omega)$ whenever $\omega \in \tilde{\Omega}$. Thus,

$$\operatorname{ess\,osc}(f, A) = \inf_{\tilde{A}\equiv A} D(f(A)) = \inf_{\tilde{A}\equiv A, \tilde{A}\subset\tilde{\Omega}\cap A} D(f(A)) = \inf_{\tilde{A}\equiv A, \tilde{A}\subset\tilde{\Omega}\cap A} D(\tilde{f}(A)) = \operatorname{ess\,osc}(\tilde{f}, A)$$

for each $A \in \Sigma$. Similary we show that

$$\operatorname{ess\,dist}(f, A, B) = \operatorname{ess\,dist}(\tilde{f}, A, B) \text{ whenever } A, B \in \Sigma .$$

Thus, $\operatorname{ess\,dist}(\cdot, A)$ and $\operatorname{ess\,osc}(\cdot, A, B)$ are well defined on $L^c_\infty(\mu, X)$.

In the case that we consider $L^c_\infty(\Sigma_K, X)$ we write osc instead of ess osc and dist instead of ess dist.

Let $A \in \Sigma$, $\pi \in \Pi(A)$, and $\varepsilon > 0$. An $f \in L^c_\infty(\mu, X)$ is said to be *(π, ε)-compatible on A* if, for each $B \in \pi$, $\operatorname{ess\,osc}(f, B) \le \varepsilon$.

$F \subset L^c_\infty(\mu, X)$ is called *(π, ε)-compatible on A* if every $f \in F$ has this property.

A sequence $(f_n : n \in \mathbb{N}) \subset L^c_\infty(\mu, X)$ is called *totally ε-incompatible on A* if, for every $\pi \in \Pi(A)$, no subsequence is (ε, π)-compatible on A, *i.e.*, if

$$\forall \pi \in \Pi(A), N \in \mathcal{P}_\infty(\mathbb{N}), \exists B \in \pi, n \in N \text{ such that } \operatorname{ess\,osc}(f_n, B) > \varepsilon .$$

Now we are in the position to state the main result of this section.

Theorem 3.1. *Let $(f_n : n \in \mathbb{N}) \subset L^c_\infty(\mu, X)$. Then at least one of the following two cases happens:*

Case 1: *There exists an increasing $(n_k) \subset \mathbb{N}$ such that $(f_{n_k} : k \in \mathbb{N})$ has the following property: For all $\varepsilon > 0$ there is a $\pi^\varepsilon \in \Pi$ for which (f_{n_k}) is $(\pi^\varepsilon, \varepsilon)$-compatible.*

Case 2: *There exists an $\varepsilon > 0$, an increasing $(n_k) \subset \mathbb{N}$, and a tree of sets with positive μ-measure $(A(k,j) : k \in \mathbb{N}_0, j \in \{1, \dots, 2^k\}) \subset \Sigma$ such that $\operatorname{ess\,osc}(f_{n_k}, A(k,i)) \le \varepsilon/4$ and $\operatorname{ess\,dist}(f_{n_k}, A(k, 2j-1), A(k,2j)) \ge \varepsilon$ whenever $k \in \mathbb{N}, i \in \{1, \dots, 2^k\}$ and $j \in \{1, \dots, 2^{k-1}\}$.*

By a tree of sets (case 2) we mean that for each $k \in \mathbb{N}$ and $i \in \{1, \dots, 2^k\}$ it follows that $A(k+1, 2i-1) \cup A(k+1, 2i) \subset A(k,i)$ and $A(k+1, 2i-1) \cap A(k+1, 2i) = \emptyset$ almost everywhere.

Remark. If we assume that $(f_n) \subset L^c_\infty(\mu, X)$ has a common relatively compact range then it is easy to see that case 1 implies that (f_n) contains a weak Cauchy subsequence. On the other hand, under this assumption, one could deduce from case 2 that (f_n) contains a subsequence which is equivalent to the ℓ_1−unit basis. Thus, Theorem (3.1) can be seen as a "vector valued version" of Rosenthal's ℓ_1 theorem.

The following Lemma (3.2) collects some frequently used statements (we omit the trivial proof), while Lemma (3.3) represents the central step of the proof of Theorem (3.1).

Lemma 3.2.

a) *Let $f \in L^c_\infty(\mu, X)$.*

i) If $A_1, A_2, B_1, B_2 \in \Sigma$, $A_1 \subset A_2$, and $B_1 \subset B_2$, then
$\operatorname{ess\,osc}(f, A_1) \le \operatorname{ess\,osc}(f, A_2)$ *and* $\operatorname{ess\,dist}(f, A_1, B_1) \ge \operatorname{ess\,dist}(f, A_2, B_2)$.

ii) If $A_n, B_n \in \Sigma$ for $n \in \mathbb{N}$ then
$\operatorname{ess\,dist}(f, \bigcup_{n\in\mathbb{N}} A_n, \bigcup_{n\in\mathbb{N}} B_n) = \inf_{n\in\mathbb{N}, m\in\mathbb{N}} \operatorname{ess\,dist}(f, A_n, B_m)$.

iii) If $A, B, C \in \Sigma$ and $\mu(C) > 0$, then

$$\operatorname{ess\,dist}(f, A, B) \le \operatorname{ess\,dist}(f, A, C) + \operatorname{ess\,dist}(f, B, C) + \operatorname{ess\,osc}(f, A) + \operatorname{ess\,osc}(f, B) + \operatorname{ess\,osc}(f, C)\,.$$

iv) If $A_n \in \Sigma$ for $n \in \mathbb{N}$ then
$\operatorname{ess\,osc}(f, \bigcup_{n\in\mathbb{N}} A_n) \le 2 \sup_{n\in\mathbb{N}} \operatorname{ess\,osc}(f, A_n) + \sup_{n\in\mathbb{N}, m\in\mathbb{N}} \operatorname{ess\,dist}(f, A_n, A_m)$.

b) *Let $A, \tilde{A} \in \Sigma$, $F, \tilde{F} \subset L^c_\infty(\mu, X)$, and $\varepsilon > 0$.*

i) If $|F| < \infty$, then there exists a $\pi \in \Pi(A)$ such that F is (π, ε)-compatible on A.

ii) If, for $\pi \in \Pi(A)$, F is (π, ε)-compatible on A and if $\tilde{A} \subset A$, then F is also $(\pi|_{\tilde{A}}, \varepsilon)$-compatible on $\tilde{A}$.

iii) *If F and $\tilde{F}$ are (π,ε)-, respectively $(\tilde{\pi},\varepsilon)$-compatible on A, then $F \cup \tilde{F}$ is $(\pi \vee \tilde{\pi},\varepsilon)$-compatible on A, where $\pi \vee \tilde{\pi} := \{A \cap \tilde{A} \mid A \in \pi \text{ and } \tilde{A} \in \tilde{\pi}\}$.*

iv) *If $\pi \in \Pi(A)$ and $\pi^B \in \Pi(B)$ for each $B \in \Pi$ and if F is (π^B,ε)-compatible on B for each $B \in \pi$, then F is $(\bigcup_{B\in\pi} \pi^B,\varepsilon)$-compatible on A.*

c) *Let $A_n \in \Sigma$ for $n \in \mathbb{N}$, and let $\varepsilon > 0$. If $(f_n{:}\, n \in\mathbb{N}) \subset \mathrm{L}^c_\infty(\mu,X)$ is totally ε-incompatible on $A := \bigcup_{n\in\mathbb{N}} A_n$, then there exists an $n_0 \in \mathbb{N}$ and a subsequence of $(f_n{:}\, n \in\mathbb{N})$ which is totally ε-incompatible on A_{n_0}.*

Lemma 3.3. *Let $A \in \Sigma$ and $\varepsilon > 0$, and let $(f_n{:}\, n \in\mathbb{N}) \subset \mathrm{L}^c_\infty(\mu,X)$ be totally ε-incompatible on A.*

Then there exist $n \in \mathbb{N}$, $M \in \mathcal{P}_\infty(\mathbb{N})$, and $A_1, A_2 \in \Sigma \cap A$ $(= \{A \cap B \mid B \in \Sigma\})$ such that

a) *$\operatorname{ess\,dist}(f_n, A_1, A_2) > \varepsilon/4$ (in particular $A_1 \cap A_2 = \emptyset$ almost everywhere),*

b) *$(f_m{:}\, m \in M)$ is totally ε-incompatible on A_1 as well as on A_2.*

Proof. Let $(f_n{:}\, n \in\mathbb{N}) \subset \mathrm{L}^c_\infty(\mu,X)$ be totally ε-incompatible on A.

We suppose that the assertion of the lemma is not true, *i.e.*, that

(1) $\quad \forall n \in \mathbb{N}, M \in \mathcal{P}_\infty(\mathbb{N})$ and $A_1, A_2 \in \Sigma \cap A$ with $\operatorname{ess\,dist}(f_n, A_1, A_2) \geq \varepsilon/4 \quad :$

$[(f_m{:}\, m \in M)$ is totally ε-incompatible on $A_1 \quad \Rightarrow$

$\exists \tilde{M} \in \mathcal{P}_\infty(M), \pi \in \Pi(A_2) : \quad (f_m{:}\, m \in \tilde{M})$ is (π,ε)-compatible on $A_2]$

Under this assumption, we first choose inductively for each $k \in \mathbb{N}_0$ an $m_k \in \mathbb{N}_0$, an $M_k \in \mathcal{P}_\infty(\mathbb{N})$ and $B_k \in \Sigma$ such that

(2)(k) $m_k \in M_{k-1}$, $m_k > m_{k-1}$, and $M_k \subset M_{k-1}$, if $k > 0$,

(3)(k) $B_k \subset B_{k-1}$ if $k > 0$,

(4)(k) $\operatorname{ess\,osc}(f_{m_k}, B_k) \leq \varepsilon$ if $k > 0$,

(5)(k) $(f_m{:}\, m \in M_k)$ is totally ε-incompatible on B_k, and

(6)(k) there exists a $\pi \in \Pi(A \setminus B_k)$, such that $(f_m{:}\, m \in M_k)$ is (π,ε)-compatible on $A \setminus B_k$.

For $k = 0$, let $m_0 := 0$, $M_0 := \mathbb{N}$, and $B_0 := A$. Then (2)(0), (3)(0), and (4)(0) are empty conditions, while (5)(0) is just the assumption and (6)(0) is trivial.

If we assume that for a $k > 0$, $(m_\ell : 0 \leq \ell < k)$, $(M_\ell : 0 \leq \ell < k)$, and $(B_\ell : 0 \leq \ell < k)$ have already been chosen, we set

(7) $\qquad m_k := \min\{m \in M_{k-1} \mid m > m_{k-1} \text{ and } \operatorname{ess\,osc}(f_m, B_{m_{k-1}}) > \varepsilon\}$,

remarking that such an m_k exists because of $(5)(k-1)$.

Since $f_{m_k} \in \mathrm{L}^c_\infty(\mu, X)$, we find a $\tilde{\pi} \in \Pi(B_{k-1})$ for which f_{m_k} is $(\tilde{\pi}, \varepsilon/16)$-compatible on B_{k-1}. From (3.2)(c) and (5)$(k-1)$ we deduce that there exists a $\tilde{B}_k \in \tilde{\pi}$ and an $\tilde{M}_k \in \mathcal{P}_\infty(M_{k-1})$, such that $(f_m{:}m \in \tilde{M}_k)$ is totally ε-incompatible on $\tilde{B}_k$.

Setting

$$B_k := \bigcup\{B \mid B \in \tilde{\pi}'\}\,, \text{ where } \tilde{\pi}' := \{\tilde{B} \in \tilde{\pi} \mid \operatorname{ess\,dist}(f_{m_k}, \tilde{B}_k, \tilde{B}) \le \varepsilon/4\}\,,$$

we deduce (3)(k) and the fact that $(f_m{:}m \in \tilde{M}_k)$ is ε-incompatible on B_k (since $\tilde{B}_k \subset B_k$).

From (3.2.)(a)(iii) and (iv) we have

$$\begin{aligned}(8)\qquad \operatorname{ess\,osc}(f_{m_k}, B_k) &\le \sup_{B,\tilde{B}\in\tilde{\pi}'} \operatorname{ess\,dist}(f_{m_k}, B, \tilde{B}) + 2 \sup_{B\in\tilde{\pi}'} \operatorname{ess\,osc}(f_{m_k}, B)\\ &\le 2 \sup_{B\in\tilde{\pi}'} \operatorname{ess\,dist}(f_{m_k}, B, \tilde{B}_k) + 5 \sup_{B\in\tilde{\pi}'} \operatorname{ess\,osc}(f_{m_k}, B)\\ &\le \varepsilon/2 + 5\varepsilon/16 < \varepsilon\quad,\end{aligned}$$

which implies (4)(k).

Defining $A_2 := \bigcup_{B\in\tilde{\pi}\setminus\tilde{\pi}'} B = B_{k-1} \setminus B_k$ and applying (3.2)(a)(ii), we have

$$\operatorname{ess\,dist}(f_{m_k}, A_2, \tilde{B}_k) = \inf_{B\in\tilde{\pi}\setminus\tilde{\pi}'} \operatorname{ess\,dist}(f_{m_k}, B, \tilde{B}_k) \ge \varepsilon/4\,.$$

Now we are in a position to apply (1) to $M := \tilde{M}_k$, $n := m_k$, $A_1 := \tilde{B}_k$, and A_2 as defined above; we deduce that we can choose $M_k \in \mathcal{P}_\infty(\tilde{M}_k)$ and $\pi \in \Pi(A_2)$ such that $(f_m{:}m \in M_k)$ is (π, ε)-compatible on A_2.

Since $A \setminus B_k = (A \setminus B_{k-1}) \cup (B_{k-1} \setminus B_k) = (A \setminus B_{k-1}) \cup A_2$, we deduce (6)$(k)$ from (6)$(k-1)$ and the choice of M_k. Property (2)(k) follows from (7) and the fact that M_k was chosen to be a subset of M_{k-1}, while (5)(k) follows from the fact that $\tilde{B}_k \subset B_k$ and that subsequences of totally incompatible sequences inherit this property. This finishes the induction step.

Now we note that, by (3)(k) and the choice of B_0,

$$A = B_0 = \bigcup_{k\in\mathbb{N}} (B_{k-1} \setminus B_k) \cup \bigcap_{k\in\mathbb{N}} B_k\,.$$

From (6)(k) and (2)(k) we deduce that, for each $k \in \mathbb{N}$, there exists a $\pi_k \in \Pi(B_{k-1} \setminus B_k)$ such that $(f_{m_j} : j \in \mathbb{N})$ is (π_k, ε)-compatible on $B_{k-1} \setminus B_k$. (3)(k) and (4)(k) imply that $(f_{m_j}{:}j \in \mathbb{N})$ is $(\{\bigcap_{j\in\mathbb{N}} B_j\}, \varepsilon)$-compatible on $B_\infty := \bigcap_{j\in\mathbb{N}} B_j$. From (3.2)(b)(iv) we deduce that $(f_{m_j}{:}j \in \mathbb{N})$ is $(\bigcup_{k\in\mathbb{N}} \pi_k \cup \{B_\infty\}, \varepsilon)$-compatible on A, which contradicts the assumption of this lemma and finishes the proof. ∎

Proof of Theorem (3.1). We start the proof by showing that, for a sequence (f_n) in $L^c_\infty(\mu, X)$ which does not satisfy case 1, there exists an $\tilde{\varepsilon} > 0$ and a subsequence which is totally $\tilde{\varepsilon}$-incompatible.

We prove this by showing that if $(f_n : n \in \mathbb{N})$ satisfies the property

(1) $\quad \forall N \in \mathcal{P}_\infty(\mathbb{N}), \varepsilon > 0 \, \exists M \in \mathcal{P}_\infty(N), \pi \in \Pi \, : \, (f_m : m \in M)$ is (π, ε)-compatible,

then case 1 is satisfied.

For this, we choose inductively for each $k \in \mathbb{N}$ an $N_k \in \mathcal{P}_\infty(\mathbb{N})$ and a $\pi_k \in \Pi$ such that $N_k \subset N_{k-1}$ (where $N_0 := \mathbb{N}$) and such that for each $B \in \pi_k$ and each $m \in N_k$, $\operatorname{ess\,osc}(f_m, B) \leq \frac{1}{k}$ (this can be done by applying, in every induction step, (1) to $N := N_{k-1}$ and $\varepsilon := \frac{1}{k}$).

Then we choose an increasing sequence $(n_k) \subset \mathbb{N}$ with $n_k \in N_k$ for $k \in \mathbb{N}$. To show that $(f_{n_k} : k \in \mathbb{N})$ satisfies the assertion of case 1, let $\varepsilon > 0$ be arbitrary and choose $k_0 \in \mathbb{N}$ with $\varepsilon > \frac{1}{k_0}$. Since $(f_{n_\ell} : \ell \geq k_0)$ is (π_{k_0}, ε)-compatible, we find a $\pi \in \Pi$ which is finer than π_{k_0} and for which $(f_{n_k} : k \in \mathbb{N})$ is (π, ε)-compatible.

To prove (3.1), we assume that $(f_n : n \in \mathbb{N})$ does not satisfy case 1 and we show that it satisfies case 2.

By the above remark, we can assume that there exists an $\tilde{\varepsilon} > 0$ such that $(f_n : n \in \mathbb{N})$ is totally $\tilde{\varepsilon}$-incompatible.

By induction, we choose for each $k \in \mathbb{N}_0$ an $n_k \in \mathbb{N}_0$, an $N_k \in \mathcal{P}_\infty(\mathbb{N})$, and a family $(A(n,j) : 1 \leq j \leq 2^n) \subset \Sigma$ such that

(2)(k) $n_k \in N_{k-1}$, $n_k > n_{k-1}$, and $N_k \subset N_{k-1}$ if $k > 0$,

(3)(k) $A(k, 2i-1) \cup A(k, 2i) \subset A(k, i)$ and
$A(k, 2i-1) \cap A(k, 2i) = \emptyset$ almost everywhere if $k > 0$ and $i \in \{1, \ldots, 2^{k-1}\}$,

(4)(k) $\operatorname{ess\,osc}(f_{n_k}, A(k,i)) \leq \tilde{\varepsilon}/16$ and $\operatorname{ess\,dist}(f_{n_k}, A(k, 2j-1), A(k, 2j)) \geq \tilde{\varepsilon}/4$ if $k > 0$, $i \in \{1, \ldots, 2^k\}$ and $j \in \{1, \ldots, 2^{n-1}\}$, and

(5)(k) for each $j \in \{1, \ldots, 2^k\}$, $(f_m : m \in N_k)$ is totally $\tilde{\varepsilon}$-incompatible on $A(k,j)$.

If $k = 0$, we put $A(0,1) := \Omega$, $n_0 := 0$ and $N_0 := \mathbb{N}$. Then (2)(0),(3)(0), and (4)(0) are empty, while (5)(0) is just the assumption.

We suppose that for a $k > 0$, n_{k-1}, N_{k-1} and $(A(k-1, j) : j \leq 2^{k-1})$ has been chosen.

First we want to verify the following:

(6) Let $M \in \mathcal{P}_\infty(N_{k-1})$ and $j \in \{1, \ldots, 2^{k-1}\}$ be arbitrary.

Then there exists an increasing sequence $(\tilde{n}_\ell : \ell \in \mathbb{N}) \subset \mathbb{N}$ and for each $\ell \in \mathbb{N}$ an $\tilde{M}_\ell \in \mathcal{P}_\infty(M)$ and $\tilde{A}(1,\ell), \tilde{A}(2,\ell) \in \Sigma \cap A(k-1, j)$ such that for each $\ell \in \mathbb{N}$:

a) $\tilde{n}_\ell \in \tilde{M}_{\ell-1}$ $(\tilde{M}_0 := M)$, $\tilde{M}(\ell) \subset \tilde{M}(\ell - 1)$,

b) the conditions (a) and (b) of Lemma (3.3) hold for

$$n = \tilde{n}_\ell, M := \tilde{M}_\ell \text{ and } A_1 := \tilde{A}(1,\ell), A_2 := \tilde{A}(2,\ell) .$$

We prove this by applying (3.3) successively for each $\ell \in \mathbb{N}$ to the sequence $(f_m : m \in \tilde{M}_{\ell-1},\ m > \tilde{n}_{\ell-1})$ and $A := A(k-1,j)$.

Applying (6) successively to each $j \leq 2^{k-1}$, we find sequences $(\tilde{n}(j,\ell) : \ell \in \mathbb{N}) \subset \mathbb{N}$, $(\tilde{A}(j,1,\ell) : \ell \in \mathbb{N})$, $(\tilde{A}(j,2,\ell) : \ell \in \mathbb{N}) \subset \Sigma \cap A(k-1,j)$, and decreasing $(\tilde{M}_\ell^{(j)} : \ell \in \mathbb{N}_0) \in \mathbb{N}$, such that

(7) the condition (6) holds for each $j \in \{1,\dots,2^{k-1}\}$

and, moreover,

$$\begin{aligned} N^{k-1} &= M_0^{(1)} \supset (\tilde{n}(1,\ell) : \ell \in \mathbb{N}) = M_0^{(2)} \supset (\tilde{n}(2,\ell) : \ell \in \mathbb{N}) \\ &= M_0^{(3)} \dots \supset (\tilde{n}(2^{k-1}-1,\ell) : \ell \in \mathbb{N}) = M_0^{(2^{k-1})} \supset (\tilde{n}(2^{k-1},\ell) : \ell \in \mathbb{N}) . \end{aligned}$$

Now we are in a position to choose $n_k \in \{\tilde{n}(2^{k-1},\ell) \,|\, \ell \in \mathbb{N}\}$ with $n_k > n_{k-1}$ and we set $\tilde{N}_k := \{\tilde{n}(2^{k-1},\ell) \,|\, \ell \in \mathbb{N}\}$. By (7), we find for each $j \in \{1,\dots,2^{k-1}\}$ an $\ell_j \in \mathbb{N}$ such that $n_k = \tilde{n}(j,\ell_j)$.

Then we choose, for each $j \in \{1,\dots,2^{k-1}\}$, a $\pi_j^1 \in \Pi(\tilde{A}(j,1,\ell_j))$ and a $\pi_j^2 \in \Pi(\tilde{A}(j,2,\ell_j))$ such that f_{n_k} is $(\pi_j^i,\tilde{\varepsilon})$-compatible on $\tilde{A}(j,i,\ell_j)$ for $i = 1,2$. Applying (3.1)(c) 2^k times, we deduce from (7) that we can choose $A(k,2j-1) \in \pi_j^1$ and $A(k,2j) \in \pi_j^2$ respectively and $N_k \in \mathcal{P}_\infty(\tilde{N}_k)$, such that $(f_m : m \in N_k)$ is totally $\tilde{\varepsilon}$-incompatible on each $A(k,j)$ with $j \in \{1,\dots,2^k\}$. Thus, we deduce (5)(k) and the first part of (4)(k). (2)(k) and (3)(k) follow from this choice also, while the second part of (4)(k) follows from (7). Thus, we have completed the proof of the induction step. The assertion of the theorem follows from (2)(k), (3)(k), (4)(k) and (5)(k) if we take $\varepsilon := \tilde{\varepsilon}/4$ and note that from (5)(k) it follows that $\mu(A(k,i)) > 0$ for $k \in \mathbb{N}$ and $i \in \{1,\dots,2^k\}$. ∎

4. Proof of Theorem (2.5)

In order to proof that each subsequence of an $X \tilde{\otimes} Y$-limited set contains a further subsequence which has the countable partition property we need the following

Lemma 4.1. *Let K be compact and let Σ_K be the σ-algebra of the Borel sets of K. Let $A \subset C(K,X)$ have a common X-limited range $A(K) := \{f(\xi) \,|\, \xi \in K\}$.*

Then the following are equivalent:

a) *Every sequence $(f_n) \subset A$ enjoys, as a sequence in $L_\infty^c(\Sigma_K, X)$, the condition of case 1 of Theorem (3.1).*

b) For each positive and finite Borel measure μ on K, $T_\mu(A)$ is limited in $\mathrm{L}_1(\mu, X)$, where $T_\mu : \mathrm{L}^c_\infty(\Sigma_K, X) \to \mathrm{L}_1(\mu, X)$, $f \mapsto f$.

In particular, we deduce that a set $A \subset \mathrm{C}(K, X)$ which is limited in $\mathrm{C}(K, X)$ satisfies condition (a).

Proof of (a) $\Rightarrow$ (b).

Let μ be a finite measure on Σ_K. It is enough to show that an arbitrary sequence $(f_n) \subset A$ contains a subsequence whose image under T_μ is limited in $\mathrm{L}_1(\mu, X)$.

Thus, let $(f_n) \subset A$. By taking a subsequence, we may assume by (a) that there is an increasing sequence of countable Σ_K-partitions (π_k) of K, $\pi_k = (B(k,m) : m \in \mathbb{N})$, and a family $(x(n,k,m) : n,k,m \in \mathbb{N}) \subset A(K))$ such that

$$\|f_n - \sum_{m\in\mathbb{N}} \chi_{B(k,m)} x(n,k,m)\|_\infty \leq \frac{1}{k}, \text{ for each } n,k \in \mathbb{N} . \tag{1}$$

Thus (note that $\| T_\mu \| = \mu(K)$),

$$(T_\mu(f_n) : n \in \mathbb{N}) \subset \bigcap_{k\in\mathbb{N}} \{f_n^{(k)} \mid n \in \mathbb{N}\} + \mathrm{B}_{\mu(K)/k}(\mathrm{L}_1(\mu, X)) ,$$

where

$$f_n^{(k)} := \sum_{m\in\mathbb{N}} \chi_{B(k,m)} x(n,k,m) \text{ for each } n,k \in \mathbb{N} .$$

Thus, by (1.2) it is sufficient to show that, for fixed $k \in \mathbb{N}$, the sequence $(f_n^{(k)})$ is limited in $\mathrm{L}_1(\mu, X)$. In order to see this, we choose for an arbitrary $\varepsilon > 0$ an $m_0 = m_0(\varepsilon)$ such that

$$\sum_{m\geq m_0} \mu(B(m,k)) < \varepsilon/(1 + \sup_{m,n\in\mathbb{N}} \| x(n,m,k) \|) .$$

Since $A(K)$ is limited in X we deduce that $A_\varepsilon := \{\sum_{m=1}^{m_0} \chi_{B(k,m)} x(n,k,m) \mid n \in \mathbb{N}\}$ is limited in $\mathrm{L}_1(\mu, X)$ and since $(f_n^{(k)})_{n\in\mathbb{N}} \subset A_\varepsilon + \varepsilon \mathrm{B}_1(\mathrm{L}_1(\mu, X))$ the assertion follows from (1.2).

Proof of $\neg$(a) $\Rightarrow$ $\neg$(b).

Using Theorem (3.1), it is enough to show that, for a sequence $(f_n) \subset A$ admitting an $\varepsilon > 0$ and a tree of sets $(A(n,j) : n \in \mathbb{N}_0, 1 \leq j \leq 2^n)$ such that the conditions of case 2 are satisfied, there is a finite Borel measure μ on K for which $(T_\mu(f_n) : n \in \mathbb{N})$ is not limited in $\mathrm{L}_1(\mu, X)$.

For this we set $\tilde{K} := \bigcap_{n\in\mathbb{N}} \bigcup_{j=1}^{2^n} \overline{A(n,j)}^K$ and $C(n,j) := \tilde{K} \cap \overline{A(n,j)}^K$ for $n \in \mathbb{N}_0, j \leq 2^n$. From the property of $(A(n,j) : n \in \mathbb{N}_0, 1 \leq j \leq 2^n)$ and the compactness of K we deduce that no $C(n,j)$ is empty.

Since f_n is continuous for each $n \in \mathbb{N}$, we deduce from the assumptions that

$$(2) \qquad \sup_{\xi,\tilde{\xi}\in C(n,j)} \|f_n(\xi) - f_n(\tilde{\xi})\| \leq \frac{\varepsilon}{4} \text{ and } \inf_{\xi\in C(n,2i),\tilde{\xi}\in C(n,2i-1)} \|f_n(\xi) - f_n(\tilde{\xi})\| \geq \varepsilon$$

whenever $n \in \mathbb{N}$, $j \leq 2^n$, and $i \leq 2^{n-1}$.

Moreover, for $n \in \mathbb{N}$ and $i \leq 2^{n-1}$ we have

$$C(m,2i) \cup C(n,2i-1) = C(n,2i) \text{ and } C(n,2i) \cap C(n,2i-1) = \emptyset .$$

For each $n \in \mathbb{N}$ and $j \leq 2^n$ we choose a $\xi(n,j) \in C(n,j)$ and an accumulation point $\mu \in \mathrm{M}(\tilde{K})$ of the sequence $(\mu_n) := (2^{-n}\sum_{i=1}^{2^n} \delta_{\xi(n,j)} : n \in \mathbb{N})$ in $\sigma(\mathrm{M}(\tilde{K}), \mathrm{C}(\tilde{K}))$ where δ_ξ denotes the Dirac measure for $\xi \in K$. We observe that for $n \in \mathbb{N}$, $m \geq n$, and $j \in \{1,2,\dots,2^n\}$,

$$\mu_m(C(n,j)) = 2^{-m}|\{i \leq 2^m \mid C(m,i) \subset C(n,j)\}| = 2^{-n} .$$

Since $C(n,j)$ is clopen in $\tilde{K}$ (note that $\tilde{K} \setminus C(n,j) = \bigcup_{i\in\{1,\dots 2^n\}\setminus\{j\}} C(n,i)$) for each $n \in \mathbb{N}$ and $j \leq 2^n$), we deduce that $\mu(C(n,j)) = 2^{-n}$ for each $n \in \mathbb{N}$ and $j \leq 2^n$.

Taking $\pi_n := \{C(n,j) \mid j \leq 2^n\} \cup \{K \setminus \tilde{K}\}$, and taking $E_n : \mathrm{L}_1(\mu, X) \to \mathrm{L}_1(\mu, X)$ to be the conditional expectation with respect to the σ-algebra generated by π_n we have

$$\begin{aligned}
&\|E_n \circ T_\mu(f_n) - E_{n-1} \circ T_\mu(f_n)\|_1 \\
&\quad = \| \sum_{i=1}^{2^{n-1}} 2^{n-1}\Big[2\chi_{C(n,2i)} \int_{C(n,2i)} f_n \mathrm{d}\mu + 2\chi_{C(n,2i-1)} \int_{C(n,2i-1)} f_n \mathrm{d}\mu \\
&\qquad\qquad - \chi_{C(n-1,i)} \int_{C(n-1,i)} f_n \mathrm{d}\mu\Big] \|_1 \\
&\quad = \sum_{i=1}^{2^{n-1}} \| \int_{C(n,2i)} f_n \mathrm{d}\mu - \int_{C(n,2i-1)} f_n \mathrm{d}\mu \| \\
&\quad \geq \sum_{i=1}^{2^{n-1}} 2^{-n}\Big[\| f_n(\xi(n,2i)) - f_n(\xi(n,2i-1)) \| \\
&\qquad\qquad - \big(\sup_{\xi,\tilde{\xi}\in C(n,2i)} \| f_n(\xi) - f_n(\tilde{\xi}) \| + \sup_{\xi,\tilde{\xi}\in C(n,2i-1)} \| f_n(\xi) - f_n(\tilde{\xi}) \| \big)\Big] \\
&\quad \geq \sum_{i=1}^{2^{n-1}} (\varepsilon - 2\varepsilon/4)2^{-n} = \varepsilon/4 ,
\end{aligned}$$

Thus, E_n is not uniformly convergent on $(T_\mu(f_n))$; since by the martingale convergence theorem [3] E_n is pointwise convergent we deduce from (1.1) that $T_\mu(A)$ cannot be limited in $\mathrm{L}_1(\mu, X)$. ∎

Now we are in the position to prove the following part of Theorem (2.5):

Claim. A limited set $A \subset X \check{\otimes} Y$ has property (2.5.2).

Let $A \subset X \check{\otimes} Y$ be limited and let (f_n) be a subsequence of A. Since $X \check{\otimes} Y$ can be seen as subspace of $C(K_X, Y)$ as well as a subspace of $C(K_Y, X)$ we deduce from Lemma (4.1) that there is an infinite $N \subset \mathbb{N}$ and, for each $\varepsilon > 0$, there are countable Σ_X- and Σ_Y- partitions $\pi^{(1)}$ and $\pi^{(2)}$ of K_X and K_Y respectively, such that the sequence $(f_n : n \in N)$ is $(\pi^{(1)}, \varepsilon/2)$ compatible on K_X (viewed in $L^c_\infty(\Sigma_X, Y)$) and $(\pi^{(2)}, \varepsilon/2)$ compatible on K_Y (viewed in $L^c_\infty(\Sigma_Y, X)$). Thus, for each $n \in N$, $B^{(1)} \in \pi^{(1)}$ and $B^{(2)} \in \pi^{(2)}$, and $\xi(1), \tilde{\xi}(1) \in B^{(1)}$ and $\xi(2), \tilde{\xi}(2) \in B^{(2)}$ we deduce that

$$\begin{aligned} &|z_n(\xi^{(1)}, \xi^{(2)}) - z_n(\tilde{\xi}^{(1)}, \tilde{\xi}^{(2)})| \\ &\quad \le |z_n(\xi^{(1)}, \xi^{(2)}) - z_n(\xi^{(1)}, \tilde{\xi}^{(2)})| + |z_n(\xi^{(1)}, \tilde{\xi}^{(2)}) - z_n(\tilde{\xi}^{(1)}, \tilde{\xi}^{(2)})| \\ &\quad \le \| z_n(\cdot, \xi^{(2)}) - z_n(\cdot, \tilde{\xi}^{(2)}) \| + \| z_n(\xi^{(1)}, \cdot) - z_n(\tilde{\xi}^{(1)}, \cdot) \| \le \varepsilon \ , \end{aligned}$$

which implies the assertion. ∎

The second assertion of Theorem (2.5) follows from the following Proposition applied to $V := X \otimes Y$, $\|.\| := \|.\|_\vee$ and $|||.||| := \|.\|_\wedge$:

Proposition 4.2. *Let $V \subset X$ be a dense subspace of X and let $|||\cdot|||$ be a norm on V which is finer then $\|\cdot\|$. We denote the completion of V corresponding to $|||\cdot|||$ by $\tilde{X}$.*

Then every X $(= (X, \|\cdot\|))$ limited set A is "almost bounded corresponding to $|||\cdot|||$." By this we mean that for each $\varepsilon > 0$ there is a $|||\cdot|||$-bounded set $A^{(\varepsilon)} \subset V$ such that

$$A \subset B_\varepsilon(X, \|.\|) + A^{(\varepsilon)} \ .$$

Proof. We have to show the following:

Let $A \subset X$ be $\|\cdot\|$-bounded, $(x_n) \subset A$ and $\varepsilon > 0$, such that

$$r_n := \inf\{|||y||| \mid y \in V \cap (x_n + B_\varepsilon(X, \|\cdot\|))\} \underset{n \to \infty}{\longrightarrow} \infty \ . \tag{1}$$

Then A is not limited in $(X, \|\cdot\|)$.

For $n \in \mathbb{N}$ define

$$A_n := \overline{(r_n/2) \cdot B_1(\tilde{X}, |||\cdot|||)}^{\|\cdot\|} \quad \text{and} \quad B_n := x_n + B_{\varepsilon/2}(X, \|\cdot\|) \ .$$

We first show that $A_n \cap B_n = \emptyset$ for each $n \in \mathbb{N}$:

Let $y \in A_n$, then there is a $\tilde{y} \in (r_n/2) \cdot B_1(\tilde{X}, |||\cdot|||)$ with $\|\tilde{y} - y\| < \varepsilon/4$, and a $\hat{y} \in V$ with $|||\hat{y} - \tilde{y}||| < \min(\varepsilon/4, r_n/2)$. Thus, $|||\hat{y}||| \le |||\tilde{y}||| + |||\hat{y} - \tilde{y}||| < r_n$, and we conclude from (1) that $\|\hat{y} - x_n\| > \varepsilon$ and finally

$$\|x_n - y\| \ge \|x_n - \hat{y}\| - \|\hat{y} - \tilde{y}\| - \|\tilde{y} - y\| > \varepsilon - 2\varepsilon/4 = \varepsilon/2 \ ,$$

which implies the assertion.

Since A_n and B_n are convex and $\|\cdot\|$-closed and since A_n is absolutely convex, we find, by the separation theorem, for each $n \in \mathbb{N}$ an $x_n^* \in X^*$, with $\|x_n^*\| = 1$, and an $a_n \geq 0$ such that

$$\langle x_n^*, y\rangle \leq a_n \leq \langle x_n^*, x\rangle\,, \quad \text{whenever } y \in A_n \text{ and } x \in B_n\,. \tag{2}$$

For $n \in \mathbb{N}$ we choose $y_n \in \mathrm{B}_{\varepsilon/2}(X, \|\cdot\|)$ with $\langle x_n^*, y_n\rangle \geq \varepsilon/4$ and we conclude that

$$\begin{aligned} \langle x_n^*, x_n\rangle &= \langle x_n^*, x_n - y_n\rangle + \langle x_n^*, y_n\rangle \\ &\geq a_n + \varepsilon/4 \geq \varepsilon/4\,. \\ &[x_n - y_n \in B_n \text{ and } a_n \geq 0]\,. \end{aligned} \tag{3}$$

Thus, we are finished once we observe that (x_n^*) is a weak*-zero sequence in $(X^*, \|\cdot\|)$.

To this end, we first observe that by (2), $\sup_{n\in\mathbb{N}} |a_n| \leq \sup_{n\in\mathbb{N}, x\in A} |\langle x_n^*, x\rangle| < \infty$, and secondly that for each $v \in V$ it follows from (1) and (2):

$$|\langle x_n^*, v\rangle| = \frac{2|||v|||}{r_n} |\langle x_n^*, \frac{r_n}{2|||v|||} v\rangle| \leq \frac{2|||v|||}{r_n} \cdot a_n \underset{n\to\infty}{\longrightarrow} 0\,;$$

Since V is dense in X and (x_n^*) is bounded in $(X^*, \|\cdot\|)$ this implies the assertion. ∎

5. Proof of Theorem 2.6

In order to prove Theorem (2.6) we first need to prove the following lemmas (5.1), (5.2) and (5.3).

Lemma 5.1. *Let (μ_n) be a bounded sequence in $\mathrm{M}(K, X)$ whose elements have pairwise disjoint supports and let $\delta > 0$.*

Then there exists a subsequence (n_k) of $\mathbb{N}$ and a normalized sequence (h_k) of nonnegative elements of $\mathrm{C}(K)$ such that the following properties hold:

(5.1.1) For each $k \in \mathbb{N}$ there is a neighborhood O_k of $\mathrm{supp}((h_1 \cdot \ldots \cdot h_{k-1}).\mu_{n_k})$ with $O_k \cap O_{k'} = \emptyset$ for $k, k' \in \mathbb{N}$ with $k \neq k'$.

(5.1.2) $\|\mu_{n_k} - (h_1 \cdot \ldots \cdot h_{k-1}).\mu_{n_k}\| \leq \delta$.

(5.1.3) $\lim_{\ell\to\infty} \sup_{k\geq\ell+1} \|(h_1 \cdot \ldots \cdot h_\ell) \cdot (1 - h_{\ell+1} \cdot \ldots \cdot h_{k-1}).\mu_{n_k}\| = 0$.

Remark. Suppose $(\mu_n) \subset \mathrm{M}(K, X^*)$ satisfies the conditions of Lemma (5.1) and is, moreover, $\sigma(\mathrm{M}(K, X^*), \mathrm{C}(K, X),)$-zero convergent and let $\delta > 0$. If (n_k), (h_k) and O_k are as in Lemma (5.1) then it follows from (5.1.3) that the sequence $(\tilde{\mu}_k)$ where $\tilde{\mu}_k := h_1 \cdot h_2 \ldots h_{k-1}.\mu_k$ is still

$\sigma(\mathrm{M}(K,X^*),\mathrm{C}(K,X),)$-zero convergent. Moreover, if $(f_n)\subset \mathrm{B}_1(\mathrm{C}(K,X))$ then (5.1.2) implies that $|\langle\mu_{n_k},f_{n_k}\rangle-\langle\tilde{\mu}_{n_k},f_{n_k}\rangle|\leq\delta$.

Proof of (5.1). (The proof uses ideas from the proof of a result of Pełczyński [**10**, Lemma 1, p.643])

W.l.o.g we may assume that $\|\mu_n\|\leq 1$ for $n\in\mathbb{N}$. First we choose a sequence $(m_k)\subset\mathbb{N}$ with

$$\sum_{k=1}^{\infty}\frac{1}{m_k}\leq\delta\ . \tag{1}$$

Then we choose inductively, for each $k\in\mathbb{N}$, $h_k\in\mathrm{C}(K)$ with $\|h_k\|=1$ and $0\leq h_k\leq 1$, an open set $O_k\subset K$, a strictly increasing sequence $(n_j^{(k)})$ in $\mathbb{N}$, and an $n_k\in\mathbb{N}_0$ such that the following properties are satisfied:

(2)(k) $(n_j^k : j\in\mathbb{N})\subset(n_j^{k-1} : j>m_k)$ and $n_k\in(n_j^{k-1} : 1\leq j\leq m_k)$ if $k>1$,

(3)(k) $\overline{O_k}\cap\overline{O_{k'}}=\emptyset$ whenever $1\leq k'<k$,

(4)(k) $\overline{\{h_k>0\}}\cap\overline{O_k}=\emptyset$,

(5)(k) $\mathrm{supp}(h_1\cdot h_2\cdot\ldots\cdot h_{k-1}.\mu_{n_k})\subset O_k$,

(6)(k) $\|h_1\cdot h_2\cdot\ldots\cdot h_{k-1}(1-h_k).\mu_{n_j^k}\|\leq 1/m_k$, and

For $k=1$ we set $n_j^{(1)}:=j$ if $j\in\mathbb{N}$, $h_1:=1$, $O_1:=\emptyset$, and $n_1:=0$, where $\mu_0:=0$, and we deduce easily that the desired conditions are satisfied.

We assume now that for $k>1$ and for all $r\in\{1,\ldots,k-1\}$, $(n_j^{(r)})$, h_r, and n_r have been chosen.

We define $A_0:=\bigcup_{r<k}\overline{O_r}$ and for each $i\in\{1,2,\ldots m_k\}$ $A_i:=\mathrm{supp}((h_1\cdot h_2\ldots h_{k-1}).\mu_{n_i^k})$. From (4)($r$), $r<k$, for each $i\in\{1,2,\ldots m_k\}$ we deduce that $A_{i_0}\cap A_i=\emptyset$. Since, by assumption, the supports of $\mu_{n_i^{k-1}}$, $1\leq i\leq m_k$, are pairwise disjoint, we can appply the normality of K in order to find, for each $0\leq i\leq m_k$, an open set $G_i\subset K$ such that

$$A_i\subset G_i\quad\text{and}\quad\overline{G_i}\cap\overline{G_j}=\emptyset\quad\text{for } i,j\in\{0,1,\ldots,m_k\}\text{ with } i\neq j\ . \tag{7}$$

For $i\in\{1,2,\ldots,m_k\}$ we set $M_i:=\{j>m_k\mid|\mu_{n_j^{k-1}}|(\overline{G_i})\leq 1/m_k\}$. Since $\mu_{n_j^{k-1}}$ is of norm not greater than one, we deduce from (7) that $1\geq\sum_{i=1}^{m_k}|\mu_{n_j^{k-1}}|(\overline{G_i})$ for each $j\in\mathbb{N}$. Thus, for each $j\in\mathbb{N}$, at least one of the summands must not be greater than $1/m_k$, and it follows that there exists an $i_0\in\{1,2,\ldots,m_k\}$ for which M_{i_0} is infinite.

We take $n_k:=n_{i_0}^{k-1}$ and $n_j^k:=n_{\ell(j)}^{(k)}$, for $j\in\mathbb{N}$, where $(\ell_j)\subset\mathbb{N}$ is increasing and consists of the elements of $M_{i_0}^{(j_0)}$. With this choice, (2)(k) follows. Since A_{i_0} is closed and since G_{i_0} is open and contains A_{i_0}, there is an open set $O_k\subset K$ with $A_{i_0}\subset O_k\subset\overline{O_k}\subset G_{i_0}$.

Now we choose an h_k satisfying $0 \le h_k \le 1$, $h_k|_{G_{i_0}^c} = 1$, and such that h_k vanishes on a neighborhood of $\overline{O_k}$.

With this choice, we deduce (4)(k) and (5)(k).

Moreover, since $(n_j^k) = (n_\ell^{k-1} : \ell \in M_{i_0})$, for each $j \in \mathbb{N}$, we deduce that

$$\| h_1 \cdot h_2 \cdot \ldots \cdot h_{k-1}(1-h_k).\mu_{n_j^k} \| \le |h_1 \cdot h_2 \cdot \ldots \cdot h_{k-1}.\mu_{n_j^k}|(\{h_k \neq 1\}) \le |\mu_{n_j^k}|(G_{i_0}) \le 1/m_k$$

which verifies (6)(k) and finishes the induction step.

Now (5.1.1) follows from (3)(k) and (5)(k). Finally, we deduce (5.1.2) and (5.1.3) respectively from the following inequalities, both following from (6)(j), (2)(j) and (1):

$$\| \mu_{n_k} - (h_1 h_2 \cdot \ldots \cdot h_{k-1}).\mu_{n_k} \| \le \sum_{j=1}^{k-1} \| (1-h_j) h_1 h_2 \cdot \ldots \cdot h_{j-1}.\mu_{n_k} \| \le \sum_{j=1}^{k-1} \frac{1}{m_j} < \delta$$

and

$$\| (h_1 \cdot \ldots \cdot h_\ell) \cdot (1 - h_{\ell+1} \cdot \ldots \cdot h_{k-1}).\mu_{n_k} \|$$
$$\le \sum_{j=\ell+1}^{k-1} \| (1-h_j) h_1 \cdot h_2 \cdot \ldots \cdot h_{j-1}.\mu_{n_k} \| \le \sum_{j=\ell+1}^{\infty} \frac{1}{m_j} \underset{\ell \to \infty}{\longrightarrow} 0 \,. \qquad \blacksquare$$

Lemma 5.2. *Let K be a compact space such that $C(K)$ is a Grothendieck space. Let $(f_n) \subset C(K,X)$ be a bounded sequence with the following properties (a), (b), and (c):*

a) *The sequence $(f_n(\xi) : n \in \mathbb{N})$ is limited in X for each $\xi \in K$.*

b) *Every subsequence of (f_n) satisfies the property of case 1 in Theorem (3.1) (we view $C(K,X)$ as a subspace of $L_\infty^c(\Sigma_K, X)$).*

c) *(f_n) is not limited in $C(K,X)$.*

Then it follows that there is a subsequence $(\tilde{f}_n)$ of (f_n), a sequence (C_n) of closed subsets of K and a sequence (O_n) of open subsets of K such that

d) *$\emptyset \neq C_n \subset O_n$ and $O_n \cap O_{n'} = \emptyset$ if $n, n' \in \mathbb{N}$, with $n \neq n'$, and*

e) *for each sequence $(h_n) \subset C(K)$ with $0 \le h_n \le 1$, $h_n|_{C_n} = 1$ and $h_n|_{O_n^c} = 0$, for $n \in \mathbb{N}$, the sequence $(h_n \tilde{f}_n : n \in \mathbb{N})$ is not limited in $C(K,X)$.*

Proof. By the assumptions in (b) and (c), we can assume that there is an $\varepsilon > 0$ and a normalized w*-zero sequence (μ_n) in $C(K,X)^* = M(K,X^*)$ with

$$\varepsilon \le \langle \mu_n, f_n \rangle \qquad \text{for all } n \in \mathbb{N} \tag{1}$$

and that, moreover, there is a countable Σ_K-partition $\pi = (B_m)$ such that (f_n) is $(\varepsilon/6, \pi)$-compatible on K. Choosing a $\xi_m \in B_m$, for each $m \in \mathbb{N}$, and setting $x(n,m) := f_n(\xi_m)$, we deduce that

$$\| f_n - \sum_{m \in \mathbb{N}} \chi_{B_m} x(n,m) \|_\infty \le \varepsilon/6 \qquad \text{for each } n \in \mathbb{N}. \tag{2}$$

Since from the Grothendieck property of $C(K)$ we deduce that $\langle \mu_n(B), x\rangle = \langle \mu_n, x\rangle(\chi_B) \underset{n\to\infty}{\longrightarrow} 0$, for $x \in X$ and $B \in \Sigma_K$, it follows that

$$\mu_n(B) \underset{n\to\infty}{\longrightarrow} 0 \qquad \text{in } \sigma(X^*, X) \text{ for any } B \in \Sigma_K \ . \tag{3}$$

Now choose inductively, for each $k \in \mathbb{N}$, n_k and m_k both in $\mathbb{N}$ such that

(4)(k) $n_k < m_k$ and if $k > 1$, then $m_{k-1} < n_k$,

(5)(k) $|\mu_{n_k}|(\bigcup_{m\geq m_k} B_m) < \frac{\varepsilon}{6c}$,

where $c := 1 + \sup\{\| x(n,m)\| \mid n, m \in \mathbb{N}\}$, and

(6)(k) $|\langle \mu_{n_k}, \sum_{m\leq m_{k-1}} \chi_{B_m} x(n_k, m)\rangle| < \frac{\varepsilon}{6}$, if $k > 1$.

If $k = 1$ we choose $n_1 = 1$ and an $m_1 > n_1$ large enough so that (5)(1) holds (note that $\| \mu_j \| = 1$ and that the sets B_j's are pairwise disjoint).

If m_{k-1} and n_{k-1} are already chosen for $k > 1$, we deduce from (3) and the limitedness of $(x(n,m) : n \in \mathbb{N}, m \leq m_{k-1})$ in X (assumption (a)) that there is an $n_k > m_{k-1}$ such that

$$|\langle \mu_{n_k}, x(n_k, m)\chi_{B_m}\rangle| < \frac{\varepsilon}{m_{k-1} 6c} \text{ for each } m \leq m_{k-1} \ , \tag{7}$$

which implies (6)(k). As in the first induction step, we can choose an $m_k > n_k$ satisfying (5)(k).

Setting $D_k := \bigcup_{j=m_{k-1}+1}^{m_k} B_j$, for each $k \in \mathbb{N}$ (with $m_0 := 0$), we deduce that

$$\begin{aligned} \langle \mu_{n_k}, \chi_{D_k} f_{n_k}\rangle &\geq \langle \mu_{n_k}, f_{n_k}\rangle - |\langle \mu_{n_k}, \chi_{D_k^c} f_{n_k}\rangle| \\ &\geq \varepsilon - \varepsilon/6 - |\langle \mu_{n_k}, \chi_{D_k^c} \sum_{m\in\mathbb{N}} \chi_{B_m} x(n_k, m)\rangle| \\ &\quad \text{[by (1), (2) and } \| \mu_{n_k} \| = 1\text{]} \\ &\geq 5\varepsilon/6 - |\langle \mu_{n_k}, \sum_{m\leq m_{k-1}} \chi_{B_m} x(n_k, m)\rangle| \\ &\qquad - |\langle \mu_{n_k}, \sum_{m>m_k} \chi_{B_m} x(n_k, m)\rangle| \\ &\geq 5\varepsilon/6 - 2\varepsilon/6 = \varepsilon/2 \\ &\quad \text{[by (5)}(k) \text{ and (6)}(k)\text{]} \ . \end{aligned} \tag{8}$$

Since for any $f \in C(K)$ and $x \in X$ the sequence $(\langle f.\mu_{n_k}, x\rangle : k \in \mathbb{N})$ is weak* convergent in $M(K)$ to zero, and thus weakly convergent, since $(\chi_{D_k} : k \in \mathbb{N})$ is limited in $L_\infty(\Sigma_K)$ (we recall that $L_\infty(\Sigma_K)$ is a Grothendieck space with the Dunford-Pettis property and that $(\chi_{D_k} : k \in \mathbb{N})$ is weakly convergent to zero in $L_\infty(\Sigma_K)$), and, finally, since $L_\infty(\Sigma_K)$ is a subspace of $C(K)^{**}$, we deduce that

$$\langle \mu_{n_k}|_{D_k}, f \cdot x\rangle = \langle f.\mu_{n_k}, x\rangle(\chi_{D_k}) \underset{k\to\infty}{\longrightarrow} 0 \text{ whenever } f \in C(K),\ x \in X \ ,$$

which implies that $(\mu_{n_k}|_{D_k} : k \in \mathbb{N})$ converges in $\sigma(\mathrm{M}(K,X^*),\mathrm{C}(K,X))$ to zero. Since μ_{n_k}, is regular for $k \in \mathbb{N}$ there is for each $k \in \mathbb{N}$ a compact set $\tilde{D}_k \subset D_k$ such that $|\mu_{n_k}|(D_k \setminus \tilde{D}_k) \leq \frac{\varepsilon}{6k\tilde{c}}$, where $\tilde{c} := \sup_{n\in\mathbb{N}}(1+\|f_n\|)$.

Thus, the sequence (ν_k), with $\nu_k := \mu_{n_k}|_{\tilde{D}_k}$ $(k \in \mathbb{N})$, has pairwise disjoint supports, converges w* to zero, and satisfies, by (8),

$$\langle \nu_k, f_{n_k}\rangle \geq \langle \mu_{n_k}, \chi_{D_k} f_{n_k}\rangle - |\langle \mu_{n_k}, \chi_{D_k\setminus\tilde{D}_k} f_{n_k}\rangle| \geq \frac{\varepsilon}{2} - \frac{\varepsilon}{6} = \frac{\varepsilon}{3}\,. \tag{9}$$

By Lemma (5.1) and the following remark we find a subsequence (k_ℓ) of $\mathbb{N}$ and a w*-zero sequence $\tilde{\nu}_\ell$ in $\mathrm{M}(K,X^*)$ such that the supports C_ℓ of $\tilde{\nu}_\ell$ have pairwise disjoint neighborhoods O_ℓ and such that for $\tilde{f}_\ell := f_{n_{k_\ell}}$ it follows that $\langle \tilde{\nu}_\ell, \tilde{f}_\ell\rangle \geq \varepsilon/6$. For an arbitrary sequence $(h_\ell) \subset \mathrm{C}(K)$ with $0 \leq h_\ell \leq 1$, $h_\ell|_{C_\ell} = 1$, and $h_\ell|_{O_\ell^c} = 0$, we deduce that

$$\begin{aligned}\langle h_\ell \tilde{f}_\ell, \tilde{\nu}_\ell\rangle &= \langle \tilde{f}_\ell, \tilde{\nu}_\ell\rangle \\ &\geq \langle f_{n_{k_\ell}}, \nu_{k_\ell}\rangle - \|f_{n_{k_\ell}}\|\cdot\|\nu_{k_\ell} - \tilde{\nu}_{k_\ell}\| \\ &\geq \varepsilon/6\,,\end{aligned}$$

which implies that $(h_\ell \tilde{f}_\ell : \ell \in \mathbb{N})$ is not limited in $\mathrm{C}(K,X)$ and so the assertion follows. ∎

Lemma 5.3. *Let K_1 and K_2 be compact spaces with $\mathrm{C}(K_1)$ and $\mathrm{C}(K_2)$ being Grothendieck spaces.*

We consider a sequence $(f_n) \subset \mathrm{C}(K_1\times K_2)$ and an $\varepsilon > 0$ with the following properties (a) and (b):

a) *For $n \in \mathbb{N}$ and $\theta = 1,2$, there are closed sets $C_n^\theta \subset K_\theta$ and open sets $O_n^\theta \subset K_\theta$ with*

$$C_n^{(\theta)} \subset O_n^{(\theta)}\,,\ O_n^{(\theta)} \cap O_{n'}^{(\theta)} = \emptyset\,,\ \text{and } \mathrm{supp} f_n \subset C_n^{(1)}\times C_n^{(2)} \text{ for } n,n' \in \mathbb{N} \text{ with } n \neq n'\,.$$

b) *There is a sequence $(\tilde{f}_n) \subset \mathrm{C}(K_1)\otimes\mathrm{C}(K_2)$ with*

$$\|\tilde{f}_n - f_n\|_\vee < \varepsilon \text{ and } \sup_{n\in\mathbb{N}} \|\tilde{f}_n\|_\wedge < \infty \text{ for } n \in \mathbb{N}\,.$$

Then for any weak-zero sequence (μ_n) in $\mathrm{B}_1(\mathrm{M}(K_1\times K_2))$ it follows that*

$$\limsup_{n\to\infty} |\langle \mu_n, f_n\rangle| \leq \varepsilon\,.$$

Proof. To avoid ambiguities, we denote the usual norm on $\mathrm{C}(K_1\times K_2)$ by $\|\cdot\|_\vee$. For $n \in \mathbb{N}$ and $\theta = 1,2$, let $C_n^{(\theta)}$ and $O_n^{(\theta)}$ be as assumed in (a) and choose $g_n^{(\theta)} \in \mathrm{C}(K_\theta)$ with

$$0 \leq g_n^{(\theta)} \leq 1\,,\ g_n^{(\theta)}|_{C_n^{(\theta)}} = 0 \text{ and } g_n^{(\theta)}|_{(O_n^{(\theta)})^c} = 1\,. \tag{1}$$

We define the operators $P_n^{(\theta)}$, for $\theta = 1,2$ and $n \in \mathbb{N}$ by $P_n^{(\theta)} :\to \mathrm{C}(K_\theta)$, $h \mapsto g_n^{(\theta)}h$, and let $\check{P}_n$ and $\hat{P}_n$ be their respective products in the injective and projective tensor spaces. We observe that for $f_1 \in \mathrm{C}(K_1)$ and $f_2 \in \mathrm{C}(K_2)$ we have

$$\check{P}_n(f_1 \otimes f_2) = P_n^{(1)}(f_1) \otimes P_n^{(2)}(f_2) = (g_n^{(1)}f_1) \otimes (g_n^{(2)}f_2) = (g_n^{(1)} \otimes g_n^{(2)}) \cdot (f_1 \otimes f_2) \,.$$

Thus, $\check{P}_n(f) = (g_n^{(1)} \otimes g_n^{(2)}) \cdot f$ for $f \in \mathrm{C}(K_1 \times K_2)$.

By condition (b) and the definition of the projective tensor norm, we can choose for each $n \in \mathbb{N}$, finite families

$$(x(n,i) : i \le \ell(n)) \subset \mathrm{B}_1(\mathrm{C}(K_1)) \,,$$
$$(y(n,i) : i \le \ell(n)) \subset \mathrm{B}_1(\mathrm{C}(K_2)) \,, \quad \text{and}$$
$$(a(n,i) : i \le \ell(n)) \subset \mathbb{R}$$

such that each $\tilde{f}_n$ defined by

$$\tilde{f}_n := \sum_{i=1}^{\ell(n)} a(n,i)x(n,i) \otimes y(n,i) \quad \text{for } n \in \mathbb{N}$$

satisfies $\| f_n - \tilde{f}_n \|_\vee \le \varepsilon$ and such that

$$\sum_{i=1}^{\ell(n)} |a(n,i)| \le 2 \sup_{n' \in \mathbb{N}} \| \tilde{f}_{n'} \|_\wedge < \infty \quad \text{for } n \in \mathbb{N} \,.$$

From assumption (a), from (1) and from the definition of $\tilde{f}_n$, for each $n \in \mathbb{N}$ we deduce that

$$\check{P}_n(f_n) = (g_n^{(1)} \otimes g_n^{(2)})f_n = f_n \text{ and} \tag{2}$$
$$\hat{P}_n(\tilde{f}_n) = \check{P}_n(\tilde{f}_n) = \sum_{i=1}^{\ell(n)} a(n,i)(x(n,i)g_n^{(1)}) \otimes (y(n,i)g_n^{(2)}) \,.$$

Thus, since $\| \check{P}_n \| = 1$,

$$\| \check{P}_n(\tilde{f}_n) - \check{P}_n(f_n) \|_\vee \le \varepsilon \,. \tag{3}$$

Since $\mathrm{C}(K_\theta)$, $\theta = 1,2$, have the Grothendieck property, from the fact that the elements of $(g_n^{(\theta)} : n \in \mathbb{N})$ have pairwise disjoint supports, we deduce that the sets

$$A^{(1)} := \{g_n^{(1)}x(n,i) \mid i \le \ell(n),\ n \in \mathbb{N}\} \text{ and } A^{(2)} := \{g_n^{(2)}y(n,i) \mid i \le \ell(n),\ n \in \mathbb{N}\}$$

are limited in $\mathrm{C}(K_1)$ and $\mathrm{C}(K_2)$ respectively. By (2.7) the absolute convex hull of the set $2\sup_{n \in \mathbb{N}} \| f_n \|_\wedge \cdot (A^{(1)} \otimes A^{(2)})$, is limited in $\mathrm{C}(K_1) \check{\otimes} \mathrm{C}(K_2)$ and contains by (2) the sequence

$(\check{P}_n(\tilde{f}_n))$. Thus, for any normalized weak*-zero sequence (μ_n) in $\mathrm{M}(K, X^*)$ it follows from (3) that

$$\begin{aligned}\limsup_{n\to\infty} |\langle \mu_n, f_n\rangle| &= \limsup_{n\to\infty} |\langle \mu_n, \check{P}_n(f_n)\rangle| \\ &\le \limsup_{n\to\infty} |\langle \mu_n, \check{P}_n(\tilde{f}_n)\rangle| + \varepsilon = \varepsilon\ ,\end{aligned}$$

which finishes the proof. ∎

Proof of Theorem (2.6). In order to prove (2.6) we show that if $(f_n) \subset \mathrm{C}(K_1 \times K_2)$ is not limited and has the countable partition property, then there is an $\varepsilon > 0$ such that for each sequence $(\tilde{f}_n) \subset \mathrm{C}(K_1) \otimes \mathrm{C}(K_2)$ with $\| f_n - \tilde{f}_n \|_\vee < \varepsilon$ it follows that $\limsup_{n\to\infty} \| \tilde{f}_n \|_\wedge = \infty$.

W.l.o.g. we can and shall assume that (f_n) is bounded in $\mathrm{C}(K_1 \times K_2)$.

We first observe that the conditions (b) and (c) of Lemma (5.2) hold by taking $K := K_1$ and $X := \mathrm{C}(K_2)$. Moreover, (a) of (5.2) is satisfied, which can be seen as follows:

For a fixed $\xi \in K_1$ we have to show that $(f_n(\xi, .)) \subset \mathrm{C}(K_2)$ has a weak Cauchy subsequence. Since (f_n), viewed in $\mathrm{L}_\infty(\Sigma_{K_2}, \mathrm{C}(K_1))$, satisfies case 1 of Theorem (3.1), also $(f_n(\xi, .))$ has this property (viewed in $\mathrm{L}_\infty(\Sigma_{K_2}, \mathbb{R})$). Thus the assertion follows from the remark succeeding Theorem (3.1).

Thus all of the assumptions of Lemma (5.2) are satisfied and we can choose an $N_1 \in \mathcal{P}_\infty(\mathbb{N})$ and, for each $n \in N_1$, an open set $O_n^{(1)} \subset K_1$ and a closed $C_n^{(1)} \subset K_1$ satisfying (d) and (e) of (5.2). We now choose, for each $n \in N_1$, a $g_n^{(1)} \in \mathrm{C}(K_1)$ with

$$0 \le g_n^{(1)} \le 1\ ,\ g_n^{(1)}|_{C_n^{(1)}} = 0\ ,\ \text{and}\ g_n^{(1)}|_{(O_n^{(1)})^c} = 1\ . \tag{1}$$

By taking $K := K_2$ and $X := \mathrm{C}(K_1)$, we may proceed as above to show that the sequence $(g_n^{(1)} f_n : n \in N_1)$ also satisfies the assumptions of Lemma (5.2). Thus, we can choose an $N_2 \in \mathcal{P}_\infty(N_1)$ and, for each $n \in N_2$, an open set $O_n^{(2)} \subset K_2$ and a closed $C_n^{(2)} \subset K_2$ such that (d) and (e) of (5.2) are satisfied and we can choose, for each $n \in N_2$, a $g_n^{(2)} \in \mathrm{C}(K_2)$ with

$$0 \le g_n^{(2)} \le 1\ ,\ g_n^{(2)}|_{C_n^{(2)}} = 0\ ,\ \text{and}\ g_n^{(1)}|_{(O_n^{(2)})^c} = 1\ . \tag{2}$$

By (e) of (5.2), we deduce the existence of $N_3 \in \mathcal{P}_\infty(N_2)$ and a normalized w*-zero sequence $(\mu_n : n \in N_3) \subset \mathrm{M}(K_1 \times K_2)$ with

$$\varepsilon := \frac{1}{2} \inf_{n\in N_3} \langle \mu_n, (g_n^{(1)} \otimes g_n^{(2)}) f_n\rangle > 0\ . \tag{3}$$

We have shown that $((g_n^{(1)} \otimes g_n^{(2)}) f_n : n \in N_3)$ satisfies condition (a) of (5.3) and for the above chosen $O_n^{(\theta)}$ and $C_n^{(\theta)}$, $n \in N_3$ and $\theta = 1, 2$, but it does not satisfy the assertion of (5.3). This implies that (b) of (5.3) cannot be satisfied, and finishes the proof. ∎

6. Examples

Finally, we want to construct three bounded subsets A, B, and C of $\ell_\infty\check{\otimes}\ell_\infty$ which demonstrate the following:

- The assertion of Proposition (2.2)(b) is not true for general Banach spaces X and Y (which is shown by the examples A and B).
- Neither (2.6.1) nor (2.6.2) are superfluous in Theorem (2.6), *i.e.*, both are necessary to imply limitedness (C and B respectively).
- Not every limited subset of $\ell_\infty\check{\otimes}\ell_\infty$ can be obtained in the way described by Proposition (2.7)(b), *i.e.*, not every limited set is almost (in the sense of (1.2)) contained in a finite sum of products of limited sets in ℓ_∞ (this is shown by the set A).

Examples. For each $n \in \mathbb{N}$ we define the following elements of $\ell_\infty\check{\otimes}\ell_\infty$ $(= \mathrm{C}(\beta\mathbb{N}\times\beta\mathbb{N}) = \mathrm{C}(\beta\mathbb{N}, \ell_\infty)$, where $\beta\mathbb{N}$ denotes the Stone Čech compactification of $\mathbb{N}$):

$$x_n := \sum_{i=1}^{n} \chi_{\{i\}} \otimes \chi_{\{i\}} \qquad \left(= \sum_{i=1}^{n} \chi_{\{(i,i)\}}\right),$$

$$y_n := \sum_{i=1}^{n} \chi_{\{i\}} \otimes \sum_{j=i}^{n} \chi_{\{j\}} \qquad \left(= \sum_{i=1}^{n} \chi_{\{i\}\times\{i,i+1,\ldots n\}}\right), \text{ and}$$

$$z_n := \sum_{i=1}^{2^{n-1}} \chi_{A(n,i)} \otimes \chi_{\{i\}} \qquad \left(= \sum_{i=1}^{2^{n-1}} \chi_{\overline{A(n,i)}} \otimes \chi_{\{i\}}\right),$$

where $(A(n,i) : n \in \mathbb{N}_0, i \in \{1,2,\ldots,2^n\}) \subset \mathcal{P}_\infty(\mathbb{N})$ satisfies the conditions of a tree of sets.

Then for $A := \{x_n \,|\, n \in \mathbb{N}\}$, $B := \{y_n \,|\, n \in \mathbb{N}\}$, and $C := \{z_n \,|\, n \in \mathbb{N}\}$ the following hold:

a) For each $D \in \{A, B, C\}$, the sets $D^{(1)} := \{f(\xi,\cdot) \,|\, \xi \in \beta\mathbb{N},\ f \in D\}$ and $D^{(2)} := \{f(\cdot,\xi) \,|\, \xi \in \beta\mathbb{N},\ f \in D\}$ are limited in ℓ_∞.

b) A satisfies (2.6.1) and (2.6.2),
B satisfies (2.6.1) but not (2.6.2), and
C satisfies (2.6.2) but not (2.6.1) of Theorem (2.6).

In particular, A is limited in $\ell_\infty\check{\otimes}\ell_\infty$ but the sets B and C are not.

Proof of 4.5.5. (1) $A^{(1)}$, $A^{(2)}$, $B^{(1)}$, $B^{(2)}$, $C^{(1)}$ and $C^{(2)}$ are limited in ℓ_∞:

To see this, $A^{(1)}$, $A^{(2)}$, $B^{(1)}$, $B^{(2)}$, and $C^{(1)}$ are subsets of the weakly conditionally compact set $\mathrm{B}_1(\mathrm{c}_\mathrm{o})$. Moreover,

$$C^{(2)} = \{\chi_{\overline{A(n,j)}} \,|\, n \in \mathbb{N}_0,\ j \le 2^n\} \cup \{0\},$$

and so each subsequence of $C^{(2)}$ contains either a sequence with pairwise disjoint supports or a monotone decreasing subsequence. This implies that $C^{(2)}$ is conditionally weakly compact and, thus, limited in ℓ_∞.

(2) A and B satisfy condition (2.6.1):

To see this, we observe that A and B are, for both embeddings into $\mathrm{L}^c_\infty(\Sigma_{\beta\mathbb{N}}, \ell_\infty)$, measurable with respect to the σ-algebra generated by the countable partition

$$\pi := \{\{i\} \mid i \in \mathbb{N}\} \cup \{\beta\mathbb{N} \setminus \mathbb{N}\} .$$

(3) A and C satisfy condition (2.6.2):

For this we first make the following observation:

Let K_1 and K_2 be two compact spaces and let $g_1, g_2, \dots, g_k \in \mathrm{C}(K_1)$ and $h_1, h_2, \dots, h_k \in \mathrm{C}(K_2)$. Then

$$\begin{aligned}
&2^{-k} \sum_{A\subset\{1,\dots,k\}} \Big(\sum_{j\in A} g_j - \sum_{j\notin A} g_j\Big) \otimes \Big(\sum_{j\in A} h_j - \sum_{j\notin A} h_j\Big) \\
&\quad = 2^{-k} \sum_{A\subset\{1,\dots,k\}} \sum_{i,j=1}^{k} (\chi_A(i)g_i - \chi_{A^c}(i)g_i) \otimes (\chi_A(j)h_j - \chi_{A^c}(j)h_j) \\
&\quad = 2^{-k} \sum_{i,j=1}^{k} \sum_{A\subset\{1,\dots,k\}} (\chi_A(i)g_i - \chi_{A^c}(i)g_i) \otimes (\chi_A(j)h_j - \chi_{A^c}(j)h_j) \\
&\quad = 2^{-k} \sum_{i=1}^{k} \sum_{A\subset\{1,\dots,k\}} (\chi_A(i)g_i - \chi_{A^c}(i)g_i) \otimes (\chi_A(i)h_i - \chi_{A^c}(i)h_i) \\
&\qquad + 2^{-k} \sum_{i,j=1, i\neq j}^{k} \sum_{A\subset\{1,\dots,k\}} (\chi_A(i)g_i - \chi_{A^c}(i)g_i) \otimes (\chi_A(j)h_j - \chi_{A^c}(j)h_j) \\
&\quad = 2^{-k} \sum_{i=1}^{k} \Big(\sum_{i\in A\subset\{1,\dots,k\}} + \sum_{i\notin A\subset\{1,\dots,k\}} \Big)(g_i \otimes h_i) \\
&\qquad + 2^{-k} \sum_{i,j=1, i\neq j}^{k} \Big(\sum_{\substack{A\subset\{1,\dots,k\} \\ i,j\in A}} - \sum_{\substack{A\subset\{1,\dots,k\} \\ i\in A, j\notin A}} - \sum_{\substack{A\subset\{1,\dots,k\} \\ j\in A, i\notin A}} + \sum_{\substack{A\subset\{1,\dots,k\} \\ i,j\notin A}} \Big)(g_i \otimes h_j) \\
&\quad = \sum_{i=1}^{k} g_i \otimes h_i .
\end{aligned}$$

If, moreover, $(g_i : i = 1, \dots, k)$ and $(h_j : j = 1, \dots, k)$ are norm aligned and have pairwise disjoint supports, we deduce that

$$\| \sum_{i=1}^{k} g_i \otimes h_i \|_\wedge \leq 2^{-k} |\mathcal{P}(\{1, 2, \dots, k\})| = 1 .$$

Now we observe that, for each $n \in \mathbb{N}$, x_n and z_n can be represented in the above form with pairwise disjoint and normed $g_1, \ldots, g_k \in C(\beta\mathbb{N})$ and $h_1, \ldots, h_k \in C(\beta\mathbb{N})$ and so the assertion follows.

(4) C does not satisfy (2.6.1):

Let $N \in \mathcal{P}_\infty(\mathbb{N})$ and let γ and γ' be two distinct branches of $(A(n,j) : n \in N, j \leq 2^n)$, *i.e.*, $\gamma = ((n, j(n,\gamma)) : n \in N)$ with $j(n,\gamma) \leq 2^n$ and $A(n, j(n,\gamma)) \subset A(m, j(m,\gamma))$ whenever $m < n$.

Then there exist

$$\xi \in A_\gamma := \bigcap_{n \in N} \overline{A(n, j(n,\gamma))} \quad \text{and}$$

$$\xi' \in A_{\gamma'} := \bigcap_{n \in N} \overline{A(n, j(n,\gamma'))}$$

and an $n \in N$ with $\| f_n(\xi) - f_n(\xi') \| = 2$ (take $n \in N$ with $j(n,\gamma) \neq j(n,\gamma')$).

Since there are uncountably many branches, we deduce the assertion.

(5) B does not have property (2.6.2).

We introduce the following notations:

$\| \cdot \|_p$, where $1 \leq p \leq \infty$, denotes the usual ℓ_p norm on ℓ_p and ℓ_p^n $(n \in \mathbb{N})$.

For $i, j \in \mathbb{N}$, let

$$a(i,j) := \begin{cases} 0 & \text{if } i = j \\ \frac{1}{j-i} & \text{if } i \neq j \end{cases}$$

and let M and M_n $(n \in \mathbb{N})$ be the matrices

$$M := (a(i,j) : i,j \in \mathbb{N}) \qquad \text{and} \qquad M_n := (a(i,j) : i,j \in \{1,2,\ldots,n\}) .$$

For $n \in \mathbb{N}$ and $1 \leq p \leq \infty$ define

$$P_p^{(n)} : \ell_p \to \mathbb{R}^n , \quad x \mapsto x^{(n)} := (x_1, x_2, \ldots, x_n) \text{ for } x = (x_i) \in \ell_p, \text{ and}$$

$$Q_p^{(n)} : \mathbb{R}^n \to \ell_p , \quad x \mapsto (x_1, x_2, \ldots, x_n, 0, 0, \ldots) .$$

Now we recall the following result of Schur [**9**, Theorem 294, p.212]:

The mapping

$$T : \ell_2 \to \mathbb{R}^{\mathbb{N}} , \quad x = (x_i) \mapsto M \circ x = \Big(\sum_{j=1}^{\infty} a(i,j) x_j : i \in \mathbb{N}\Big) ,$$

is well defined, takes its values in ℓ_2, and is a bounded linear operator on ℓ_2.

From this result we deduce that for each $n \in \mathbb{N}$ the operator

$$T_n := \frac{1}{n} Q_1^{(n)} \circ M^{(n)} \circ P_\infty^{(n)} : \quad \ell_\infty \to \ell_1$$

satisfies

$$
\begin{aligned}
(6) \qquad \|T_n\| &= \sup_{x,y\in B_1(\ell_\infty)} |\langle x, T_n(y)\rangle| \\
&= \frac{1}{n} \sup_{x,y\in B_1(\ell_\infty^n)} |\langle x, M_n \circ y\rangle| \\
&= \frac{1}{n} \sup_{x,y\in B_1(\ell_\infty^n)} |\langle \frac{x}{\|x\|_2}, M_n \circ \frac{y}{\|y\|_2}\rangle| \|x\|_2 \|y\|_2 \\
&\le \sup_{x,y\in B_1(\ell_2)} |\langle x, M \circ y\rangle| = \|T\|_2 \\
&[\|x\|_2 \le \sqrt{n}\|x\|_\infty \text{ for } x \in \mathbb{R}^n] .
\end{aligned}
$$

Finally, we need the following inequality which can be obtained by passing to integrals of minorizing functions:

$$
(7) \qquad \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{1}{j-i} \ge (n-2)\ln(n-2) \text{ whenever } n \in \mathbb{N} \text{ with } n > 2 .
$$

We have, for each $n \in \mathbb{N}$ and each $\tilde{y} \in (y_n + B_{1/4}(\ell_\infty \tilde{\otimes} \ell_\infty)) \cap \ell_\infty \otimes \ell_\infty$, that

$$
(8) \qquad \tilde{y}(i,j) := \langle \delta_{(i,j)}, \tilde{y}\rangle \begin{cases} \ge 3/4 & \text{if } j \in \{i, i+1, \dots, n\} \\ \le 1/4 & \text{if } j \in \{1, 2, \dots, i-1\} \end{cases}
$$

for any $i, j \in \{1, 2, \dots, n\}$. Viewing T_n as an element of $(\ell_\infty \hat{\otimes} \ell_\infty)^* = L(\ell_\infty, \ell_\infty^*)$, we deduce that

$$
\begin{aligned}
(9) \qquad n\langle T_n, \tilde{y}\rangle &= n \sum_{i,j\in\{1,\dots,n\}} \tilde{y}(i,j) T_n(\chi_{\{i\}} \otimes \chi_{\{j\}}) \\
&= \sum_{i,j\in\{1,\dots,n\}} \tilde{y}(i,j) a(i,j) \\
&= \sum_{i=1}^{n} \sum_{j=i+1}^{n} \frac{1}{j-i} \tilde{y}(i,j) + \sum_{i=1}^{n} \sum_{j=1}^{i-1} \frac{1}{j-i} \tilde{y}(i,j) \\
&= \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{1}{j-i} (\tilde{y}(i,j) - \tilde{y}(j,i)) \\
&\ge \frac{1}{2} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \frac{1}{j-i} \ge \frac{n-2}{2} \ln(n-2) .
\end{aligned}
$$

In order to show the assertion, let $A^{(1/4)} \subset \ell_\infty \otimes \ell_\infty$ with $\{y_n \mid n \in \mathbb{N}\} \subset A^{(1/4)} + B_{1/4}(\ell_\infty \tilde{\otimes} \ell_\infty)$. For each $n \in \mathbb{N}$, we can choose $\tilde{y}_n \in A^{1/4}$ with $\|\tilde{y}_n - y_n\|_\vee \le 1/4$, which implies by (9) that

$$
\limsup_{n\to\infty} |\langle T_n, y_n\rangle| \le \frac{1}{2} \limsup_{n\to\infty} \frac{n-2}{n} \ln(n-2) = \infty .
$$

By (6), (T_n) is bounded in $(\ell_\infty \hat{\otimes} \ell_\infty)^*$, this implies that $(\tilde{y}_n)$ is unbounded in $\ell_\infty \hat{\otimes} \ell_\infty$, which finishes the proof. ∎

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Lower and upper 2-estimates for order bounded sequences and Dunford-Pettis operators between certain classes of Banach lattices

FRANK RÄBIGER*

Mathematisches Institut
Universität Tübingen
Auf der Morgenstelle 10
D-7400 Tübingen, F.R.G.

Abstract. We introduce the class of weak Schur spaces, *i.e.*, Banach lattices in which relatively weakly compact sets and almost order bounded sets coincide. There follows a detailed study of Banach lattices in which every semi-normalized, order bounded, weakly null sequence contains a subsequence satisfying a lower, resp. an upper, 2-estimate. From the previous results we obtain conditions under which non-Dunford-Pettis operators between certain classes of Banach lattices fix a copy of ℓ_2.

Introduction

By a result of Rosenthal ([**17**], Thm. 1) every non-Dunford-Pettis operator from $L_1[0,1]$ into itself fixes a copy of ℓ_2. Mainly, Rosenthal's proof is based on the following two properties of $E = L_1[0,1]$:

a) a subset of E is relatively weakly compact if and only if it is almost order bounded,

b) every semi-normalized, order bounded, weakly null sequence in E contains a subsequence equivalent to the unit vector basis of ℓ_2.

In the present paper we study Banach lattices which satisfy a) or b) and extend Rosenthal's result to a more general situation. In Section 1 we introduce weak Schur spaces, *i.e.*, Banach lattices in which the classes of relatively weakly compact sets and almost order bounded sets coincide. We will see that certain Orlicz and Lorentz function spaces belong to this class. The main results are contained in Sections 2 and 3. Section 2 is devoted to Banach lattices in which every semi-normalized, order bounded, weakly null sequence contains a subsequence which satisfies a lower, resp. an upper, 2-estimate (property (L_2), resp. (U_2)). It turns out that property (L_2), an apparently stronger condition, and the order continuity of the norm are equivalent. As a consequence we obtain that a bounded sequence in a weak Schur space contains

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a subsequence which is either convergent or satisfies a lower 2-estimate. This generalizes the main result in Aldous' and Fremlin's paper [1]. Concerning property (U_2) we present a sufficient condition from which we deduce that every p-concave Banach lattice, $1 \leq p < \infty$, has both properties (U_2) and (L_2). In Section 3 we generalize Rosenthal's theorem about non-Dunford-Pettis operators on $L_1[0,1]$ ([17]) and extend related results due to Bourgain, Ghoussoub and Rosenthal ([5], [7]). For instance, we show that every non-Dunford-Pettis operator from a p-concave weak Schur space, $1 \leq p < \infty$, into a weak Schur space fixes a copy of ℓ_2. This holds in particular for certain Orlicz and Lorentz function spaces.

Our terminology is standard. Unexplained notation can be found in the books of Lindenstrauss and Tzafriri ([12], [13]) and Schaefer's monograph [18]. However, we want to explain some frequently used terms and fix some notations.

Let E and F be Banach spaces. An operator from E into F is a continuous, linear mapping $T : E \to F$. $L(E,F)$ denotes the set of all operators from E into F. The dual of E is denoted by E^*. The weak topology on E and the weak* topology on E^* are the topologies $\sigma(E,E^*)$ and $\sigma(E^*,E)$ respectively. B_E is the closed unit ball of E. A sequence $(x_n) \subset E$ is called semi-normalized if (x_n) is bounded and $\inf_n \|x_n\| > 0$. The spaces ℓ_p^Γ, $1 \leq p \leq \infty$, and c_0^Γ with index set Γ are defined as usual. If $\Gamma = \mathbb{N}$ (resp. $\Gamma = \{1,\ldots,n\}$) we write ℓ_p (resp. ℓ_p^n) instead of ℓ_p^Γ etc.. A sequence $(x_n) \subset E$ is said to be equivalent to the unit vector basis (u.v.b.) of ℓ_p, $1 \leq p < \infty$, if there exists an isomorphism from ℓ_p into E which maps the n-th unit vector e_n of ℓ_p into x_n, $n \in \mathbb{N}$. The ℓ_p-sum $\oplus_{\ell_p}(E_n)$, $1 \leq p \leq \infty$, of a sequence (E_n) of Banach spaces is the space $\{(x_n) : x_n \in E_n \text{ for each } n \in \mathbb{N} \text{ and } (\|x_n\|) \in \ell_p\}$ equipped with the canonical norm. By $L_p[0,1]$, $1 \leq p \leq \infty$, we denote as usual the Lebesgue spaces on $[0,1]$.

Let E and F be Banach lattices. $E_+ := \{x \in E : x \geq 0\}$ is the positive cone of E. If $x,y \in E$ and $x \leq y$ we set $[x,y] := \{z \in E : x \leq z \leq y\}$. A set $A \subset E$ is called order bounded if it is contained in an order interval $[x,y]$. Absolute value, positive and negative part of $x \in E$ are given by $|x| := \sup(-x,x)$, $x_+ := \sup(x,0)$ and $x_- := \sup(-x,0)$ respectively. $x,y \in E$ are called disjoint if $\inf(|x|,|y|) = 0$. By a disjoint sequence $(x_n) \subset E$ we mean a sequence of mutually disjoint vectors. The terms countably order complete, ideal, band, band projection and order continuous norm have the same meaning as in [18]. E is called an abstract L_p-space (AL_p-space), $1 \leq p < \infty$, if for disjoint vectors $x,y \in E$ we always have $\|x+y\|^p = \|x\|^p + \|y\|^p$. E is called an abstract M-space (AM-space) if $\|x+y\| = \max(\|x\|,\|y\|)$ holds for arbitrary disjoint vectors $x,y \in E$. An operator $T \in L(E,F)$ is called positive if $TE_+ \subseteq F_+$ and T is called regular if it is the difference of positive operators.

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1. Weak Schur Spaces

In this section we introduce a class of Banach lattices which, concerning weakly compact subsets, behave like AL-spaces (cf. Proposition 1.2).

Definition 1.1. A Banach lattice E is called a *weak Schur space* if every disjoint, weakly null sequence in E converges in norm.

It follows from the definition that weak Schur spaces cannot contain a closed sublattice isomorphic to c_0. Hence weak Schur spaces are weakly sequentially complete ([**3**],Thm.14.12).

It is well known that a subset of an AL-space is relatively weakly compact if and only if it is almost order bounded (cf. [**14**], Bsp.II.7, Satz II.2). Our first statement shows that this property characterizes weak Schur spaces. Recall that a subset A of a Banach lattice E is *almost order bounded* if for every $\varepsilon > 0$ there exists an $x \in E_+$ such that $A \subset [-x, x] + \varepsilon B_E$.

Proposition 1.2. *Let E be a Banach lattice. Then the following assertions are equivalent:*

a) *E is a weak Schur space.*

b) *$A \subset E$ is relatively weakly compact if and only if A is almost order bounded.*

c) *Every normalized, disjoint sequence in E contains a subsequence equivalent to the unit vector basis of ℓ_1.*

Proof. Both a) and b) imply that E has order continuous norm (cf. [**3**], Thm.12.13, Thm.12.9). The equivalence of a) and b) is now a consequence of results of Kühn ([**9**], Folgerung) and Meyer-Nieberg ([**14**], Satz II.2). Since in a Banach lattice every disjoint, weak Cauchy sequence converges weakly to zero (cf. [**3**], Sec.14, Exerc.17) the equivalence of a) and c) follows from Rosenthal's ℓ_1-theorem (cf. [**12**], 2.e.5). ∎

Several well known Banach function spaces belong to the class of weak Schur spaces. Concerning the definition of Lorentz and Orlicz spaces we refer to [**13**], p.120 (cf. also Example 2.6).

1.3 Examples. The following Banach lattices are weak Schur spaces:

a) AL-spaces (cf. remark before Proposition 1.2).

b) Banach lattices with the Schur property, *i.e.*, weakly null sequences converge in norm.

c) Lorentz function spaces $L_{W,1}(\Omega, \Sigma, \mu)$ over a finite measure space (Ω, Σ, μ) (this was shown in [**6**], Thm.5.1, for the interval [0,1] and Lebesgue measure λ and can be proved for

arbitrary finite measure spaces by similar arguments). In particular $L_{p,1}(\Omega, \Sigma, \mu)$ is a weak Schur space for every $1 \leq p < \infty$ (cf. [13], 2.b.8).

d) Orlicz function spaces $L_M[0,1]$ on $[0,1]$ where the Orlicz function M satisfies the Δ_2-condition at ∞, *i.e.*, $\overline{\lim}_{t\to\infty} \frac{M(2t)}{M(t)} < \infty$, and the complementary function M^* defined by $M^*(u) := \sup_{t>0}(tu - M(t))$, $0 \leq u < \infty$, satisfies $\lim_{t\to\infty} \frac{M^*(2t)}{M^*(t)} = \infty$ (cf. [11] and Proposition 1.2).

e) The space $E = (\oplus \ell_\infty(E_n))^*$ with $E_n = \ell_{p_n}^n$, $n \in \mathbb{N}$, and $\lim_n p_n = \infty$ (cf. [15], §9, Bsp.4 and Proposition 1.2).

f) The ℓ_1-sum of a family of weak Schur spaces.

2. Lower and upper 2-estimates for order bounded sequences in Banach lattices

One crucial point in the proof of Rosenthal's result on non-Dunford-Pettis operators on $L_1[0,1]$ ([17], Theorem 1) is the fact that a semi-normalized, order bounded, weakly null sequence in $L_1[0,1]$ contains a subsequence equivalent to the unit vector basis (u.v.b.) of ℓ_2.

Definition 2.1. Let E be a Banach lattice. We say E has *property* (U_2) (resp. (L_2)) if every semi-normalized, order bounded, weakly null sequence in E contains a subsequence (x_n) which satisfies an upper 2-estimate (resp. a lower 2-estimate), *i.e.*, there exists $c > 0$ such that for all $n \in \mathbb{N}$ and arbitrary scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$ we have $\| \sum_{k=1}^n \alpha_k x_k \| \leq c(\sum_{k=1}^n \alpha_k^2)^{1/2}$ (resp. $c(\sum_{k=1}^n \alpha_k^2)^{1/2} \leq \| \sum_{k=1}^n \alpha_k x_k \|$).

Obviously, a Banach lattice E has both properties (U_2) and (L_2) if and only if every semi-normalized, order bounded, weakly null sequence in E contains a subsequence equivalent to the u.v.b. of ℓ_2. This is in particular the case for $E = L_1[0,1]$ and, by a remark of Rosenthal ([17], p.II.4), more generally for $E = L_1(\Omega, \Sigma, \mu)$ where (Ω, Σ, μ) is a finite measure space. An easy application of the Kakutani representation theorem for AL-spaces (cf. [18], II.8.5) yields the following result:

Proposition 2.2. *Every AL-space satisfies the properties (U_2) and (L_2).*

We will see next that Banach lattices with order continuous norm are exactly those with property (L_2) and that this property is equivalent to an apparently stronger condition. Let us state at first the following lemma.

Lemma 2.3. *Let E be a Banach lattice with order continuous norm and let (x_n) be a semi-normalized, almost order bounded, weakly null sequence in E. Then there exist $x^* \in E_+^*$ and a subsequence (x_{n_k}) of (x_n) such that $\inf_k \langle |x_{n_k}|, x^* \rangle > 0$.*

Proof. Every disjoint sequence in an order interval of E converges in norm to zero ([3], Thm. 12.13). This implies that every almost order bounded, disjoint sequence in E is norm

convergent to zero. In particular, this is the case for the solid hull $so\{x_n : n \in \mathbb{N}\}$ of (x_n). As a consequence of [9], Satz, we obtain that every positive, weakly null sequence in $so\{x_n : n \in \mathbb{N}\}$ converges in norm. Hence $(|x_n|)$ cannot be weakly null from which the assertion follows. ∎

Given a Banach lattice E and $x^* \in E_+^*$ we set $p_{x^*}(x) := \langle |x|, x^* \rangle$, $x \in E$. p_{x^*} defines a seminorm on E. The completion of $E/\ker p_{x^*}$ endowed with the norm induced by p_{x^*} is an AL-space which we denote by (E, x^*). $j_{x^*} : E \to (E, x^*)$ is the map induced by the quotient map $q : E \to E/\ker p_{x^*}$ (cf. [18], II.8.Example 1).

Theorem 2.4. *Let E be a Banach lattice. Then the following assertions are equivalent:*

a) *E has property (L_2).*

b) *Every semi-normalized, almost order bounded, weakly null sequence in E contains a subsequence which satisfies a lower 2-estimate.*

c) *E has order continuous norm.*

Proof. a) ⇒ c): Assume that E has property (L_2) and does not have order continuous norm. Then, by our assumption and Theorem 12.13 in [3], there exists a normalized, disjoint sequence (x_n) in an order interval $[-x, x] \subset E$, $x \in E_+$, which satisfies a lower 2-estimate. In particular we obtain $c\sqrt{n} \leq \|\sum_{k=1}^n x_k\| \leq \|x\|$ for some fixed constant $c > 0$ and all $n \in \mathbb{N}$ which is impossible. c) ⇒ b): Let (x_n) be a semi-normalized, almost order bounded, weakly null sequence in E. Choose $x^* \in E_+^*$ and (x_{n_k}) as in Lemma 2.3. Then by a result of Aldous and Fremlin ([1],Thm.6) a subsequence of $(j_{x^*} x_{n_k})$ hence a subsequence of (x_{n_k}) satisfies a lower 2-estimate. b) ⇒ a) is trivial. ∎

The next corollary extends the main result in Aldous'and Fremlin's [1].

Corollary 2.5. *Let E be a weak Schur space and (x_n) be a bounded sequence in E. Then (x_n) contains a subsequence which is either convergent or satisfies a lower 2-estimate.*

Proof. We assume that (x_n) has no convergent subsequence. If (x_n) has no weak Cauchy subsequence we can find by Rosenthal's ℓ_1-theorem (cf. [12], 2.e.5) a subsequence equivalent to the u.v.b. of ℓ_1 which certainly satisfies a lower 2-estimate. Otherwise there is a weak Cauchy subsequence (y_n) of (x_n). Since E is weakly sequentially complete (y_n) converges weakly to some $y \in E$. $(y_n - y)$ is weakly null and almost order bounded. Hence, by Theorem 2.4, a subsequence $(y_{n_k} - y)$ satisfies a lower 2-estimate. From Lemma 2 in [1] we obtain that there exists $k_0 \in \mathbb{N}$ such that $(y_{n_k})_{k \geq k_0}$ satisfies a lower 2-estimate and the assertion follows. ∎

Remark. Without proof we mention the following result (cf. [16]). For a Banach lattice E every bounded sequence in E has a subsequence which either converges in norm or satisfies a

lower 2-estimate if and only if every semi-normalized disjoint sequence in E has a subsequence which satisfies a lower 2-estimate.

We are now concerned with property (U_2). It can be easily seen that ℓ_∞ does not have this property (since it contains subspaces isomorphic to ℓ_p, $p < 2$). From this and [18], II.5.14, we obtain that every countably order complete Banach lattice which satisfies (U_2) has order continuous norm. The following example, communicated to the author by D. Leung, shows that Banach lattices with order continuous norm, even weak Schur spaces, do not necessarily have property (U_2).

2.6 Example. Let $1 < \alpha < 3/2$. Let $W : (0,1] \to \mathbb{R}_+$ be given by $W(t) := \frac{1}{t(-\ln t)^\alpha}$ if $0 < t \le e^{-\alpha}$ and $W(t) := \alpha^{-\alpha} e^\alpha$ if $e^{-\alpha} < t \le 1$. W is a non-increasing, integrable function and satisfies $\lim_{t\to 0} W(t) = \infty$. $E = L_{W,1}[0,1]$ is the Lorentz function space of all measurable functions f on $[0,1]$ such that $\|f\|_{W,1} := \int_0^1 f^*(t)W(t)\,dt < \infty$ (where f^* denotes the decreasing rearrangement of f (cf. [13], p.117, p.120)). E has order continuous norm. On the other hand the sequence (r_n) of Rademacher functions is order bounded in E, converges weakly to zero (cf. [13], 2.c.10) and is not equivalent to the unit vector basis of ℓ_2 (cf. [13], 2.b.4 (Hint: Let $\|\cdot\|_M$ be the Orlicz norm in [13], 2.b.4. Then for $0 < a \le e^{-\alpha}$ we have

$$\|\chi_{[0,a)}\|_{W,1}\|\chi_{[0,a)}\|_M^{-1} = \frac{(-\ln a)^{1-\alpha}}{\alpha - 1}\Big(\ln\big(1 + \frac{e-1}{a}\big)\Big)^{1/2} \ge \frac{(-\ln a)^{1-\alpha}}{\alpha - 1}(-\ln a)^{1/2}$$
$$= \left(\frac{(-\ln a)}{\alpha - 1}\right)^{(3/2)-\alpha}$$

where $\chi_{[0,a)}$ is the characteristic function of $[0,a)$. Hence $\lim_{a\to 0}\|\chi_{[a,0)}\|_{W,1}\|\chi_{[0,a)}\|_M^{-1} = \infty)$). Since for any subsequence (r_{n_k}) of (r_n) and arbitrary scalars $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$ we always have $(\sum_{k=1}^m \alpha_k r_k)^* = (\sum_{k=1}^m \alpha_k r_{n_k})^*$ it follows that no subsequence of (r_n) can be equivalent to the u.v.b. of ℓ_2.

In our next result we present a necessary condition for a Banach lattice to have property (U_2). Let us introduce at first the following notation. If E is a Banach lattice and $u \in E_+$ we denote by E_u the order ideal $I_u := \bigcup_{n\in\mathbb{N}} n[-u,u]$ generated by u equipped with the complete norm $\|x\|_u := \inf\{\lambda > 0 : x \in \lambda[-u,u]\}$. E_u is an AM-space with unit u and the canonical injection $i_u : E_u \to E$ is continuous and positive (cf. [18], II.7.2, Cor.).

Theorem 2.7. *Let E be a Banach lattice. Suppose that for each $u \in E_+$ there exists $p_u \in [1,\infty)$ such that $i_u : E_u \to E$ factors with positive factors through an AL_{p_u}-space (i.e., there exists an AL_{p_u}-space F and positive operators $T_1 \in L(E_u, F)$ and $T_2 \in L(F,E)$ such that $i_u = T_2 \circ T_1$). Then E has property (U_2).*

Proof. Let (x_n) be a semi-normalized, weakly null sequence in E contained in an order interval $[-u,u]$, $u \in E_+$. Let F be an AL_p-space with $1 \leq p < \infty$ and let $0 \leq T_1 \in L(E_u, F)$ and $0 \leq T_2 \in L(F,E)$ be such that $i_u = T_2 \circ T_1$. We may assume that the sequence $(T_1 x_n)$ is weakly convergent to some element $y \in F$. Since (x_n) is weakly null in E we obtain $T_2 y = 0$. We set $y_n := T_1 x_n - y$, $n \in \mathbb{N}$. Then (y_n) is an order bounded, weakly null sequence in F not converging in norm to zero. Denote by G the separable, closed sublattice of F generated by (y_n) and $T_1 u$. By the Kakutani representation theorem there is a finite measure space (Ω, Σ, μ) and a lattice isomorphism T from G onto $L_p(\Omega, \Sigma, \mu)$ with $T(T_1 u) = e$ where e denotes the function constant one (cf. [**13**], 1.b.2). (Ty_n) converges weakly to zero in $L_p(\Omega, \Sigma, \mu)$ and is contained in the interval $[-e, e]$ hence it is weakly null in $L_r(\Omega, \Sigma, \mu)$ for $r := \max\{p, 2\}$. $L_r(\Omega, \Sigma, \mu)$ has an unconditional basis (– this follows from [**10**], p.128, Cor., and the existence of an unconditional basis for $L_r[0,1]$ (cf. [**13**], 2.c.5)) and, by passing to a subsequence if necessary, we may assume that (Ty_n) is an unconditional basic sequence in $L_r(\Omega, \Sigma, \mu)$ (cf. [**12**], 1.a.12). Then, since $L_r(\Omega, \Sigma, \mu)$ has type 2 (cf. [**13**], p.73), (Ty_n) satisfies an upper 2-estimate in $L_r(\Omega, \Sigma, \mu)$ hence in $L_p(\Omega, \Sigma, \mu)$. It follows that (y_n) and hence $(T_2 y_n)$ satisfy an upper 2-estimate. Since $T_2 y_n = x_n$, $n \in \mathbb{N}$, the assertion is proved. ∎

Remarks. a) Actually, we have shown in the proof of Theorem 2.7: If (x_n) is a semi-normalized, weakly null sequence in an order interval $[-u,u]$ of a Banach lattice E, $u \in E_+$, and if $i_u : E_u \to E$ factors through an AL_p-space with positive factors for some $1 \leq p < \infty$ then a subsequence of (x_n) satisfies an upper 2-estimate.

b) As an immediate consequence of Theorem 2.7 and Theorem 2.4 we obtain the following folklore result: Every semi-normalized, order bounded, weakly null sequence in an AL_p-space, $1 \leq p < \infty$, contains a subsequence equivalent to the u.v.b. of ℓ_2.

c) Property (L_2) implies that every semi-normalized, almost order bounded, weakly null sequence contains a subsequence which satisfies a lower 2-estimate (Theorem 2.4). There is no analogue for property (U_2). Precisely, in every $L_p[0,1]$, $1 \leq p < 2$, there is a semi-normalized, almost order bounded, weakly null sequence such that no subsequence satisfies an upper 2-estimate:

Indeed, let $p < r < 2$. There is an isometry from $L_r[0,1]$ into $L_p[0,1]$ ([**13**], 2.f.5). Hence there is an isometry S from ℓ_r into $L_p[0,1]$. Let $x_n := Se_n$, $n \in \mathbb{N}$, where (e_n) denotes the u.v.b. of ℓ_r. (x_n) is a normalized, weakly null sequence. We show that (x_n) is almost order bounded: If $p = 1$ this follows from Proposition 1.2. Otherwise, we consider $S' : L_q[0,1] \to \ell_s$ where $p^{-1} + q^{-1} = r^{-1} + s^{-1} = 1$. Let (y_n) be a bounded disjoint sequence in $L_q[0,1]$. Then (y_n) is contained in a sublattice of $L_q[0,1]$ which is isometric to ℓ_q. We have $q > s$, hence,

by Pitt's theorem, every operator from ℓ_q into ℓ_s is compact ([**12**], 2.c.3). Consequently $S'y_n$ is norm null for every bounded disjoint sequence (y_n) in $L_q[0,1]$. This implies by results due to Meyer-Nieberg (cf. [**3**], Thm.18.13, Thm.18.9) that SB_{ℓ_r} and in particular (x_n) is almost order bounded. We conclude with the observation that every subsequence of (x_n) is equivalent to the u.v.b. of ℓ_r. Since $r < 2$, no subsequence of x_n satisfies an upper 2-estimate.

Our next result is a consequence of Theorem 2.7 and a factorization theorem due to Krivine. Concerning the notion of p-convexity and p-concavity of operators and Banach lattices for $1 \leq p \leq \infty$ we refer to [**13**], Section 1.d.

Proposition 2.8. *Every p-concave Banach lattice E, $1 \leq p < \infty$, has the properties (L_2) and (U_2).*

Proof. Let E be a p-concave Banach lattice for some fixed $1 \leq p < \infty$. From the definition of p-concavity (cf. [**13**], 1.d.3) it follows easily that E does not contain a sublattice isomorphic to c_0. Hence E has order continuous norm (cf. [**18**], II.5.15, II.5.14) and satisfies (L_2) by Theorem 2.4. On the other hand for given $u \in E_+$ the operator $i_u : E_u \to E$ is ∞-convex (cf. [**13**], 1.d.9, and use the fact that E_u is an AM-space). Hence, by [**13**], 1.d.5, i_u is also p-convex. We are now in the position to apply Krivine's factorization theorem (cf. [**13**], 1.d.11) and obtain that i_u factors through an AL_p-space. A look at the proof of Krivine's result shows that the factors can be chosen positive if the original operator (in our case i_u) is positive. Now we may apply Theorem 2.7 which completes the proof. ∎

2.9 Examples. a) By a remark in Johnson et al. ([**8**], p.168) an Orlicz function space $L_M[0,1]$ on $[0,1]$ is p-concave for some $1 \leq p < \infty$ if and only if the Orlicz function $M : [0,\infty) \to \mathbb{R}_+$ satisfies the Δ_2-condition at ∞, *i.e.*, $\overline{\lim}_{t\to\infty} \frac{M(2t)}{M(t)} < \infty$.

b) Let (Ω, Σ, μ) be a measure space, $1 \leq p < \infty$ and $1 \leq q < \infty$. Then the Lorentz function space $L_{p,q}(\Omega, \Sigma, \mu)$ (cf. [**13**], 2.b.8) is r-concave for $r > \max\{p, q\}$ (cf. [**13**], 2.g.6, 2.g.18, 1.d.5).

c) Let $1 \leq p < \infty$ and E be a Banach lattice which satisfies a lower p-estimate (cf. [**13**], 1.f.4) then E is r-concave for all $r > p$ (cf. [**13**], 1.f.7).

3. Non-Dunford-Pettis operators

An operator T between Banach spaces X and Y is called a *Dunford-Pettis operator* if T maps weakly convergent sequences into norm convergent sequences. In the following we are interested in the behaviour of non-Dunford-Pettis operators defined on weak Schur spaces. We

will generalize results of Bourgain ([5]) and Rosenthal ([17], Theorem 1) and will extend a result of Ghoussoub and Rosenthal ([7], Theorem IV.2).

Let us first mention a characterization of Dunford-Pettis operators defined on weak Schur spaces which generalizes Proposition 1 in Bourgain's paper [4]. Recall that every Banach lattice with order continuous norm contains a topological orthogonal system (t.o.s.), *i.e.*, there exists a disjoint family $(u_\gamma)_{\gamma\in\Gamma}$ of positive vectors such that the ideal generated by $\{u_\gamma : \gamma \in \Gamma\}$ is dense (cf. [18], III.5.Example 2). In particular every weak Schur space contains a t.o.s..

Proposition 3.1. *Let E be a weak Schur space and let T be an operator from E into a Banach space X. Then the following assertions are equivalent:*

a) T is a Dunford-Pettis operator.

b) For every t.o.s. *$(u_\gamma)_{\gamma\in\Gamma}$ of E, all operators $T\circ i_\gamma$, $\gamma \in \Gamma$, are compact (where $i_\gamma : E_{u_\gamma} \to E$ denotes the canonical injection).*

c) There exists a t.o.s. *$(u_\gamma)_{\gamma\in\Gamma}$ of E such that $T \circ i_\gamma$ is compact for every $\gamma \in \Gamma$.*

Proof. Order intervals in E are weakly compact (cf. [18], II.5.10). Then a) $\Rightarrow$ b) follows by Eberlein's theorem. b) $\Rightarrow$ c) is trivial.

c) $\Rightarrow$ a): Suppose T is not a Dunford-Pettis operator. Then there exists a weakly null sequence (x_n) in E such that $\|Tx_n\| = 5$ for all $n \in \mathbb{N}$. Let $u \in E_+$ be such that $(x_n) \subseteq [-u,u] + \|T\|^{-1}B_E$ (we may assume $T \neq 0$). There exist $n_0 \in \mathbb{N}$ and $\gamma_1,\dots,\gamma_k \in \Gamma$ such that $\|u - \sum_{j=1}^k \inf(u, n_0 u_{\gamma_j})\| \leq \|T\|^{-1}$ (cf. [18], p.169). We define $v := n_0 \sum_{j=1}^k u_{\gamma_j}$; then $(x_n) \subseteq [-v,v] + 2\|T\|^{-1}B_E$. Let P_j, $1 \leq j \leq k$, be the band projection from E onto the band generated by u_{γ_j} and let $Q_j : E_v \to E_{u_{\gamma_j}}$ be the operator induced by P_j. Let $i_v : E_v \to E$ and $i_j : E_{u_{\gamma_j}} \to E$, $1 \leq j \leq k$, be the canonical injections. Then $i_v = \sum_{j=1}^k i_j \circ Q_j$, hence by our assumption $T \circ i_v$ is compact. Now we choose $y_n \in [-v,v]$ and $z_n \in 2\|T\|^{-1}B_E$ such that $x_n = y_n + z_n$, $n \in \mathbb{N}$. Since $[-v,v]$ is weakly compact we may assume, if necessary by passing to a subsequence, that (y_n) converges weakly to $y \in [-v,v]$. (x_n) is a weakly null sequence, hence (Tz_n) converges weakly to $-Ty$. Since $\|Tz_n\| \leq 2$ for all $n \in \mathbb{N}$ we obtain $\|Ty\| \leq 2$. The compactness of $T \circ i_v$ implies that (Ty_n) converges in norm to Ty. Hence, there exists $n_1 \in \mathbb{N}$ such that $\|Ty_n\| < 3$ for all $n \geq n_1$. Consequently, $5 = \|Tx_n\| \leq \|Ty_n\| + \|Tz_n\| < 5$ for all $n \geq n_1$ which is a contradiction. This completes the proof. ∎

Remark. If E is a Banach lattice with order continuous norm the implications a) $\Rightarrow$ b) $\Rightarrow$ c) in Proposition 3.1 remain true.

The following lemma is an easy consequence of Proposition 3.1 and the fact that order intervals in Banach lattices with order continuous norm are weakly compact. The proof is left

to the reader.

Lemma 3.2. *Let E be a weak Schur space and let T be a non-Dunford-Pettis operator from E into a Banach space X. Furthermore let $(u_\gamma)_{\gamma\in\Gamma}$ be a t.o.s. of E. Then there exist $\gamma_0 \in \Gamma$ and a weakly null sequence $(x_n) \subset [-u_{\gamma_0}, u_{\gamma_0}]$ such that $\inf_n \|Tx_n\| > 0$.*

Next we introduce a new class of operators. An operator T from a Banach lattice E into a Banach lattice F is called *almost order bounded* if T maps order bounded sets into almost order bounded sets. We denote with $L^{ao}(E,F)$ the set of almost order bounded operators from E into F. $L^{ao}(E,F)$ is a closed linear subspace of $L(E,F)$ containing the regular operators. We are now in a position to prove the main result of this section. It generalizes a result of Rosenthal ([**17**], Theorem 1).

Theorem 3.3. *Let E be a weak Schur space and let $(u_\gamma)_{\gamma\in\Gamma}$ be a t.o.s. of E. Suppose that for every $\gamma \in \Gamma$ there exists $1 \le p_\gamma < \infty$ such that the canonical injection $i_\gamma : E_{u_\gamma} \to E$ factors through an AL_{p_γ}-space with positive factors. Furthermore let F be a Banach lattice with order continuous norm and let $T \in L^{ao}(E,F)$. If T is a non-Dunford-Pettis operator then T fixes a copy of ℓ_2.*

Proof. By Lemma 3.2 there exist $\gamma \in \Gamma$ and a weakly null sequence $(x_n) \subset [-u_\gamma, u_\gamma]$ such that $\inf_n \|Tx_n\| > 0$. Since $i_\gamma : E_{u_\gamma} \to E$ factors through an AL_{p_γ}-space with positive factors for some $1 \le p_\gamma < \infty$ there exists a subsequence (y_n) of (x_n) which satisfies an upper 2-estimate (cf. Remark a) following Theorem 2.7). On the other hand a subsequence of (Ty_n) satisfies a lower 2-estimate by Theorem 2.4. This proves the theorem. ∎

Remark. In order to see that Theorem 3.3 is a generalization of Rosenthal's Theorem 1 in [**17**] we mention that every operator from $L_1[0,1]$ into $L_1[0,1]$ is regular (cf. [**18**], IV.1.5).

If E is a Banach lattice with order continuous norm and F is a weak Schur space, then $L^{ao}(E,F) = L(E,F)$. Hence we obtain from Theorem 3.3:

Corollary 3.4. *Suppose E satisfies the assumptions of Theorem 3.3. Let F be a weak Schur space. Then every non-Dunford-Pettis operator $T \in L(E,F)$ fixes a copy of ℓ_2.*

Concerning the definition of rearrangement invariant function spaces and of Boyd indices we refer to [**13**], 2.a.1, 2.b.1.

Corollary 3.5. *Suppose E is a Banach lattice satisfying one of the following properties:*

a) *E is a weak Schur space and p-concave for some $1 \le p < \infty$.*

b) *E is a rearrangement invariant function space on $[0,1]$ which is weak Schur and has an upper Boyd index $q_E < \infty$.*

Then E satisfies the conditions of Theorem 3.3. In particular if F is an order continuous Banach lattice then every non-Dunford-Pettis operator $T \in L^{ao}(E,F)$ fixes a copy of ℓ_2. If in addition F is a weak Schur space then every non-Dunford-Pettis operator $T \in L(E,F)$ fixes a copy of ℓ_2.

Proof. a) In the proof of Proposition 2.8 we verified that for $1 \leq p < \infty$ every p-concave Banach lattice satisfies the factorization condition of Theorem 3.3.

b) The conditions on E imply $L_p[0,1] \subseteq E$ for some $1 \leq p < \infty$ (cf. [**13**], 2.b.3). If u is the function constant one then $E_u = L_\infty[0,1]$ and from our first observation it follows that $i_u : E_u \to E$ factors through $L_p[0,1]$ with positive factors. ∎

3.6 Examples. The following Banach lattices E satisfy condition a) or b) of Corollary 3.5:

a) E is an AL-space (cf. Example 1.3 and observe that E is 1-concave).

b) $E = L_M[0,1]$ is an Orlicz function space on $[0,1]$ where M satisfies the Δ_2-condition at ∞ and the complementary function M^* satisfies $\lim_{t\to\infty} \frac{M^*(2t)}{M^*(t)} = \infty$ (cf. Example 2.9 a) and Example 1.3 d)).

c) E is a Lorentz function space $L_{W,1}(\Omega,\Sigma,\mu)$ on a finite measure space (Ω,Σ,μ) and p-concave for some $1 \leq p < \infty$ (cf. Example 1.3 c)). This is in particular the case for $E = L_{r,1}(\Omega,\Sigma,\mu)$, $1 \leq r < \infty$ (cf. Example 1.3 c), Example 2.9 b)).

We conclude with a generalization of the main result in Bourgain's [**5**] and an extension of a result of Ghoussoub and Rosenthal ([**7**], Theorem IV.2).

Proposition 3.7. *Let E be an AL-space, F a Banach lattice with order continuous norm and $T \in L^{ao}(E,F)$. If T is a non-Dunford-Pettis operator then T fixes a copy of $\oplus_{\ell_1}(\ell_2)$.*

Proof. We may assume that E is separable. By the Kakutani representation theorem (cf. [**18**], II.8.5) E is lattice isometric to a space $L_1(\Omega,\Sigma,\mu)$ where (Ω,Σ,μ) is a finite measure space. Without loss of generality we may assume $E = L_1[0,1]$ (cf. [**10**], p.128, Cor.). Recall the definition of the AL-space (F,x^*) and of the canonical map $j_{x^*} : F \to (F,x^*)$, $x^* \in F_+^*$ (cf. remarks following Lemma 2.3). Lemma 3.2 and Lemma 2.3 imply the existence of $x^* \in F_+^*$ such that $j_{x^*} \circ T : E \to (F,x^*)$ is a non-Dunford-Pettis operator. Let G be the closed (separable) sublattice of (F,x^*) generated by $(j_{x^*} \circ T)E$. G is lattice isometric to $L_1[0,1] \oplus \ell_1^\Gamma$ for some index set Γ (cf. [**10**], p.128, Cor.). Let P be the corresponding projection from G onto $L_1[0,1]$. Then $P \circ j_{x^*} \circ T : L_1[0,1] \to L_1[0,1]$ is a non-Dunford-Pettis operator. By Bourgain's theorem ([**5**]) $P \circ j_{x^*} \circ T$ fixes a copy of $\oplus_{\ell_1}(\ell_2)$, hence T does the same. ∎

Remark. If E is an AL-space and F is a Banach lattice not containing c_0 then every operator from E into F is regular (cf. [**18**], IV.1.5, II.5.15). Hence every non-Dunford-Pettis operator

from an AL-space into a Banach lattice not containing c_0 fixes a copy of $\oplus_{\ell_1}(\ell_2)$. This was shown using completely different methods by Ghoussoub and Rosenthal (cf. [7], Thm.IV.2).

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Embedding ℓ^1 into Tensor Products of Banach Spaces

DENNY H. LEUNG

Department of Mathematics
The University of Texas at Austin
Austin, Texas 78712

Abstract. It is known that if E and F are Banach spaces neither of which contains a copy of ℓ^1, and if either E' or F' has the Radon-Nikodým property, then the ε-tensor product $E\widetilde{\otimes}_\varepsilon F$ does not contain a copy of ℓ^1. We show that the converse of this assertion fails by showing that $JH\widetilde{\otimes}_\varepsilon JH$ fails to contain ℓ^1.

1. Introduction.

Let E and F be Banach spaces. Even if neither E nor F contains a copy of ℓ^1, it still may happen that $E\widetilde{\otimes}_\varepsilon F$ does. Indeed, by Example 3.3.3 in [4], we have $\ell^1 \hookrightarrow JT\widetilde{\otimes}_\varepsilon JT$, while it is well known that JT, the James Tree space, fails to contain a copy of ℓ^1. However, if it is assumed additionally that either E' or F' has the Radon-Nikodým property (RNP), then it is true that $\ell^1 \not\hookrightarrow E\widetilde{\otimes}_\varepsilon F$ (or even more generally $K_{w^*}(E', F)$) [4, Theorem 3.3.1]. In this note, we show that $\ell^1 \not\hookrightarrow JH\widetilde{\otimes}_\varepsilon JH$, where JH denotes the James Hagler space [2]. Thus the non-containment of ℓ^1 in a ε-tensor product does not imply that either factor has a RNP dual.

Throughout we consider only real Banach spaces. Let E be a Banach space. The unit ball of E is denoted by U_E and the dual space by E'. The identity operator on E is written 1_E or simply 1. The closed linear span of a set A in E is denoted by $[A]$. We say that E contains a copy of ℓ^1 ($\ell^1 \hookrightarrow E$ in symbols) if E has a subspace isomorphic to ℓ^1. By Rosenthal's theorem [3], this happens if and only if there is a bounded sequence in E with no weakly Cauchy subsequence. For Banach spaces E and F, we denote by $K_{w^*}(E', F)$ the space of all compact weak*-weakly continuous operators from E' into F, with the operator norm. The ε-tensor product $E\widetilde{\otimes}_\varepsilon F$ is the closure of the finite rank operators in $K_{w^*}(E', F)$. In case either E or F has the approximation property, then $K_{w^*}(E', F) = E\widetilde{\otimes}_\varepsilon F$. If $U_{E'}$ and $U_{F'}$ are given their respective weak* topologies, then it is easy to see that $R \mapsto \widehat{R} : K_{w^*}(E', F) \to C(U_{E'} \times U_{F'})$, $\widehat{R}(x', y') = \langle Rx', y' \rangle$ is an isometric embedding. Hence a bounded sequence $(R_n) \subset K_{w^*}(E', F)$ converges weakly to 0 if and only if $\langle R_n x', y' \rangle \to 0$ for all $x' \in E'$ and $y' \in F'$.

If M is an infinite set, we let $\mathcal{P}_\infty(M)$ denote the set of all infinite subsets of M.

We would also like to recall a few facts concerning the spaces JT and JH. Let $T = \bigcup_{n=0}^{\infty}\{0,1\}^n$ be the dyadic tree. The elements of T are called *nodes*. If φ is a node of the form $\varphi = (\varepsilon_i)_{i=1}^n$, we say that φ has *length* n and write $|\varphi| = n$. The length of the empty node is defined to be 0. For $\varphi, \psi \in T$ with $\varphi = (\varepsilon_i)_{i=1}^n$ and $\psi = (\delta_i)_{i=1}^m$, we say that $\varphi \leq \psi$ if $n \leq m$ and $\varepsilon_i = \delta_i$ for $1 \leq i \leq n$. Also $\phi \leq \varphi$ for all $\varphi \in T$.

A *segment* is a subset of T of the form $\{\psi : \xi \leq \psi \leq \varphi\}$ for some $\xi, \varphi \in T$. A *branch* is a maximal linearly ordered subset of T. The set of all branches is denoted by Γ. For $\gamma \in \Gamma$ and $n \in \mathbb{N}$, e_n^γ denotes the node of length n that belongs to γ. If $x : T \to \mathbb{R}$ is a finitely supported function and S is a segment. We define (with slight abuse of notation) $S(x) = \sum_{\varphi \in S} x(\varphi)$. Similarly, if $\gamma \in \Gamma$, we define $\gamma(x) = \sum_{\varphi \in \gamma} x(\varphi)$.

The James tree space JT is defined as the completion of the set of all finitely supported functions $x : T \to \mathbb{R}$ under the norm:

$$\|x\| = \sup\left\{ \left(\sum_{i=1}^n |S_i(x)|^2 \right)^{1/2} : S_1, \ldots, S_n \text{ are pairwise disjoint segments} \right\}.$$

To define JH, we need to notion of an admissible family of segments. A pairwise disjoint family of segments $S_1, \ldots, S_n$ is *admissible* if all segments "begin" and "end" at the same level, *i.e.*, if there are integers k and m such that the shortest element in S_i has length k and the longest element in S_i has length m for $i \leq n$. The James Hagler space JH is defined as the completion of the set of all finitely supported functions $x : T \to \mathbb{R}$ under the norm:

$$\|x\| = \sup\left\{ \sum_{i=1}^n |S_i(x)| : S_1, \ldots, S_n \text{ is an admissible family of segments} \right\}.$$

Note that in both JT and JH, all S and γ extend uniquely to norm 1 functionals. We call them *segment functionals* and *branch functionals*, respectively.

For $\xi \in T$, we define the function e_ξ on T by $e_\xi(\varphi) = \delta_{\xi\varphi}$ for all $\varphi \in T$. Then $(e_\varphi)_{\varphi \in T}$ in the lexicographic order on T is a basis for both JH and JT. The biorthogonal functionals of $(e_\varphi)_{\varphi \in T}$ are called the *node functionals*. With a harmless abuse of notation, we will use the same symbol, say φ, to denote both a node and the node functional corresponding to it. It can be shown that the linear span of the node functionals and the branch functionals is dense in JH' and JT', in their respective topologies. For all m, Q_m denotes the norm 1 (in both JH and JT) projection given by $Q_m(\sum_\varphi a_\varphi e_\varphi) = \sum_{|\varphi|>m} a_\varphi e_\varphi$.

2. ℓ^1 embeds in $JT\widetilde{\otimes}_\varepsilon JT$.

As is pointed out in the introduction, the fact that neither E nor F contains ℓ^1 does not imply that $\ell^1 \not\hookrightarrow E\widetilde{\otimes}_\varepsilon F$. In fact $\ell^1 \hookrightarrow JT\widetilde{\otimes}_\varepsilon JT$ by Example 3.3.3 in [4]. The proof given there,

however, is indirect. We wish to give a simple proof here which exhibits explicitly a copy of ℓ^1 in $JT\widetilde{\otimes}_\varepsilon JT$.

Example. [4] $\ell^1 \hookrightarrow JT\widetilde{\otimes}_\varepsilon JT$.

Proof. For $n \in \mathbb{N}$, define a subset A_n of T by $A_n = \{\varphi = (\varepsilon_i)_{i=1}^n : \varepsilon_n = 0\}$. Now let $R_n : JT' \to JT$ be defined by $R_n(x') = \sum_{\varphi \in A_n} \langle e_\varphi, x' \rangle e_\varphi$. It is easy to see that $(R_n) \subset JT\widetilde{\otimes}_\varepsilon JT$ is a normalized sequence. We claim that (R_n) is equivalent to the canonical ℓ^1 basis. In fact, let $(a_n) \in \ell^1$. Without loss of generality, we may assume that $\sum_{n\in N} |a_n| \geq \frac{1}{2} \sum_{n\in\mathbb{N}} |a_n|$, where $N = \{n : a_n \geq 0\}$. Choose $\gamma = (e_n^\gamma) \in \Gamma$ so that $e_n^\gamma \in A_n$ if and only if $n \in N$. A direct computation reveals that $\langle R_n\gamma, \gamma \rangle = 1$ if $n \in N$ and 0 otherwise. Thus

$$\Big\| \sum a_n R_n \Big\| \geq \Big| \Big\langle \sum a_n R_n \gamma, \gamma \Big\rangle \Big| = \Big| \sum_{n\in N} a_n \langle R_n\gamma, \gamma \rangle \Big| \geq \tfrac{1}{2} \sum |a_n| \,.$$

This completes the proof since the reverse estimate is trivial. ∎

3. $\ell^1 \not\hookrightarrow JH\widetilde{\otimes}_\varepsilon JH$.

Suppose on the contrary that (R_n) is a sequence in $JH\widetilde{\otimes}_\varepsilon JH$ equivalent to the ℓ^1 basis. Since R_n' maps JH' into JH and since $\ell^1 \not\hookrightarrow JH$, $(R_n'x')$ has a weakly Cauchy subsequence for all $x' \in JH'$. Fix m. Because $(1 - Q_m)'$ has finite rank, there exists $M \in \mathcal{P}_\infty(\mathbb{N})$ such that $(R_n'(1 - Q_m)'x')_{n\in M}$ is weakly Cauchy for all $x' \in JH'$. This shows, together with the remark in §1, that $((1 - Q_m)R_n)_{n\in M}$ is weakly Cauchy. Now we choose sets $M_1 \supset M_2 \supset \cdots$ in $\mathcal{P}_\infty(\mathbb{N})$ such that $((1 - Q_m)R_n)_{n\in M_m}$ is weakly Cauchy for all m. Then if (n_m) is an increasing sequence with $n_m \in M_m$ for all m, we see that the sequence $(R_{n_{2k-1}} - R_{n_{2k}})_{k=1}^\infty$ is equivalent to ℓ^1. Furthermore, $\lim_k (1 - Q_m)(R_{n_{2k-1}} - R_{n_{2k}}) = 0$ weakly for all m. Similar considerations may also be applied to the sequence (R_n'). Relabeling, we may assume that $\lim_n (1 - Q_m)R_n = \lim_n (1 - Q_m)R_n' = 0$ weakly for all m. The goal we are aiming at is to show that the sequence (R_n) must be unbounded under the assumptions we have put on it. It is helpful to separate out parts of our arguments into the following lemmas.

Lemma 3.1. *Let $\varepsilon > 0$ and a decreasing sequence of sets $(M_k) \subset \mathcal{P}_\infty(\mathbb{N})$ be given. Suppose there exist ξ and a sequence of pairwise distinct elements (γ_k) in Γ such that*

$$\lim_{n\in M_{k+1}} |\langle R_n\xi, \gamma_k \rangle| \text{ exists and is } > \varepsilon \text{ for all } k.$$

Then (R_n) is unbounded.

Proof. Fix i. Since $\gamma_1, \ldots, \gamma_i$ are distinct, there exists m such that $(Q_m'\gamma_k)_{k=1}^i$ is isometrically equivalent to the $\ell^\infty(i)$ basis ([3, proof of Lemma 9]). Then

$$\varepsilon < \Big| \lim_{n\in M_{k+1}} \langle R_n\xi, \gamma_k \rangle \Big| = \Big| \lim_{n\in M_{k+1}} \langle Q_m R_n\xi, \gamma_k \rangle \Big| = \Big| \lim_{n\in M_{k+1}} \langle R_n\xi, Q_m'\gamma_k \rangle \Big| \,,$$

$1 \le k \le i$. Thus, for large enough n_0 in M_{i+1}, $|\langle R_{n_0}\xi, Q'_m\gamma_k\rangle| > \varepsilon$. But then $x' \equiv \sum_{k=1}^{i} \operatorname{sgn}(\langle R_{n_0}\xi, Q'_m\gamma_k\rangle)Q'_m\gamma_k$ has norm 1 while

$$|\langle R_{n_0}\xi, x'\rangle| = \sum_{k=1}^{i} |\langle R_{n_0}\xi, Q'_m\gamma_k\rangle| > i\varepsilon \ .$$

Hence $\|R_{n_0}\| > i\varepsilon$. Since i is arbitrary, the lemma is proved. ∎

Lemma 3.2. *Let $\varepsilon > 0$ and a decreasing sequence of sets $(M_k) \subset \mathcal{P}_\infty(\mathbb{N})$ be given. Suppose there exist two sequences of pairwise distinct elements (ξ_k) and (γ_k) in Γ such that*

$$a_{ij} \equiv \lim_{n\in M_{j+1}} |\langle R_n\xi_j, \gamma_i\rangle| \ \text{ exists for all } \ i,j \ ,$$

$a_{ii} > \varepsilon$ for all i, and $\lim_j a_{ij} = 0 = \lim_i a_{ij}$ for all i, j. Then (R_n) is unbounded.

Proof. By considering subsequences of (ξ_k) and (γ_k), we may assume additionally that $a_{ij} < \varepsilon 2^{-i-j-1}$ for all $i \neq j$. Fix l. As in the proof of the previous lemma, let m be such that $(Q'_m\xi_k)_{k=1}^{l}$ and $(Q'_m\gamma_k)_{k=1}^{l}$ are isometrically equivalent to the $\ell^\infty(l)$ basis. Then

$$\Big|\lim_{n\in M_{j+1}} \langle R_n\xi_j, \gamma_i\rangle\Big| = \Big|\lim_{n\in M_{j+1}} \langle R_n\xi_j, Q'_m\gamma_i\rangle\Big| = \Big|\lim_{n\in M_{j+1}} \langle R_nQ'_m\xi_j, Q'_m\gamma_i\rangle\Big|$$

for $1 \le i, j, \le l$, since $(1 - Q_m)R_n$ and $(1 - Q_m)R'_n$ converge to 0 weakly as $n \to \infty$. Thus there exists $n_0 \in M_{l+1}$ so large that

$$|\langle R_{n_0}Q'_m\xi_j, Q'_m\gamma_i\rangle| \begin{cases} < \varepsilon 2^{i-j-1} & \text{for } i \neq j \\ > \varepsilon & \text{for } i = j \end{cases} \qquad 1 \le i,j \le l \ .$$

Let $b_{ij} = \langle R_{n_0}Q'_m\xi_j, Q'_m\gamma_i\rangle$, $1 \le i, j \le l$ and let

$$x' = \sum_{j=1}^{l} (\operatorname{sgn} b_{jj})Q'_m\xi_j \quad , \quad y' = \sum_{i=1}^{l} Q'_m\gamma_i \ .$$

Then $\|x'\| = \|y'\| = 1$ while

$$\begin{aligned} |\langle R_{n_0}x', y'\rangle| &= \Big|\sum_{j=1}^{l}\sum_{i=1}^{l} (\operatorname{sgn} b_{jj})\langle R_{n_0}Q'_m\xi_j, Q'_m\gamma_i\rangle\Big| \\ &\ge \sum_{i=1}^{l} (\operatorname{sgn} b_{ii})\langle R_{n_0}Q'_m\xi_j, Q'_m\gamma_i\rangle - \sum_{\substack{i\neq j \\ 1\le i,j\le l}} |\langle R_{n_0}Q'_m\xi_j, Q'_m\gamma_i\rangle| \\ &\ge \sum_{i=1}^{l} |\langle R_{n_0}Q'_m\xi_j, Q'_m\gamma_i\rangle| - \sum_{i\neq j} \varepsilon 2^{-i-j-1} > \varepsilon l - \varepsilon/2 \ . \end{aligned}$$

Since l is arbitrary, this shows that (R_n) is unbounded. ∎

Let (a_n) be a sequence of real numbers, the *oscillation* of the sequence (a_n), $\mathrm{osc}(a_n)$, is given by $\limsup a_n - \liminf a_n$. We now claim that for every $M \in \mathcal{P}_\infty(\mathbb{N})$, there are elements $\xi, \gamma \in \Gamma$ such that $\mathrm{osc}(\langle R_n\xi, \gamma\rangle)_{n\in M} > 0$. Otherwise, let $M \in \mathcal{P}_\infty(\mathbb{N})$ be such that $\mathrm{osc}(\langle R_n\xi, \gamma\rangle)_{n\in M} = 0$ for all $\xi, \gamma \in \Gamma$. For every $x' \in JH'$, $(R_n x')$ is a bounded sequence of JH and hence has a weakly Cauchy subsequence. By a simple diagonalization argument, we may assume that $(R_n\varphi)_{n\in M}$ is weakly Cauchy for all $\varphi \in T$, since T is countable. Similarly, we may assume that $(R_n'\varphi)_{n\in M}$ is weakly Cauchy for all $\varphi \in T$. Then, for all $(x', y') \in (\Gamma\cup T)\times(\Gamma\cup T)$, we have $\mathrm{osc}(\langle R_n x', y'\rangle)_{n\in M} = 0$. But since $[\Gamma \cup T] = JH'$ and (R_n) is bounded, we obtain that (R_n) is weakly Cauchy, contrary to the assumption that (R_n) is equivalent to the ℓ^1 basis. Thus the claim is proved.

By applying some standard maneuvers, it can be shown that the claim above can actually be achieved uniformly (see, *e.g.*, [1, p.209, Step 1]). That is, we may assume:

(*) there exists $\varepsilon > 0$ such that for all $M \in \mathcal{P}_\infty(\mathbb{N})$, there are $\xi, \gamma \in \Gamma$ such that $\mathrm{osc}(\langle R_n\xi, \gamma\rangle)_{n\in M} > \varepsilon$.

We now choose a decreasing sequence of sets $(M_k) \subset \mathcal{P}_\infty(\mathbb{N})$, and sequences (ξ_k) and (γ_k) in Γ satisfying:

(a) $(\langle R_n\xi_{k-1}, \gamma_{k-1}\rangle)_{n\in M_k}$ converges with $\lim_{n\in M_k} |\langle R_n\xi_{k-1}, \gamma_{k-1}\rangle| > \varepsilon/2$ for $k \geq 2$;

(b) $(R_n\xi_{k-1})_{n\in M_k}$ and $(R_n'\gamma_{k-1})_{n\in M_k}$ are weakly Cauchy for $k \geq 2$; and

(c) $\mathrm{osc}(\langle R_n\xi_k, \gamma_k\rangle)_{n\in M_k} > \varepsilon$ for all k.

The choices are made inductively. For $k = 1$, apply (*) to $M_1 = \mathbb{N}$ to obtain $\xi_1, \gamma_1 \in \Gamma$ such that $\mathrm{osc}(\langle R_n\xi_1, \gamma_1\rangle)_{n\in M_1} > \varepsilon$. If the choices are made up to $k \geq 1$, let $M' \in \mathcal{P}_\infty(M_k)$ be such that $(\langle R_n\xi_k, \gamma_k\rangle)_{n\in M'}$ converges with $\lim_{n\in M'} |\langle R_n\xi_k, \gamma_k\rangle| > \varepsilon/2$ and $(R_n\xi_k)_{n\in M'}$ and $(R_n'\gamma_k)_{n\in M'}$ are weakly Cauchy. This is possible since $\mathrm{osc}(\langle R_n\xi_k, \gamma_k\rangle)_{n\in M_k} > \varepsilon$ and $(R_n\xi_k)_{n\in M'}$ and $(R_n'\gamma_k)_{n\in M'}$ are bounded sequences in JH and hence have weakly Cauchy subsequences. Now apply (*) to M' to obtain $M_{k+1} \in \mathcal{P}_\infty(M')$ and $\xi_{k+1}, \gamma_{k+1} \in \Gamma$ such that $\mathrm{osc}(\langle R_n\xi_{k+1}, \gamma_{k+1}\rangle)_{n\in M_{k+1}} > \varepsilon$. This completes the inductive definition.

Note that conditions (b) and (c) imply that $\xi_i \neq \xi_j$ and $\gamma_i \neq \gamma_j$ if $i \neq j$. Let

$$a_{ij} \equiv \lim_{n\in M_{j+1}} |\langle R_n\xi_j, \gamma_i\rangle| \quad \text{for all} \quad i, j\ .$$

The limits exist because of (b). Moreover, $a_{ii} > \varepsilon/2$ by (a). If $\lim_j a_{ij} = 0 = \lim_i a_{ij}$ for all i, j, then Lemma 3.2 implies that (R_n) is unbounded, obtaining a contradiction. Thus we may assume that either (A) there is a j_0 such that $\lim_i a_{ij_0} \neq 0$ or (B) there is a i_0 such that $\lim_j a_{i_0 j} \neq 0$. We will treat case (A), the argument in the other case is essentially the same.

By considering a subsequence, we may assume there exists $\delta > 0$ such that $a_{ij_0} > \delta$ for all i. Let $\xi = \xi_{j_0}$. For $i \geq j_0$, $M_{i+1} \subset M_{j_0+1}$, hence

$$a_{ij_0} = \lim_{n \in M_{j_0+1}} |\langle R_n \xi_{j_0}, \gamma_i \rangle| = \lim_{n \in M_{i+1}} |\langle R_n \xi, \gamma_i \rangle| \, .$$

Therefore, $\lim_{n \in M_{i+1}} |\langle R_n \xi, \gamma_i \rangle| > \delta$ for all $i \geq j_0$. Thus (R_n) is unbounded by Lemma 3.1, again reaching the desired contradiction.

We have thus proved

Theorem 3.3. *$JH \widetilde{\otimes}_\varepsilon JH$ does not contain a copy of ℓ^1.*

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A remark on the paper "Martingale inequalities in rearrangement invariant function spaces" by W.B. Johnson and G. Schechtman

PAWEL HITCZENKO*

Department of Mathematics
Texas A&M University
College Station, Texas 77843

1. Introduction and terminology

Our main purpose here is to provide an alternative proof of the convex function inequality in rearrangement invariant function spaces with finite upper Boyd index. This inequality and the good λ inequality were the main tools used in the proof of Burkholder-Davis-Gundy inequalities for convex functions (cf. [1] or [2]). Both these inequalities were recently generalized by Johnson and Schechtman [6] to rearrangement invariant function spaces which are "far from L_∞." Their proof of the convex function inequality was based on a technique due to Neveu. We would like to point out here, that this inequality itself may be viewed as a consequence of the good λ inequality (via a comparison result which, we believe, is of independent interest).

The investigation of Johnson and Schechtman was partially motivated by recent results of de la Peña [3] concerning comparison of L_Φ norms of martingales and sums of independent random variables under certain convexity assumptions on Φ. We would like to indicate still another proof of de la Peña's result, a proof which is a direct consequence of Hoffmann-Jörgensen's inequality. This will be done in the last section. The next section contains a proof of the convex function inequality.

Now, let us recall necessary definitions and fix some notation. We will work on a measure space $(0,1)$ equipped with its Borel σ-algebra $\mathcal{B}$ and Lebesgue measure μ. Given an increasing sequence $(\mathcal{A}_n)$ of sub-σ-algebras of $\mathcal{B}$ and two adapted sequences (f_n) and (g_n) of random variables, we say that they are tangent if for each $n \geq 1$ the conditional distributions of f_n and g_n given $\mathcal{A}_{n-1}$ are identical a.s. (*i.e.*, if for each $t \in R$, $\mu(f_n > t \mid \mathcal{A}_{n-1}) = \mu(g_n > t \mid \mathcal{A}_{n-1})$ a.s.).

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* On leave from Academy of Physical Education, Warsaw, Poland.

For any sequence of random variables there exists a sequence (g_n) (which can be assumed to be defined on the same measure space $((0,1),\mathcal{B},\mu)$ and adapted to $(\mathcal{A}_n)$) such that (f_n) and (g_n) are tangent and (g_n) has the following additional property: There exists a σ-algebra $\mathcal{G}$ such that:

(a) the conditional distribution of g_n given $\mathcal{G}$ is the same as the conditional distribution of g_n given $\mathcal{A}_{n-1}$, $n \geq 1$,

(b) (g_n) is a sequence of $\mathcal{G}$-conditionally independent random variables.

We will refer to this special sequence (g_n) as to a decoupled version of (f_n). For more information on tangent sequences see [8] and references therein.

If (f_n) is any sequence of random variables then f_n^* denotes $\max_{1\leq k\leq n}|f_k|$ and f^* stands for $\sup_n |f_n|$. By $I(A)$ we denote the indicator function of the set A.

Our definition of rearrangement invariant function space is that of Lindenstrauss-Tzafriri [9]. The upper Boyd index of such a space X is the smallest p with the property that for all $0 < c < 1$ one has:

$$\|D_c\| \leq c^{1/p} ,$$

where D_c is a dilation operator on X defined by $D_c x(t) = x(t/c)$, $x \in X$, and $x(s)$ is understood as 0 if $s > 1$ (see [9] for more details).

2. Convex function inequality in rearrangement invariant function spaces

This section contains a proof of the following result due to Johnson and Schechtman [6].

Proposition 2.1. *Let X be a rearrangement invariant function space on $(0,1)$ with finite upper Boyd index $p < \infty$. Let $(\mathcal{A}_n)$ be an increasing sequence of sub-σ-algebras of the Borel σ-algebra. Then there is a constant c (depending only on p) such that for every adapted sequence (Z_n) of non-negative random variables we have that:*

$$\Big\| \sum E Z_n \mid \mathcal{A}_{n-1} \Big\|_X \leq c \Big\| \sum Z_n \Big\|_X .$$

Our proof relies on a comparison result for tangent sequences of non-negative random variables. Before we state that inequality let us list two simple lemmas. The first one is a version of the good λ inequality and is contained in [6], while the second can be found in [4].

Lemma 2.2. *Let f and g be non-negative random variables on $(0,1)$ and let X be a rearrangement invariant function space on $(0,1)$ with upper Boyd index $p < \infty$. Suppose that $\beta > 0$, $\delta > 0$, $\varepsilon > 0$ satisfy $\beta\varepsilon^{1/p} < 1$ and*

$$\mu(f > \beta\lambda\,,\, g \leq \delta\lambda) \leq \varepsilon\mu(f > \lambda) , \qquad \lambda > 0 .$$

Then:

$$\|f\|_X \leq \frac{\beta}{\delta(1-\beta\varepsilon^{1/p})}\|g\|_X \ .$$

Lemma 2.3. *If (f_n) and (g_n) are tangent sequences of random variables on $(0,1)$, then:*

$$\mu(f^* > \lambda) \leq 2\mu(g^* > \lambda) \ , \qquad \lambda > 0 \ .$$

In particular,

$$\|f^*\|_X \leq 2\|g^*\|_X$$

for any rearrangement invariant function space X on $(0,1)$.

We are now ready for a comparison result:

Lemma 2.4. *Assume X is a rearrangement invariant function space on $(0,1)$ with upper Boyd index $p < \infty$. Then there is a constant c such that for all tangent sequences (f_n) and (g_n) of non-negative random variables one has:*

$$\Big\|\sum f_n\Big\|_X \leq c\Big\|\sum g_n\Big\|_X \ .$$

Proof (sketch). The proof is the same as the proof of Theorem 2 in [4] (which itself was a modification of an argument given in [1]). For $n \geq 1$ we write:

$$\begin{aligned} h_n &= \max\{f^*_{n-1}, g^*_{n-1}\} \\ f'_n &= f_n I(f_n \leq 2h_n) \\ f''_n &= f_n I(f_n > 2h_n) \\ g'_n &= g_n I(g_n \leq 2h_n) \ . \end{aligned}$$

Then, the random variables $\sum f'_n$ and $(\sum g'_n) \vee h^*$ satisfy the good λ inequality with $\varepsilon = 3\delta/(\beta - 1 - \delta)$ whenever $\beta - 1 > \delta > 0$. Thus, in view of Lemma 2.2 we have:

$$\Big\|\sum f'_n\Big\|_X \leq c\Big\{\Big\|\sum g'_n\Big\|_X + \|h^*\|_X\Big\} \leq c\Big\{\Big\|\sum g_n\Big\|_X + \|f^*\|_X + \|g^*\|_X\Big\}$$

and, by Lemma 2.3

$$\|f^*\|_X \leq 2\|g^*\|_X \leq 2\Big\|\sum g_n\Big\|_X \ .$$

To estimate $\|\sum f''_n\|_X$ we observe that on the set $\{f_n > 2h_n\}$ we have

$$f_n + 2h_n \leq 2f_n \leq 2h_{n+1} \ ,$$

so that

$$f''_n \leq 2(h_{n+1} - h_n) \ ,$$

or

$$\sum f_n'' \leq 2h^*$$

which gives

$$\Big\| \sum f_n'' \Big\|_X \leq 2\|h^*\|_X \leq 6 \Big\| \sum_n g_n \Big\|_X ,$$

as desired. For details see [4, proof of Theorem 2].

Proof of Proposition 2.1. Given a sequence (Z_n) we denote by (Y_n) a decoupled version of (Z_n). Then:

$$\Big\| \sum EZ_n \mid \mathcal{A}_{n-1} \Big\|_X = \Big\| \sum EY_n \mid \mathcal{A}_{n-1} \Big\|_X = \Big\| \sum EY_n \mid \mathcal{G} \Big\|_X = \Big\| E\Big(\sum Y_n\Big) \mid \mathcal{G} \Big\|_X$$

and, by the norm diminishing property of the conditional expectation (cf. [9, Theorem 2.a.4]) the last quantity does not exceed

$$\Big\| \sum Y_n \Big\|_X .$$

From the previous lemma we infer that:

$$\Big\| \sum Y_n \Big\|_X \leq c \Big\| \sum Z_n \Big\|_X$$

which completes the proof.

Remark 2.5. A slight modification of the argument of Burkholder [1, proof of Theorem 15.1] yields that if (d_n) and (e_n) are tangent sequences of martingale differences on $(0,1)$ and X is a rearrangement invariant function space on $(0,1)$ with upper Boyd index $p < \infty$ then

$$\Big\| \Big(\sum_{k=1}^n d_k \Big)^* \Big\|_X \approx \Big\| \Big(\sum_{k=1}^n e_k \Big)^* \Big\|_X .$$

This gives another proof of the square function inequality in rearrangement invariant function spaces.

3. Comparison inequality of de la Peña – a convex case

Throughout this section, given a sequence (X_n) of random variables we denote by (W_n) a sequence of independent random variables such that for each $n \geq 1$, X_n and W_n are equidistributed. We will show another proof of the following result due to de la Peña [3]:

Proposition 3.1. *Assume that $\Phi : R_+ \to R_+$ is a convex, increasing function which is 0 at 0 and satisfies the growth condition $\Phi(2t) \leq c\Phi(t)$, for some positive constant c and all positive t's. Then there is a constant K (depending only on c) such that:*

(a) if (X_n) is a martingale difference sequence and $\Phi(\sqrt{t})$ is convex then:

$$E\Phi\left(\left(\sum_{k=1}^{n} W_k\right)^*\right) \leq KE\Phi\left(\left(\sum_{k=1}^{n} X_k\right)^*\right) .$$

(b) if (X_n) is a sequence of non-negative random variables then:

$$E\Phi\Big(\sum W_k\Big) \leq KE\Phi\Big(\sum X_k\Big) .$$

Remark 3.2. In his proof of this result de la Peña used Klass' two sided bounds for $E\Phi(\sum W_k)$. He also showed, by different method, an analogous result for concave functions rather than convex. More precisely, he obtained the reversed inequality in (a) under the assumption that Φ is convex and $\Phi(\sqrt{t})$ is concave and reversed inequality in (b) for concave Φ's. Later on, Johnson and Schechtman [7] deduced de la Peña's results from their comparison theorems for rearrangement invariant norms of sums of independent and disjoint functions. As we will show below, under the assumption of convexity the result can be derived from Hoffmann-Jörgensen's inequality [5].

Proof of Proposition 3.1. It follows from Burkholder-Davis-Gundy inequalities that (a) and (b) are equivalent, so it suffices to establish part (b). To this end, observe that by a well-known consequence of Hoffmann-Jörgensen's inequality [5], we have:

$$E\Phi\Big(\sum W_k\Big) \leq c\Big\{\Phi\Big(E\sum W_k\Big) + \sum E\Phi(W_k)\Big\}$$

(for $\Phi(t) = t^p$, $p \geq 1$, this is just Rosenthal's inequality). Since W_k and X_k are equidistributed, $EW_k = EX_k$ and $E\Phi(W_k) = E\Phi(X_k)$, $k \geq 1$. Thus, by the superadditivity of Φ the second term of the right-hand side above is no greater than:

$$\sum E\Phi(X_k) = E\sum \Phi(X_k) \leq E\Phi\Big(\sum X_k\Big) .$$

The first term, in view of Jensen's inequality does not exceed:

$$\Phi\Big(\sum EX_k\Big) = \Phi\Big(E\sum X_k\Big) \leq E\Phi\Big(\sum X_k\Big) .$$

This completes the proof.

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Twisted types and uniform stability

FOUAD CHAATIT

Department of Mathematics
The University of Texas at Austin
Austin, Texas 78712

1. Introduction

Types on Banach spaces were first introduced by J.L. Krivine and B. Maurey in [9]. Whereas it is well known that the set $\mathcal{T}(X)$, of all types on a Banach space X, is separable for the weak topology [9] or the strong topology [13] whenever X is a separable stable Banach space, the situation is not as clear for the uniform topology. Indeed R. Haydon showed in [7] that there exists a separable stable Banach space whose types are not separable in the uniform topology. In section 2 we introduce, following ideas implicitly contained in [9], the notion of uniformly stable Banach spaces. While this condition is sufficient for the uniform separability of types, we must say that we are not giving any general criteria for checking whether the set of types on a given Banach space is uniformly separable. Indeed, R. Haydon (unpublished) showed that the set of types on L^1 is uniformly separable but L^1 is not uniformly stable. However the notion of uniform stability covers a pretty large class of Banach spaces. In section 3, after recalling some facts about random probability measures and uniformly integrable sequences in L^1, we prove again the result of R. Haydon concerning the uniform separability of types on L^1. In section 4 we prove that every stable Banach space whose norm is uniformly smooth is uniformly stable. Thus for $1 < p < \infty$, ℓ^p, L^p as well as the Schatten classes C^p, are uniformly stable, as are the spaces $L^p(X)$ whenever X is both stable and uniformly smooth. In the fifth section we obtain that for $1 < p < \infty$ an ℓ_p-sum of uniformly stable Banach spaces is uniformly stable. Actually we show that uniform stability is preserved by taking a sum over a stable Banach space with a 1-unconditional basis and uniformly Frechet-differentiable norm. We also show that if E is a uniformly stable Banach space then $L^p(E)$ is also uniformly stable for $1 < p < \infty$.

This work constitutes part of the author's "thèse de troisième cycle" prepared at University of Paris VII under the guidance of Professor Bernard Maurey, to whom the author is deeply indebted.

Let us now recall some basic definitions. If X is a Banach space and $x \in X$ the map

$$\tau_x : X \to \mathbb{R} \quad ; \quad y \mapsto \tau_x(y) = \|x + y\|$$

is called a *degenerate type*. The closure of the set $\{\tau_x; x \in X\}$ in the product topology of $\mathbb{R}^X$ is defined to be the set $\mathcal{T}(X)$ of all *types* on X. Thus if $\tau \in \mathcal{T}(X)$ then there exists a bounded net (x_i) in X (called an *approximating net of* τ) such that $\|y + x_i\| \to \tau(y)$ for all $y \in X$. In the case where X is separable there exists a sequence (x_n) in X such that $\|y + x_n\| \to \tau(y)$ for all $y \in X$.

If τ is a type and λ is a nonzero scalar then the type $\lambda\tau$ is defined by $(\lambda\tau)(y) = |\lambda|\tau(\lambda^{-1}y)$ and $0\tau(y) = \|y\|$.

If $\mathcal{U}$ is an ultrafilter on $\mathbb{N}$, $(x_i)_i$ an approximating net of the type τ_1 and $\mathcal{V}$ is an ultrafilter on $\mathbb{N}$ and $(y_j)_j$ an approximating net of the type τ_2, then the *convolution* of τ_1 and τ_2 is defined, for stable Banach spaces, to be the type

$$\tau_1 * \tau_2(y) = \lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} \|x_i + y_i + y\| \ .$$

(A Banach space is stable if $\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} \|x_i + y_j\| = \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} \|x_i + y_j\|$ for every bounded sequence (x_i) and (y_j) and every nontrivial ultrafilter $\mathcal{U}$ and $\mathcal{V}$.)

As in [9] the *weak topology* on $\mathcal{T}(X)$ is defined to be that induced by the product topology of $\mathbb{R}^X$, that is to say, the topology of pointwise convergence on X.

The *strong topology* is given by the pseudometrics d_M where

$$d_M(\sigma, \tau) = \sup\{|\sigma(y) - \tau(y)| \ ; \ y \in X \quad \|y\| \leq M\}$$

if σ and τ are types on X.

The *uniform topology* is given by the metric d where $d(\sigma, \tau) = \sup\{|\sigma(y) - \tau(y)|; y \in X\}$. Let $(x_n)_n$ be a semi-normalized basic sequence. A basic sequence $(e_n)_n$ is said to be a *spreading model* of (x_n) if for all $k \in N$ and all $\varepsilon > 0$ there exists N so that if $N < n_1 < n_2 < \cdots < n_k$ and $(a_i)_{i=1}^k \subseteq \mathbb{R}$ with $\sup_i |a_i| \leq 1$, then

$$\left| \left\| \sum_{i=1}^k a_i x_{n_i} \right\| - \left\| \sum_{i=1}^k a_i e_i \right\| \right| < \varepsilon \ .$$

(See [3] for more information on spreading models.)

For the other basic definitions we refer to [6] or [9]. Finally we should note that uniform separability of types is not an isomorphic invariant. Indeed in [8] R. Haydon and B. Maurey have obtained an equivalent norm on ℓ_2 for which the types are not strongly and thus not uniformly separable. For the same reason it will follow that uniform stability is not an isomorphic invariant.

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2. "Twisted types" and uniformly stable Banach spaces

In this section we introduce, following ideas contained in [9], the notions of "twisted types" and uniformly stable Banach spaces.

Let X be a separable Banach space and $y \in X$. The map $\sigma_y : X \to \mathbb{R}$; $x \mapsto \|x+y\| - \|y\|$ will be called a *degenerate twisted type*. The closure of the set $\{\sigma_y; y \in X\}$ for the pointwise convergence topology in $\mathbb{R}^X$, is the set of all twisted types, and will be denoted by K.

Thus if σ is a "twisted type" there exists a sequence $(y_n)_{n\in\mathbb{N}}$, not necessarily bounded in X, and a nontrivial ultrafilter $\mathcal{U}$ on $\mathbb{N}$, such that

$$\sigma(x) = \lim_{n,\mathcal{U}} \|x + y_n\| - \|y_n\| \ .$$

We should notice that the inequality

$$|\sigma_y(x)| = \big|\|x+y\| - \|y\|\big| \leq \|x\|$$

is preserved for twisted types. As a consequence K, the set of all twisted types, is compact in $\mathbb{R}^X$ since it is a closed subset of

$$\prod_{x\in X} [-\|x\|, \|x\|] \ .$$

Similarly the inequality

$$|\sigma_y(x) - \sigma_y(x')| \leq \|x - x'\|$$

is preserved by all twisted types. Twisted types are then 1-Lipschitzian functions and since X is separable, K can be embedded in the set $\mathbb{R}^D$ where D is any countable dense subset of X. Hence K is metrizable.

Example of a twisted type:

For every $y \in X$, $\|y\| = 1$, if the norm is Gateaux-differentiable at y then the differential of the norm at y is a twisted type. Indeed

$$\begin{aligned}
\lim_{n\to\infty} \|x + ny\| - \|ny\| &= \lim_{n\to\infty} n\left[\left\|\frac{x}{n} + y\right\| - \|y\|\right] \\
&= \lim_{n\to\infty} n\left[D_y \cdot \frac{x}{n} + \frac{\|x\|}{n}\varepsilon\Big(\frac{\|x\|}{n}\Big)\right] \\
&= D_y \cdot x \\
&\text{since } \varepsilon\Big(\frac{\|x\|}{n}\Big) \to 0 \text{ when } n \to \infty \ .
\end{aligned}$$

∎

Uniformly stable Banach spaces:

Definition 1. A separable Banach space X is said to be *uniformly stable* if for every bounded sequence $(x_n)_{n\in\mathbb{N}}$ in X and every sequence $(y_m)_{m\in\mathbb{N}}$ in X we have

$$\lim_{n\to\infty}\lim_{m\to\infty}\|x_n+y_m\|-\|y_m\| = \lim_{m\to\infty}\lim_{n\to\infty}\|x_n+y_m\|-\|y_m\|$$

whenever those limits exist.

Finite dimensional spaces are easily seen to be uniformly stable. We note also that if X is uniformly stable then X is stable.

Now if $\tau(x)=\lim_{n\to\infty}\|x+a_n\|$ and $\sigma(x)=\lim_{m\to\infty}\|x+b_m\|-\|b_m\|$ then we define $[\sigma,\tau]=\lim_{n\to\infty}\lim_{m\to\infty}\|a_n+b_m\|-\|b_m\|$.

The bracket defines a function $[\cdot,\cdot]$ from $\mathcal{T}(X)\times K$ into $\mathbb{R}$, where $\mathcal{T}(X)$ is the set of all types on X, whenever X is a uniformly stable Banach space. Indeed if $\tau(x)=\lim_{n\to\infty}\|x+a'_n\|$ and $\sigma(x)=\lim_{m\to\infty}\|x+b'_m\|-\|b'_m\|$ then

$$\begin{aligned}
\lim_{n\to\infty}\lim_{m\to\infty}\|a_n+b_m\|-\|b_m\| &= \lim_{n\to\infty}\sigma(a_n)\\
&= \lim_{n\to\infty}\lim_{m\to\infty}\|a_n+b'_m\|-\|b'_m\|\\
\text{(by uniform stability)}\quad &= \lim_{m\to\infty}\lim_{n\to\infty}\|a_n+b'_m\|-\|b'_m\|\\
&= \lim_{m\to\infty}\tau(b'_m)-\|b'_m\|\\
&= \lim_{m\to\infty}\lim_{n\to\infty}\|a'_n+b'_m\|-\|b'_m\|\\
\text{(using uniform stability again)}\quad &= \lim_{n\to\infty}\lim_{m\to\infty}\|a'_n+b'_m\|-\|b'_m\|\,.
\end{aligned}$$

Proposition 2. *Let X be a uniformly stable Banach space, then the set $\mathcal{T}(X)$ of all types on X, equipped with the uniform topology, is separable.*

Proof. We have already seen that the bracket $[\cdot,\cdot]$ defines a mapping from $\mathcal{T}(X)\times K$ into $\mathbb{R}$, whenever X is uniformly stable.

As $\mathcal{T}(X)$ (respectively, K) is metrizable and $\widetilde{X}$, the set of all degenerate types (respectively, $\widetilde{\widetilde{X}}$ the set of all degenerate twisted types) is dense in $\mathcal{T}(X)$ (respectively, in K) the mapping $[\cdot,\cdot]$ is separately continuous.

Embedding $\mathcal{T}(X)$ in $\mathcal{C}(K)$, the Banach space of continuous functions on K with uniform

norm, using the mapping $u : \tau \mapsto u(\tau) = [\tau, \cdot]$ we obtain then:

$$\begin{aligned} \|u(\tau_1) - u(\tau_2)\|_{\mathcal{C}(K)} &= \sup_{\sigma \in K} \left|[\tau_1, \sigma] - [\tau_2, \sigma]\right| \\ &= \sup_{\sigma_y \in \widetilde{\widetilde{X}}} \left|[\tau_1, \sigma] - [\tau_2, \sigma]\right| \\ &= \sup_{y \in X} |\tau_1(y) - \tau_2(y)| \ . \end{aligned}$$

This embedding is then isometric if $\mathcal{T}(X)$ is equipped with the uniform topology. As K is a compact metrizable set, $\mathcal{C}(K)$ is separable as well as all its subspaces. ∎

3. A result of R. Haydon

Let $\overline{\mathbb{R}} = [-\infty, +\infty]$ and $\mathcal{P}(\mathbb{R})$ be the set of probability measures on $\mathbb{R}$. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space such that $L^1(\Omega, \mathcal{A}, \mathbb{P})$ is separable. Recall that a random probability measure is a map $\mu : \Omega \to \mathcal{P}(\mathbb{R})$ such that $\omega \mapsto \int \varphi(u)\mu_\omega(du)$ is measurable for every bounded continuous function φ on $\mathbb{R}$. Since the dual of the separable Banach space $L^1(\Omega, \mathcal{C}(\overline{\mathbb{R}}))$ is the space of scalarly measurable bounded maps from Ω into $M(\overline{\mathbb{R}}) \equiv \mathcal{C}(\overline{\mathbb{R}})^*$, the set of random probability measures can be identified with a subset of the unit ball of $L^1(\Omega, \mathcal{C}(\overline{\mathbb{R}}))^*$. The random probability measures are then given the weak-* topology of $\mathrm{Ball}(L^1(\Omega, \mathcal{C}(\overline{\mathbb{R}}))^*)$, which is metrizable since $L^1(\Omega, \mathcal{C}(\overline{\mathbb{R}}))$ is separable.

If $(y_m)_{m \in \mathbb{N}}$ is an arbitrary sequence in L^1 and $\mathcal{V}$ is a nontrivial ultrafilter on $\mathbb{N}$, then $\nu = w\text{-}^*\lim_{m,\mathcal{V}} \delta_{y_m}$ is a random probability measure on $\overline{\mathbb{R}}$. Recall also that if $(x_n)_{n \in \mathbb{N}}$ is a sequence of uniformly integrable functions and $\mathcal{U}$ is a nontrivial ultrafilter over $\mathbb{N}$, then $\mu = w\text{-}^*\lim_{n,\mathcal{U}} \delta_{x_n}$ is a random probability measure on $\mathbb{R}$ and the function $\omega \mapsto \int |u|\mu_\omega(du)$ is in L^1 (prop. II.1 of [**10**]). Moreover the equality

$$\text{(E)} \qquad \lim_{n,\mathcal{U}} \mathbb{E} f(x_n) = \mathbb{E} \int f(\omega, u)\mu_\omega(du) \ ,$$

where $f(x_n)$ is the function $\omega \mapsto f(\omega, x_n(\omega))$ and $f \in L^1(\Omega, \mathcal{C}(\overline{\mathbb{R}}))$ can be extended to those functions $f : \Omega \to \mathcal{C}(\overline{\mathbb{R}})$ such that there exists $g \in L^1(\Omega)$ and a constant k with $|f(\omega, u)| \leq g(\omega) + k|u|$ [**11**].

Remark. We can then, in particular, represent a twisted type σ on L^1 in terms of a random probability measure by the following:

$$\begin{aligned} &\text{if} \qquad \sigma(x) = \lim_{m,\mathcal{U}} \|x + y_m\| - \|y_m\| \\ &\text{then} \qquad \sigma(x) = \mathbb{E} \int_{\overline{\mathbb{R}}} (|x(\omega) + v| - |v|)\nu_\omega(dv) \\ &\text{where} \qquad \nu = \lim_{m,\mathcal{U}} \delta_{y_m} \ . \end{aligned}$$

Proposition 3. *If* $\sigma(x) = \lim_{n\to\infty} \|x + x_n\|_{L^1}$ *with* $(x_n)_{n\in\mathbb{N}}$ *uniformly integrable and if the degenerate types* σ_{y_n} *converge to* σ, *then* $(y_n)_{n\in\mathbb{N}}$ *is also uniformly integrable.*

To prove this proposition we will need the following

Lemma 4. *If* $\sigma(x) = \lim_{n\to\infty} \|x + x_n\|_{L^1}$ *where* $(x_n)_n$ *is a sequence of uniformly integrable functions in* L^1 *then*

$$\frac{1}{2k}\{\sigma * -\sigma\}^{\circledast k}(0) \to 0 \quad as \quad k \to \infty \ .$$

Proof of the lemma. Since $(x_n)_n$ is uniformly integrable $(x_n)_n$ admits a subsequence, say (x'_n), that is weakly convergent. Then if $(y_n)_n = (x'_n - x'_{n+1})_n$ denotes the difference sequence, $(y_n)_n$ is clearly weakly null. By the weak Banach-Saks property of L^1 (see [5], p.112) one can pass to a subsequence (y'_n) of (y_n) whose arithmetic means are norm convergent (to zero since y_n converges weakly to zero).

$$(*) \qquad \begin{cases} \text{Thus for every } \varepsilon > 0 \ \left\| \frac{1}{k} \sum_{n=1}^{k} y'_n \right\| > \varepsilon \\ \text{for at most finitely many } k\text{'s.} \end{cases}$$

Now if we suppose

$$\left\| \frac{1}{2k}\{\sigma * -\sigma\}^{\circledast k} \right\| = \frac{1}{2k}\{\sigma * -\sigma\}^{\circledast k}(0) > \delta > 0$$

for some $\delta > 0$ for infinitely many k's, then

$$\lim_{\substack{n_1 \le n_2 \cdots \le n_k \\ n_1 \to \infty}} \frac{1}{2k} \|x_{n_1} - x_{n_2} + \cdots + x_{n_{2k-1}} - x_{n_{2k}}\| = \frac{1}{2k} \|e_1 - e_2 + \cdots + e_{2k-1} - e_{2k}\| > \delta$$

for infinitely many k's.

But by definition of the spreading model for every $k \in \mathbb{N}$ and $\varepsilon > 0$ (in particular $\varepsilon = \delta/2$) there exists $N \in \mathbb{N}$ such that

$$\left| \left\| \frac{e_1 - e_2 + \cdots + e_{2k-1} - e_{2k}}{2k} \right\| - \left\| \frac{1}{2k} \sum_{i=1}^{k} (x_{n_{2i-1}} - x_{n_{2i}}) \right\| \right| \le \frac{\delta}{2}$$

for $N < n_1 < n_2 < \cdots < n_{2k}$. Thus

$$\left| \left\| \frac{e_1 - e_2 + \cdots + e_{2k-1} - e_{2k}}{2k} \right\| - \left\| \frac{1}{2k} \sum_{i=1}^{k} y'_n \right\| \right| < \frac{\delta}{2} \quad \text{for} \quad n > N$$

i.e.,

$$\left\| \frac{1}{2k} \sum_{i=1}^{k} y'_n \right\| > \frac{\delta}{2}$$

for infinitely many k which contradicts $(*)$. ∎

Proof of proposition 3. Let $\sigma(x) = \lim_{n\to\infty} \|x+y_n\|_{L^1}$. We will apply the subsequence splitting lemma [4] to the sequence $(y_n)_n$. More precisely, since (y_n) is bounded in L^1, there exists a subsequence $(y_{n_j})_{j\in\mathbb{N}}$ of $(y_n), (z_j)$ uniformly integrable and $(t_j)_j$ such that $z_j \wedge t_j = 0$, t_i are mutually disjointly supported, and $y_{n_j} = z_j + t_j$. Then

$$\begin{aligned}\sigma(x) &= \lim_{j\to\infty} \|x + y_{n_j}\| \\ &= \lim_{j\to\infty} \|x + z_j + t_j\| \\ &= \lim_{j\to\infty} \|x + z_j\| + \|t_j\| \\ &= \lim_{j\to\infty} \|x + z_j\| + \lambda \,.\end{aligned}$$

If $\lambda > 0$, then for every $x \in L^1$ since

$$\lim_{k\to\infty} \frac{1}{2k}\{\sigma * -\sigma\}^{\circledast k}(x) = \lim_{k\to\infty} \lim_{n_1\to\infty} \dots \lim_{n_k\to\infty} \left\| \frac{x + z_{n_1} - z_{n_2} + \dots + z_{n_{2k-1}} - z_{n_{2k}}}{2k} \right\| + \lambda$$

then

$$\lim_{k\to\infty} \frac{1}{2k}\{\sigma * -\sigma\}^{\circledast k}(x) \geq \lambda$$

therefore

$$\inf_{x\in L^1} \lim_{k\to\infty} \frac{1}{2k}\{\sigma * -\sigma\}^{\circledast k}(x) \geq \lambda > 0$$

which contradicts the preceding lemma. ∎

Remark. After a first version of this paper was submitted for publication Professor R. Haydon showed us a short and elegant way of proving proposition 3 based on the following lemma:

Lemma. *Let (x_n) be an approximating sequence for a type τ on L^1 and let u be any element of L^1 that is a.e. nonzero (e.g. $u \equiv 1$ if we are dealing with a probability space). The sequence (x_n) is uniformly integrable if and only if*

$$\tau(Ru) + \tau(-Ru) - 2R\|u\| \to 0 \quad \text{as} \quad R \to \infty \,.$$

Proof. The sequence (x_n) is uniformly integrable if and only if $\varphi(n,R) =: \int_{|x_n|\geq R|u|} |x_n|\, d\mu$ converges uniformly to 0 as $R \to_+ \infty$. But

$$\begin{aligned}\varphi(n,R) &= \int (|x_n| - R|u|)^+ \, d\mu \\ &= \int \sup(|x_n| - R|u|, 0)\, d\mu \\ &= \int \left(\sup(|x_n|, R|u|) - R|u|\right) d\mu \\ &= \int \left(\frac{|x_n + Ru| + |x_n - Ru|}{2} - R|u| \right) d\mu \\ &= \tfrac{1}{2}\left(\|x_n + Ru\| + \|x_n - Ru\| - 2R\|u\|\right)\end{aligned}$$

so that $\varphi(u,R)$ tends to

$$\varphi(\infty,R) = \tfrac{1}{2}\big(\tau(Ru)+\tau(-Ru) = 2R\|u\|\big) \quad \text{as} \quad n \to \infty\ .$$

Elementary analysis now shows that $\varphi(\infty,R) \to 0$ as $R \to \infty$ iff $\varphi(n,R) \to 0$ as $R \to \infty$ uniformly. ∎

In the sequel we will denote by $\mathcal{T}_{ui}(L^1)$ the set of all "uniformly integrable" types on L^1, that is the set of all types on L^1 such that there exists a uniformly integrable sequence $(x_n)_{n\in\mathbb{N}}$ in L^1 with $\sigma(x) = \lim_{n\to\infty}\|x+x_n\|$. $\mathcal{U}$ and $\mathcal{V}$ will always be nontrivial ultrafilters on $\mathbb{N}$.

Proposition 5. *Let K be the set of twisted types on L^1 and let*

$$[\cdot,\cdot] : \mathcal{T}_{ui}(L^1) \times K \to \mathbb{R} \quad \textit{be given as above by}$$

$$(\sigma,\tau) \mapsto [\sigma,\tau] = \lim_{n,\mathcal{U}}\lim_{m,\mathcal{V}} \|x_n+y_m\| - \|y_m\|$$

where $(x_n)_n$ is a uniformly integrable sequence such that $\sigma = \lim_{n,\mathcal{U}} \sigma_{x_n}$ and $\tau = \lim_{m,\mathcal{V}} \tau_{y_m}$. Then $[\cdot,\cdot]$ is well defined and separately continuous.

Proof. In order that the bracket defines a mapping it suffices (see section 2) that the iterated limits commute

$$\begin{aligned}
\lim_{m,\mathcal{V}}\lim_{n,\mathcal{U}} \|x_n+y_m\| - \|y_m\| &= \lim_{m,\mathcal{V}}\lim_{n,\mathcal{U}} \mathbb{E}\big(|x_n(\omega)+y_m(\omega)| - |y_m(\omega)|\big) \\
&= \lim_{m,\mathcal{V}} \mathbb{E}\int_{\mathbb{R}} \big(|u+y_m(\omega)| - |y_m(\omega)|\big)\, d\mu_\omega(u) \\
&\qquad (\text{where} \quad \mu = \omega\text{-}^* \lim_{n,\mathcal{U}} \delta_{x_n}) \\
&= \mathbb{E}\int_{\overline{\mathbb{R}}}\Big(\int_{\mathbb{R}} \big(|u+v| - |v|\big)\, d\mu_\omega(u)\Big)\, d\nu_\omega(v) \\
&\qquad (\text{where} \quad \nu = \omega\text{-}^* \lim_{m,\mathcal{V}} \delta_{y_m})\ .
\end{aligned}$$

Let $h(\omega,v) = \int_{\mathbb{R}} \big||u+v| - |v|\big|\, d\mu_\omega(u)$ then

$$\begin{aligned}
\int_{\overline{\mathbb{R}}} |h(\omega,v)|\, d\nu_\omega(v) &\le \int_{\overline{\mathbb{R}}}\int_{\mathbb{R}} |u|\, d\mu_\omega(u)\, d\nu_\omega(v) \\
&\le \int_{\mathbb{R}} |u|\, d\mu_\omega(u) \\
&< \infty
\end{aligned}$$

for $\mathbb{P}$-almost all ω, since, as recalled at the beginning of this section, $\omega \mapsto \int |u|\, d\mu_\omega(u)$ is a non-negative, integrable function. Thus for $\mathbb{P}$-almost all ω, Fubini's theorem applies and therefore

$$\lim_{m,\mathcal{V}}\lim_{n,\mathcal{U}} \|x_n+y_n\| - \|y_m\| = \mathbb{E}\int_{\mathbb{R}}\Big(\int_{\overline{\mathbb{R}}} \big(|u+v| - |v|\big)\, d\nu_\omega(v)\Big)\, d\mu_\omega(u)\ .$$

Now, letting $g(\omega, u) = \int_{\overline{\mathbb{R}}}(|u+v|-|v|)\,d\nu_\omega(v)$, we have $|g(\omega,u)| \leq |u|$ so the equation (E) holds and,

$$\begin{aligned}\lim_{m,\mathcal{V}}\lim_{n,\mathcal{U}} \|x_n + y_m\| - \|y_m\| &= \lim_{n,\mathcal{U}} \mathbb{E}\int_{\overline{\mathbb{R}}}\left(|x_n(\omega)+v| - |v|\right)d\nu_\omega(v)\\ &= \lim_{n,\mathcal{U}}\lim_{m,\mathcal{V}} \mathbb{E}\left(|x_n(\omega)+y_m(\omega)| - |y_m(\omega)|\right)\\ &= \lim_{n,\mathcal{U}}\lim_{m,\mathcal{V}} \|x_n+y_n\| - \|y_m\|\ .\end{aligned}$$

The separate continuity of $[.,.]$ can then easily be checked as in proposition 2 of section 2. ∎

An immediate consequence of this proposition is that the set $\mathcal{T}_{ui}(L^1)$, when given the uniform topology, is separable. The proof of this follows from the proof of proposition 2 of section 2.

Now we are ready to prove the following theorem, due to R. Haydon:

Theorem 6. *The set $\mathcal{T}(L^1)$ of all types on L^1 is separable when equipped with the uniform topology.*

Proof. For every type σ on L^1 there exists $\lambda \in \mathbb{R}$ and a uniformly integrable type τ on L^1 such that $\sigma = \tau * \lambda\tau_1$ where τ_1 is the type defined by the canonical basis of ℓ_1. This can be seen using the subsequence splitting lemma again. More precisely if $(x_n)_n$ is a bounded sequence in $L^1[0,1]$ which is an approximating sequence for σ, then by the subsequence splitting lemma we may assume that $x_n = z_n + t_n$ where

(i) $(z_n)_{n\in\mathbb{N}}$ is uniformly integrable

(ii) $z_n \wedge t_n = 0$ for all n and $t_n \wedge t_m = 0$ for all $n \neq m$

(iii) $\lim_j \|t_j\|$ exists and equals $\lambda \in \mathbb{R}$.

Thus if (e_j) is the unit vector basis of ℓ_1, embedded into L_1 and generating a type τ_1, we have

$$\begin{aligned}\sigma(x) &= \lim_{n\to\infty} \|x + x_n\|\\ &= \lim_{n\to\infty} \|x + z_n + t_n\|\\ &= \lim_{n\to\infty} \|x + z_n\| + \|t_n\|\\ &= \lim_{n\to\infty} \|x + z_n\| + \lambda\\ &= \lim_{n\to\infty} \|x + z_n\| + \|\lambda_n e_n\|\\ &= \lim_{n\to\infty} \|x + z_n + \lambda e_n\|\ .\end{aligned}$$

Therefore $\sigma = \tau * \lambda\tau_1$.

Now, as the set $\mathcal{T}_{ui}(L^1)$ is separable for the uniform topology, if D is the set of the types $\sigma_{n,m} = e_n * r_m\tau_1$ where $\{e_n\}$ is dense in $\mathcal{T}_{ui}(L^1)$ and (r_m) runs over $\mathbb{Q}$ then D is a countable dense subset of $\mathcal{T}(L^1)$. ∎

Remark. Here we have used the "twisted types" to check the separability of $\mathcal{T}_{ui}(L^1)$, and this enables us to show the separability of $\mathcal{T}(L^1)$. On the other hand in [**13**], Y. Raynaud has noticed that if E is a reflexive Banach space and $\mathcal{T}_0(E)$ the set of all types whose approximating sequence is weakly null over an ultrafilter, then the separability of $\mathcal{T}_0(E)$ in the uniform topology suffices to ensure the separability of $\mathcal{T}(E)$. Our proof of Haydon's result may be viewed as an adaptation to the L^1 setting of the observation of [**13**].

4. From stability to uniform stability

We note that the finite dimensional spaces ℓ_p^n for $1 < p < \infty$ are uniformly stable. Also we notice that c_0 is not uniformly stable since it is not even stable [**9**]. Notice that ℓ_1 is not uniformly stable. Indeed, if $(e_n)_n$ is the canonical basis of ℓ_1 and $y = \sum_{i=1}^{\infty} x_i e_i$ is a fixed member of ℓ_1 such that $x_i \geq 0$ for every $i \in \mathbb{N}$, and if we put $z_n = -e_n$ and $y_m = my$ then

$$\|z_n + y_m\|_{\ell_1} - \|y_m\|_{\ell_1} = |mx_n - 1| - mx_n \ .$$

So $\lim_{n\to\infty} \lim_{m\to\infty} |mx_n - 1| - mx_n = -1$ while $\lim_{m\to\infty} \lim_{n\to\infty} |mx_n - 1| - mx_n = 1$, and ℓ_1 is not uniformly stable. Consequently L^1 is not uniformly stable.

It would be interesting to investigate the behavior of a $(1+\varepsilon)$-isomorphic copy of ℓ_1 concerning uniform stability. More precisely does there exists some $\varepsilon > 0$ such that $d(X, \ell_1) < 1 + \varepsilon$ implies X is not uniformly stable? Here

$$d(E,F) = \inf\{\|T\| \ \|T^{-1}\| \ \text{ s.t. } \ T : E \to F \text{ and } T \ \text{ isomorphism onto } \}$$

is the usual "Banach-Mazur distance."

The following theorem clarifies one aspect of the relationship between the "smoothness' of the norm and uniform stability.

Theorem 7. *Let X be a stable Banach space whose norm is uniformly smooth. Then X is uniformly stable.*

Proof. Let us consider a bounded sequence $(x_n)_n$ in X and an arbitrary sequence $(y_m)_m$ in X. If $(y_m)_{m\geq 1}$ is bounded we just use the stability of X. If $(y_m)_{m\geq 1}$ is not bounded then by passing to a subsequence we may suppose that $\|y_m\| \to \infty$. Note that we are assuming here that

$$\lim_{n\to\infty} \lim_{m\to\infty} \|x_n + y_m\| - \|y_m\| = A \ \text{ exists}$$

and

$$\lim_{m\to\infty} \lim_{n\to\infty} \|x_n + y_m\| - \|y_m\| = B \ \text{ exists.}$$

Thus to have $A = B$ it suffices to prove the equality for subsequences of $(x_n)_n$ or $(y_m)_m$. Now

$$\begin{aligned}\|x_n + y_m\| - \|y_m\| &= \|y_m\| \left[\left\| \frac{x_n}{\|y_m\|} + \frac{y_m}{\|y_m\|} \right\| - 1 \right] \\ &= \|y_m\| \left[\left\langle D_{\left(\frac{y_m}{\|y_m\|}\right)} , \frac{x_n}{\|y_m\|} \right\rangle + \frac{\|x_n\|}{\|y_m\|} \varepsilon\left(\frac{x_n}{\|y_m\|}\right) \right] \\ &= \left\langle D_{\left(\frac{y_m}{\|y_m\|}\right)} , x_n \right\rangle + \|x_n\| \varepsilon\left(\frac{x_n}{\|y_m\|}\right) .\end{aligned}$$

Here let us notice that we could have taken a further subsequence of $(y_m)_{m\in\mathbb{N}}$ such that $(D_{(y_m/\|y_m\|)})_{m\in\mathbb{N}}$ converges weakly and a subsequence of $(x_n)_{n\in\mathbb{N}}$ that converges weakly, by the reflexivity of X. Then we have

$$\begin{aligned}\lim_{m\to\infty} \lim_{n\to\infty} \left\langle D_{\left(\frac{y_m}{\|y_m\|}\right)} , x_n \right\rangle &= \lim_{m\to\infty} \left\langle D_{\left(\frac{y_m}{\|y_m\|}\right)} , w\text{-}\lim x_n \right\rangle \\ &= \left\langle w\text{-}\lim D_{\left(\frac{y_m}{\|y_m\|}\right)} , w\text{-}\lim x_n \right\rangle \\ &= \lim_{n\to\infty} \left\langle w\text{-}\lim D_{\left(\frac{y_m}{\|y_m\|}\right)} , x_n \right\rangle \\ &= \lim_{n\to\infty} \lim_{m\to\infty} \left\langle D_{\left(\frac{y_m}{\|y_m\|}\right)} , x_n \right\rangle .\end{aligned}$$

On the other hand we have

$$\lim_{m\to\infty} \lim_{n\to\infty} \|x_n\| \varepsilon\left(\frac{x_n}{\|y_m\|}\right) = 0 .$$

And it is well-known that uniform smoothness of X is equivalent to uniform Frechet differentiability of its norm [6] so

$$\lim_{n\to\infty} \lim_{m\to\infty} \|x_n\| \varepsilon\left(\frac{x_n}{\|y_m\|}\right) = 0$$

which ends the proof of the theorem. ∎

Professor R. Haydon has recently informed us that the norm of a uniformly stable Banach space is Frechet smooth at any point of Gateaux smoothness.

Corollary 8. *Let $1 < p < \infty$. The spaces ℓ_p, L^p, and $L^p(X)$ whenever X is stable and uniformly smooth, are uniformly stable.*

Proof. The case of ℓ_p and L^p for $1 < p < \infty$ is obvious. Now if X is a stable Banach space, then $L^p(X)$ is stable [9]. On the other hand, it is well-known that when $1 < p < \infty$ if X is uniformly smooth then $L^p(X)$ is also uniformly smooth. So the last theorem applies. ∎

For the next corollary recall that if H is a Hilbert space and $1 \le p < \infty$ then the C_p class denotes the Banach space of compact linear operators T on H for which the eigenvalues of $(T^*T)^{1/2}$ are p^{th}-power summable with the pth root of the sum of the pth power of these eigenvalues as a norm.

Corollary 9. *The Banach spaces C_p for $1 < p < \infty$, are uniformly stable.*

Proof. For $1 < p < \infty$ C.A. McCarthy [12] showed that C_p are uniformly convex and since the dual of C_p is $C_{p/(p-1)}$, C_p is also uniformly smooth.

More recently, Y. Raynaud [14] and J. Arazy [1] have shown that C_p are stable. Thus C_p fulfills the assumption of the last theorem. ∎

5. A stability result

In this section we shall prove a stability result concerning the notion of uniform stability. This provides a method of constructing uniformly stable Banach spaces.

Let us first recall that if E is a Banach space with a basis $(e_i)_{i\in\mathbf{N}}$ and $(E_i)_{i\in\mathbf{N}}$ is a sequence of Banach spaces then the E-sum of $(E_i)_{i\in\mathbf{N}}$ is the Banach space

$$X = \left(\sum \oplus E_i\right)_E = \left\{(x^i)_{i\in\mathbf{N}} \in \prod_{i\in\mathbf{N}} E_i \text{ such that } \sum_1^\infty \|x^i\|_{E_i} e_i \text{ converges in } E\right\}$$

where the norm of $x \in X$ is defined by

$$\|x\|_X = \left\| \sum_{i=1}^\infty \|x^i\|_{E_i} e_i \right\|_E .$$

Theorem 10. *Let $(E_i)_{i\in\mathbf{N}}$ be a sequence of uniformly stable Banach spaces. Let E be a stable Banach space with a 1-unconditional basis $(e_i)_i$ and a uniformly Frechet-differentiable norm. Then $X = (\sum \oplus E_i)_E$ is uniformly stable.*

Proof. Let us consider $\|x_n + y_m\| - \|y_m\|$ where $(x_n)_{n\in\mathbf{N}}$ is a bounded sequence in X and $(y_m)_{m\in\mathbf{N}}$ is a sequence in X.

If $(y_m)_{m\in\mathbf{N}}$ is bounded then we can commute the limits over $\|x_n + y_m\| - \|y_m\|$ using a result of J. Bastero [2] which asserts the stability of X.

So we may assume that $(y_m)_{m\in\mathbf{N}}$ is not bounded and by passing to a subsequence we may suppose that $\|y_m\| \to \infty$ when $m \to \infty$. Then if $z_m = y_m/\|y_m\|$ we have

$$\|x_n + y_m\| - \|y_m\| = \|y_m\| \left[\left\| \frac{x_n}{\|y_m\|} + z_m \right\| - \|z_m\| \right] .$$

Thus if z_m^i is the ith coordinate of z_m we have

$$\begin{aligned}\|x_n + y_m\| - \|y_m\| &= \|y_m\| \left[\left\| \sum_{i=1}^{\infty} \left\| \frac{x_n^i}{\|y_m\|} + z_m^i \right\|_{E_i} e_i \right\|_E - \left\| \sum_{i=1}^{\infty} \|z_m^i\|_{E_i} e_i \right\|_E \right] \\ &= \|y_m\| \left[\left\| \sum_{i=1}^{\infty} \left(\left\| \frac{x_n^i}{\|y_m\|} + z_m^i \right\|_{E_i} - \|z_m^i\|_{E_i} + \|z_m^i\|_{E_i} \right) e_i \right\|_E \right. \\ &\qquad \left. - \left\| \sum_{i=1}^{\infty} \|z_m^i\|_{E_i} e_i \right\|_E \right] \\ &= \|y_m\| \left[\left\| \sum_{i=1}^{\infty} \left(\left\| \frac{x_n^i}{\|y_m\|} + z_m^i \right\|_{E_i} - \|z_m^i\|_{E_i} \right) e_i + \sum_{i=1}^{\infty} \|z_m^i\| e_i \right\|_E \right. \\ &\qquad \left. - \left\| \sum_{i=1}^{\infty} \|z_m^i\| e_i \right\|_E \right] .\end{aligned}$$

So if we put

$$\delta_{nm} = \sum_{i=1}^{\infty} \left(\left\| \frac{x_n^i}{\|y_m\|} + z_m^i \right\|_{E_i} - \|z_m^i\|_{E_i} \right) e_i \ , \quad \text{then} \quad \|\delta_{nm}\| \leq \left\| \frac{x_n}{y_m} \right\|$$

and $\|\frac{x_n}{y_m}\| \to 0$ as $m \to \infty$ uniformly with respect to n, then using the differentiability of the norm of E we get an element ξ_m^* of E^* such that:

$$\begin{aligned}\|x_n + y_m\| - \|y_m\| &= \|y_m\| \left[\langle \xi_m^*, \delta_{nm} \rangle_{E^* \times E} + o(\delta_{nm}) \right] \\ &= \langle \xi_m^*, \Delta_{nm} \rangle_{E^* \times E} + \|y_m\| o(\delta_{nm})\end{aligned}$$

where $\Delta_{nm} = \|y_m\| \delta_{nm} = \sum_{i=1}^{\infty} (\|x_n^i + y_m^i\| - \|y_m^i\|) e_i$. ∎

We will need in the sequel the following splitting lemma of B. Maurey (for a proof see [2]).

Lemma 11. *Let E be a Banach space with an unconditional basis $(e_i)_{i \in \mathbb{N}}$. Suppose that E does not contain c_0. Let $(x_n)_{n \in \mathbb{N}}$ be a bounded sequence in E. Then there exists a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ of $(x_n)_{n \in \mathbb{N}}$ such that:*

$$x_{n_k} = y_k + w_k \quad , \quad y_k \wedge w_k = 0$$

$(y_k)_k$ converges in E and $(w_k)_k$ is a block basic sequence of $(e_i)_i$.

We apply this lemma to the sequence $(\tilde{x}_n)_n = (\sum_{i=1}^{\infty} \|x_n^i\| e_i)_n$, and by passing to a subsequence we can write $\tilde{x}_n = \tilde{x}_n^1 + \tilde{x}_n^2$ where $\tilde{x}_n^1$ converges in E, $\tilde{x}_n^2$ is a block basis sequence, and $\tilde{x}_n^1$ and $\tilde{x}_n^2$ are disjointly supported. Let

$$x_n^1 \text{ be the vector of } \left(\sum \oplus E_i \right)_E \text{ defined by } \tilde{x}_n^1$$

$$x_n^2 \text{ be the vector of } \left(\sum \oplus E_i \right)_E \text{ defined by } \tilde{x}_n^2 \ .$$

Claim. We may suppose that x_n^1 is uniformly finitely supported. Indeed $\tilde{x}_n^1$ is convergent to $\tilde{x}_\infty = \sum_{i=1}^\infty \tilde{x}_\infty^i e_i$ in E. So for every $\varepsilon > 0$ there exists $n_1 \in \mathbb{N}$ such that for every $m \geq n_1$

$$\|\tilde{x}_m^1 - \tilde{x}_\infty\| < \frac{\varepsilon}{3}$$

also there exists $\nu \in \mathbb{N}$ such that

$$\left\|\tilde{x}_\infty - \sum_{i=1}^{\nu} x_\infty^i e_i\right\| < \frac{\varepsilon}{3} .$$

On the other hand, since E doesn't contain c_0, as $\tilde{x}_n^1 \to \tilde{x}_\infty$ there exists $n_2 \in \mathbb{N}$ such that for every $m \geq n_2$

$$\left|\|x_\infty^i e_i\| - \|x_m^i\|\right| < \frac{\varepsilon}{3\nu} \quad \text{for} \quad 1 \leq i \leq \nu .$$

Then for every $\varepsilon > 0$ there exists ν and $M = \max(n_1, n_2)$ such that for every $n \geq M$

$$\left\|\tilde{x}_n^1 - \sum_{1}^{\nu} \|x_n^i\| e_i\right\|_E < \varepsilon$$

which proves the claim.

Now

$$\langle \xi_m^*, \Delta_{nm} \rangle = \left\langle \xi_m^*, \sum_{i=1}^{\infty} (\|x_n^i + y_m^i\|_{E_i} - \|y_m^i\|) \ e_i \right\rangle_{E^* \times E}$$

$$= \underbrace{\left\langle \xi_m^*, \sum_{i=1}^{\nu} (\|x_n^i + y_m^i\| - \|y_m^i\|) \ e_i \right\rangle_{E^* \times E}}_{(1)} + \underbrace{\left\langle \xi_m^*, \sum_{i=\nu+1}^{\infty} (\|x_n^{(2)i} + y_m^i\| - \|y_m^i\|) \ e_i \right\rangle_{E^* \times E}}_{(2)} .$$

On the finite sum part (1) we can commute the limits over n and m respectively, as a consequence of the uniform stability of each E_i, for $1 \leq i \leq \nu$.

So

$$\lim_{n\to\infty} \lim_{m\to\infty} \sum_{i=1}^{\nu} (\|x_n^i + y_m^i\|_{E_i} - \|y_m^i\|_{E_i}) e_i = \lim_{m\to\infty} \lim_{n\to\infty} \sum_{i=1}^{\nu} (\|x_n^i + y_m^i\|_{E_i} - \|y_m^i\|_{E_i}) e_i$$

and these limits are norm limits in E. Then notice that

$$\lim_{m\to\infty} \lim_{n\to\infty} \left\langle \xi_M^* \ , \ \sum_{i=1}^{\nu} (\|x_n^i + y_m^i\|_{E_i} - \|y_m^i\|_{E_i}) e_i \right\rangle_{E^* \times E}$$

$$= \left\langle w.\lim_{m\to\infty} \xi_m^* \ , \ \lim_{m\to\infty} \lim_{n\to\infty} \sum_{i=1}^{\nu} (\|x_n^i + y_m^i\|_{E_i} - \|y_m^i\|_{E_i}) e_i \right\rangle_{E^* \times E} .$$

On the remaining part of the sum we observe that

$$\left| \|x_n^{(2)i} + y_m^i\| - \|y_m^i\| \right| \leq \|x_n^{(2)i}\| .$$

But E is reflexive and $\tilde{x}_n^2 = \sum_{i=\nu+1}^{\infty} \|x_n^{(2)i}\| e_i$ is a block basis sequence; so it converges weakly to zero and then

$$\lim_{n\to\infty} \lim_{m\to\infty} \langle |\xi_m^*|, \tilde{x}_n^2 \rangle = \langle w\text{-} \lim_{m\to\infty} |\xi_m^*| \,,\, w\text{-} \lim_{n\to\infty} \tilde{x}_n^2 \rangle$$
$$= 0$$
$$\left(= \lim_{m\to\infty} \lim_{n\to\infty} \langle |\xi_m^*| \,,\, \tilde{x}_n^2 \rangle\right) . \qquad \blacksquare$$

Corollary 12. *For $1 < p < \infty$, $(\sum \oplus \ell_n^1)_{\ell_p}$ is uniformly stable.*

Thus uniform smoothness is not a necessary condition of uniform stability.

Theorem 13. *Let $1 < p < \infty$. If E is a uniformly stable Banach space then $L^p(E)$ is also uniformly stable.*

Proof. Let $(x_n)_{n\in\mathbb{N}}$ be a bounded sequence in $L^p(E)$ and let $(y_m)_{m\in\mathbb{N}}$ be an unbounded sequence in $L^p(E)$. (If $(y_m)_m$ is bounded the result follows just from the stability of $L^p(E)$ [9].) Again, we consider a subsequence of (y_m), also denoted by $(y_m)_n$ such that $\|y_m\| \to \infty$. Then

$$\|x_n + y_m\| - \|y_m\|$$
$$= \|y_m\| \left[\left(\int \left\| \frac{x_n(\omega)}{\|y_m\|} + z_m(\omega) \right\|_E^p d\mathbb{P}(\omega) \right)^{1/p} - \left(\int \|z_m(\omega)\|_E^p \, d\mathbb{P}(\omega) \right)^{1/p} \right]$$

where $z_m = \frac{y_m}{\|y_m\|}$ as before.

$$= \|y_m\| \left[\left(\int \left(\left\| \frac{x_n(\omega)}{\|y_m\|} + z_m(\omega) \right\|_E - \|z_m(\omega)\|_E + \|z_m(\omega)\|_E \right)^p d\mathbb{P}(\omega) \right) \right.$$
$$\left. - \left(\int \|z_m(\omega)\|_E^p \, d\mathbb{P}(\omega) \right)^{1/p} \right] .$$

Then putting $\delta_{nm}(\cdot) = \| \frac{x_n(\cdot)}{\|y_m\|} + z_m(\cdot)\|_E - \|z_m(\cdot)\|_E$ and $\xi_m(\cdot) = \|z_m(\cdot)\|_E$ and $\xi_m^*(\cdot) \in L^{p^*}$, the support function of ξ_m, we get that the above is

$$= \|y_m\| \left[\langle \xi_m^*, \delta_{nm} \rangle_{L^{p/p-1} \times L^p} + o(\delta_{nm}) \right]$$
$$= \langle \xi_m^*, \Delta_{nm} \rangle_{L^{p/p-1} \times L^p} + \|y_m\| o(\delta_{nm})$$

where $\Delta_{nm}(\cdot) = \|y_m\| \delta_{nm}(\cdot) = \|x_n(\cdot) + y_m(\cdot)\|_E - \|y_m(\cdot)\|_E$.

Let $\tilde{x}_n = \|x_n(\cdot)\|_E$. $(\tilde{x}_n)$ is a bounded sequence of L^p, so $\tilde{x}_n^p$ is bounded in L^1. By passing to a subsequence and applying the splitting lemma we can write $\tilde{x}_n = \tilde{x}_n^1 + \tilde{x}_n^2$ where $\tilde{x}_n^1$ is of pth-power uniformly integrable, $(\tilde{x}_n^2)_n$ are mutually disjointly supported and $\tilde{x}_n^1 \wedge \tilde{x}_n^2 = 0$. To this splitting are then associated the decompositions

$$x_n = x_n^1 + x_n^2 \quad \text{and} \quad \Delta_{nm} = \Delta_{nm}^1 + \Delta_{nm}^2 .$$

Thus

$$\langle \xi_m^*, \Delta_{nm} \rangle_{L^{p^*} \times L^p} = \langle \xi_m^*, \Delta_{nm}^1 \rangle_{L^{p^*} \times L^p} + \langle \xi_m^*, \Delta_{nm}^2 \rangle_{L^{p^*} \times L^p} .$$

But since (x_n^2) are mutually disjointly supported we have:

$$\lim_{m \to \infty} \lim_{n \to \infty} \langle \xi_m^*, \Delta_{nm}^2 \rangle_{L^{p^*} \times L^p} = 0$$

and by the reflexivity of L^p we must also have:

$$\lim_{n \to \infty} \lim_{m \to \infty} \langle \xi_m^*, \Delta_{nm}^2 \rangle_{L^{p^*} \times L^p} = 0 .$$

Similarly by passing to a subsequence of ξ_m^* we can also split $\xi_m^* = \xi_m^{*1} + \xi_m^{*2}$ and have, for the same reason as before:

$$\lim_{m \to \infty} \lim_{n \to \infty} \langle \xi_m^{*2}, \Delta_{nm}^1 \rangle_{L^{p^*} \times L^p} = \lim_{n \to \infty} \lim_{m \to \infty} \langle \xi_m^{*2}, \Delta_{nm}^1 \rangle_{L^{p^*} \times L^p} = 0$$

where ξ_m^{*1} is the part of ξ_m^* which is of the p^{*th} power uniformly integrable and the ξ_m^{*2}'s are mutually disjointly supported.

Now we can suppose that $\|\xi_m^{*1}\|_\infty \leq K$ and $\|\Delta_{nm}^1\|_\infty \leq \|\tilde{x}_n^1\|_\infty \leq K$ where K is a constant. Indeed if $A_m^R = \{\omega \in \Omega : |\xi_m^{*1}(\omega)| > R|u|\}$

$$\lim_{R \to \infty} \sup_m \left| \int_{A_m^R} \xi_m^{*1}(\omega) \Delta_{nm}^1(\omega) \, d\mathbb{P}(\omega) \right|$$

$$\leq \lim_{R \to \infty} \sup_m \left(\int_{A_m^R} |\xi_m^{*1}(\omega)|^{p^*} \, d\mathbb{P} \right)^{1/p^*} \left(\int_\Omega |\tilde{x}_n^1|^p \, d\mathbb{P} \right)^{1/p}$$

which equals 0 by uniform integrability of $|\xi_m^{*1}|^{p^*}$ and boundedness of x_n. The same argument works for $\|\tilde{x}_n^1\|_\infty \leq K$ using the uniform integrability of $|\tilde{x}_n^1|^p$ and the boundedness of ξ_n^*.

On the other hand using the representation of [15] for the stable function $f(x,y) = \|x+y\|_E - \|y\|_E$ through a reflexive Banach space F and its dual F^* we get:

For every $\omega \in \Omega$ $\quad \Delta_{nm}^1(\omega) = \langle U_{x_n^1(\omega)}, V_{y_m(\omega)} \rangle_{F \times F^*}$

where $U : K.\,\text{Ball}(E) \longrightarrow \text{Ball } F$

and $V : E \longrightarrow C.\,\text{Ball}(F^*)$

are bounded maps and C is a constant.

Then

$$\langle \xi_m^{*1}, \Delta_{nm}^1 \rangle_{L^{p^*} \times L^p} = \langle U_{x_n^1}, \xi_n^{*1} . V_{y_m} \rangle_{L^p(F) \times L^{p^*}(F^*)}$$

with $\quad \|U_{x_n^1}\|_{L^p(F)} \leq \text{ cste}$

and $\quad \|\xi_m^{*1} V_{y_m}\|_{L^{p^*}(F^*)} \leq \text{ cste}$

Therefore the reflexivity of $L^p(F)$ gives the desired commutation of limits for $\langle \xi_m^{*1}, \Delta_{nm}^1 \rangle$. ∎

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Addresses:

Professor A. Dold
Mathematisches Institut, Universität Heidelberg
Im Neuenheimer Feld 288
W-6900 Heidelberg
Federal Republic of Germany

Professor B. Eckmann
Mathematik, ETH-Zentrum
CH-8092 Zürich, Switzerland

Professor F. Takens
Mathematisch Instituut
Rijksuniversiteit Groningen
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